

AN ELEMENTARY TREATISE ON MICROECONOMICS



*Being a First Course in Demand, Supply, Cost and the
Behaviour of Markets*

S. SHIR, M. ROUD

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S . S H I R , M . R O U D

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First edition

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*To the one whom Wisdom deliberately chased, but
who was always the faster one in the chase. RU ·
MGT-30 · Shoheliya Khanom Rimi*

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TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches An Elementary Treatise on Microeconomics one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.

PREFACE

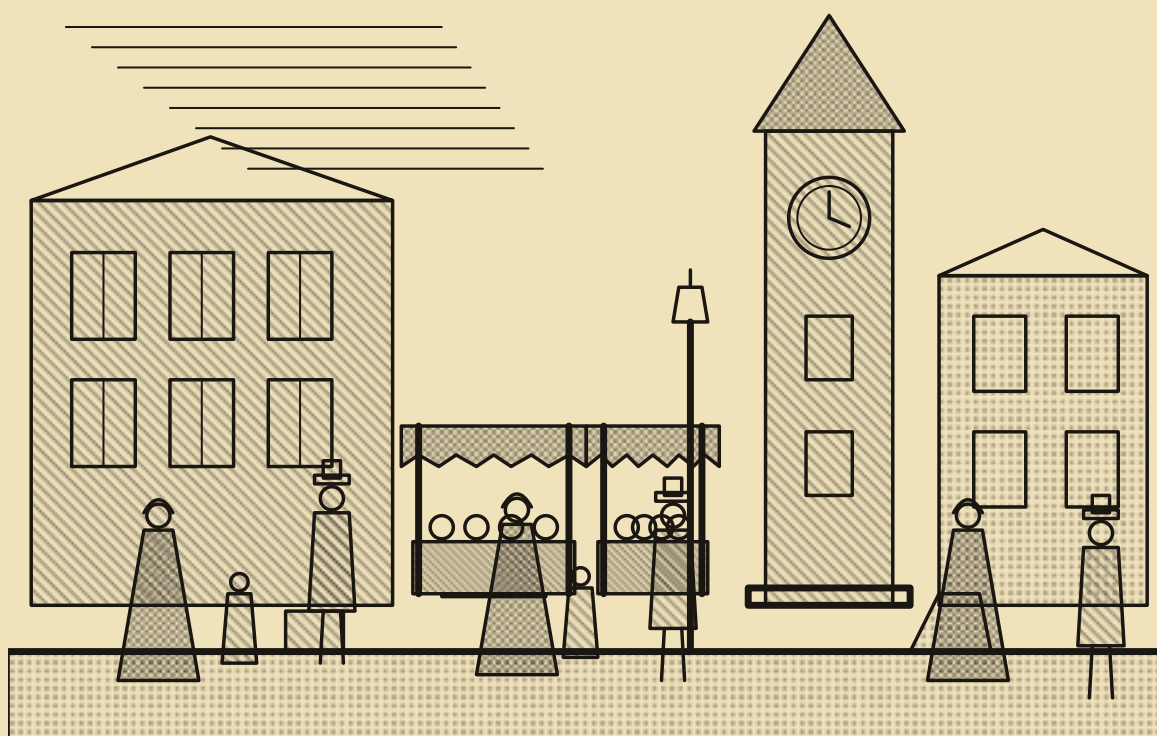
This treatise was written for the reader who has never studied economics and wishes to begin properly. We have tried to say each thing once, plainly, and in its right place.

The book moves from the nature of economic choice, through the forces of demand and supply and the conduct of the consumer, to the costs of the producer and the behaviour of firms in the market. A reader who takes the chapters in order will find that each rests upon the one before.

Where a word has a special meaning, it is explained at its first use and gathered into the glossary at the end. The questions at the close of each chapter are answered at the back of the book; we advise the reader to attempt them first.

If this book spares a single student an hour of needless confusion, its labour is repaid.

S. Shir, M. Roud



A market square in the 1880s, with stalls, shoppers and a clock tower

CHAPTER I

What Economics Is

Why can no one have everything they want?



This chapter introduces economics as the study of choice when wants are many and means are few. It separates the study of single shops from the study of a whole country. It teaches you to tell a statement of fact from a statement of opinion, and to see why every economic law carries a condition. It also sets the habits this book uses in every chapter.

This chapter needs no earlier knowledge.

A market square shows many people facing one problem: choice.

The seller also faces limits, so both sides of the stall must choose.

The two choices look different but share one shape.

This puzzle is the subject of the book; the chapter plan follows.

You will need: nothing. The chapter starts from things you already do: spending, saving, buying and selling. Each subject word is explained at the moment it first appears, in plain English, with a small example.

Picture a market square on a Saturday morning in a town of the 1880s. Stalls sell rice, cloth, fruit, pens and notebooks. Hundreds of people walk between them, and each carries a short list of things to buy and a limited amount of money. One father wants shoes for his daughter and a new lamp for the house. He can pay for only one of the two today. Which does he choose, and how does he decide?

Now consider the other side of the stalls. A cloth seller has one table, ten hours of daylight and a fixed amount of money to spend on stock. She can buy more cotton or more silk for tomorrow, but she cannot buy as much of both as she would like. Whatever she chooses, she gives up something else. She must also decide the price at which to sell. A price that is too high leaves her cloth unsold, and a price that is too low leaves her with too little money.

The father's choice is about a lamp, and the seller's choice is about silk. They look unrelated, yet they share one shape. Wants are many, the means to satisfy them are few, and so someone must choose. A harder question follows. When thousands of such choices are made each day, why does the square not fall into disorder? Why do shoes, lamps and cloth appear at the stalls in roughly the quantities that people wish to buy, at prices they can pay?

No single person in the square plans all of this, and that fact makes the puzzle worth studying. This book builds its answer step by step, and it begins with the plainest question of all: what does this subject study? Section 1.1 gives the meaning of the subject. Section 1.2 divides it into two branches. Section 1.3 separates statements of fact from statements of opinion. Section 1.4 explains the kind of law that economists can state, and the condition that every such law carries.

* * *

1.1 THE MEANING AND SCOPE OF ECONOMICS

GO SLOWLY HERE.

Begin with your own week: wants that cost more than your money.

Three plain words carry the idea: want, good and resource.

Economics is the study of choice under limited resources.

Economics. The study of how people, firms and governments choose to use limited resources to satisfy wants.

Begin with something you know well. Imagine a student who receives Tk 100 for the week. She wants a notebook at Tk 60, a meal in a café at Tk 60, and a bus ticket at Tk 30. Together these wants cost Tk 150, which is more than Tk 100. She cannot have all three, so she must choose, and each choice means giving up something else. This small case holds the whole subject in one picture.

Three plain words are needed. A want is a wish to have something, such as a notebook. A good is anything that satisfies a want. Some books also say commodity or product; this book always says good, and uses service only for help given by a person, such as a lesson. A resource is anything used to make goods: land, people's work, tools and money. Keep these three words in mind, because the next definition uses all of them.

The student's problem has a name in the subject. **Economics** means the study of how people, firms and governments choose to use limited resources to satisfy wants. A firm is any organisation that makes goods to sell, whether a small bookshop or a large factory; some books say business or enterprise, but this book always says firm. A government is the body that makes rules and raises money for a country. Our student is a person, the cloth seller is a firm, and a government that must choose

between building a school and building a road faces the same problem on a larger scale.

IN SHORT

Economics is about choice. Wants are many, resources are limited, and so every person, firm and government must choose.

Figure 1 shows the idea in a picture. Read it from left to right. The tall bar on the left stands for wants, and the shorter bar on the right stands for the means to satisfy them. The dashed line marks the top of the means. The distance between the dashed line and the top of the wants is the gap that cannot be filled, and the word *choice* sits between the bars because the gap forces it.

Read the first figure as two bars with a gap between them.

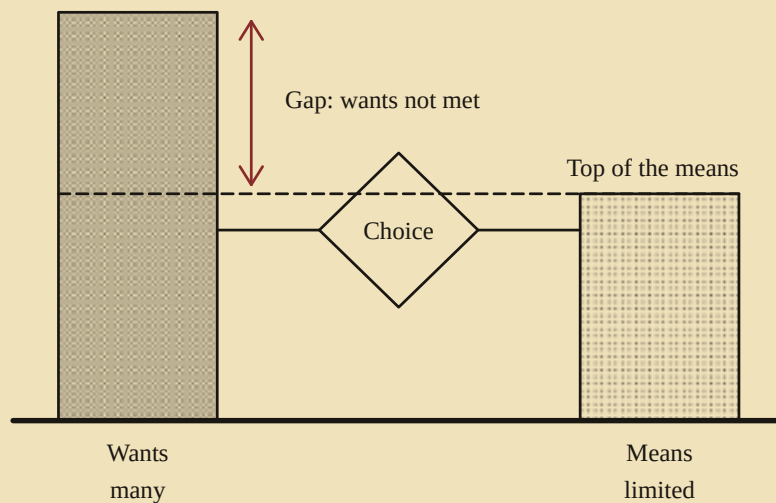


FIG. 1.1. *Economics as choice among wants and limited means*

The *scope* of a subject is the range of things it studies. Economists usually group the activities they study into four. Consumption is the use of goods to satisfy wants, as when a family eats bread. Production is the making of goods, as when a printer makes books. Exchange is the buying and selling by which goods pass from one person to another. Distribution is the sharing of the income from production among those who helped to make the goods, such as workers and owners. The four are linked: nothing is consumed until it is produced and exchanged, and the income shared in distribution pays for the next round of consumption.

Economics covers four connected activities.

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhanmondi, Dhaka, founded on 4 April 1996 by Nurul Meghna. In 2025 its sales were Tk 14,400,000, its profit was Tk 750,000, and it had 7 staff. The shop buys books from printers, which is exchange. Readers take the books home to read them, which is consumption. The shop's income pays its staff and its owner, which is distribution. Printers who make the books are engaged in production.

The running case shows all four activities in one small shop.

Is economics a science? A science is organised knowledge, built by gathering facts and testing explanations against them. Economics does this: it collects facts about prices, output and choices, and it offers explanations of cause and effect. An art, in the older meaning of the word, is a practical skill used to reach an aim. When an adviser tells a government how to lower the price of food, the adviser uses economics as an art. Most writers call economics a social science, because

Economics is a science that explains and an art that advises.

Recap of the section in a few lines.

GO SLOWLY HERE.

A street map and a country map show different views of one land.

Microeconomics is the close view of single consumers and firms.

Microeconomics. The branch of economics that studies choices made by individual consumers and firms.

Market. Word in use. In daily life, a place with stalls where people buy and sell. In economics, any arrangement in which buyers and sellers deal with each other, in one place or by telephone or screen.

A microeconomist asks questions about one firm, such as this shop.

Macroeconomics is the wide view of the whole economy.

Macroeconomics. The branch of economics that studies the whole economy, including output, prices and employment.

Economy. Word in use. In daily life, a saving, as in an economy of effort. In economics, the whole system by which the people of a country make, share and use goods.

The same shop is a very small part of the national picture.

Read the second figure from left to right: one shop, then the whole country.

it studies people, who differ from one another, change their minds and cannot be placed in a laboratory. One exception to the comparison with physics follows from this: economic results are less exact.

Before moving on. Economics is the study of choice. It exists because wants are many and resources are limited. It covers consumption, production, exchange and distribution, and it is both a science, which explains, and an art, which advises. You should now be able to say, in one sentence, why a student with Tk 100 and wants worth Tk 150 faces an economic problem.

1.2 MICRO AND MACRO

Start with maps. A street map of Dhanmondi shows each lane, shop and house, and it helps you find one bookshop. A map of the whole country does not show the lanes, but it shows the roads that join the cities, and it helps you plan a long journey. Neither map is wrong, and neither replaces the other. They are two ways of looking at the same land. Economics uses the same two views, and each has its own name.

The close view is called **microeconomics**. Microeconomics means the study of the choices made by individual consumers and individual firms. A consumer is a person who buys a good to use it, not to sell it again. The word begins with the Greek word for small.¹ In microeconomics we ask how one consumer decides what to buy, how one firm decides how much to make and at what price, and how the choices of many consumers and firms meet in a market.

Case file, illustrative. Meghna and Sons is a micro unit. In 2025 its sales were Tk 14,400,000, its profit was Tk 750,000, and it had 7 staff. A microeconomist who studies this shop asks how its owner chose the price of its books, how many staff to employ, and how many copies to keep in stock. Each of these questions concerns one firm.

The wide view is called **macroeconomics**. Macroeconomics means the study of the whole economy, including output, prices and employment. Output is the total amount of goods made in a country in a period of time. The price level is the general level of prices across the country. Employment is the number of people who have paid work. A macroeconomist does not ask about one shop. The questions are of this kind: why did the price level of the whole country rise, or why did total employment fall?

Case file, illustrative. Seen through the macro view, the same shop almost disappears. The macro view follows the price level and the total employment of the whole country, and it does not follow the shop's own books. The shop's 7 staff are counted inside the national total of jobs, together with the staff of every other firm.

Figure 2 places the two views side by side. On the left, the picture is close: one shop, with its sales, profit and staff. On the right, the picture is wide: many shops and houses make up a country, and our shop is only one of them, marked

¹ *Mikros* is Greek for small and *makros* is Greek for large. Many scientific words are built from Greek parts in this way.

by hatching. Under each picture, the label names what that view measures. Read the figure as a change of distance, not a change of subject.

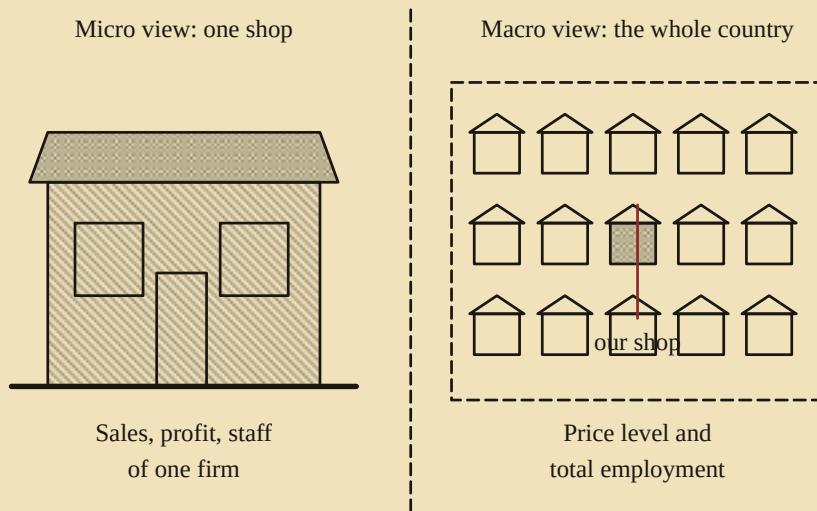


FIG. 1.2. *Micro and macro viewpoints on one shop*

The table below compares the two branches. Read it across. Each row names one feature, then gives the micro answer in the middle column and the macro answer in the right column.

MICRO AND MACRO COMPARED

FEATURE	MICROECONOMICS	MACROECONOMICS
Unit studied	One consumer, one firm or one market	The whole economy
Typical question	How does one firm choose its price?	Why did the price level of the country change?
Case file example	Sales of Tk 14,400,000 and 7 staff at one shop	The price level and total employment of the country
Main concern	Single prices and quantities	Totals and averages

The branches differ in the size of the unit, yet they are linked. Total employment in the country is the sum of the jobs in every firm, so macro totals come from micro choices. The link runs the other way as well. When the price level of the country rises, a shop pays more for its supplies, and that changes the shop's choices. One exception needs care. Size alone does not decide the branch: a very large firm is still one firm, so a question about it belongs to microeconomics.

What is the scope of microeconomics? It covers how a consumer chooses among goods, how a firm chooses what to make and how to make it, how prices form where buyers and sellers meet, and how the people who supply work, land and money are paid. This book teaches microeconomics. Macroeconomics appears here only so that you can see where the boundary lies.

Before moving on. Microeconomics studies single consumers and firms. Macroeconomics studies the whole economy. The branches differ in the unit they study, and each depends on the other. You should now be able to say how a question about one shop differs from a question about the price level of a whole country.

A table sets the two branches side by side, row by row.

The branches differ in size but are linked in both directions.

This book teaches microeconomics, so its scope is the choices of single units.

Recap of the section in a few lines.

1.3 POSITIVE AND NORMATIVE STATEMENTS

Two friends argue about textbooks, and only one dispute can be settled by checking.

Positive economics deals with statements that facts can test.

Positive economics. Statements about what is, which can be tested against facts.

The shop's own record gives a clean example of a positive statement.

Normative economics deals with what ought to be, which depends on values.

Normative economics. Statements about what ought to be, which depend on values and cannot be tested by facts alone.

One question separates the two kinds of statement.

A table sets the two kinds of statement side by side.

Two friends sit in the shop and discuss books. The first says, "The shop sold 41,000 copies in 2025." The second says, "Textbooks should be cheaper." Both sound like ordinary remarks, but they are different in kind. Could a person settle either one by looking at records? Think about this before you read on.

The first remark is an example of **positive economics**. Positive economics means statements about what is, which can be tested against facts. To test the remark, a person counts the copies recorded in the shop's sales books. The records will show the statement to be true or false. A positive statement may be false, because positive means testable, not correct. If the records show 39,000 copies, the statement fails the test but is still positive.

Case file, illustrative. Statement: "The shop sold 41,000 copies in 2025." This is positive. The shop's sales books can confirm it or reject it, and no opinion is needed to decide.

The second remark is an example of **normative economics**. Normative economics means statements about what ought to be, which depend on values and cannot be tested by facts alone. A value is a belief about what is good or fair. The remark "Textbooks should be cheaper" gives no fact to check. A student who wants low prices and a bookseller who wants a fair return can agree on every fact and still disagree on this point. Words such as should, ought, fair and better often signal a normative statement.

To tell them apart, ask one question: could records or measurement show this statement to be true or false? If the answer is yes, the statement is positive. If the answer depends on what we think is right, the statement is normative. Many statements mix the two. "Textbook prices are too high because paper costs more" has a positive part, the claim about cost, and a normative part, the words too high. Separate the parts and test each one on its own.

The table below compares the two kinds of statement. Read each row across, from the feature on the left to the answers for positive and normative statements.

POSITIVE AND NORMATIVE STATEMENTS COMPARED

FEATURE	POSITIVE	NORMATIVE
Speaks about	What is	What ought to be
Can facts test it?	Yes	Not by facts alone
Case file example	The shop sold 41,000 copies in 2025	Textbooks should be cheaper
Signal words	Is, was, rose, fell	Should, ought, fair, better

PAUSE AND CHECK 1.P1 Classify each statement as positive or normative, and give a reason: (a) The shop employs 7 staff. (b) The shop should employ an eighth person.

Recap of the section in a few lines.

Before moving on. A positive statement says what is and can be tested against facts. A normative statement says what ought to be and depends on values. Positive does not mean true. You should now be able to say which kind of statement a given sentence is, and what test shows it.

1.4 ECONOMIC LAWS

Water boils at 100 degrees Celsius at sea level, and a stone released from the hand falls to the ground. These statements hold in every place, without exception. Now consider this one: when the price of a good rises, consumers usually buy less of it. Is that equally certain? Many students find this step confusing, because both statements are called laws. The difference is the subject of this section.

An **economic law** means a statement of how people usually behave, true when other conditions stay the same. Take the statement about price. When the price of a notebook rises, most consumers buy fewer notebooks. The word usually is part of the law. Some consumers may still buy the same number, but the usual pattern is clear enough to guide a seller or a government.

The condition has a name. **Ceteris paribus** is a Latin phrase meaning all other things stay the same.² Economists use it to study one change at a time. To test how the price of notebooks affects purchases, we imagine that incomes, tastes and the prices of other goods stay fixed. Otherwise we could not tell which change caused the result. The phrase is a method of thinking. It does not claim that other things never change; in real life they change all the time.

IN SHORT

An economic law is a usual pattern, and it holds ceteris paribus: all other things staying the same.

How do economic laws differ from natural laws? A law of nature, such as the fall of a stone, describes things that do not choose, so it can be exact and the same everywhere. An economic law describes people, who have different tastes and change their minds. For this reason, an economic law states a tendency, not a certainty. People also cannot be placed in a laboratory and held still, so economists must use the words ceteris paribus to separate one cause from many.

Because resources are limited, every economy faces the same three basic problems. It must decide what to produce, which means which goods, and how many of each. It must decide how to produce, which means with which resources and methods. It must decide for whom to produce, which means how the goods will be shared among people. The market square in our opening puzzle answers these questions every day without any one person planning the answers.

Before moving on. An economic law states how people usually behave, and it holds ceteris paribus. It is a tendency, not a certainty, because people choose. Every economy must decide what, how and for whom to produce. You should now be able to say why the words ceteris paribus must accompany an economic law.

WORKED CASE: SORTING QUESTIONS ABOUT ONE SHOP

The worked case below applies the two sorting tests of this chapter. The first test asks whether a statement is positive or normative. The second asks whether a question _____ belongs _____ to _____ microeconomics

² Latin was the language of ancient Rome. Scholars in Europe used it for many centuries, so many technical phrases still come from it.

GO SLOWLY HERE.

A law of nature is exact, so what can an economic law claim?

An economic law states a usual pattern of behaviour under a condition.

Economic law. A statement of how people usually behave, true when other conditions stay the same.

Ceteris paribus names the condition that every economic law carries.

Ceteris paribus. A Latin phrase meaning all other things stay the same, used to study one change at a time.

Economic laws are less exact than natural laws because people choose.

Every economy faces three basic problems.

Recap of the section in a few lines.

This case applies the chapter's two sorting tests to five items from the shop.

or macroeconomics. It is an exercise in reasoning, and the steps are shown in full.

The situation lists five items drawn from the shop's case file.

Case file, illustrative. Situation: five items are taken from the file of Meghna and Sons. (a) "The shop sold 41,000 copies in 2025." (b) "Textbooks should be cheaper." (c) "What decides the total number of jobs in the whole country?" (d) "How did the shop's owner choose the number of staff?" (e) "The shop's profit in 2025 was Tk 750,000."

The question asks for two sorting decisions per item.

Question: for each statement, decide whether it is positive or normative. For each item, decide whether its subject is micro or macro. A question that asks for information is not a statement, so for items (c) and (d) only the second decision applies.

Items (a) and (e) can be checked against records, so they are positive.

Reasoning, part one. Item (a) can be checked against the shop's sales books, so it is positive. Its subject is one shop, so it is micro. Item (e) is also a statement of fact: the shop's accounts show its profit, and the figure is Tk 750,000. It is positive, and it is micro, because it describes one firm.

Item (b) states a value, and item (c) asks about a total.

Reasoning, part two. Item (b) says what ought to be, and no record can settle it, so it is normative. It concerns the price of one kind of good, so it is a micro matter. Item (c) asks about the total number of jobs in the whole country, so its subject is macro. It is a question, and so it is neither positive nor normative.

Item (d) is about one firm, so it is a micro question.

Reasoning, part three. Item (d) asks how one owner chose the number of staff. The unit is one firm, so the question is micro. The answer would be a matter of fact, such as the shop's sales and wages, though the item itself is a question.

The conclusions are collected in a table.

Conclusion: the table below gives the result. Read each row across, from the item to its two sorting decisions.

SORTING THE FIVE ITEMS

ITEM	POSITIVE OR NORMATIVE	MICRO OR MACRO
(a) 41,000 copies sold	Positive	Micro
(b) Textbooks should be cheaper	Normative	Micro
(c) Total jobs in the country	Not a statement	Macro
(d) The owner's choice of staff	Not a statement	Micro
(e) Profit of Tk 750,000	Positive	Micro

A check confirms each decision with the test from the chapter.

Check: for each statement, ask whether records could show it to be true or false. Items (a) and (e) pass, and item (b) fails. For each subject, ask whether the unit is one firm or a total. Items (a), (b), (d) and (e) concern single units, and item (c) concerns a total. The decisions agree.

* * *

SUMMARY

The chapter's main ideas, in a few lines.

Economics is the study of how people, firms and governments choose to use limited resources to satisfy wants. It has two branches: microeconomics studies single consumers and firms, and macroeconomics studies the whole economy. Positive statements say what is and can be tested against facts, and normative statements say what ought to be and depend on values. An economic law states how people usually behave, *ceteris paribus*. Every economy must decide what, how and for whom to produce.

THINK IT THROUGH

1.T1 In a town, the price of notebooks rose in March, and notebook sales fell. In the same month, the school gave a free notebook to every student. A newspaper says the price rise alone caused the fall. Using the words *ceteris paribus*, explain why this claim is weak.

1.T2 A friend says, "Economic laws are useless, because they are not exact." Using Section 1.4, write a reply of three sentences.

WHERE PEOPLE GO WRONG

Many people believe that economics is the study of money. Money appears in many economic questions, but it is a tool, not the subject. The subject is choice under limited resources, and choices are made even where no money changes hands, as when a student chooses how to spend a free afternoon.

Some students think that micro means small things and macro means large things. This is a common confusion. The words refer to the unit studied. A large firm with thousands of staff is still one firm, so questions about it are micro. A question about the total number of jobs in the country is macro, however the jobs are spread.

Another confusion is that positive statements are true and normative statements are worthless. Positive statements can be false, as the case of 39,000 copies showed. Normative statements are not worthless: choices about what ought to be are the reason many people study economics. Facts alone cannot settle them by facts alone.

The last confusion concerns *ceteris paribus*. Some students read it as a claim that nothing else ever changes. In fact, the phrase tells us to imagine other things unchanged, so that one cause can be studied on its own. Without it, an economist cannot tell whether a fall in sales came from a higher price or from some other change.

Many people think economics is the study of money, but it studies choice.

Micro does not mean small things, and macro does not mean large things.

A positive statement is not always true, and a normative statement is not worthless.

Ceteris paribus is a thinking device, not a claim about the world.

IN HISTORY

Alfred Marshall (1842 to 1924) was an English economist who taught at the University of Cambridge. He began to write his book *Principles of Economics* in 1881 and published it in 1890. For many years it was the leading economics textbook in England. In it, Marshall described economics as the study of people in "the ordinary business of life".

The phrase matters for two reasons. First, it makes people, not money, the centre of the subject. Second, it treats economics as part of ordinary life. Buying a notebook, hiring an assistant and choosing a price are all examples of that business. The choices of Meghna and Sons are of exactly this kind, as are the choices of the father and the cloth seller in our opening puzzle.

Marshall's wording is still quoted in textbooks today. This chapter's definition adds one idea that his phrase does not state directly: the focus on limited resources, which explains why choice is needed at all.

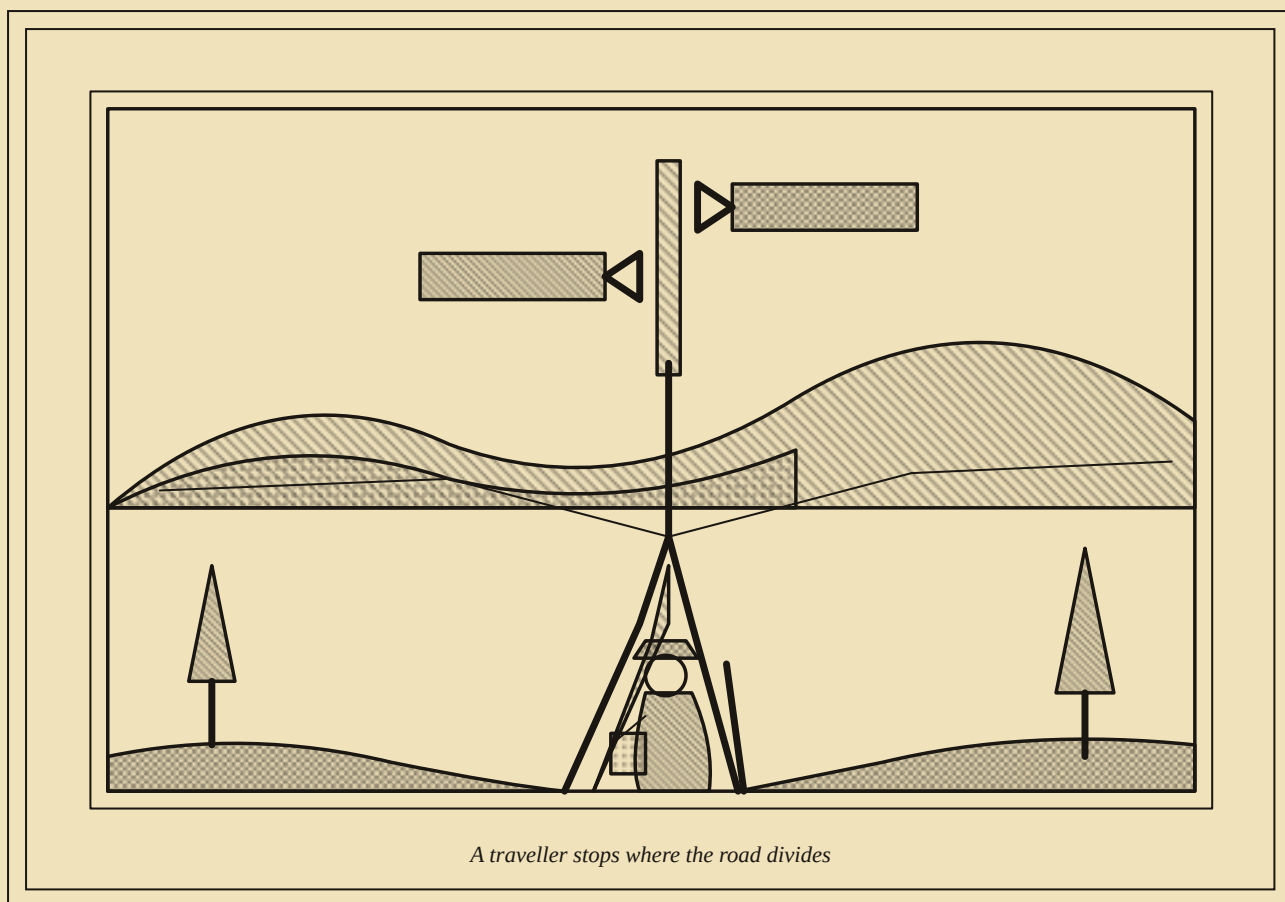
Alfred Marshall's 1890 book gave economics a definition that centres on people.

The definition moves the subject from money towards everyday human conduct.

The definition still serves as a starting point, with later additions.

REVIEW QUESTIONS

- 1.1 Classify each statement as positive or normative, and give a reason: (a) Meghna and Sons had 7 staff in 2025. (b) Bookshops ought to pay their staff more. (c) The shop's profit in 2025 was Tk 750,000. (d) Fewer people should buy books online.
- 1.2 Separate each question into micro or macro: (a) How should one bookshop choose its staff? (b) Why did prices across the country rise? (c) How does a consumer choose between two goods? (d) What decides total employment in the country?
- 1.3 In your own words, explain why a student with Tk 100 and wants worth Tk 150 faces an economic problem.
- 1.4 Name the four activities in the scope of economics and give one example of each from a bookshop.
- 1.5 What does *ceteris paribus* mean, and what is one difference between an economic law and a law of nature?



A traveller stops where the road divides

CHAPTER II

Scarcity, Choice and Opportunity Cost

Every road taken leaves another road behind.



This chapter shows why every choice gives something up, how to measure what is given up, and how a single curve can picture the choices open to a person, a firm or a country.

YOU WILL NEED

Chapter 1, 'What Economics Is', section 1.1 'The Meaning and Scope of Economics', for the meaning of economics; Chapter 1, section 1.4 'Economic Laws', for laws and the same-conditions rule; and Appendix A1, 'Maths You Will Use', for ratios and simple graphs.

A traveller must choose one road, and the other road is lost.

The traveller's trouble is not a lack of roads but a lack of time.

The same pattern appears in every shop, household and country.

A traveller walks along a country road at the end of an afternoon. Ahead, the road divides in two. A signpost stands on each branch. One road leads to a market town, where the traveller could sell goods and buy supplies. The other leads to a river village, where an old friend waits and a good meal is promised. Night is coming, and there is strength for only one more journey today.

The traveller does not lack roads, and does not lack good reasons. Both roads are worth taking. What the traveller lacks is time, because the evening allows only one walk. Whichever road is chosen, something valuable is left behind. No clever planning removes that loss. It can only be understood, measured and weighed against the gain.

The same pattern appears everywhere. A family with Tk 20,000 to spend cannot buy everything on its list. A shopkeeper with one free afternoon cannot both count stock and visit a supplier. A government with one budget cannot build every road and school it would like. This chapter gives the pattern a name and a measure. Section 2.1 explains why choice cannot be avoided. Section 2.2 shows how to measure what choice gives up. Section 2.3 draws the idea as a curve.

* * *

2.1 WANTS AND SCARCITY

WANTS

GO SLOWLY HERE.

Everyday life shows how one wish leads to the next.

Want. Something a person would like to have, however little or much the person can pay for it.

A want is anything a person would like to have.

Wants differ among people, but wishing never ends.

Individual wants can be satisfied for a moment, but not for good.

Begin with your own week. This morning you wished for a good breakfast, and you had it. By noon you wished for a new pair of shoes. By evening you wished for a holiday by the sea. Each wish was met or set aside, and each was soon replaced by another. You never reached a day with no wish left. This simple observation is the first step in economics.

Economists give such wishes a name. A **want** means something a person would like to have, however little or much the person can pay for it. A bicycle is a want. So are a meal, a university degree, an afternoon of rest and a quiet street. The definition does not ask whether the want is wise or sensible. A want need not be a physical good; it may be a service, such as a haircut, or a condition, such as free time.

Wants differ from person to person and from place to place. A farmer in Mymensingh, a banker in London and a student in Dhaka will list different wants. Wants also change with age and with income. Yet one feature is shared. When one want is satisfied, another appears, often a larger one. A person who owns a bicycle wishes for a motorcycle; a person with a motorcycle wishes for a car. Economists therefore treat wants, taken together, as without a limit.

There is an exception that confuses many students. A single want can be satisfied. After a full meal, you do not wish for another. This does not contradict the point above. The want for that meal has ended for now, yet it returns within

hours, and other wants stand ready to take its place. What has no end is the whole list.

Before moving on. A want is something a person would like to have, whatever the price. One want may be met for a time, but the list as a whole keeps growing. This is half of the picture. The other half follows.

SCARCITY AND RESOURCES

Think of a market stall. The owner has a small table, a few hours and a purse of money. The customers wish for many goods, but the stall can offer only what the table can hold, what the hours allow and what the purse can buy. The owner's wishes are larger than the stall. The same is true for every household, firm and country, for the same reason: the things used to make goods and services are limited.

Economists call these things resources. Resources are the inputs used to make goods and services: land and raw materials, the work of people, tools and machines, money, and time. A bakery's resources include its oven, its flour, its bakers and its hours. A country's resources include its farmland, its rivers, its factories and its workers. Not one of them is unlimited at a given time.

When unlimited wants meet limited resources, the result is **scarcity**. Scarcity means the condition that resources are limited while wants are unlimited, so choices must be made. A family cannot buy everything on its list, and a country cannot build everything its citizens wish for. Scarcity is the reason economics exists. If every want could be met at once, nothing would need to be chosen and nothing would need to be studied.

Scarcity has two common misreadings. The first is that scarcity means rarity. Air is common, but a pure breath of it in a crowded city may be scarce, and the same is true of clean water. The second is that scarcity belongs only to poor people. A billionaire also has limited time and must choose. Scarcity affects rich and poor, and it applies to every society.¹

Scarcity forces choice. Faced with limited resources, a person, a firm or a government must decide which wants to meet and which to leave. A choice may be made carefully or carelessly, openly or without thought. Doing nothing is also a choice, because the hours and the money go somewhere. The question is never whether to choose, but how well.

Each decision maker meets scarcity from a different side. A household has a fixed income and must divide it among food, rent, school fees and savings. A firm has a limited budget, a limited number of workers and a limited number of working hours, and must decide what to make and how much. A government has limited tax income and must divide it among roads, schools, hospitals and defence. The three decision makers differ in size, yet all three must choose, and every choice leaves some wants unmet.

Before moving on. Resources are limited; wants are not. Scarcity is the condition that results, and it forces choices on everyone. You should now be able to say why a billionaire and a farmer both face scarcity, though their choices differ.

* * *

Before moving on: a want is a wish, and the whole list of wants has no end.

GO SLOWLY HERE.

A market stall shows the limit that meets unlimited wishes.

Resources are the things used to produce goods and services.

Scarcity. The condition that resources are limited while wants are unlimited, so choices must be made.

Scarcity is the gap between unlimited wants and limited resources.

Scarcity is not the same as rarity, and it is not the same as poverty.

Choice. Word in use. In daily life, the thing selected, or the freedom to select. In economics, picking one use of limited resources and so giving up the others.

Choice is the act of selecting among uses of limited resources.

Households, firms and governments each face scarcity in their own way.

Before moving on: scarcity is the meeting of limited resources with unlimited wants.

¹ Robbins, whom you will meet in the section 'In History', argued that scarcity of means with alternative uses is the common feature of every economic problem.

2.2 OPPORTUNITY COST

WHAT IS GIVEN UP

GO SLOWLY HERE.

Every choice has a price in what is left behind, even when no money changes hands.

Cost. Word in use. In daily life, the price paid for a good. In this book, price is what the consumer pays, and cost is what is given up or spent to make or obtain something.

Opportunity cost. The value of the best alternative given up when a choice is made.

Opportunity cost is the value of the best alternative that is given up.

Opportunity cost often has no price tag, yet it is real.

Trade-off. Giving up some of one thing in order to have more of another.

A trade-off is giving up some of one thing to gain more of another.

The owner, the employee and the consumer each see the cost differently.

When the alternative has no value, the opportunity cost is zero.

You should now be able to say what the opportunity cost of any choice is.

Opportunity cost can be turned into a number when outputs are known.

GO SLOWLY HERE.

Return to the traveller. If the traveller walks to the market town, the evening with the old friend is lost. If the traveller walks to the river village, the market sale is lost. In each case the lost evening has a value, even though no money is paid for it. Economists want a word for this loss, so that choices can be compared fairly.

The word is **opportunity cost**, which means the value of the best alternative given up when a choice is made. Notice the words 'best alternative'. When the traveller has three roads to choose from, only the most valuable of the two roads left behind counts. The opportunity cost of an hour spent at a cinema is the best other use of that hour, such as paid work or study, and not all the possible uses added together.

Opportunity cost is not always measured in money. A student who spends Friday evening at a wedding gives up the evening's study. A shopkeeper who keeps the shop open on a holiday gives up a day of rest. Money can measure the cost when the alternative has a money value, but time, effort and enjoyment count equally.

PAUSE AND CHECK 2.P1 You have a free Saturday morning. You can sleep late, study for three hours or earn Tk 300 helping at a shop. If you choose to study, what is the opportunity cost?

A related word describes the act itself. A **trade-off** means giving up some of one thing in order to have more of another. A farmer who plants more jute plants less rice. A student who studies an extra hour gives up an hour of sleep. Opportunity cost measures the size of what is given up; the trade-off is the exchange itself.

Opportunity cost looks different from each viewpoint. For the owner of a shop, the cost of opening on a holiday is the rest and family time given up. For an employee, the cost of an extra shift is the evening at home. For a consumer, the cost of buying a book is the other goods the same money could have bought. In each case the question is the same: what is the best use that is left behind?

There is an exception. If the best alternative has no value, the opportunity cost is zero. A shop table that stands empty overnight and has no other use costs nothing to use for a night's storage. Such cases are rare, because most resources have some other use, and economists therefore look for the alternative before they call anything free.

You should now be able to say that the opportunity cost of a choice is the best alternative left behind, and that a trade-off is the exchange of some of one thing for more of another.

MEASURING OPPORTUNITY COST

When the choice involves producing goods, opportunity cost can be measured exactly. In words: the opportunity cost of one unit of a good equals the quantity of the other good given up, divided by the quantity of the first good gained. In symbols, opportunity cost per unit = quantity given up ÷ quantity gained.

Case file, illustrative. Rafiq has 40 hours a week for two jobs. In one hour he can shelve 20 books or pack 10 online parcels. If he spends all 40 hours on shelving, he completes 800 books and 0 parcels. If he spends all 40 hours on packing, he completes 400 parcels and 0 books.

Given: the record above. *Find:* the opportunity cost of one extra parcel, in books. *Formula:* opportunity cost per unit = quantity given up ÷ quantity gained. *Steps.* One hour of packing gains 10 parcels. The same hour, spent on shelving, would have produced 20 books. So 10 parcels cost 20 books, and one parcel costs $20 \div 10 = 2$ books. *Answer:* each extra parcel gives up 2 books. *Check:* going from 800 books to 0 books gives up 800 books, and $800 \div 400$ parcels is also 2 books for each parcel. The two methods agree.

The opposite choice has the reverse cost. One extra book gives up $10 \div 20 = 0.5$ of a parcel. In general the opportunity cost of A in terms of B is the inverse of the opportunity cost of B in terms of A. A common slip is to forget this, and to give 2 books where 0.5 of a parcel is wanted. Always say which good is being measured, and in which unit.

Case file, illustrative. Nurul Meghna, who founded Meghna and Sons in 1996, has Tk 500,000 to spend on one of two uses: new titles for the shop, or an online site. The opportunity cost of choosing the new titles is the online site, and the opportunity cost of choosing the online site is the new titles.

Before moving on. Opportunity cost is the best alternative given up. When outputs are known, divide the quantity given up by the quantity gained. Rafiq's extra parcel costs 2 books. Nurul's choice of one use of Tk 500,000 costs the other.

* * *

2.3 THE PRODUCTION POSSIBILITY CURVE

DRAWING THE CHOICES

Rafiq's choice is not limited to all shelving or all packing. He may share his 40 hours in any proportion. A picture can show every such choice together. Before drawing it, test one more case from his record. Suppose he gives 20 hours to shelving and 20 hours to packing. Shelving gives $20 \times 20 = 400$ books. Packing gives $20 \times 10 = 200$ parcels. That is a third combination: 400 books and 200 parcels.

Plot the combinations and join them. The result is a **production possibility curve**, which means a line showing the largest combinations of two goods that can be produced with all resources used fully. Like the demand schedule in Chapter 4, it holds other things the same: Rafiq's hours, his speed and his tools stay fixed while the choices change. Each point on the line is one way of dividing the 40 hours.

To read the figure, look at the axes. The horizontal axis shows the number of books shelved; the vertical axis shows the number of parcels packed. A point is one combination of the two. Point A is 800 books and 0 parcels. Point C is 0 books and 400 parcels. Point B is 400 books and 200 parcels. The line joining them is straight because Rafiq's speeds do not change as he moves his hours.

Recall: economics defined, Chapter 1, section 1.1 'The Meaning and Scope of Economics'.

The steps for Rafiq follow the usual order: Given, Find, Formula, Steps, Answer, Check.

The opportunity cost of books is the reverse ratio.

Opportunity cost can also be a choice between two uses of money.

Before moving on: the best alternative given up, measured in the units of the other good.

A picture can show every choice open to Rafiq at once.

Recall: laws and the same-conditions rule, Chapter 1, section 1.4 'Economic Laws'.

Production possibility curve. A line showing the largest combinations of two goods that can be produced with all resources used fully.

The production possibility curve joins the largest combinations of two goods.

The axes show books and parcels, and each point is one combination.

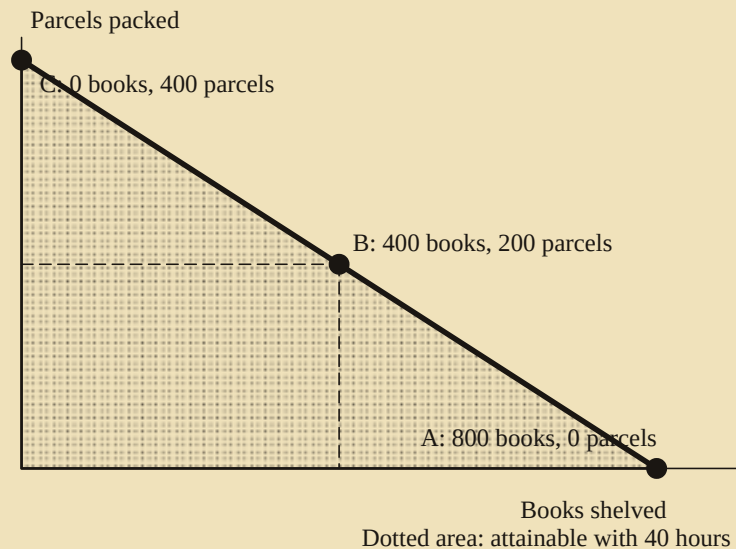


FIG. 2.1. Rafiq's production possibility curve with three points

Points on, inside and outside the curve have three different meanings.

The slope of the line shows the opportunity cost.

Real curves usually bow outward, and the cost rises as you move along.

More resources or better methods move the whole curve outward.

A whole country can be pictured with the same curve.

READING THE CURVE

The curve divides the picture into three regions. A point on the curve is attainable and uses all 40 hours. A point inside the curve is attainable but wastes something. For example, 400 books and 100 parcels needs only $20 + 10 = 30$ hours, so 10 hours are left idle. A point outside the curve cannot be reached with present resources. For example, 600 books and 300 parcels needs $30 + 30 = 60$ hours, which is more than 40.

The curve also shows opportunity cost. Moving along the line from A toward C, each extra parcel gives up 2 books, as we found earlier. The steepness of the line therefore carries the price of one good in terms of the other. A trade-off is visible as a move along the curve: more of one good, less of the other.

Rafiq's curve is a straight line because every hour of his time is equally good at each job. Most real curves bow outward, like a bowl turned on its side. The reason is that resources differ. A farm field suited to rice grows jute poorly, so moving it to jute costs more and more rice. This book returns to the idea when it studies costs of production. For now, remember that a straight line is a simplification chosen for easy arithmetic.

GROWTH SHIFTS THE CURVE

The curve is drawn for fixed resources and fixed methods. If either improves, the curve moves. Suppose Rafiq's week grows from 40 to 50 hours. All shelving now gives 1,000 books and all packing gives 500 parcels, so the whole line moves outward. Faster work has the same effect. A combination that was out of reach becomes possible. Economists call this growth. It does not remove scarcity, because wants rise as well.

The same picture serves a whole country. Imagine a small country that uses all of its farmland, workers and machines to produce food and medicines. A curve shows the largest combinations of the two. A country at a point on the curve can have more medicines only by giving up some food. A country at a point inside the curve, perhaps because many workers are idle, can gain more of both goods without giving up anything. This explains why economists care about unemployment: it moves a country inside its curve.

Before moving on. The production possibility curve shows the largest combinations of two goods. Points on it use all resources; points inside waste some; points outside are out of reach. Its slope shows opportunity cost, and growth moves the whole curve outward.

Before moving on: the curve shows the attainable, and growth moves it.

* * *

WORKED EXAMPLES: CHANGING THE SETTING

The first example is Rafiq above, with clean numbers. The next three change the setting. Each uses the order Given, Find, Formula, Steps, Answer, Check.

Four examples, each a little less tidy than the last.

Given: a farmer has 3 acres. On one acre, she can grow 2,500 kg of rice or 1,500 kg of jute. *Find:* the opportunity cost of 1 kg of jute, in rice. *Formula:* quantity given up ÷ quantity gained. *Steps.* One acre of jute gives up 2,500 kg of rice and gains 1,500 kg of jute. So 1 kg of jute costs $2,500 \div 1,500 = 1.67$ kg of rice. *Answer:* about 1.67 kg of rice. *Check:* the reverse cost of 1 kg of rice is $1,500 \div 2,500 = 0.6$ kg of jute, and 1.67×0.6 is close to 1, as it should be.

Rung 2: a farm with less tidy numbers gives a ratio that is not whole.

Given: in one hour, a bakery can make 6 dozen rolls or 3 dozen cakes. *Find:* the opportunity cost of one cake, in rolls. *Steps.* Convert to single items: 6 dozen = 72 rolls and 3 dozen = 36 cakes. One hour of cakes gives up 72 rolls and gains 36 cakes, so one cake costs $72 \div 36 = 2$ rolls. *Answer:* 2 rolls. *Check:* working in dozens gives $6 \div 3 = 2$ as well. The slip to avoid is to compare dozens of one good with single items of the other.

Rung 3: the units change from dozens to single items.

Given: a workshop can produce these combinations with all resources fully used.

Rung 4: a table combines costs, points inside and outside, and a rising ratio.

COMBINATIONS OF CHAIRS AND TABLES A WORKSHOP CAN MAKE

CHAIRS	TABLES
0	30
10	26
20	18
30	0

Find: the opportunity cost of an extra chair at each step, and whether 20 chairs with 20 tables is attainable. *Steps.* From 0 to 10 chairs, tables fall from 30 to 26, so 4 tables are given up for 10 chairs: 0.4 tables per chair. From 10 to 20 chairs, 8 tables are given up: 0.8 per chair. From 20 to 30 chairs, 18 tables are given up: 1.8 per chair. At 20 chairs the most tables possible is 18, so 20 tables is out of reach. *Answer:* the cost rises at each step, and the point is unattainable. *Check:* the sum of the three losses is $4 + 8 + 18 = 30$, the full number of tables.

The table is read row by row, and the cost is found between rows.

* * *

SUMMARY

Wants are without limit; resources are limited. This meeting is scarcity, and it forces choice. Opportunity cost is the value of the best alternative given up, and a trade-off is the exchange of some of one thing for more of another. Opportunity cost is found by dividing the quantity given up by the quantity gained. The production possibility curve shows the largest combinations of two goods; its slope shows opportunity cost, and growth moves it outward.

The chapter's main results in a few lines.

THINK IT THROUGH

2.T1 You can study for one hour or work in a shop and earn Tk 120 an hour. Explain the cost of studying that hour, and say why the answer may differ for another student.

2.T2 Rafiq's week grows to 50 hours. State his new maximum books and parcels, and say whether 600 books and 300 parcels are now attainable.

WHERE PEOPLE GO WRONG

Four common misunderstandings, each corrected.

Many students believe that opportunity cost is all the alternatives added together. It is only the best one. They also believe that opportunity cost is the money paid; yet a free evening used for study still has a cost. A third error is to think that scarcity means poverty or rarity, when it describes the meeting of limited resources and unlimited wants for everyone. The last error is to think that a point inside the curve is a good choice because it is attainable. It is attainable, but it wastes resources that could give more of both goods.

IN HISTORY

In 1932 an English economist defined economics as the study of choice under scarcity.

Lionel Robbins (1898 to 1984) was an English economist who taught at the London School of Economics. In 1932 the publisher Macmillan of London issued his book *An Essay on the Nature and Significance of Economic Science*. Earlier writers had defined economics as the study of wealth or of business. Robbins argued instead that economics studies human behaviour as a relationship between ends and scarce means that have alternative uses.

The definition made scarcity and choice the centre of the subject.

The new definition changed the question. It did not matter whether an activity concerned money; any choice among uses of limited means had an economic side. The book became one of the most cited works on the nature of the subject, and its idea shapes this chapter: wants are many, means are limited, and choice follows. Not every economist accepted the definition, and some argued that it left out matters such as the study of national output.

A fifth error concerns money already spent.

A fifth error is to treat a sunk loss as a cost of the next decision. Money already spent and not recoverable cannot be given up again, so only the alternatives still open count when you choose.

REVIEW QUESTIONS

2.1 Define want, and explain why the list of wants as a whole has no end even though one want can be satisfied.

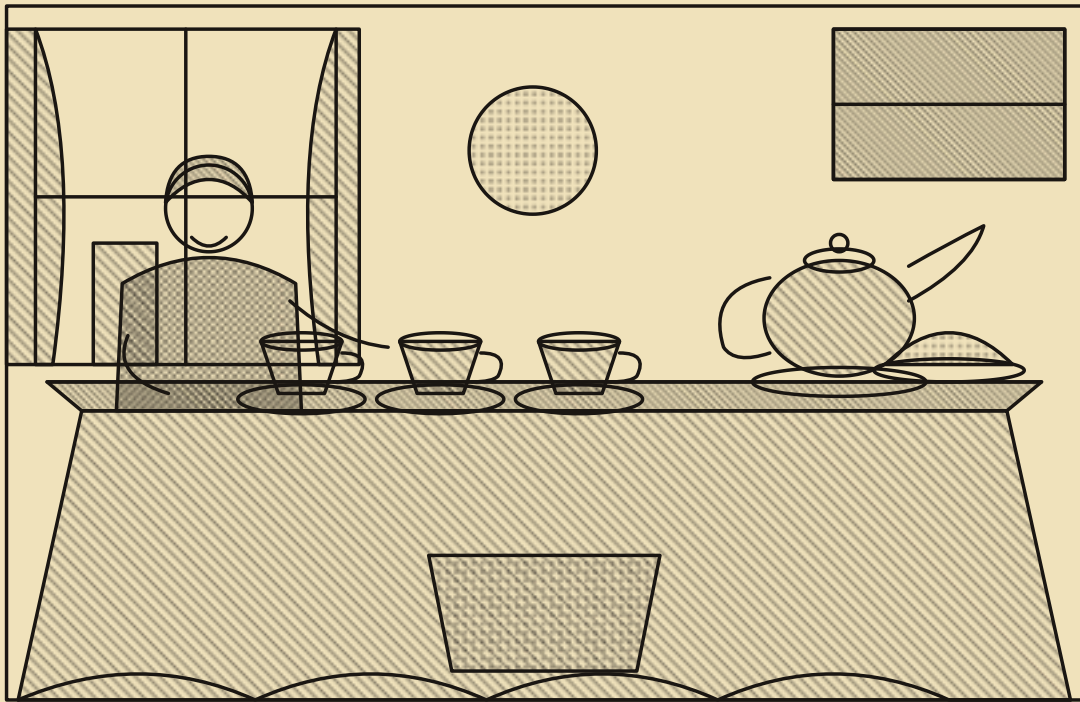
2.2 Explain scarcity in your own words, and give one example for a rich person and one for a poor person.

2.3 A student can spend Saturday studying, working for Tk 400 or resting. Studying is chosen. What is the opportunity cost?

2.4 In one hour a tailor can make 8 shirts or 4 pairs of trousers. Find the opportunity cost of one pair of trousers, in shirts, and of one shirt, in trousers.

2.5 Using Rafiq's record, find his combination of books and parcels if he spends 30 hours on shelving and 10 hours on packing. Is it on, inside or outside his curve?

- 2.6 A production table shows 0 cars with 50 buses, 5 cars with 40 buses, and 10 cars with 20 buses. Find the opportunity cost of a car in each step and say what happens to it.
- 2.7 Give one reason why a real production possibility curve bows outward, and one event that would shift it outward.



A Victorian tea table, with the diner finishing her third cup

CHAPTER III

Utility and Diminishing Marginal Utility

Why does the same cup of tea please a little less each time?



This chapter explains how economists describe satisfaction, how they count it with an imaginary unit, and why each extra unit of a good adds less than the one before. A tea table and a student buying exercise books show the rule at work, and the chapter ends with the limits of the rule.

Two earlier ideas and one appendix help with this chapter.

A diner enjoys her first cup of tea and refuses her fourth.

The tea, the table and the price stayed the same, yet the pleasure changed.

This small puzzle explains why buyers stop buying.

The chapter moves slowly, in small steps, because the ideas are abstract.

Every purchase begins with a want that the buyer hopes to satisfy.

GO SLOWLY HERE.

Utility. The satisfaction a person gets from using a good or service. Economists call the satisfaction from a good its utility.

Two people can gain different utility from the same good.

You will need two ideas from earlier chapters: opportunity cost, from Chapter 2 'Scarcity, Choice and Opportunity Cost', section 2.2 'Opportunity Cost'; and the meaning of an economic law, from Chapter 1 'What Economics Is', section 1.4 'Economic Laws'. Appendix A1 'Maths You Will Use' helps with adding columns of numbers and with reading a graph.

Imagine a long table set for afternoon tea in a Victorian town house. A diner sits down, tired and cold after a walk. The first cup of hot tea is a delight, and she drinks it slowly. The second cup is still pleasant, though the delight is smaller. She drinks the third mainly because it stands in front of her. When the maid offers a fourth, the diner covers the cup with her hand and says, politely, that she has had enough.

Look at what did not change. The tea is the same tea. The cups, the table and the maid are the same. The tea is free, since the hostess pays for it. Yet the drink that gave great pleasure at the start gives almost none at the end. Why should a good thing lose its power to please as we take more of it?

The question looks small, but it is not. Every day, people decide how much of a good to buy and when to stop. A bookseller must ask how many notebooks one customer will want. If every extra notebook pleased as much as the first, a customer would buy without end, and no shop could ever fill the wish. In fact customers stop. To understand why they stop is the first step towards understanding how buyers choose, which later chapters study.

Here we build the answer in small steps. First we name the pleasure and learn to count it, using imaginary numbers. Then we state the rule by which pleasure fades, and we test it on a student buying exercise books at Meghna and Sons. Many students find this chapter slow at first, because satisfaction cannot be seen or weighed. For that reason we go slowly, one idea at a time, with small numbers.

* * *

3.1 THE IDEA OF UTILITY

Begin with your own day. Why did you eat your last meal? You were hungry, and food satisfied the hunger. Hunger is a want, and the pleasure or relief of meeting it is a kind of reward. Every time a person drinks, reads, travels or buys, a want is being met and a reward is felt. Economics needs a short name for that reward, so that we can reason about it without confusion.

Economists call the reward **utility**. Utility means the satisfaction a person gets from using a good or service. A cup of hot tea gives utility to a cold traveller. A bus ticket gives utility to a worker who must reach the office. Notice what the word does not say. Utility is not a judgement of whether a good is wise or healthy. A sweet snack may harm the teeth, yet it gives utility to the person who enjoys it, because utility counts only the satisfaction that person feels.

Utility belongs to the person, not to the good. Picture a market stall that sells green mangoes. A child who loves their sharp taste may feel great satisfaction from one mango, while an older neighbour who dislikes sour fruit feels almost none. The good is the same, but the utility differs. For this reason economists speak of the utility a particular person gets, at a particular time, and not of the utility a good has in itself. A good that pleases nobody has no utility at all, however carefully it was made.

You may know the word from daily life, where a *utility* is a public service such as the supply of water or electricity. In this book the word always has its economic sense: satisfaction. When a company that supplies water appears in an example, we call it a water company and keep the word utility for the feeling.

Now suppose the diner drinks three cups. Each cup gave her some satisfaction. If we add the satisfaction from the first cup, the second cup and the third cup, we reach **total utility**. Total utility means the whole satisfaction from all units of a good consumed. For the diner, it is everything she enjoyed from all three cups together. Total utility answers one question: how much satisfaction did she gain altogether?

Now ask a different question: how much did the third cup add? The answer is the **marginal utility** of the third cup. Marginal utility means the extra satisfaction from consuming one more unit of a good. If the diner had two cups and gains a little from a third, that little gain is the marginal utility of the third cup. A staircase gives a useful picture. Total utility is how high you stand after climbing some steps. Marginal utility is the height of the last step you climbed. They are two different measures: the height so far, and the height of one step.

The word *marginal* needs care for the same reason. A margin is the edge of a page. In economics the margin is the edge of what you already have, and marginal means what one more unit adds at that edge. The word does not mean small or unimportant. The marginal utility of a cup may be large, as it is for the first cup after a long walk.

PAUSE AND CHECK 3.P1 The diner drinks one cup, then a second cup that pleases her a little. After the second cup, is her total satisfaction higher or lower than after the first? Give a reason.

MEASURING SATISFACTION

Can satisfaction be counted? Nobody owns a machine that reads how happy a cup of tea makes a person. William Stanley Jevons, an English economist who wrote in the 1870s, found it useful to imagine such a machine.¹ Economists therefore agree to count satisfaction in an imaginary unit.

The unit is the **util**. A util means an imaginary unit used to count satisfaction in economic examples. If the first cup gives 10 utils and the second gives 8 utils, we say that the first cup pleased the diner more. The numbers are not real measurements. They are a tool that lets us reason exactly about more and less.

The table below shows the diner's tea in utils. The numbers are imaginary and chosen to be small. Each row is one cup. The middle column gives the marginal utility of that cup, and the last column gives the total utility after that cup.

Utility. Word in use. In daily life, a utility is a public service such as water or electricity. In economics, utility is the satisfaction a good gives.

The everyday word and the economic word differ.

Total utility. The whole satisfaction from all units of a good consumed.

Adding the satisfaction from every unit gives total utility.

GO SLOWLY HERE.

Marginal utility. The extra satisfaction from consuming one more unit of a good.

Marginal utility looks only at the last unit, not at the whole.

Marginal. Word in use. In daily life, marginal often means small or at the edge, as in a marginal improvement. In economics, marginal means extra: the change caused by one more unit.

In economics, marginal means extra.

Satisfaction cannot be seen, so economists count it with an imaginary unit.

Util. An imaginary unit used to count satisfaction in economic examples.

The imaginary unit is the util.

A small table shows five cups of tea, with the extra and the whole counted in utils.

¹ Recall: opportunity cost, Chapter 2, 'Scarcity, Choice and Opportunity Cost', section 2.2 'Opportunity Cost'. We use it later in this chapter, when we ask what a buyer forgoes.

SATISFACTION FROM FIVE CUPS OF TEA (IMAGINARY UTILS)

CUP	MARGINAL UTILITY (UTILS)	TOTAL UTILITY (UTILS)
1	10	10
2	8	18
3	5	23
4	2	25
5	0	25

Read a row to see what one cup added and what all cups gave.

Counting in utils turns a feeling into a list that can be compared.

Utils compare one person's own moments, not different people.

The section ends by recalling the three ideas and the unit.

Three economists in three countries found the same rule within a few years.

Look again at the marginal utility column of the tea table.

GO SLOWLY HERE.

Law of diminishing marginal utility. As a person consumes more units of a good, the satisfaction added by each extra unit falls.

The law states that each extra unit adds less than the one before.

To read a row, take cup 3. Its marginal utility is 5, so the third cup added 5 utils. Its total utility is 23, so the three cups together gave 23 utils. Notice the link between the columns: the total after cup 3 equals the total after cup 2 plus the marginal utility of cup 3, because 18 plus 5 is 23. Each total is the sum of all the marginal utilities so far.

Why count at all? Without numbers, we could say only that the diner liked the first cup more than the second. With numbers, we can say how much more, and we can say how fast the pleasure fades. That speed is exactly what the law in the next section describes. The numbers must be treated as a tool of reasoning, in the way that a map is a tool: it does not equal the land, but it helps a traveller decide.

One limit is important. A util belongs to one person at one time. We may say that the diner liked cup 1 more than cup 2. We may not say that her 10 utils equal 10 utils felt by another person, because no common scale exists.

Before moving on. Utility is satisfaction. Total utility is the satisfaction from all units; marginal utility is the extra from one more unit. We count satisfaction in utils, which are imaginary but useful. You should now be able to say, in your own words, why the total after the fourth cup can be higher than after the third even though the fourth cup added only a little.

3.2 THE LAW OF DIMINISHING MARGINAL UTILITY

In the early 1870s, an economist in England, one in Austria and one in Switzerland each published the same observation about satisfaction. None knew of the work of the others. The section called In History, at the end of this chapter, tells that story. Here we explain the observation itself, and we use the tea table to see it.

THE LAW STATED

Return to the table of five cups. Read down the marginal column: 10, 8, 5, 2, 0. Each number is smaller than the one before. The cups are the same cups, and the diner is the same diner. Yet each extra cup adds less. This pattern is so regular that the economists give it a name and call it a law.

The pattern is the **law of diminishing marginal utility**. The law of diminishing marginal utility says that as a person consumes more units of a good, the satisfaction added by each extra unit falls. The word *diminishing* means becoming smaller. Notice that the law speaks about the marginal utility, the extra, and not about the total.²

² Recall: laws, Chapter 1, 'What Economics Is', section 1.4 'Economic Laws'. An economic law describes a regular tendency in human choice, not a rule that nature enforces.

THE LAW IN SHORT

Keep consuming one good, and the extra satisfaction from each new unit becomes smaller.

This point is the one that confuses students most. When marginal utility falls from 10 to 8 to 5, the total still rises, from 10 to 18 to 23. A smaller step is still a step upward. The total utility rises as long as the marginal utility is above zero. At the fifth cup the marginal utility is zero, so the total stops rising and stays at 25. If a sixth cup were forced upon the diner, it would give her discomfort, and the marginal utility would fall below zero.

WHY THE LAW HOLDS

Why should the extra satisfaction shrink? Begin with the bridge from daily life. A person who is very thirsty has a strong want for water. The first glass meets the strongest part of that want, and the relief is great. After the glass, the want is smaller. The second glass meets what remains, so the relief is smaller. Each unit uses up part of a want that has a limit, and a smaller want leaves less satisfaction to gain.

Return to the puzzle at the start of the chapter. If marginal utility did not fall, the diner would drink cup after cup, and no shop would ever need to attract new customers, because each customer would buy forever. Because marginal utility falls, every buyer reaches a point where one more unit is not worth having. When the unit is not free, a buyer must also compare the satisfaction gained with what is forgone, which is the opportunity cost of the purchase. A later chapter joins these two ideas.

Look at the same fact from two viewpoints. For the consumer, it is a feeling: the fourth cup is not worth much. For the owner of a shop such as Meghna and Sons, it is a pattern in sales: a customer buys the first notebook readily, hesitates over the second and rarely wants a long row of identical ones. The owner cannot see the utils, but the owner can see the customer stop. The law gives a reason for what the owner observes. An assistant who serves the customers sees the same fact from a third position: a customer who comes in for one book rarely leaves with ten, and the assistant learns to expect it.

THE SCHEDULE AND THE CURVE

A table that lists the utility of each unit is called a schedule, and the same numbers drawn as a line are called a curve. We now use the shop's own example. The next paragraph is a record of a case, and its figures are fixed.

Case file, illustrative. A student buys exercise books at Meghna and Sons. The marginal utility of each book, in utils, is: 1st book 40, 2nd book 30, 3rd book 20, 4th book 10, 5th book 0. The total utility after each book is 40, 70, 90, 100 and 100.

The law is about the extra, so the total can still rise while the extra falls.

A want is limited, and each unit satisfies part of it.

The law helps explain why buyers do not buy without end.

The same fact looks different from the consumer's side and the seller's side.

A table of utility for one person is a schedule; a line through the numbers is a curve.

The case file records the student's utility for five books.

The case table sets the numbers side by side.

THE STUDENT'S UTILITY FROM EXERCISE BOOKS (UTILS)		
BOOK	MARGINAL UTILITY (UTILS)	TOTAL UTILITY (UTILS)
1st	40	40
2nd	30	70
3rd	20	90
4th	10	100
5th	0	100

PAUSE AND CHECK 3.P2 Using the tea table, how much did the fourth cup add to the diner's total satisfaction? Which column gives the answer?

Before reading a chart, learn what its two axes show.

Now we draw the numbers. Every chart in this book has two lines at right angles, called axes.³ On the horizontal axis we place the number of books. On the vertical axis we place utils. A point on the chart means: at this number of books, the utils have this value. The first chart shows total utility.

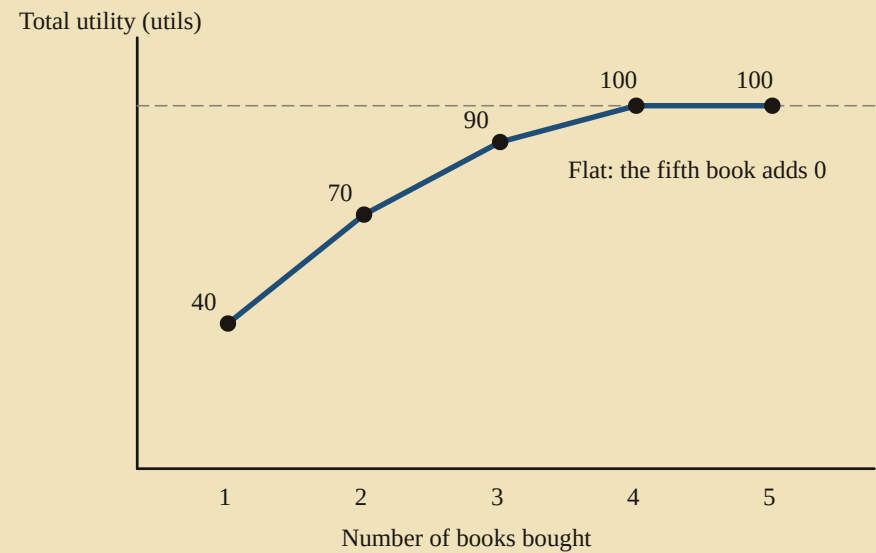


FIG. 3.1. Total utility rises with each book and then becomes flat

The total line climbs quickly, then more slowly, then stops climbing.

Read the chart from left to right. The line climbs steeply from the first book to the second, because 40 utils are added. It climbs less steeply to the third, because 30 and 20 are added. It becomes flat after the fourth book, because the fifth adds nothing. A flat line means the total has stopped rising.

³ See Appendix A1 'Maths You Will Use', section A1.5 'Reading a Graph', for axes, points and slope.

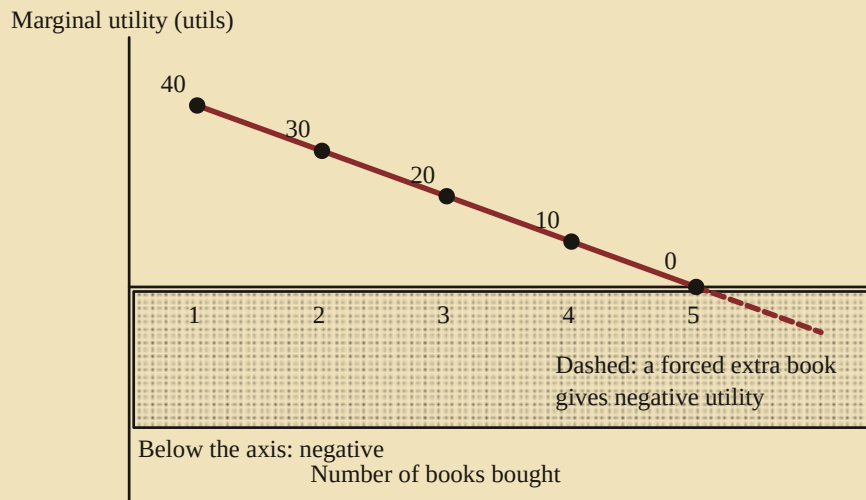


FIG. 3.2. *Marginal utility falls with each book and reaches zero*

The second chart shows the extra from each book instead of the whole. Its points go down: 40, 30, 20, 10, 0. A line through them slopes downward from left to right, and that downward slope is the law in picture form. The line reaches the horizontal axis at the fifth book, where marginal utility is zero. The dashed part below the axis shows what would happen if a further book were forced upon the student: marginal utility would be negative, and total utility would fall.

Before moving on. The law says the extra satisfaction from each unit falls. The marginal column goes down while the total column goes up and then stops. At the point where marginal utility is zero, total utility is at its highest. You should now be able to say why the total can rise while the extra falls.

The second chart shows the same facts as the extra from each book.

The two charts are the same story told twice.

3.3 LIMITS OF THE LAW

The law holds when certain conditions are met. The units must be alike in size and quality, as cups of the same tea are. The consumer's tastes must stay the same during the time of consumption. The consumption must follow without long gaps, because after a rest the want returns and the first unit pleases again. Economists say these conditions together as: other things remain equal.

Some cases look like exceptions. A collector of stamps may value each new stamp more as the set nears completion. Yet the unit that matters is then the complete set, and the law holds for sets. A person who is addicted to a drug may crave more each time, but this is a harm and not a normal want. Treat such cases as limits of the law, not as proofs that it fails. A third case is a good that is useful only in a set, such as a left shoe: the first shoe alone adds little, and the second adds much. Here again the sensible unit is the pair.

Before moving on. The law needs alike units, steady tastes and no long gaps. Apparent exceptions usually change the unit.

A law holds only under stated conditions.

Apparent exceptions usually disappear when the unit is defined with care.

The section closes with its conditions in two lines.

WORKED CASE: READING A UTILITY SCHEDULE

A table gives some numbers and hides others. Two links let us find the hidden ones. In words: total utility after a unit equals the total before it plus the marginal utility of that unit. In symbols, $TU(n) = TU(n - 1) + MU(n)$, where $TU(n)$ is the total utility after n units, $TU(n - 1)$ is the total after one fewer unit, and

Two links between the columns let us calculate any missing number.

The first example uses the student's own numbers and finds a total.

The second example starts from totals and finds the extra from each unit.

Four common slips spoil these calculations.

The chapter's main ideas in a few lines.

Two short problems that use the tea table.

A common error is to believe that falling marginal utility means falling total utility.

Utility is not the same as usefulness or goodness.

The law applies to one period and one person, not to a lifetime.

MU(n) is the marginal utility of the nth unit. Rearranged, $MU(n) = TU(n) - TU(n - 1)$.⁴

Example 1. *Given:* the marginal utilities of the first three exercise books are 40, 30 and 20 utils. *Find:* the total utility after three books. *Formula:* total utility is the sum of the marginal utilities so far. *Steps:* $40 + 30 = 70$ utils; $70 + 20 = 90$ utils. *Answer:* total utility after three books is 90 utils. *Check:* the case table shows 90 in the row of the 3rd book, so the answer is sensible.

Example 2. *Given:* a different person drinks glasses of sugarcane juice, and total utility after each glass is 12, 22, 30, 36, 40, 42 and 42 utils. *Find:* the marginal utility of each glass. *Formula:* $MU(n) = TU(n) - TU(n - 1)$, and for the first glass $MU(1) = TU(1)$. *Steps:* glass 1: 12; glass 2: $22 - 12 = 10$; glass 3: $30 - 22 = 8$; glass 4: $36 - 30 = 6$; glass 5: $40 - 36 = 4$; glass 6: $42 - 40 = 2$; glass 7: $42 - 42 = 0$. *Answer:* the marginal utilities are 12, 10, 8, 6, 4, 2 and 0 utils. *Check:* the list falls at every step, as the law predicts, and the total stops rising when marginal utility reaches zero.

Common slips. Students often read the marginal column as if it were the total column. They forget to write utils beside the numbers. They subtract in the wrong order, which gives a negative number for a gain. And they treat the first row differently from the others: for the first unit, marginal utility and total utility are the same number.

SUMMARY

Utility is the satisfaction a good gives. Total utility is the whole satisfaction from all units, and marginal utility is the extra from one more unit. We count satisfaction in imaginary utils. The law of diminishing marginal utility says that each extra unit adds less than the one before. Total utility rises as long as marginal utility is above zero, and it stops rising when marginal utility reaches zero.

THINK IT THROUGH

3.T1 Why might the diner's fifth cup of tea give her zero utility? Give two reasons from the diner's own situation.

3.T2 In the tea table, suppose the diner could drink a sixth cup that gave her minus 3 utils. What would her total utility be after six cups, and what should she do?

WHERE PEOPLE GO WRONG

Error 1: "If marginal utility falls, the total falls." This is wrong. A smaller extra is still an extra, so the total rises while marginal utility is above zero. The total falls only when marginal utility is negative.

Error 2: "Utility means a good is useful or good for you." This is wrong. Utility is the satisfaction a person feels, whether or not the good is wise. Economics describes the feeling and leaves the moral judgement to others.

Error 3: "The law means the first unit is always the best." This is wrong. The law describes units consumed in one period. After a long rest, the want returns and the next unit pleases again.

⁴ Appendix A1 'Maths You Will Use', section A1.3 'Rearranging an Equation', shows how to move a term across the equals sign.

Error 4: "Ten utils for me equal ten utils for you." This is wrong. A util is an imaginary unit for one person's own comparison, and no common scale links two people.

Utils are a tool, not a measurement of the world.

IN HISTORY

In 1854, a Prussian civil servant named Hermann Heinrich Gossen published a book in Germany that stated the idea of diminishing satisfaction and drew its curves. Few people bought it, and Gossen died in 1858, largely unknown. In 1871, William Stanley Jevons in England published *The Theory of Political Economy*, and in the same year Carl Menger in Austria published *Grundsätze der Volkswirtschaftslehre*. In 1874, Léon Walras, who taught at Lausanne in Switzerland, published the first part of *Éléments d'économie politique pure*.

Four writers found the rule, and three of them did so within four years.

The three writers did not know of one another's work when they began. Historians call these years from 1871 to 1874 the marginal revolution, because each author built a theory of value on marginal utility. Jevons learned of Gossen's book after publishing his own, and in a later edition he acknowledged that Gossen had come first.

The revolution was the work of independent minds in different countries.

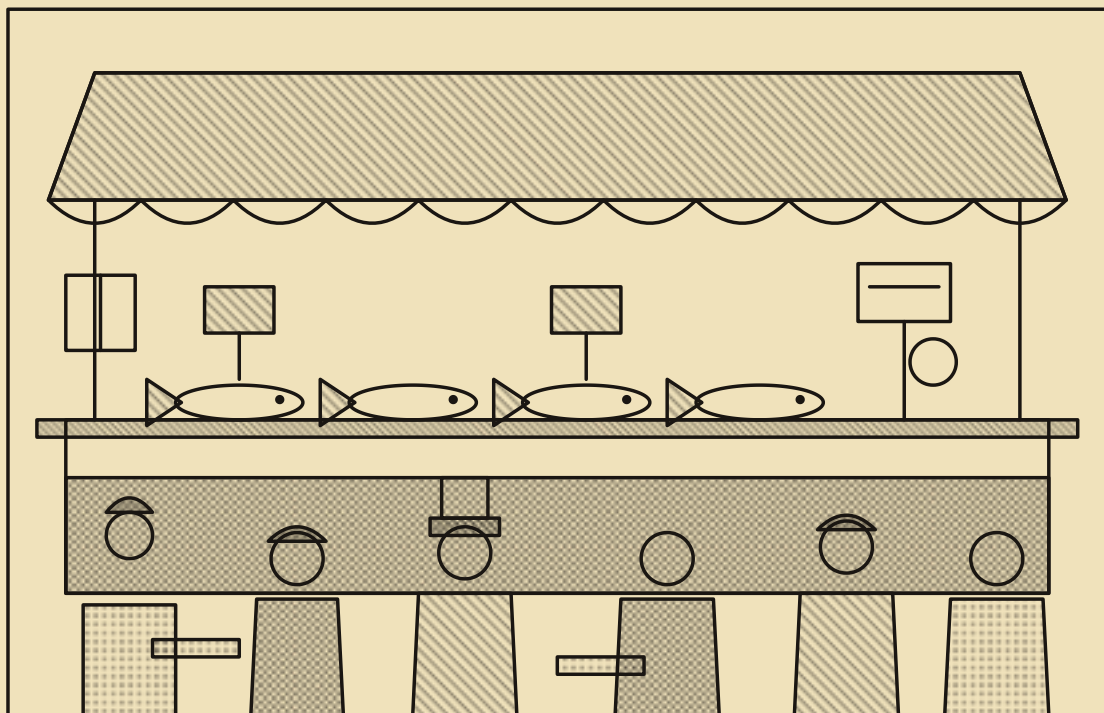
The story carries a lesson for a student. Gossen stated the idea first and was ignored, while the three later writers were read and remembered. An idea becomes part of a science only when other people understand it, test it and use it. The law you have learned passed that test, and today it is the starting point of the theory of how buyers choose.

The history teaches that a good idea can wait for its readers.

REVIEW QUESTIONS

- 3.1 Define utility, and give one example from your own day.
- 3.2 Complete the table. Marginal utilities for four units are 50, 35, 20 and 5 utils. Find the total utility after each unit.
- 3.3 Total utilities for five units are 20, 36, 48, 54 and 56 utils. Find the marginal utility of each unit.
- 3.4 State the law of diminishing marginal utility, and name one condition under which it holds.
- 3.5 In the case table, what is the marginal utility of the 5th exercise book? What is the opportunity cost of buying that book, in words, if the student could spend the money on another good?
- 3.6 A student says the total falls after the third cup because the marginal utility falls. Explain the mistake.

Six questions that check the chapter's ideas.



A Victorian fishmonger's stall, with price tablets and a crowd of buyers

CHAPTER IV

Demand and the Demand Curve

Why does a crowd gather round a stall when the price tablet changes?



This chapter builds the central diagram of the book. It explains what buyers are willing and able to buy at each price, why a lower price brings a larger quantity, how to draw the demand curve, and what moves or shifts it. It ends with related goods and a few unusual cases.

Two earlier ideas and one appendix help with this chapter.

A fishmonger changes a price tablet, and the crowd responds.

The crowd does not obey the fishmonger's orders, yet it behaves in a regular way.

The chapter gives that pattern a name, a table and a picture.

You will need two ideas from earlier chapters: the fall of satisfaction with each extra unit, from Chapter 3 'Utility and Diminishing Marginal Utility', section 3.2 'The Law of Diminishing Marginal Utility'; and the habit of holding other conditions fixed, from Chapter 1 'What Economics Is', section 1.4 'Economic Laws'. Appendix A1 'Maths You Will Use' helps with reading a graph and with rearranging an equation.

Picture a fishmonger's stall in a Victorian market town, early on a Saturday morning. Along the front of the stall stand small tablets, each marking the price of one kind of fish. A crowd of buyers moves slowly past. At nine o'clock the fishmonger lowers the tablet for the herring. Within minutes the crowd thickens, baskets fill, and the tray empties faster than before. At noon the tablet is raised again, and the crowd moves on to the cheaper trays.

Nobody ordered the buyers to do this. Each acted for private reasons, and each had a different purse and a different taste. Yet the crowd as a whole behaved in a way that the fishmonger could predict: a lower price brought more buyers, and a higher price sent them away. How can millions of separate choices form a pattern that is so regular? And can we describe the pattern exactly, so that any firm could use it?

In this chapter we build the answer in small steps. We name what buyers do, record it in a table, and draw it as a curve. Then we ask what makes the curve move, and we meet some goods that behave in unusual ways. Many students find the central diagram hard at first, because two different changes both seem to alter the same picture. We therefore go slowly, and we separate the two changes with care.

* * *

4.1 QUANTITY DEMANDED AND THE LAW OF DEMAND

Begin with a question: how many notebooks would you buy at each price?

GO SLOWLY HERE.

Demand. The quantities of a good that buyers are willing and able to buy at each price in a period.

Demand is the whole set of prices and quantities, not one number.

Demand. Word in use. In daily life, demand means a strong request or order, as in a demand for payment. In economics, demand is the whole list of quantities buyers will buy at each price.

In this book, demand alone always means the whole list.

Begin with a question from your own life. A plain 200-page notebook costs Tk 60. How many would you buy this week? Now suppose the price falls to Tk 20. Would you buy the same number? Most people answer that they would buy more at the lower price, and fewer at the higher price. The number of units that buyers choose to buy at one particular price in a period is called the quantity demanded. It is a single number, tied to a single price.

Economists use the word **demand** for something larger. Demand means the quantities of a good that buyers are willing and able to buy at each price in a period. Notice the two conditions, willing and able. A wish without money is not demand, and money without a wish is not demand either. For the notebook, demand is the full list: so many at Tk 60, so many at Tk 50, so many at Tk 40, and so on. One entry in the list is a quantity demanded. The whole list is demand.

The everyday word needs care here. In daily speech people say that "demand rose" when they mean that more was bought. This book keeps two words apart. Quantity demanded means one quantity at one price. Demand alone means the whole list of prices and quantities, which the next pages record in a table and then draw as a line.

The simplest way to record demand is a table. A **demand schedule** is a table showing the quantity demanded at each price. We now use the case of the shop. The next paragraph is a record, and its figures are fixed.

Case file, illustrative. Meghna and Sons records the weekly demand for its 200-page notebook. At a price of Tk 60 the quantity demanded is 20 notebooks; at Tk 50, 30; at Tk 40, 40; at Tk 30, 50; at Tk 20, 60. The quantity demanded equals 80 minus the price in taka.

WEEKLY DEMAND SCHEDULE FOR THE 200-PAGE NOTEBOOK

PRICE (TK)	QUANTITY DEMANDED (NOTEBOOKS A WEEK)
60	20
50	30
40	40
30	50
20	60

To read a row, take the line for Tk 40. It says that at a price of Tk 40, buyers choose to buy 40 notebooks a week. Read down the table and notice the pattern: as the price falls by Tk 10 each time, the quantity demanded rises by 10 each time. The rule in the case file, quantity equals 80 minus the price, gives the same numbers: 80 minus 40 is 40.

The pattern has a name. The **law of demand** says that when the price of a good rises, the quantity demanded falls, if other conditions stay the same. The final clause matters. Income, tastes and the prices of other goods must stay fixed, or we cannot tell whether a change in the quantity bought came from the price.¹ The law also works in the other direction: when the price falls, the quantity demanded rises.

Why should buyers behave in this way? Two reasons are enough for now. The first links to the last chapter.² Each extra notebook gives a buyer less satisfaction than the one before, so a buyer will take a further notebook only if its price is lower. The second reason is that a lower price brings in new buyers who could not afford the good before, and it tempts existing buyers to find new uses. A cheap notebook may become a diary, a ledger and a gift.

PAUSE AND CHECK 4.P1 At a price of Tk 50 the quantity demanded is 30 notebooks. Without drawing anything, say what the quantity demanded is at Tk 30, and state in one sentence the law of demand that your answer shows.

A table is exact, but a picture shows the pattern at a glance. A **demand curve** is a line on a graph showing the quantity demanded at each price. To read the graph, notice its two axes, which are lines at right angles.³ The vertical axis shows the price in taka. The horizontal axis shows the quantity demanded in notebooks a week. A point on the graph means: at this price, buyers choose this quantity.

Demand schedule. A table showing the quantity demanded at each price.

A demand schedule is a table of prices and the quantities demanded at them.

The case file records the weekly demand for the shop's notebook.

The same record is laid out as a table.

Each row pairs one price with one quantity demanded.

GO SLOWLY HERE.

Law of demand. When the price of a good rises, the quantity demanded falls, if other conditions stay the same.

The pattern in the table is the law of demand.

Two reasons explain why a lower price brings a larger quantity demanded.

Demand curve. A line on a graph showing the quantity demanded at each price.

A demand curve draws the table as a line on a graph.

¹ Recall: ceteris paribus, which means other things remaining equal, Chapter 1, 'What Economics Is', section 1.4 'Economic Laws'.

² Recall: diminishing utility, Chapter 3, 'Utility and Diminishing Marginal Utility', section 3.2 'The Law of Diminishing Marginal Utility'.

³ See Appendix A1 'Maths You Will Use', section A1.5 'Reading a Graph', for axes, points and slope.

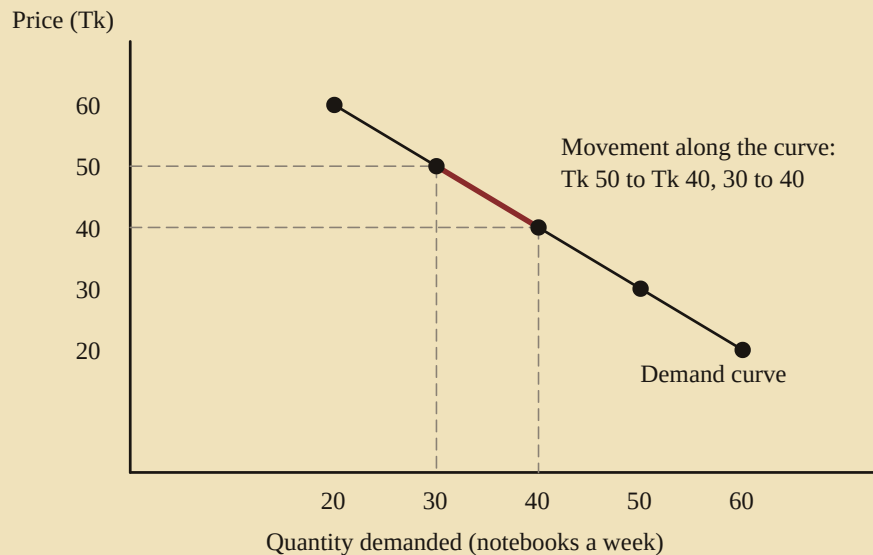


FIG. 4.1. The demand curve for the notebook, with a movement along it from Tk 50 to Tk 40

The line slopes downward, because a lower price goes with a larger quantity.

The section closes with the four ideas in a few lines.

Two different changes can alter the picture, and they must be kept apart.

GO SLOWLY HERE.

A change in the price of the good moves us along one curve.

Five other conditions decide where the whole curve lies.

Read the line from the top left to the bottom right. At the top, the price is high and the quantity is small. At the bottom, the price is low and the quantity is large. We call it a curve although in this case it is a straight line, because the word covers lines of every shape. The downward slope is the law of demand drawn as a picture: lower price, larger quantity demanded. In the picture, the short bold part between the points for Tk 50 and Tk 40 will be used in the next section.

Before moving on. Quantity demanded is one number at one price. Demand is the whole list of prices and quantities, which the demand schedule records and the demand curve draws. The law of demand says that, other conditions staying the same, a lower price brings a larger quantity demanded. You should now be able to say why the curve slopes downward.

4.2 MOVEMENT AND SHIFT

The fishmonger faced two different events. At nine o'clock his own price fell, and more fish sold. Suppose that on another Saturday a rival stall across the street cut its price, and fewer buyers came to him even though his tablet had not changed. Both events changed how much fish he sold. Are they the same kind of change? They are not, and the difference is the key to this chapter.

First, the price of the good itself changes. The notebook falls from Tk 50 to Tk 40. Buyers respond by buying more, and the quantity demanded rises from 30 to 40 notebooks a week. On the graph, the point moves along the same line, from the point for Tk 50 to the point for Tk 40. We call this a movement along the curve. The curve itself does not change position, because the whole list of prices and quantities is the same list as before.

Second, something other than the price changes. The curve was drawn with other conditions held fixed, and these conditions are called the determinants of demand. When one of them changes, buyers want different quantities at every price, and the curve must be redrawn in a new position. The table below lists the usual determinants, with an example for the notebook. We introduce the table in words first: each row names one determinant, and the middle column gives a notebook example.

THE DETERMINANTS OF DEMAND, WITH A NOTEBOOK EXAMPLE

DETERMINANT	EXAMPLE FOR THE NOTEBOOK	EFFECT ON DEMAND
Buyers' incomes	Families have more money to spend	Rises
Prices of related goods	A rival notebook becomes cheaper	Falls
Tastes	Schools ask pupils to keep notebooks	Rises
Number of buyers	A new college opens nearby	Rises
Expectations	Buyers expect a price rise next month	Rises now

To read a row, take the second line: when a rival notebook becomes cheaper, some buyers leave, and demand for our notebook falls. The word *falls* here means that fewer notebooks are wanted at every price, not only at one price.

When a determinant changes, the whole curve moves to a new position. We call this a shift of the curve, which some books call a change in demand. This book uses the word shift. The case file gives a precise example.

Case file, illustrative. The price of a rival notebook falls by Tk 10. Weekly demand for the shop's notebook moves down by 5 at every price. At Tk 60 the quantity demanded becomes 15; at Tk 50, 25; at Tk 40, 35; at Tk 30, 45; at Tk 20, 55.

The new quantities are 5 smaller than the old ones in every row, so the curve moves to the left. A move to the left means a smaller quantity at every price. A move to the right means a larger quantity at every price. The picture shows the old curve and the new curve, and the arrow shows the direction.

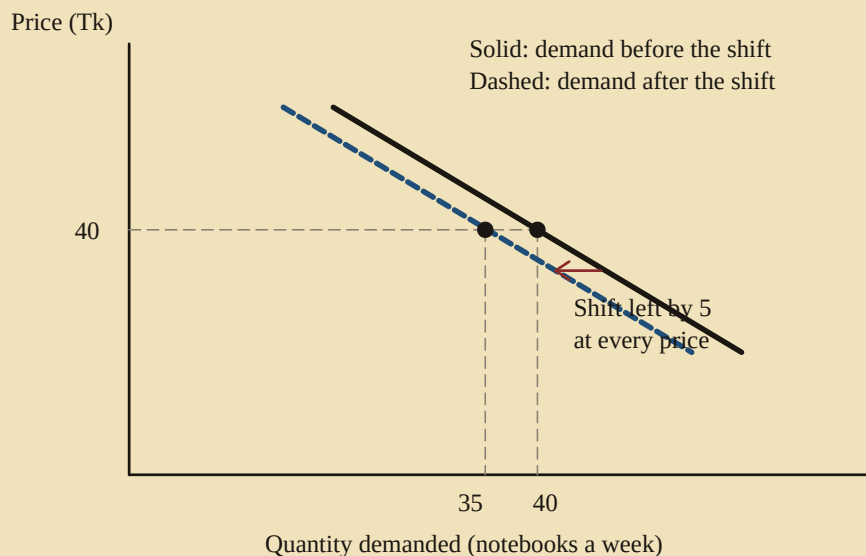


FIG. 4.2. The demand curve after a shift to the left, caused by a cheaper rival notebook

Read the picture along a horizontal line at Tk 40. The old curve meets this line at 40 notebooks. The new curve meets it at 35. The horizontal gap between the two curves is the same at every price: 5 notebooks. Prices on the vertical axis did not change, so the shift is not a movement along either curve.

PAUSE AND CHECK 4.P2 Meghna and Sons raises the price of the notebook from Tk 40 to Tk 50. Is this a movement along the curve or a shift? Give one reason.

Before moving on. Ask one question: what changed? If the price of the good itself changed, there is a movement along the curve. If a determinant changed, there is a shift of the whole curve. You should now be able to say, for any change, which of the two it is.

Read each row as a change of condition and the resulting change of demand.

GO SLOWLY HERE.

A change in a determinant shifts the whole curve.

The case file records a shift caused by a rival's price.

The new curve lies to the left of the old curve.

Both curves slope downward, and the gap between them is 5 notebooks.

A short test separates the two changes.

Some changes in demand come from the price of another good.

Substitute goods. Goods that can replace each other, so a price rise for one raises demand for the other.

Substitute goods can replace each other.

Complementary goods. Goods used together, so a price rise for one lowers demand for the other.

Complementary goods are used together.

Read each row as a price change for the first good and its effect on the second.

Normal good. A good for which demand rises when buyers' incomes rise.

A normal good is bought more when incomes rise.

Normal. Word in use. In daily life, normal means usual or ordinary. In economics, a normal good is one whose demand rises with income.

The economic word normal has a narrow meaning.

Giffen good. A rare, very basic good for which quantity demanded rises when its price rises.

A Giffen good breaks the law of demand, but it is rare.

4.3 RELATED AND UNUSUAL GOODS

A shift can come from a change in the price of another good, and the effect depends on how the two goods are related. The rival notebook in the case file is the first kind of relation. A shopper who finds the rival cheaper may switch to it, and our notebook loses buyers. The second kind of relation works the other way. We look at both kinds now, then at income, and then at four unusual cases.

Two notebooks of the same size and quality are **substitute goods**. Substitute goods are goods that can replace each other, so a price rise for one raises demand for the other. If the rival notebook becomes dearer, some buyers move to ours, and demand for ours shifts to the right. The case file showed the reverse: a price fall for the rival shifted our demand to the left. Tea and coffee, or buses and trains, are other pairs.

Some goods are not rivals but partners. **Complementary goods** are goods used together, so a price rise for one lowers demand for the other. A pen and a notebook are complementary goods. If pens become dearer, fewer people write, and fewer notebooks are wanted at every price. Tea and sugar, and a lamp and its oil, are other pairs. Substitutes move in opposite directions, and complements move in the same direction, as the table below summarises in words first.

HOW A PRICE RISE FOR ONE GOOD CHANGES DEMAND FOR ANOTHER

RELATION	EXAMPLE PAIR	PRICE OF FIRST GOOD	DEMAND FOR SECOND GOOD
Substitute goods	Our notebook and a rival notebook	Rises	Rises
Complementary goods	Pen and notebook	Rises	Falls

To read the second row, take the pair of pen and notebook. When the price of the pen rises, demand for the notebook falls. Be careful with direction: the effect on the second good is a shift of its curve, not a movement along it.

Income is another determinant. A **normal good** is a good for which demand rises when buyers' incomes rise. Notebooks are a normal good, because families with more money buy more of them. Most goods behave in this way. A few goods behave in the opposite way, and demand for them falls as incomes rise, because buyers move to better goods; we do not name them here.

The word needs care. A normal good is not a usual good, nor an average one. It is any good whose demand rises when buyers' incomes rise. A luxury can be a normal good, and so can a basic good.

The law of demand has a famous exception. A **Giffen good** is a rare, very basic good for which quantity demanded rises when its price rises. Alfred Marshall, an English economist, wrote in 1890 that Sir Robert Giffen, a Victorian statistician, had noticed such behaviour among very poor households that spent most of their money on bread. When bread became dearer, the argument runs, they could no longer afford meat, and so they bought more bread. Whether this really happened is debated, and clear modern cases are few.

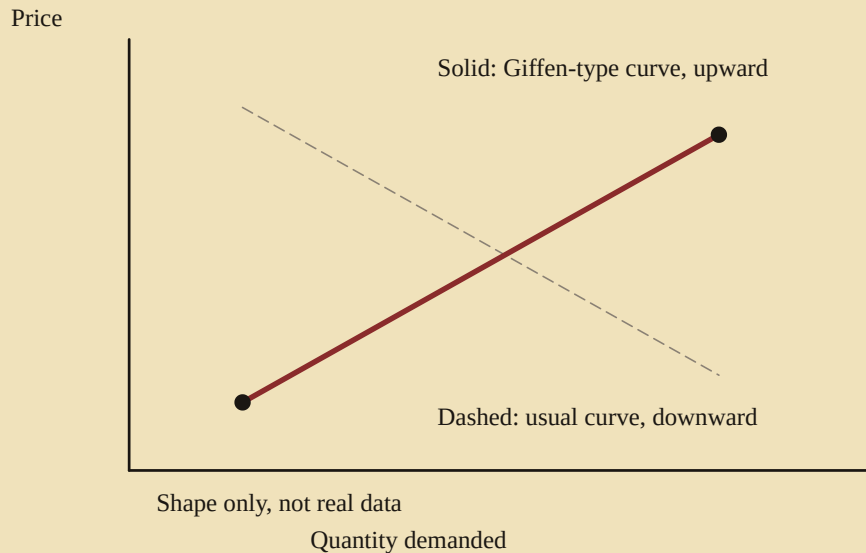


FIG. 4.3. A Giffen-type curve slopes upward, in contrast with the usual curve

The picture contrasts the usual curve, which slopes downward, with a Giffen-type curve, which slopes upward. On the upward line, a higher price goes with a larger quantity demanded. The numbers on this picture are not real data; it shows only the shape. Note one fact that keeps the law of demand valuable: the Giffen good is a rare, special case, and the law holds for almost every good that you will meet.

Two further cases depend on what other buyers do. The **snob effect** is a fall in demand for a good because many other people now buy it. A buyer who values owning something rare may leave a good once it becomes common. A scarf that a few people wore as a mark of taste loses its appeal when every street shows it.

The **bandwagon effect** is the opposite: a rise in demand for a good because many other people buy it. A mobile phone that all of a buyer's friends use becomes more attractive to that buyer, because it is easier to share messages and games. Both effects shift the curve, since they come from other buyers' choices and not from the good's own price.

Before moving on. Substitute goods replace each other, and complementary goods are used together. A normal good is bought more as incomes rise. A Giffen good is a rare exception to the law of demand, and the snob and bandwagon effects show that the choices of other buyers can shift demand. You should now be able to name one pair of each kind.

WORKED CASE: USING THE NOTEBOOK DEMAND SCHEDULE

The case file gives a rule: the quantity demanded equals 80 minus the price in taka. In symbols, $Q = 80 - P$, where Q is the quantity demanded in notebooks a week and P is the price in taka. Rearranged, $P = 80 - Q$.⁴

Example 1. *Given:* $Q = 80 - P$. *Find:* the quantity demanded when the price is Tk 45. *Formula:* quantity demanded equals 80 minus the price. *Steps:* $Q = 80 - 45 = 35$ notebooks a week. *Answer:* at Tk 45, buyers choose 35 notebooks a week. *Check:* 35 lies between 30 (at Tk

The Giffen-type curve slopes upward, against the usual downward slope.

Snob effect. A fall in demand for a good because many other people now buy it.

The snob effect reduces demand when many others buy.

Bandwagon effect. A rise in demand for a good because many other people buy it.

The bandwagon effect raises demand when many others buy.

The section closes with the pairs and the exceptions.

Two links let us calculate any quantity or price from the demand rule.

The first example is a clean calculation of a quantity.

⁴ Appendix A1 'Maths You Will Use', section A1.3 'Rearranging an Equation', shows how to move a term across the equals sign.

The second example finds a price and then follows a shift.

Classification of a change as a movement or a shift is a matter of asking what changed.

Four common slips spoil these answers.

The chapter's main ideas in a few lines.

Two problems that apply the chapter to new settings.

A shift is often confused with a movement.

Demand is not the same as quantity demanded.

50) and 40 (at Tk 40) in the table, as it should for a price between Tk 50 and Tk 40.

Example 2. *Given:* the old rule $Q = 80 - P$, and the new rule after the rival's price fall, $Q = 75 - P$. *Find:* the price at which buyers choose 25 notebooks under each rule, and the change in the quantity demanded at Tk 40. *Formula:* $P = 80 - Q$ for the old rule, and $P = 75 - Q$ for the new rule. *Steps:* old rule, $P = 80 - 25 = \text{Tk } 55$; new rule, $P = 75 - 25 = \text{Tk } 50$; at Tk 40 the old quantity is $80 - 40 = 40$ and the new quantity is $75 - 40 = 35$, a change of $40 - 35 = 5$. *Answer:* 25 notebooks are chosen at Tk 55 before the shift and at Tk 50 after it, and at Tk 40 the quantity demanded is 5 notebooks lower. *Check:* the shift is 5 at every price, as the case file states.

Worked classification. *Situation:* three events occur in one month. The shop lowers its notebook price from Tk 50 to Tk 40. A rival lowers its price by Tk 10. A new college opens nearby. *Question:* which events are movements and which are shifts? *Reasoning:* in the first event, the price of the good itself changed. In the second, the price of a related good changed. In the third, the number of buyers changed. *Conclusion:* the first is a movement along the curve; the second and third are shifts. *Check:* only the first event changes the price on the vertical axis for our own notebook.

Common slips. Students often call a price change a shift. They mix the sides of the formula, using $Q = 80 - P$ when they need $P = 80 - Q$. They forget units, and they treat a fall in demand as a fall in the quantity demanded at only one price.

SUMMARY

Demand is the set of quantities buyers are willing and able to buy at each price, and a quantity demanded is one entry in that set. The law of demand says that, other conditions staying the same, a lower price brings a larger quantity demanded, so the demand curve slopes downward. A change in the price of the good is a movement along the curve. A change in a determinant is a shift of the curve. Substitute and complementary goods, normal goods, and the Giffen, snob and bandwagon cases show how demand depends on other goods, incomes and other buyers.

THINK IT THROUGH

4.T1 A jeweller finds that a certain luxury ring sells more when the price rises. Give two reasons why a buyer might behave in this way, and say whether the case breaks the law of demand.

4.T2 A bus fare rises, and at the same time the price of petrol falls. For bus travel, say which change is a movement along the curve and which is a shift, and give the direction of the shift.

WHERE PEOPLE GO WRONG

Error 1: "A fall in the price of the good shifts its curve." This is wrong. A fall in the price of the good moves the point along the same curve. Only a change in a determinant shifts the curve.

Error 2: "Demand and quantity demanded mean the same." This is wrong. Quantity demanded

is one number at one price. Demand alone is the whole list of prices and quantities.

Error 3: "Sales rose after the price rose, so the law of demand fails." This may be wrong. Incomes, tastes or the price of a rival may also have changed, and these shift the curve. The law speaks only of the effect of the good's own price when other conditions stay the same.

The law needs other things to stay equal.

Error 4: "A buyer who wishes for a good demands it." This is wrong. Demand needs both the will and the ability to pay.

Wanting is not the same as demanding.

IN HISTORY

In 1838, the French mathematician Antoine Augustin Cournot, who lived from 1801 to 1877, published *Recherches sur les principes mathématiques de la théorie des richesses*, which is known in English as *Researches into the Mathematical Principles of the Theory of Wealth*. In its fourth chapter, titled in English "Of the law of demand", Cournot wrote demand as a function of price: the quantity sold depends on the price. He treated it as a fact of experience that the function falls as the price rises, and he drew it as a curve with price and quantity as its two axes.

A French mathematician drew the first demand curve.

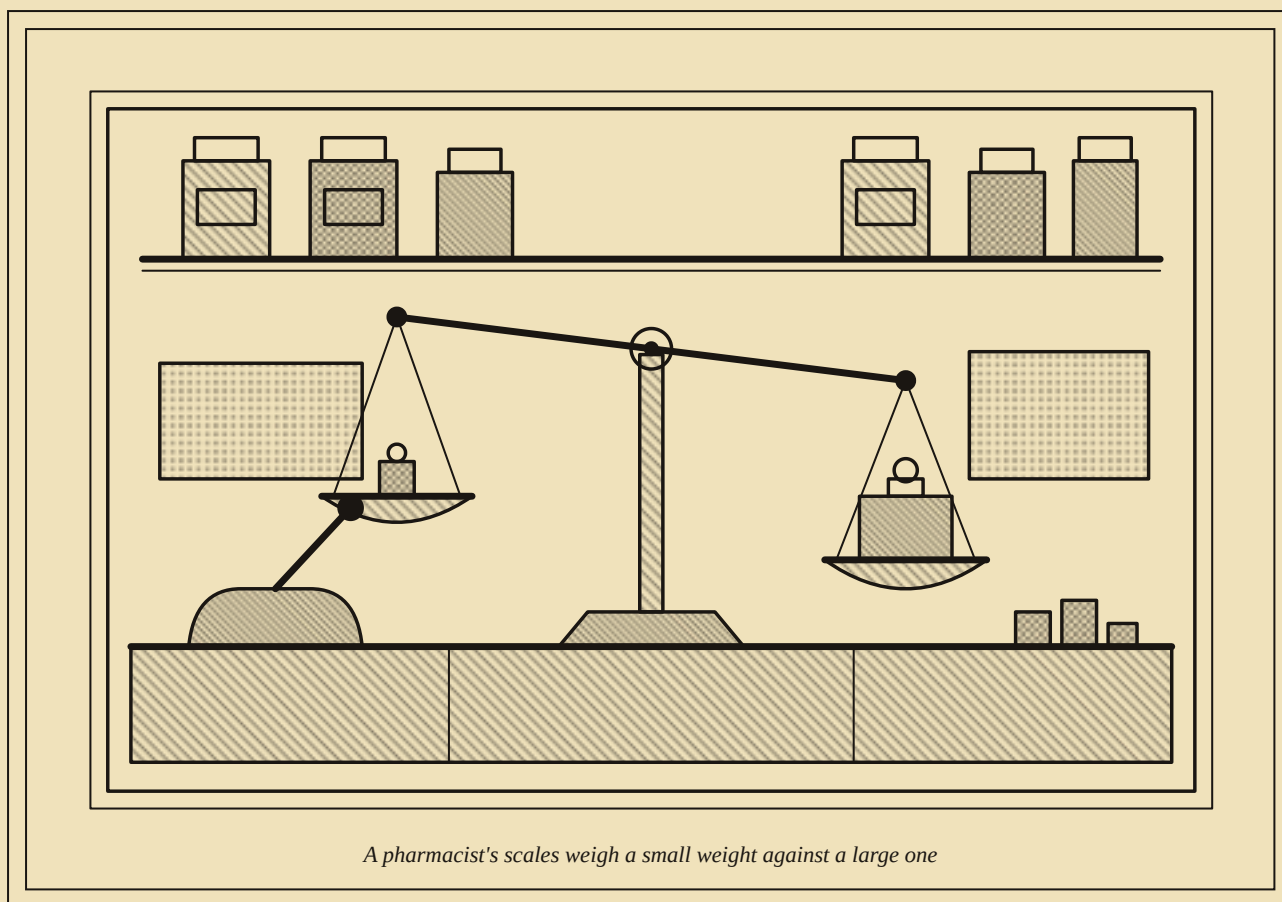
Cournot's book received hardly any response when it appeared, and the reading public of the 1830s had little interest in economics written with algebra and diagrams. Later economists, among them Léon Walras of Lausanne, drew on his ideas. Today every textbook, including this one, draws the demand curve in the way that Cournot first did.

The work was ignored for years, but its method became standard.

REVIEW QUESTIONS

- 4.1 Plot the demand schedule of the notebook from the case file on a graph, with price on the vertical axis. Say what the slope of the line shows.
- 4.2 After the rival's price fall, the quantities demanded at Tk 60, 50, 40, 30 and 20 are 15, 25, 35, 45 and 55. Plot the new curve on the same graph as the old one. Then draw a third curve that shows a rise in demand of 10 at every price, and give its quantity at Tk 40.
- 4.3 Classify each change as a movement along the curve or a shift of the curve: (a) the notebook price falls from Tk 60 to Tk 50; (b) pens become cheaper; (c) incomes of students fall; (d) the shop lowers the notebook price from Tk 40 to Tk 30.
- 4.4 Using $Q = 80 - P$, find the quantity demanded at Tk 35 and the price at which the quantity demanded is 55.
- 4.5 Name a pair of substitute goods and a pair of complementary goods. For each pair, say how a price rise for the first good changes demand for the second.
- 4.6 Define a Giffen good, and say why it is not the usual case.
- 4.7 A new college opens near the shop, and the price of the notebook is unchanged. Say what happens to demand, and explain which determinant has changed.

Seven questions that check the chapter's ideas.



CHAPTER V

Elasticity of Demand

How hard does a change in price push the consumer?



This chapter teaches you to measure how strongly consumers respond to a change in price, in their own income and in the price of a rival good, and to use the result to judge whether a lower price brings a firm more money.

YOU WILL NEED

Chapter 4, 'Demand and the Demand Curve', section 4.1 'Quantity Demanded and the Law of Demand', for demand and its curve; Chapter 4, section 4.2 'Movement and Shift', for the difference between a movement along a curve and a shift; and Appendix A1, 'Maths You Will Use', for percentages and fractions.

Two goods lose a fifth of their price, yet they sell very differently.

The law of demand gives the direction of the change but not its size.

A price cut brings more customers but less money from each one.

Picture a street in Dhanmondi, Dhaka, on a Friday evening. One stall sells packets of salt. Beside it, a shop sells fashionable shoes. On the same day, both owners cut their prices by one-fifth. The salt seller waits all evening for a rush of customers, but it does not come. Families buy the same few packets they always buy, because a household cannot use more salt merely because salt is cheaper. The shoe seller finds a queue at the door by seven o'clock.

The law of demand, which you met in Chapter 4, says that a lower price raises the quantity demanded. It holds in both shops. Yet it says nothing about how much the quantity demanded rises. Salt rose by a trifle; shoes rose by a great deal. For an owner who must decide whether to cut a price, the direction is not enough. The owner needs a number that tells how strongly consumers respond.

There is a second puzzle behind the first. A lower price attracts more customers, but each customer pays less. Whether the shop finishes the day with more money received or with less depends on which effect is larger, and that depends on the same strength of response. This chapter builds the number. Section 5.1 defines and calculates it. Section 5.2 teaches you to read it. Section 5.3 explains what makes it large or small. Section 5.4 extends the idea to income and to the price of a rival good.

* * *

5.1 PRICE ELASTICITY OF DEMAND

WHY WE NEED PERCENTAGES

Begin with a household question. Suppose your monthly rent rises by Tk 500, and in the same month the price of a pen rises by Tk 5. Both rises look small, yet they are very different. Tk 500 added to a rent of Tk 20,000 is a rise of one-fortieth. Tk 5 added to a pen that costs Tk 10 is a rise of one-half. Only the change measured against the starting level shows how large the change is. For this reason economists measure changes in percentages.

Quantity has the same difficulty. Ten extra notebooks in a week is a large rise for a shop that sells thirty and a tiny rise for a shop that sells three thousand. A measure of response must therefore compare a percentage change in quantity with a percentage change in price. Such a measure has no units. It can be compared across goods, shops and countries.

GO SLOWLY HERE.

Money amounts and unit counts hide the real size of a change.

Response. Word in use. In daily life, an answer or a reply. In economics, the change in quantity demanded that follows a change in price or in another cause.

Quantity needs the same treatment: ten more units mean different things to different shops.

This measure has a name. **Price elasticity of demand** means the percentage change in quantity demanded divided by the percentage change in price. Suppose a price rises by 10% and the quantity demanded falls by 20%. The price elasticity of demand is 20 divided by 10, which is 2. The name comes from a picture: a rubber band is elastic when a small pull stretches it far, and quantity demanded is elastic when a small change in price stretches it far.¹

Before moving on. Changes in money and in units cannot be compared, so we use percentages. Price elasticity of demand divides the percentage change in quantity demanded by the percentage change in price. A result of 2 means quantity changes twice as fast as price.

FINDING A PERCENTAGE CHANGE

You need one tool before you can calculate. In words, the percentage change equals the change in the value, divided by the value you compare it with, multiplied by 100. In symbols, percentage change = (new value - old value) ÷ old value × 100. This ordinary method works well for small changes. It has one weakness, which many students find confusing, because the answer depends on the direction of travel.

Test it with the price of a notebook. A price rising from Tk 40 to Tk 50 rises by Tk 10, and 10 divided by 40 is 25%. A price falling from Tk 50 to Tk 40 falls by the same Tk 10, but 10 divided by 50 is 20%. One gap between the same two prices has produced two percentages. If we used this method, the elasticity between Tk 40 and Tk 50 would depend on which price we chose as the start, and two people measuring the same demand schedule could report different numbers.

The remedy is the midpoint method, which divides the change by the midpoint, the average of the old and new values. In words: percentage change = change ÷ average of the old and new values × 100. In symbols, with old value A and new value B, percentage change = $(B - A) \div ((A + B) \div 2) \times 100$. The average is the same whichever value you call old, so the answer is the same in both directions. Between Tk 40 and Tk 50 the average is Tk 45, the change is Tk 10, and $10 \div 45$ gives 22.22% whether the price rises or falls.²

Before moving on. The ordinary percentage change depends on the starting point. The midpoint method removes that problem by dividing the change by the average of the old and new values. Every elasticity in this chapter uses it.

THE FORMULA AND A FIRST EXAMPLE

Now the whole formula can be stated. In words: price elasticity of demand = percentage change in quantity demanded ÷ percentage change in price, each found by the midpoint method. In symbols, $E = \% \Delta Q \div \% \Delta P$. Here E stands for price elasticity of demand, $\% \Delta Q$ for the percentage change in quantity

Elastic. Word in use. In daily life, able to stretch and then return, like a rubber band. In economics, the quantity demanded stretches a long way when the price changes.

Price elasticity of demand. The percentage change in quantity demanded divided by the percentage change in price.

Price elasticity of demand is the percentage change in quantity demanded divided by the percentage change in price.

Before moving on: the measure compares two percentages and has no units.

A percentage change is the change divided by a base value, times 100.

The ordinary method gives different answers going up and going down.

The midpoint method divides each change by the average of the old and new values.

Before moving on: use the average of old and new values as the base.

The whole formula uses two midpoint percentages and one division.

GO SLOWLY HERE.

¹ By the law of demand, price and quantity demanded move in opposite directions, so the raw ratio is negative: -20 divided by +10 gives -2. Economists usually drop the minus sign and report the size of the number, ignoring its sign. This book does the same, and the sign is always understood to be there.

² Some books divide by the starting value and accept a small difference. The midpoint method is used here because it gives one answer for each pair of points. If another book uses the starting value, expect slightly different results when the changes are large.

demanded and $\% \Delta P$ for the percentage change in price. The Greek capital letter Δ (delta) means 'change in'.

IN SHORT

Price elasticity of demand measures how strongly quantity demanded responds to price. Find each percentage change by the midpoint method, divide the quantity figure by the price figure, and report the size of the result.

The first example uses the smallest clean numbers, so only the method is new.

Each line of arithmetic is shown, with its reason.

You should now be able to turn a price change and a quantity change into one number.

The notebook schedule gives a realistic example with less tidy numbers.

Recall: demand schedule, Chapter 4, section 4.1 'Quantity Demanded and the Law of Demand'.

The averages 35 and 45 come from the quantities and the prices.

Five slips account for most wrong answers.

Before moving on: the notebook example gives 1.29.

Our first example uses clean numbers, so that the method is the only new thing. It is an everyday setting, not the running case. *Given:* a tea stall raises its price per cup from Tk 9 to Tk 11, and the quantity demanded falls from 24 cups to 16 cups an hour. *Find:* the price elasticity of demand. *Formula:* $E = \% \Delta Q \div \% \Delta P$, with midpoint percentages.

Steps. First, the quantity. The change is $16 - 24 = -8$ cups. The average is $(24 + 16) \div 2 = 20$ cups. So $\% \Delta Q = -8 \div 20 \times 100 = -40\%$. Second, the price. The change is $11 - 9 = \text{Tk } 2$. The average is $(9 + 11) \div 2 = \text{Tk } 10$. So $\% \Delta P = 2 \div 10 \times 100 = +20\%$. Third, divide: $E = -40 \div +20 = -2$, and the size is 2. *Answer:* the price elasticity of demand is 2, so quantity demanded changes twice as fast as price. *Check:* a 20% rise in price led to a 40% fall in quantity, and 40 is larger than 20, so a result above 1 is sensible.

PAUSE AND CHECK 5.P1 A good's price rises from Tk 8 to Tk 12, and the quantity demanded falls from 30 to 20. Find $\% \Delta P$, $\% \Delta Q$ and E .

You should now be able to say how a change in price and a change in quantity become one number: find each percentage change from the average of its old and new values, then divide quantity by price.

A REALISTIC EXAMPLE: THE NOTEBOOK

Case file, illustrative. Meghna and Sons, a bookshop in Dhanmondi, Dhaka, records a demand schedule for its popular 200-page notebook by the rule quantity = 80 - price, with price in taka. The price falls from Tk 50 to Tk 40, and the quantity demanded rises from 30 to 40. By the midpoint method, the percentage change in quantity is $10 / 35 = 28.57\%$, and the percentage change in price is $10 / 45 = 22.22\%$. The price elasticity of demand is 1.29.

Follow the arithmetic slowly. The two quantities are 30 and 40, so their average is 35, and the change of 10 gives $10 \div 35 = 28.57\%$. The two prices are 50 and 40, so their average is 45, and the change of 10 gives $10 \div 45 = 22.22\%$. Dividing, $28.57 \div 22.22 = 1.29$. *Check:* quantity moved by a larger percentage than price, so the result must lie above 1, and it does.

Common slips are few. Students divide price by quantity instead of quantity by price. They divide by the starting value in one place and by the average in another. They mix a figure in taka with a figure in units. They forget the units of time, comparing a weekly figure with a monthly one. And they report a negative sign in one answer and drop it in the next. Decide on the order and the sign rule before you begin.

Before moving on. The notebook's price elasticity of demand between Tk 50 and Tk 40 is 1.29. Always divide quantity by price, use midpoints for both, and keep the time period the same.

* * *

5.2 READING THE RESULT

A number such as 1.29 is useful only when you know what it says. The dividing line is 1, because 1 is the case where the two percentages are equal. Three cases follow, each with a name.

When the result is above 1, demand is **elastic demand**. It means demand for which the percentage change in quantity is larger than the percentage change in price. The notebook, at 1.29, is an example. The tea stall at 2 is another. The curve for such a good is flat, as the first figure below shows.

When the result is below 1, demand is **inelastic demand**. It means demand for which the percentage change in quantity is smaller than the percentage change in price. Salt in our opening puzzle is a likely example: a price cut of one-fifth brings only a trifling rise in the quantity demanded. The curve for such a good is steep.

When the result equals 1, demand is **unit elastic demand**. It means demand for which the percentage change in quantity equals the percentage change in price. Your pause question, with a result of 1, is an example. It is the dividing line between the other two cases.

To read the figure below, notice what the axes show. The vertical axis shows price, and the horizontal axis shows quantity demanded. Each point on a curve is one price with the quantity demanded at that price. Starting from the same point, a price fall of the same size moves the flat curve a long way to the right and the steep curve only a short way.

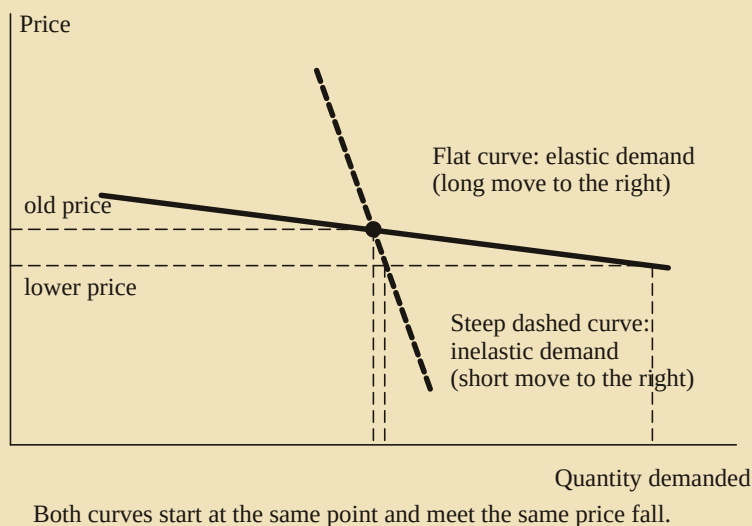


FIG. 5.1. Two demand curves through the same point: a flat curve for elastic demand and a steep curve for inelastic demand

A warning follows. Elasticity is not the same as the slope of the curve. Along one straight demand curve, the slope never changes, yet the elasticity does. On the notebook schedule, the fall from Tk 60 to Tk 50 gives 2.2, the fall from Tk 50 to Tk 40 gives 1.29, and the fall from Tk 40 to Tk 30 gives 0.78. A single good can be elastic at high prices and inelastic at low prices.

A single number is useful only if you know what it says.

Elastic demand. Demand for which the percentage change in quantity is larger than the percentage change in price.

Elastic demand has a result above 1: consumers respond strongly.

Inelastic demand. Demand for which the percentage change in quantity is smaller than the percentage change in price.

Inelastic demand has a result below 1: consumers respond weakly.

Unit elastic demand. Demand for which the percentage change in quantity equals the percentage change in price.

Unit elastic demand has a result of exactly 1: the two percentages are equal.

The axes show price and quantity demanded; each point is one pair.

Slope and elasticity are different things, as a straight line shows.

The three cases are compared in one table.

The table below compares the three cases. Read each row from left to right: the first column names the case, the second gives the size of the result, the third says which percentage is larger, and the fourth anticipates the test described next.

THE THREE CASES OF PRICE ELASTICITY OF DEMAND

CASE	SIZE OF RESULT	LARGER PERCENTAGE	EFFECT OF A PRICE CUT ON MONEY RECEIVED
Elastic demand	Above 1	Quantity	Rises
Unit elastic demand	Equal to 1	Neither	Unchanged
Inelastic demand	Below 1	Price	Falls

Total revenue. Price multiplied by quantity sold.

Total revenue is price times quantity, the money a firm receives from sales.

The total revenue test links the size of the result to the change in revenue.

The notebook's total revenue rises when its price falls.

Along one schedule, revenue rises and then falls as the price keeps falling.

Why does the result matter to a firm? Because of **total revenue**, which means price multiplied by quantity sold. If a stall sells 10 cups at Tk 5, its total revenue is Tk 50. A price cut lowers the price paid for every unit but raises the number of units sold. Which effect wins depends on the elasticity.

The total revenue test says: if demand is elastic, a fall in price raises total revenue; if demand is unit elastic, total revenue does not change; if demand is inelastic, a fall in price lowers total revenue. A rise in price gives the opposite results. The notebook confirms the first rule, as the next record shows.

Case file, illustrative. When the notebook's price falls from Tk 50 to Tk 40, total revenue rises from Tk 1,500 (Tk 50 $\times$ 30) to Tk 1,600 (Tk 40 $\times$ 40). The price elasticity of demand over this fall is 1.29, so demand is elastic.

To read the next figure, look at the table on its left and the graph on its right. In the table, each row gives a price, the quantity demanded from the rule quantity = 80 - price, and total revenue. In the graph, the horizontal axis shows price, the vertical axis shows total revenue, and each point is one row of the table. Total revenue climbs as the price falls from Tk 60 to Tk 40, and then drops. The peak lies where demand changes from elastic to inelastic.

Price	Quantity	Total revenue
Tk 60	20	Tk 1,200
Tk 50	30	Tk 1,500
Tk 40	40	Tk 1,600
Tk 30	50	Tk 1,500
Tk 20	60	Tk 1,200

Quantity = 80 - price. Boxed row: the peak.

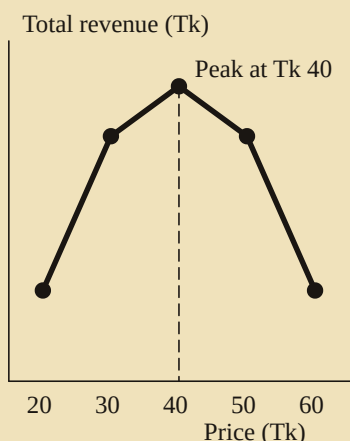


FIG. 5.2. The total revenue test: a table of prices, quantities and total revenue beside a graph of total revenue against price

Before moving on. A result above 1 is elastic demand, equal to 1 is unit elastic demand, and below 1 is inelastic demand. Total revenue is price times quantity. When demand is elastic a price cut raises it, and when demand is inelastic a price cut lowers it. The same good may be elastic in one price range and inelastic in another.

* * *

5.3 FACTORS BEHIND ELASTICITY

Why is salt inelastic and a pair of shoes elastic? Three features of a good explain most cases. None of them gives a number; each tells you which side of 1 to expect.

The first feature is the availability of substitutes, which are other goods that can be used for the same purpose. A consumer who finds the price of one brand of tea too high can buy another brand, so demand for one brand is elastic. Demand for tea in general has fewer substitutes, so it is less elastic. The more narrowly a good is described, the more substitutes it has.

The second feature is the difference between a necessity, which a household buys whatever the price, and a luxury, which it can easily do without. Salt, rice and basic medicines are necessities. Fashionable shoes and restaurant meals are luxuries. A word of caution: the same good can be a necessity for one household and a luxury for another, so this is a tendency, not a rule.

The third feature has two parts. Time: consumers need time to find substitutes and change habits, so demand is usually less elastic soon after a price change than long after. Share of income: a good that takes a large part of a household's income, such as rent, makes consumers watch its price closely, while a pen takes so little that its price rarely matters. Both parts shift the result toward elastic as time passes or the share rises.

Before moving on. Demand tends to be more elastic when close substitutes exist, when the good is a luxury, when consumers have time to adjust, and when the good takes a large share of income. Each is a guide to which side of 1 to expect.

* * *

5.4 INCOME AND CROSS ELASTICITY

The method is not limited to the good's own price. Quantity demanded also responds to consumers' income and to the price of other goods. Each response has its own measure, found in the same way.

Income elasticity means the percentage change in quantity demanded divided by the percentage change in consumers' income. A change in income shifts the demand curve rather than moving along it, which is why this measure is separate from the price measure. The sign matters here and is kept. A positive result means that more income raises the quantity demanded; a negative result means that it lowers it.

Case file, illustrative. A rise of 10% in income lifts the quantity of fiction demanded by 15%. Income elasticity = $15\% \div 10\% = 1.5$.

Before moving on: the result is read against 1, and the revenue test follows from it.

Three questions help predict whether demand is elastic or inelastic.

Many close substitutes make demand elastic.

Necessities tend to be inelastic and luxuries elastic.

Time and the share of income also matter.

Before moving on: substitutes, necessity and luxury, time and share of income.

The same idea can measure responses to causes other than the good's own price.

Recall: movement and shift, Chapter 4, section 4.2 'Movement and Shift'.

Income elasticity. The percentage change in quantity demanded divided by the percentage change in consumers' income.

Income elasticity compares the response of quantity to a change in income.

A 10% rise in income lifts fiction by 15%.

Cross elasticity. The percentage change in quantity demanded of one good divided by the percentage change in the price of another.

Cross elasticity links the quantity of one good to the price of another.

A 10% dearer rival notebook lifts the shop's notebook by 4%.

Before moving on: income elasticity and cross elasticity use the same ratio.

Cross elasticity means the percentage change in quantity demanded of one good divided by the percentage change in the price of another. A positive result means the goods are substitutes, since a dearer rival sends consumers to the good. A negative result means the goods are used together, so a dearer partner reduces the use of both.

Case file, illustrative. A rise of 10% in the price of a rival notebook raises the quantity demanded of the shop's notebook by 4%. Cross elasticity = $4\% \div 10\% = 0.4$.

Before moving on. Each measure divides a percentage change in quantity demanded by a percentage change in its cause. Income elasticity uses consumers' income; cross elasticity uses the price of another good. Their signs carry meaning.

* * *

WORKED EXAMPLES: CHANGING THE SETTING

Two further examples change the period, the sign and the setting.

Rung 3: the periods differ, so one figure must be converted before dividing.

The steps convert first, then calculate.

Rung 4: a bus fare combines schedule reading, elasticity and the revenue test.

The first two examples are above: the tea stall and the notebook. The next two reshape the problem. Both follow the same order: Given, Find, Formula, Steps, Answer, Check.

Given: a rice seller raises the price from Tk 80 to Tk 100 a kilogram. At Tk 80, households buy 30 kg a week; at Tk 100, they buy 100 kg a month. Take a month as four weeks. *Find:* the price elasticity of demand. *Formula:* $E = \% \Delta Q \div \% \Delta P$, with midpoint percentages. The two quantities are in different periods, and this is the slip to avoid.

Steps. Convert 100 kg a month to $100 \div 4 = 25$ kg a week. Quantity: change = $25 - 30 = -5$; average = 27.5; $\% \Delta Q = -5 \div 27.5 \times 100 = -18.18\%$. Price: change = $+20$; average = 90; $\% \Delta P = 20 \div 90 \times 100 = +22.22\%$. Divide: $18.18 \div 22.22 = 0.82$. *Answer:* $E = 0.82$, so demand is inelastic. *Check:* total revenue per week is Tk $80 \times 30 =$ Tk 2,400 before and Tk $100 \times 25 =$ Tk 2,500 after. A price rise with inelastic demand raises revenue, and it did.

Given: a bus company's schedule shows 800 passengers a day at a fare of Tk 10, and 700 at a fare of Tk 12. The company asks whether raising the fare from Tk 10 to Tk 12 will increase its total revenue. *Find:* E and the change in total revenue. *Steps.* Quantity: change = -100 ; average = 750; $\% \Delta Q = -13.33\%$. Price: change = $+2$; average = 11; $\% \Delta P = +18.18\%$. $E = 13.33 \div 18.18 = 0.73$. Revenue: Tk $10 \times 800 =$ Tk 8,000 before, Tk $12 \times 700 =$ Tk 8,400 after. *Answer:* demand is inelastic and revenue rises by Tk 400. *Check:* the elasticity test and the direct multiplication agree, so the working is sound.

* * *

SUMMARY

The chapter's main results in a few lines.

Price elasticity of demand is the percentage change in quantity demanded divided by the percentage change in price, found with midpoints. Above 1 is elastic demand, equal to 1 is unit elastic demand, below 1 is inelastic demand. The total revenue test links the result to the effect of a price change on money received. Substitutes, necessity, time and share of income guide prediction. Income elasticity and cross elasticity apply the same ratio.

THINK IT THROUGH

5.T1 A pharmacy sells a medicine with no substitute, and its price elasticity of demand is 0.3. The owner raises the price by 10%. Estimate the change in quantity demanded and say what happens to total revenue.

5.T2 Explain why a firm whose demand is inelastic might still hesitate to raise its price a great deal. Think of time and substitutes.

WHERE PEOPLE GO WRONG

Many students believe that a steep curve and an inelastic good are the same thing, and that a flat curve always means elastic. Slope depends on the units on the axes; elasticity compares percentages. Along a straight line the slope is constant while elasticity changes from point to point. Students also believe that a negative result means demand is inelastic. The sign shows only that price and quantity move in opposite directions; the size alone tells you which case you have. A third error is to call a good elastic or inelastic everywhere; the result belongs to a price range, a period and a group of consumers. The last error is to use the total revenue test for a change in income. The test applies to the good's own price only.

Four common misunderstandings, each corrected.

IN HISTORY

In October 1973 the Yom Kippur War, which began on 6 October between Israel and Egypt and Syria, led Arab members of the Organization of the Petroleum Exporting Countries to announce cuts in production and an embargo on oil shipments to the United States and some other countries. The embargo lasted until March 1974. During this time the price of crude oil rose from about US\$3 a barrel to nearly US\$12, roughly four times.

In 1973 oil prices rose fourfold, yet demand fell slowly.

The rich countries did not respond with a matching fall in the quantity demanded. In the short run, households and firms could not change their cars, heating systems and machines, and there were few substitutes for petrol and heating oil. Demand was therefore strongly inelastic. In the years that followed, consumers bought smaller cars, firms improved their equipment, and new oil fields opened. Demand became more elastic as time passed, as section 5.3 would lead you to expect.

Consumers could not change quickly, so demand fell only slowly.

REVIEW QUESTIONS

5.1 Define price elasticity of demand in your own words, and say why percentages are used.

5.2 Why does the midpoint method give the same answer when a price rises as when it falls?

5.3 A price rises from Tk 20 to Tk 30 and the quantity demanded falls from 60 to 40. Calculate E and say whether demand is elastic, inelastic or unit elastic.

5.4 Using the notebook schedule, calculate E for the fall from Tk 40 to Tk 30, and apply the total revenue test.

5.5 A rise of 20% in income raises the quantity of mangoes demanded by 10%. Calculate income elasticity.

- 5.6 A rise of 5% in the price of tea lowers the quantity of sugar demanded by 3%. Calculate cross elasticity and say what the sign shows.
- 5.7 Give one reason why demand for a brand of tea is more elastic than demand for tea in general.



CHAPTER VI

Consumption and Consumers' Surplus

*Why does a consumer who pays the full price still leave
with more than she paid for?*



*This chapter explains how people use goods to satisfy their wants
and how they spend a limited income. It then shows that a
consumer often gains more from a good than the price paid, and
teaches you to measure that gain in taka.*

YOU WILL NEED

Chapter 3 'Utility and Diminishing Marginal Utility', section 3.1 'The Idea of Utility'; Chapter 4 'Demand and the Demand Curve', section 4.1 'Quantity Demanded and the Law of Demand'; Appendix A1 'Maths You Will Use'.

A woman pays Tk 60 for a water pot that she values at Tk 100.

The sale is recorded as Tk 60, yet she feels richer by Tk 40.

The puzzle: how can a gain exist that nobody pays for, and can it be measured?

On a market day in a small town, a woman passes a stall that sells water pots. Her old pot has cracked, so she needs a new one. In her own mind she has fixed a limit: she would pay Tk 100 for a good pot, and not one taka more. The stallholder asks Tk 60. She pays at once, takes the pot and walks home.

Now examine the exchange as an accountant would. The woman handed over Tk 60 and received a pot. The stallholder received Tk 60 and handed over a pot. Each side has a record, and the records agree. Yet the woman feels that she has done well, and she is right. The pot was worth Tk 100 to her, and she paid only Tk 60. Forty taka of satisfaction appears on no receipt, in no ledger and in no bank account.

Here is the puzzle. If trade is a fair exchange, how can one side gain something that nobody pays for? And if she gains, can we measure the gain? Many students find the question strange at first, because we are used to thinking that the price tells us what a good is worth. This chapter shows that the price tells us only part of the story. We begin with what consumers do. Then we measure what they gain.

* * *

6.1 CONSUMPTION AND CHOICE

Everyday acts such as eating or travelling use things that others provide.

Consumption. The use of goods and services by people to satisfy their wants.

Consumption is the use of goods and services to satisfy wants.

Buying is not consuming: the use may come later than the payment.

Only the final user consumes; a firm that uses a good to make other goods does not.

The consumer feels a want satisfied; the owner sees only a sale.

Think about your own day. You eat breakfast, travel by bus, read a message on your phone and drink water. Each act uses something that other people made or provided, and each act satisfies a want. Economics gives this ordinary activity a name, and the name lets us study it with care.

The word **consumption** means the use of goods and services by people to satisfy their wants. A student who eats a meal is engaged in consumption. So is a family that uses electricity to light a room. Here a *good* is a physical thing, such as a notebook, and a *service* is work done for a person, such as a haircut.

Two points prevent confusion. First, buying is not the same as consuming. A consumer may buy a notebook on Monday and use it over many weeks, filling one page after another. The consumption happens during the use, not at the moment of payment. A haircut is different, because the service is used as it is given.

Second, consumption is the final purpose of production. A firm that makes cloth does not consume the cloth. A tailor buys it to make shirts, and only the person who wears a shirt consumes it. A good used to make other goods is therefore not consumption in this sense.

Different people see the same act in different ways. The consumer feels a want satisfied. The owner of a bookshop sees only a sale and counts the money received. The shop records the price, but it never records the satisfaction in the mind of the person who buys. This difference between what a consumer feels and what a firm records is the key to the chapter.

Every consumer has an income, which is the money received in a period such as a month. Wants are many, but income is limited. A taka spent on one good cannot be spent again on another. So every act of consumption is also a choice: to buy one good is to decide not to buy something else with the same money.

How should a consumer choose? Chapter 3 showed that each extra unit of a good adds less utility than the one before it. This fact guides the choice. Suppose a student has Tk 100 left at the end of a month. The next Tk 20 spent on tea adds more utility than the next Tk 20 spent on a fourth magazine. A sensible student moves money from the magazine to the tea. The student stops when no further move would add utility.

The rule has limits. Real consumers do not calculate every purchase. Habit, advice from friends and the mood of the day all play a part. The idea of a sensible consumer describes the usual case, and it is not a picture of every person. It is still useful, because it explains why the amounts that consumers spend on different goods change when prices change.

Before moving on. Consumption is the use of goods and services to satisfy wants, and it happens when the good is used. Consumers have limited income, so they spend it where an extra taka adds the most utility. The price a consumer pays does not show the satisfaction received.

* * *

6.2 CONSUMERS' SURPLUS

Return to the woman with the pot. What is the source of her extra Tk 40? To answer, we first need a name for the limit she carried in her mind. Many students find this step confusing, because the limit is not written anywhere. It exists only as a decision that the consumer would make if the price rose.

Willingness to pay means the highest price a buyer would pay for one more unit of a good. For the woman, the willingness to pay for the pot was Tk 100. At any price above Tk 100 she would not buy. At Tk 100 or below she would buy.

Willingness to pay is closely tied to marginal utility, the extra utility from one more unit. The first pot is very useful to the woman. A second pot would add far less utility, so she would pay much less for it. Her willingness to pay therefore falls as she buys more units. This is the reason the demand curve slopes downward: each further unit is worth less to the same consumer.

Two cautions apply. First, willingness to pay is a limit in the mind, not a price on a tag. Nobody pays it unless the price rises to meet it. Second, it differs from person to person. A consumer with a larger income may be willing to pay more for the same good. Two consumers at the same stall can therefore hold different limits for the same pot.

PAUSE AND CHECK 6.P1 The price of a pen is Tk 12. A consumer's willingness to pay for one more pen is Tk 9. Will the consumer buy it? Explain.

Income is limited and each taka can be spent only once, so every purchase is a choice.

Spend. Word in use. In daily life, to pay out money, or to pass time. In economics, to use part of income to buy goods, so that the same money cannot buy anything else.

Recall: marginal utility, Chapter 3, section 3.1 'The Idea of Utility'.

A sensible consumer spends each taka where it adds the most utility.

Real consumers do not calculate every purchase, so the rule is a guide only.

Recap: consumption, limited income and the gap between price and satisfaction.

The woman's extra Tk 40 needs a name and a measure.

Willingness to pay. The highest price a buyer would pay for one more unit of a good.

Willingness to pay is the highest price a consumer would pay for one more unit.

Recall: marginal utility, Chapter 3, section 3.1; the demand curve, Chapter 4, section 4.1.

Willingness to pay is marginal utility measured in money, and it falls with each extra unit.

Willingness to pay is a limit in the mind, and it differs between consumers.

Consumers' surplus. The gap between what buyers are willing to pay for a good and what they actually pay.

Surplus. Word in use. In daily life, an amount left over after needs are met. In economics, the gain a consumer keeps when the price paid is below the most the consumer would pay.

Consumers' surplus is the gap between willingness to pay and the price paid.

Three of five customers buy the notebook at Tk 40, and their surplus totals Tk 30.

The customer sees a bargain, but the owner sees only a Tk 40 sale.

To measure surplus, subtract the price from willingness to pay for each unit bought, then sum.

Read the table row by row: the first buyer gains most and the last buyer gains least.

The surplus depends on the price: a lower price gives the same consumer a larger surplus.

With many customers the steps become a smooth line, and the surplus becomes a triangle.

Now the puzzle has an answer. **Consumers' surplus** means the gap between what buyers are willing to pay for a good and what they actually pay.¹ The woman's willingness to pay was Tk 100 and she paid Tk 60, so her consumers' surplus on the pot is Tk 100 minus Tk 60, which is Tk 40. The surplus is not money. It is satisfaction measured in taka, and it never appears on a receipt.

Case file, illustrative. Five regular customers of Meghna and Sons were recorded with these willingness to pay amounts for the popular 200-page notebook: Tk 60, Tk 50, Tk 40, Tk 30 and Tk 20. The market price is Tk 40. Three customers buy, because the price is at or below their willingness to pay. Consumers' surplus = $(60 - 40) + (50 - 40) + (40 - 40) = \text{Tk } 30$.

The same sale looks different from each side. The customer with a willingness to pay of Tk 60 sees a bargain of Tk 20. The owner sees a sale at Tk 40, the same as every other sale of the notebook that day. The owner cannot see the Tk 20, because it exists only in the mind of the customer.

The method has three steps. For each unit bought, subtract the price from the willingness to pay. Omit any unit whose willingness to pay is below the price, because the consumer does not buy it. Then sum the differences. The table applies the steps to the five customers of the notebook case. Its columns show the customer, the willingness to pay, whether the customer buys at Tk 40, and the surplus.

NOTEBOOK CASE AT THE PRICE OF TK 40

CUSTOMER	WILLINGNESS TO PAY (TK)	BUYS AT TK 40?	SURPLUS (TK)
A	60	Yes	20
B	50	Yes	10
C	40	Yes	0
D	30	No	0
E	20	No	0
Total			30

Read the row for Customer A first. A would pay Tk 60 and pays Tk 40, so A keeps Tk 20. Customer C buys but keeps nothing, because the willingness to pay equals the price. Customers D and E do not buy and gain no surplus. The first buyer has the largest surplus and the last buyer the smallest, which follows from a willingness to pay that falls with each unit.

One more point needs care. The surplus belongs to the consumer and the price together, not to the good alone. The same customer who keeps Tk 20 at a price of Tk 40 would keep Tk 30 at a price of Tk 30. A consumer whose willingness to pay equals the price still chooses to buy, but gains nothing. For this reason a surplus of zero does not mean that the good is useless to that consumer.

Now suppose the shop has thousands of customers instead of five. The steps in the table become so small that they form a smooth line, and that line is the demand curve. On the chart that follows, the vertical axis shows price in taka

¹ The idea is older than its name. The section 'In History', at the end of this chapter, tells how it began.

and the horizontal axis shows the quantity bought. Each point on the line shows the highest price at which consumers would still buy that quantity. The dashed line is the market price. The consumers' surplus is the hatched triangle under the demand curve and above the price line. The dotted rectangle below it is the money that consumers actually pay.

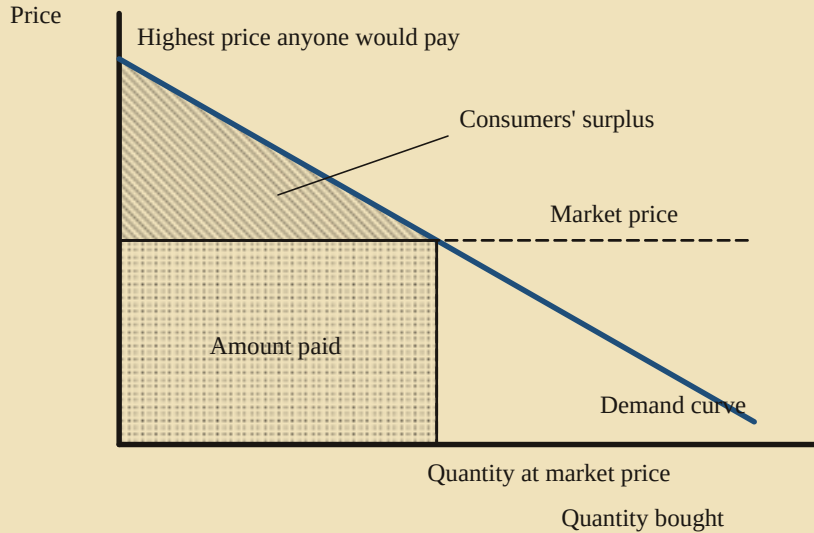


FIG. 6.1. Consumers' surplus is the triangle under the demand curve and above the price

When the demand curve is a straight line, the triangle has a simple area: half of the base multiplied by the height. The base is the quantity bought. The height is the highest price anyone would pay minus the market price. Suppose the highest price is Tk 100, the market price is Tk 60 and 40 units are bought. The surplus is $1/2 \times 40 \times (100 - 60) = 1/2 \times 40 \times 40 = \text{Tk } 800$.

Before moving on. Willingness to pay is the highest price a consumer would pay for one more unit. Consumers' surplus is the gap between that limit and the price actually paid, added over all units bought. On a chart it is the triangle under the demand curve and above the price line.

* * *

6.3 USING THE IDEA

Why measure a gain that nobody pays for? Because it shows the full value of a good to its users. A government that builds a bridge cannot judge it by the tolls alone, since the users gain more than they pay. A shopkeeper learns something too. Customers with a large surplus would have bought at a slightly higher price, so a price set too low gives away part of what the shop could earn. A price set too high loses the customers whose surplus was small.

The idea has limits, and the first concerns money. A taka means more to a poor consumer than to a rich one. Two consumers with the same surplus in taka may therefore gain different satisfaction. The measure shows the size of a gain in money. It does not show how fair the gain is, and it cannot compare the feelings of two people exactly.

The second limit is that willingness to pay is hidden. Nobody can read it from the mind of a consumer. Economists estimate the demand curve from the prices and quantities that markets record, so the triangle is an approximation. The estimate is weakest for goods that are needed to live, such as water,

ADVANCED NOTE: A BEGINNER MAY DEFER THIS PARAGRAPH.

For a straight demand curve, the surplus triangle has an area of half base times height.

Recap: willingness to pay, consumers' surplus and the triangle on the chart.

Surplus shows the full value of a good to its users, beyond the price they pay.

Surplus is measured in taka, and a taka means more to a poor consumer than to a rich one.

The surplus can only be estimated, because willingness to pay is hidden.

A fall in price is a movement along the demand curve, not a shift, and it raises surplus.

The notebook price falls to Tk 30: four customers buy and surplus rises to Tk 60.

Surplus rises for two reasons: old buyers gain more per unit, and new buyers join.

A firm cannot collect the surplus without raising the price, and a higher price reduces the quantity bought.

Recap: uses, limits and the effect of a fall in price on surplus.

because the first units give a very large surplus that no chart can measure sensibly.

Now consider a change in price. When the price of the notebook falls, consumers move along the demand curve to a larger quantity demanded. This is a movement along the curve. It is not a shift, which needs a change in something other than price, such as income. Our question is what the movement does to the surplus.

Case file, illustrative. The market price of the notebook falls from Tk 40 to Tk 30. Four of the five customers now buy. Consumers' surplus = $(60 - 30) + (50 - 30) + (40 - 30) + (30 - 30) = 30 + 20 + 10 + 0 = \text{Tk } 60$.

The surplus rose from Tk 30 to Tk 60, a rise of Tk 30. Two things explain it. First, each of the three old buyers keeps Tk 10 more, because the price fell by Tk 10, and $3 \times \text{Tk } 10$ is Tk 30. Second, a new buyer joins. In this case the new buyer adds nothing, because the willingness to pay of Tk 30 equals the new price. With thousands of customers, new buyers join in a steady stream, and each adds a thin slice to the triangle.

A manager must take care here. The surplus is a gain of the consumers, not income of the firm. The firm could collect more of it only by raising the price, and a higher price lowers the quantity demanded. Some consumers would then stop buying. So the surplus is a fact to understand, and it does not tell the manager what price to charge.

Before moving on. Surplus shows the value of a good beyond its price, and governments and firms can use it. It is measured in money, so it cannot compare the feelings of different people exactly, and it can only be estimated. A fall in price is a movement along the demand curve, and it raises the consumers' surplus.

* * *

WORKED EXAMPLES: MEASURING CONSUMERS' SURPLUS

Two examples apply the method: the first with small numbers, the second with more realistic ones.

Example 1: surplus on three cups of tea at Tk 6 a cup.

The formula subtracts the price from each willingness to pay and sums the results.

The surplus is Tk 6, and the check agrees.

Example 2: pens at a school stall, with a price fall from Tk 12 to Tk 9.

We now apply the method in full, in the order Given, Find, Formula, Steps, Answer and Check. The first example uses small numbers, so that the method is the only new thing to learn. The second is a little messier and shows a change in price.

Example 1: cups of tea. Given. A consumer is willing to pay Tk 10 for the first cup of tea, Tk 8 for the second, Tk 6 for the third and Tk 4 for a fourth. The price is Tk 6 a cup. *Find.* The consumers' surplus.

Formula. In words: consumers' surplus equals the sum, over the units bought, of the willingness to pay minus the price. In symbols: $S = (W_1 - P) + (W_2 - P) + \dots$, where S is the consumers' surplus, W is the willingness to pay for each unit bought and P is the price. *Steps.* The fourth cup is not bought, because Tk 4 is below the price of Tk 6. First cup: $10 - 6 = \text{Tk } 4$. Second cup: $8 - 6 = \text{Tk } 2$. Third cup: $6 - 6 = \text{Tk } 0$. Sum: $4 + 2 + 0 = \text{Tk } 6$.

Answer. The consumers' surplus is Tk 6. *Check.* The consumer pays $3 \times \text{Tk } 6 = \text{Tk } 18$ and would have paid up to $10 + 8 + 6 = \text{Tk } 24$. The gap is $24 - 18 = \text{Tk } 6$, which agrees with the answer.

Example 2: pens at a school stall. Given. Four consumers are willing to pay Tk 25, Tk 18, Tk 14 and Tk 9 for one pen each. The price is Tk 12, and later it falls to Tk 9. *Find.* The consumers' surplus at each price, and the rise between them.

Formula. The same as before: $S = (W_1 - P) + (W_2 - P) + \dots$ for every pen bought. *Steps at Tk 12.* The consumer willing to pay Tk 9 does not buy. $25 - 12 = \text{Tk } 13$. $18 - 12 = \text{Tk } 6$. $14 - 12 = \text{Tk } 2$. Sum: $13 + 6 + 2 = \text{Tk } 21$. *Steps at Tk 9.* All four consumers buy. $25 - 9 = \text{Tk } 16$. $18 - 9 = \text{Tk } 9$. $14 - 9 = \text{Tk } 5$. $9 - 9 = \text{Tk } 0$. Sum: $16 + 9 + 5 + 0 = \text{Tk } 30$. Rise: $30 - 21 = \text{Tk } 9$.

Answer. The surplus is Tk 21 at the price of Tk 12 and Tk 30 at the price of Tk 9, a rise of Tk 9. *Check.* The price fell by Tk 3, so each of the three old buyers gains Tk 3, and $3 \times 3 = \text{Tk } 9$. The new buyer adds Tk 0. The total agrees with the rise.

Three slips are common. The first is a wrong sign. Subtract the price from the willingness to pay, not the reverse, or the answer is negative. The second is to include a consumer who does not buy. Omit such a consumer, because the surplus from that unit is zero. The third is to forget the unit. Write Tk beside each figure, and do not mix prices from different dates.

* * *

SUMMARY

Consumption is the use of goods and services to satisfy wants. Consumers have limited income and spend it where an extra taka adds the most utility. Willingness to pay is the highest price a consumer would pay for one more unit. Consumers' surplus is the gap between willingness to pay and the price actually paid, and on a chart it is the triangle under the demand curve and above the price. A fall in price is a movement along the demand curve and it raises the surplus. In the notebook case, the surplus rose from Tk 30 to Tk 60 when the price fell from Tk 40 to Tk 30.

THINK IT THROUGH

6.T1 In the notebook case the surplus rose from Tk 30 to Tk 60 when the price fell from Tk 40 to Tk 30. Explain in your own words why the surplus rises when the price falls.

6.T2 A public bridge charges a toll of Tk 20, and 1,000 people cross each day. Many of them would have paid more. Explain why the tolls collected do not measure the benefit of the bridge to its users.

WHERE PEOPLE GO WRONG

A common belief is that the price shows what a good is worth to the consumer. It does not. The price shows only what the consumer must pay. The worth to the consumer is the willingness to pay, which is usually higher. The difference between the two is the surplus.

A second belief is that surplus is money that the consumer receives. No money changes hands. Surplus is satisfaction expressed in taka, and it stays in the mind of the consumer.

A third mistake is to include a consumer who does not buy. A person whose willingness to pay is below the price gains no surplus and adds nothing to the total. If the subtraction gives a negative number, the unit should not have been counted.

The same formula is applied twice, once at each price.

The surplus is Tk 21 at Tk 12 and Tk 30 at Tk 9, and the rise passes the check.

Three slips are common: wrong sign, buyers who do not buy, and missing units.

The main ideas of the chapter in a few lines.

A common belief is that the price shows what a good is worth to the consumer; it does not.

A second belief is that surplus is money received; it is satisfaction counted in taka.

A third mistake is to count a consumer who does not buy.

A fourth mistake is to call a fall in price a shift.

A fifth mistake is to confuse the triangle with the rectangle on the chart.

In 1844 Jules Dupuit, a French engineer, showed that people gain more from a bridge than the toll they pay.

Dupuit saw that a lower toll brings more crossings and a larger gain; Marshall later named the idea.

Governments still use the idea to compare a project's benefits with its cost to build.

A fourth mistake is to call a fall in price a shift. A fall in the price of a good moves consumers along the same demand curve to a larger quantity demanded. A shift needs a change in something else, such as income.

A fifth mistake is to confuse the two parts of the chart. The rectangle shows the money that consumers pay, and the firm receives it. The triangle shows the surplus, and the consumers keep it. A reader who adds the two together has found the total value of the units bought to consumers, which is a different quantity from the surplus.

IN HISTORY

In 1844 a French engineer named Jules Dupuit published an article on the utility of public works in a French engineering journal, the *Annales des ponts et chaussées*.² He worked on roads and bridges for the state, and he asked how the benefit of such works should be judged. He showed that people gain more from a bridge than the toll they pay.

Dupuit saw that a lower toll brings more crossings, and that those who cross at a high toll would have paid even more. He called the gain *relative utility*. The British economist Alfred Marshall, who taught at Cambridge, later gave the idea the name used today. He did so in his textbook *Principles of Economics*, first published in 1890.

Governments still use the idea. When they compare a bridge, a road or a railway with the cost to build it, they include the surplus of the people who will use it. The cost here is the builder's expense, and the benefit is more than the tolls collected.

REVIEW QUESTIONS

- 6.1 Define consumption in your own words and give one example from your own day.
- 6.2 Explain the difference between price and willingness to pay, using the notebook case.
- 6.3 In the notebook case, compute the consumers' surplus when the price is Tk 50, and again when the price is Tk 20.
- 6.4 Four consumers are willing to pay Tk 30, Tk 22, Tk 16 and Tk 10 for one pen each. Compute the consumers' surplus at a price of Tk 16 and at a price of Tk 10, and find the rise between them.
- 6.5 Using marginal utility and the demand curve, explain why the surplus on the last unit a consumer buys is close to zero.

² Dupuit, 'De la mesure de l'utilité des travaux publics', *Annales des ponts et chaussées*, 1844. An English translation appeared in *International Economic Papers* in 1952.



A Victorian draper's shop: a shopper weighs two bolts of cloth against the contents of her purse

CHAPTER VII

Indifference Curve Analysis

Can a shopper's wishes be drawn on paper?



This chapter draws a consumer's taste as a line of equal satisfaction and her purse as a line of limited spending. Where the two lines meet in the right way, we find the combination of two goods that suits her best.

YOU WILL NEED

Chapter 3 'Utility and Diminishing Marginal Utility', section 3.1 'The Idea of Utility'; Chapter 6 'Consumption and Consumers' Surplus', section 6.2 'Consumers' Surplus'; Appendix A1 'Maths You Will Use'.

A shopper's choice depends on taste and on the size of her purse.

This chapter answers both questions with two lines on one graph.

The first line shows what pleases the shopper; the second shows what she can afford.

Suppose you enter a market stall that sells only two goods, mangoes and guavas, and you have Tk 100 to spend. You could buy only mangoes. You could buy only guavas. Or you could buy some of each, in many different mixes. Every mix may cost the same, yet you will not feel the same about them. Which mix should you choose? Most people say, "The one I like best." That answer is true, but it hides two questions. What does it mean to like one mix as much as another? And how does the purse limit the choice?

Chapter 3 gave one way to think about taste: count the satisfaction, or utility, of each mix. Counting is difficult, because nobody can look inside a buyer's mind and read a number. This chapter offers a second way that needs no counting. It asks only that a shopper can say whether one mix is better than, worse than, or equal to another. From that single ability we can draw a picture of her taste, and then add a picture of her purse.

The picture has two parts. The first is a line of taste: all the mixes that please the shopper equally. The second is a line of limit: all the mixes that her money can buy. Where the two lines meet in the right way, we shall find her best choice. Section by section, we give each line its proper name, learn how to measure its steepness, and then bring the two together. Many students find a graph with several lines confusing at first, because each line answers a different question. We therefore go one line at a time.

* * *

7.1 INDIFFERENCE CURVES

GO SLOWLY HERE.

A friend's two plates of fruit can please you equally.

Economists call a particular amount of each of two goods a combination.

Indifference curve. A line showing all combinations of two goods that give a consumer the same satisfaction.

An indifference curve joins all the combinations that please equally.

Begin with an everyday case. A friend offers you a plate of fruit. Plate one holds 4 mangoes and 2 guavas. Plate two holds 3 mangoes and 4 guavas. After a moment's thought you say that you would be equally happy with either plate. On plate two you have lost one mango and gained two guavas, and your pleasure has not changed. In ordinary speech you are *indifferent* between the plates, which means you do not mind which one you receive.

Economists use the same idea with two goods of their choosing. Let the goods be notebooks and pens. A particular amount of each good, such as 4 notebooks and 14 pens, is called a *combination*. (Some books say *bundle*; this chapter uses *combination* throughout.) For any two combinations, a consumer can say: this one is better, that one is better, or they are equal. That is the only ability we need, and every adult shopper has it.

Now we can name the main tool of the chapter. An **indifference curve** is a line showing all combinations of two goods that give a consumer the same satisfaction. The two plates of fruit above would lie on one indifference curve, because you value them equally. The satisfaction in question is what Chapter 3 called utility.¹ Notice the change of method. Chapter 3 tried to count utility in numbers; here we only compare. Every

¹ Recall: utility, Chapter 3, 'Utility and Diminishing Marginal Utility', section 3.1 'The Idea of Utility'.

combination on one curve has the same utility, whatever that utility is worth in numbers.

Many students find the word *curve* confusing, because it suggests a bend, and the economist's curve is sometimes straight. Treat it as a name for any line drawn on a graph to show a relationship. An indifference curve does bend, for a reason that we meet in section 7.2. Until then, think of it only as a path through the combinations that please equally.

Now build one indifference curve from small, round numbers. The table below lists five combinations of notebooks and pens that one student values equally. Read each row from left to right: the letter names the combination, and the next two columns give its notebooks and its pens. These figures are made up to teach the idea; they are not part of the case file. Notice how the number of pens falls as the number of notebooks rises.

ONE STUDENT'S EQUALLY SATISFYING COMBINATIONS (ILLUSTRATION)

COMBINATION	NOTEBOOKS	PENS
A	3	20
B	4	14
C	6	8
D	8	5
E	12	2

A graph shows the same facts in a second form. The first time we use one, here is how to read it. The horizontal axis counts notebooks. The vertical axis counts pens. Each point is one combination: its distance along tells the notebooks, and its height tells the pens. Join the five points with a smooth line and you have an indifference curve. It falls from left to right and bends in towards the corner where the axes meet.

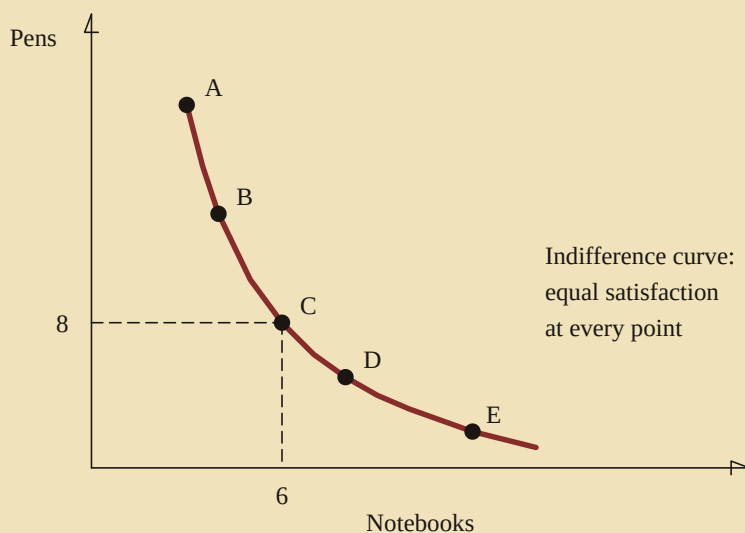


FIG. 7.1. An indifference curve through the five combinations A to E

Why does the curve fall? Suppose the student holds combination A and receives one more notebook. She now has more of one good, so she is better off. To return to her old level of satisfaction she must give up some pens. That is why the notebooks go up while the pens go down. If both numbers rose together, the

Curve. Word in use. In daily life, a bend in a road or a river. In economics, a line drawn on a graph, which may bend or may be straight.

The word curve means a line on a graph, bent or straight.

A table of five combinations shows the same satisfaction throughout.

The same five combinations drawn as points make a falling line.

The curve falls because more of one good must replace less of the other.

Indifference map. A set of indifference curves, with higher curves showing greater satisfaction.

Many curves together, higher ones better, form an indifference map.

Four properties of the curves follow from one assumption.

Recap of section 7.1.

Marginal rate of substitution. The amount of one good a consumer will give up to gain one more unit of another good.

GO SLOWLY HERE.

Moving along a curve means swapping one good for the other.

A worked example turns the swap into a number.

The rate falls at every step along the curve.

new combination would be better, so it could not lie on the same curve. This argument uses one assumption: the student always prefers more of a good to less.

PAUSE AND CHECK 7.P1 Combination P has 5 notebooks and 10 pens. Combination Q has 5 notebooks and 12 pens. Can P and Q lie on the same indifference curve? Give your reason.

One curve shows one level of satisfaction, but the student can imagine many levels, so there are many curves. An **indifference map** is a set of indifference curves, with higher curves showing greater satisfaction. Think of the contour lines on a walker's map, where each line joins places of equal height; here each line joins combinations of equal satisfaction, and a curve further from the corner of the axes is a higher hill. The student prefers 6 notebooks and 14 pens to 6 notebooks and 8 pens, because it has the same notebooks and more pens, so it lies on a higher curve.

Four properties follow from the assumption that the consumer prefers more of a good to less. First, each curve slopes downward, as the table showed. Second, a curve further from the corner where the axes meet gives greater satisfaction, because it holds more of the goods. Third, curves bend in towards that corner; section 7.2 'Marginal Rate of Substitution' explains why. Fourth, two curves never cross. The assumption has limits. If more of a good makes the consumer worse off, such as a sweet she has already eaten too many of, the first property fails. This chapter keeps to the usual case.

Before moving on. An indifference curve joins the combinations of two goods that please a consumer equally. It slopes down, and a map of such curves ranks every combination, with higher curves better and no two curves crossing. *You should now be able to say what one point on a curve shows and why a higher curve is preferred.*

7.2 MARGINAL RATE OF SUBSTITUTION

Return to the table of equally satisfying combinations. Moving from B (4 notebooks, 14 pens) to C (6 notebooks, 8 pens), the student gains 2 notebooks and gives up 6 pens, and her satisfaction stays the same. At what rate does she swap pens for notebooks? The **marginal rate of substitution** is the amount of one good a consumer will give up to gain one more unit of another good. Here she gives up 3 pens for each extra notebook, so the rate over this step is 3. The word *marginal* means "one more", as in the marginal utility of Chapter 3.²

Given. Combination B has 4 notebooks and 14 pens; combination C has 6 notebooks and 8 pens; both lie on one curve. *Find.* The rate at which pens are swapped for notebooks from B to C. *Formula.* In words: pens given up divided by notebooks gained. In symbols: $\text{rate} = F \div G$, where F is the fall in pens and G is the gain in notebooks. *Steps.* $F = 14 - 8 = 6$ pens. $G = 6 - 4 = 2$ notebooks. $\text{Rate} = 6 \div 2 = 3$ pens per notebook. *Answer.* She gives up 3 pens for each extra notebook. *Check.* Three pens for each of two notebooks is 6 pens, which agrees with the fall we measured.

Apply the same steps to each neighbouring pair in the table. The rates are 6, 3, 1.5 and 0.75. The rate falls at every step. The reason lies in the student's own wishes. When she has many pens and few notebooks, a notebook is precious to her and a pen is cheap, so she will give up many pens for one notebook. When

² Recall: marginal utility, Chapter 3, 'Utility and Diminishing Marginal Utility', section 3.1 'The Idea of Utility'.

pens are few and notebooks many, each pen is precious, so she gives up only a few. This falling rate is why an indifference curve bends in towards the corner of the axes: the curve grows flatter as we move to the right.

So far we measured the rate over a step. At a single combination, the economist measures it for a very small step, and the result equals the steepness of the curve at that point. At combination C the steepness is 2, which lies between the rate 3 on its left and the rate 1.5 on its right. This is the value that appears in the case of section 7.4. Common slips are easy to make here: dividing in the wrong order, putting the goods on the wrong axes, and forgetting to say "pens per notebook".

PAUSE AND CHECK 7.P2 Along an indifference curve the student holds more notebooks and fewer pens. Why does she give up fewer pens for each extra notebook?

Before moving on. The marginal rate of substitution is the number of pens a consumer gives up for one more notebook, with satisfaction unchanged. It falls along a curve, which is why the curve bends inwards. *You should now be able to say how to compute the rate between two combinations on one curve.*

At one combination, the rate is the steepness of the curve there.

Recap of section 7.2.

7.3 THE BUDGET LINE

Taste is only half the story. Return to the stall with Tk 100. Suppose mangoes cost Tk 20 each and guavas cost Tk 10 each. Spending all the money on mangoes buys 5. Spending it all on guavas buys 10. Three mangoes and 4 guavas cost Tk 60 and Tk 40, which is Tk 100 again. Each of these combinations is affordable, and together they follow a pattern that we can draw.

A **budget line** is a line showing every combination of two goods that a consumer can buy with a given income at given prices. At the stall, the points for 5 mangoes, 10 guavas, and 3 mangoes with 4 guavas all lie on one budget line. Points inside the line are affordable but leave some money unspent. Points beyond the line cost more than the shopper has. The word *line* here means a path of points on a graph, and this one is straight.

A fixed sum of money limits what any shopper can buy.

Budget line. A line showing every combination of two goods that a consumer can buy with a given income at given prices.

Line. Word in use. In daily life, a queue of people or a mark made with a pen. On a graph, a path of points; the budget line is straight, while an indifference curve bends.

A budget line shows every combination that the money can buy.

Case file, illustrative. A student has Tk 400 to spend on notebooks at Tk 40 each and pens at Tk 20 each. The most that can be bought is 10 notebooks or 20 pens.

Given. Income Tk 400; notebook price Tk 40; pen price Tk 20. *Find.* The most of each good, and the pens given up for each extra notebook. *Formula.* In words: the most of a good is income divided by its price; the pens given up for one notebook are the notebook price divided by the pen price. In symbols: $M = I \div P$, where M is the most of a good, I is income and P is that good's price; price ratio = notebook price $\div$ pen price. *Steps.* Notebooks: $Tk\ 400 \div Tk\ 40 = 10$. Pens: $Tk\ 400 \div Tk\ 20 = 20$. Price ratio: $Tk\ 40 \div Tk\ 20 = 2$ pens per notebook. *Answer.* She can buy at most 10 notebooks or 20 pens, and each extra notebook costs her 2 pens. *Check.* Ten notebooks cost $10 \times Tk\ 40 = Tk\ 400$, and 20 pens cost $20 \times Tk\ 20 = Tk\ 400$.

The case file gives the student's income and the two prices.

A worked example finds both ends of the line and its steepness.

To draw the line, mark 10 on the axis of notebooks and 20 on the axis of pens, and join the marks with a ruler. Each point on the line is a combination that uses exactly Tk 400. The line is straight because, at fixed prices, each extra

The line is straight and its steepness is the price ratio.

Income shifts the line sideways; a price change turns it.

Recap of section 7.3.

Both pictures on one graph show the best affordable combination.

Consumer equilibrium. The point where a consumer gets the greatest satisfaction possible from a given income.

Consumer equilibrium is the best affordable combination.

The best point is where one curve touches the line and does not cross it.

notebook always costs the same 2 pens. Its steepness is the price ratio, 2. Compare this with the curve: the curve bends because her wishes change as her holdings change, while the line is straight because prices stay fixed.

Two changes alter the line. If income rises and prices stay the same, the whole line shifts outwards, parallel to the old one, because the price ratio is unchanged: with Tk 600 she could buy 15 notebooks or 30 pens. If one price changes, the line turns around the point on the axis of the other good. If pens fell to Tk 10, she could still buy only 10 notebooks, but now 40 pens, so the line turns outwards around the notebook point and becomes steeper.

Before moving on. The budget line shows every combination that a given income buys at given prices. Its ends are income divided by each price, and its steepness is the price ratio. A rise in income shifts it; a price change turns it. *You should now be able to say where the line meets each axis and why it is straight.*

7.4 CONSUMER EQUILIBRIUM

We now hold two pictures. A map of curves shows what the student wants, and a budget line shows what she can afford. Draw both on one graph. Her aim is to reach the highest curve possible, and the budget line stops her from reaching any curve that lies beyond it. So the question becomes: which is the highest curve that still touches the budget line?

Consumer equilibrium is the point where a consumer gets the greatest satisfaction possible from a given income. The word *equilibrium* means a state of balance: once the consumer is there, she has no wish to change. For the student, it is the combination on the budget line that lies on the highest reachable curve.

In the figure below, the straight falling line is the budget line, and the two bent lines are indifference curves. The higher curve never meets the budget line, so the student cannot reach it. The lower curve crosses the line at two points, P and Q, and between them the line runs through combinations that lie on higher curves, so she can do better by moving. The best curve is the one that touches the line at a single point, marked B, without crossing. At B the curve and the line have the same steepness.

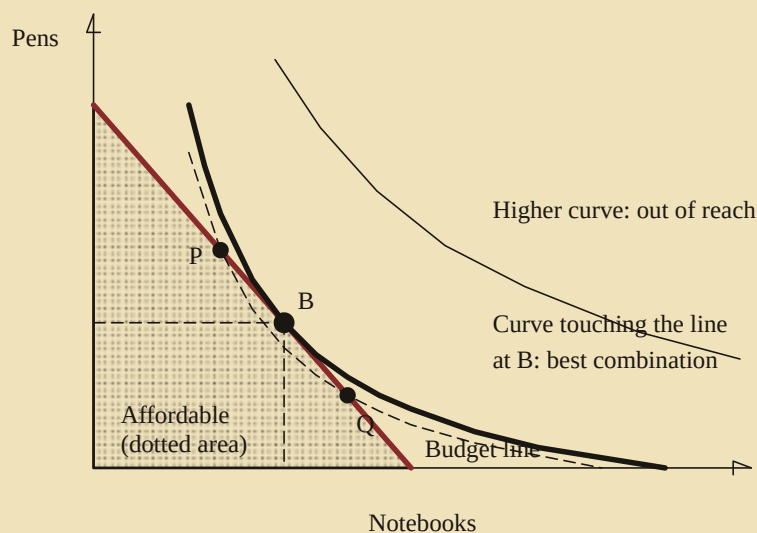


FIG. 7.2. *The budget line, a curve that crosses it, and the curve that touches it at the best combination*

The steepness of the curve is the marginal rate of substitution, and the steepness of the line is the price ratio. At the touching point the two are equal. The rule has a plain meaning. The curve's steepness says how many pens she would give up for one more notebook. The line's steepness says how many pens she must give up. If the first is larger, a notebook is worth more to her than it costs, so she buys more notebooks; her willingness to pay, measured in pens, exceeds the price.³ If the first is smaller, she buys fewer. She stops only when the two are equal.

At the best point, the rate she accepts equals the rate the market asks.

IN SHORT

At consumer equilibrium the marginal rate of substitution equals the price ratio, and the combination lies on the budget line.

The rule needs care. If the student would rather spend everything on one good, the best point lies on an axis, and the two steepnesses need not be equal. The rule also assumes that all income is spent and that the curves bend in as described in section 7.1. This chapter treats only the usual case, in which both goods are bought.

The rule has exceptions, so check its conditions.

Before moving on. Consumer equilibrium is the point on the budget line that lies on the highest reachable curve. There the curve touches the line, and the marginal rate of substitution equals the price ratio. *You should now be able to say why a combination on a crossing curve is not the best.*

Recap of section 7.4.

A WORKED CASE: THE STUDENT'S BEST COMBINATION

Case file, illustrative. A student has Tk 400 for notebooks at Tk 40 and pens at Tk 20. The most that can be bought is 10 notebooks or 20 pens. The best bundle is 6 notebooks and 8 pens (Tk 240 + Tk 160 = Tk 400), where the marginal rate of substitution is 2, equal to the price ratio $40 / 20 = 2$.

The case file states the student's best combination.

Given. Income Tk 400; notebooks Tk 40; pens Tk 20; the curve table of section 7.1; the claimed best combination of 6 notebooks and 8 pens. *Find.* Whether this combination is the best the student can afford. *Formula.* In words: spending equals notebooks times their price plus pens times their price; the best combination has its rate equal to the price ratio. In symbols: $S = (N \times \text{Tk } 40) + (K \times \text{Tk } 20)$, where S is spending, N is notebooks and K is pens.

We test the case file's claim step by step.

Steps. (1) Spending: $S = (6 \times 40) + (8 \times 20) = 240 + 160 = \text{Tk } 400$, so the combination lies on the budget line. (2) Price ratio: $40 \div 20 = 2$. (3) The rate at 6 notebooks and 8 pens is 2, as section 7.2 found. (4) The rate equals the price ratio, so the curve touches the line. (5) Test neighbours on the line. P is 4 notebooks and 12 pens: $160 + 240 = \text{Tk } 400$. The table says 4 notebooks with 14 pens is equally good as the claimed combination, and P has 2 pens fewer, so P is worse. Q is 8 notebooks and 4 pens: $320 + 80 = \text{Tk } 400$. The table says 8 notebooks with 5 pens is equally good, and Q has 1 pen fewer, so Q is worse. *Answer.* Yes: 6 notebooks and 8 pens is the best combination she can afford. *Check.* A better combination, 6 notebooks and 14 pens, costs $240 + 280 = \text{Tk } 520$, which is more than Tk 400, so she cannot buy it.

The steps check spending, steepness and the two nearest affordable neighbours.

Given. Income Tk 540; notebooks Tk 60; pens Tk 45. *Find.* The ends of the budget line and its steepness. *Formula.* The most of a good is income divided by its price; steepness is the notebook price divided by the pen price. *Steps.*

A second example uses less tidy numbers.

³ Recall: willingness to pay and consumers' surplus, Chapter 6, 'Consumption and Consumers' Surplus', section 6.2 'Consumers' Surplus'.

Four common slips spoil these calculations.

Recap of the worked case.

The chapter's tools fit together in four steps.

Indifferent does not mean uninterested.

The curve shows taste, not what she can afford.

The rate and the price ratio are different things.

The best combination is not the cheapest or the most equal.

Notebooks: $540 \div 60 = 9$. Pens: $540 \div 45 = 12$. Steepness: $60 \div 45 = 1.33$ pens per notebook. *Answer.* At most 9 notebooks or 12 pens, with 1.33 pens given up per notebook. *Check.* $9 \times 60 = \text{Tk } 540$ and $12 \times 45 = \text{Tk } 540$.

Common slips deserve a warning. Some students divide income by the wrong price, which swaps the two ends of the line. Some put the pen price on top of the price ratio, which turns the steepness upside down. Some forget the units, so that 1.33 loses its meaning of pens per notebook. Others test only one neighbour and declare a winner; test both sides of the touching point.

Before moving on. To find the best combination, compute the ends of the budget line, find the price ratio, locate the point where the rate equals it, and test the neighbours. *You should now be able to say* how each of the five steps supports the answer.

* * *

SUMMARY

An indifference curve joins combinations that please a consumer equally, and a map of such curves ranks every combination. The marginal rate of substitution measures the pens she will swap for one more notebook, and it falls along a curve. The budget line shows what her income buys at given prices, and its steepness is the price ratio. Consumer equilibrium is where the budget line touches the highest curve it can reach; there the rate equals the price ratio.

THINK IT THROUGH

7.T1 Income is Tk 400, notebooks cost Tk 40, and pens fall in price from Tk 20 to Tk 10. Show that the student can still afford 6 notebooks and 8 pens. What does that tell you about her satisfaction after the price fall?

7.T2 Explain why two indifference curves cannot cross. Use the idea that the consumer prefers more of a good to less.

WHERE PEOPLE GO WRONG

Some students think that being indifferent means not caring. The student cares a great deal. She is indifferent only between combinations on the same curve, and she strongly prefers any combination on a higher curve to one on a lower curve.

Some students read an indifference curve as a list of what the consumer can buy. It is not. The curve says nothing about money. Only the budget line shows affordability, and a point on a curve may cost more than her income.

Some students treat the marginal rate of substitution and the price ratio as one idea. The first comes from her wishes and changes along a curve. The second comes from the market and stays fixed along a budget line. They are equal only at consumer equilibrium.

Some students expect the best combination to hold equal numbers of the two goods, or to cost the least. The best combination is the one on the highest reachable curve. For the student it is 6 notebooks and 8 pens, which are not equal numbers.

IN HISTORY

The curves in this chapter are older than the 1930s. Francis Edgeworth, an economist who taught at Oxford, drew the first indifference curves in his book *Mathematical Psychics* in 1881. The Italian economist Vilfredo Pareto used them again in his *Manuale di economia politica* in the early 1900s, and he stressed that a theory of choice needs only the comparison of combinations.

Indifference curves were drawn before 1934.

In 1934 two economists at the London School of Economics, John Hicks and R. G. D. Allen, published *A Reconsideration of the Theory of Value*. It appeared in two parts in the journal *Economica*, in February and May 1934. They credited Pareto as their inspiration and rebuilt the theory of consumer choice on indifference curves, so that the argument no longer needed satisfaction to be counted in numbers. The budget line, the marginal rate of substitution and the tangency rule of this chapter are the core of that theory, and the method is still the usual way to teach consumer choice.

In 1934 Hicks and Allen rebuilt consumer theory on the curves.

REVIEW QUESTIONS

7.1 A student has Tk 200. Notebooks cost Tk 40 and pens cost Tk 20. Find the most of each good she can buy, and name two other combinations on her budget line.

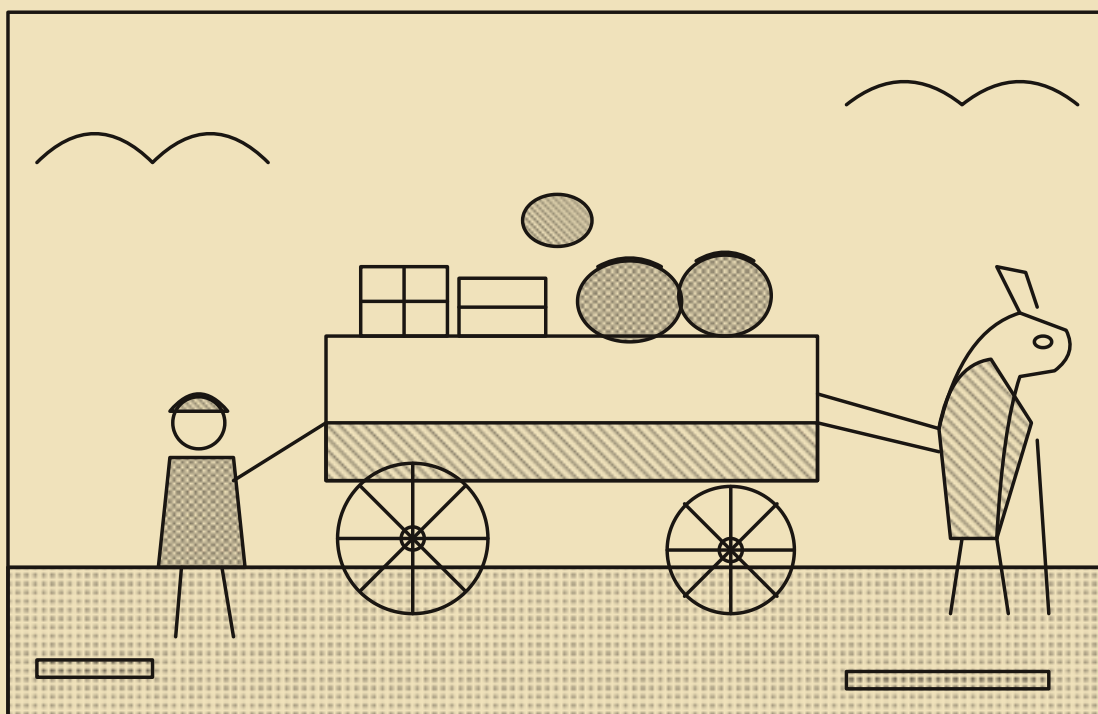
7.2 From combination C (6 notebooks, 8 pens) the student moves to combination D (8 notebooks, 5 pens) and stays equally satisfied. Find the marginal rate of substitution over this step. Is it larger or smaller than the rate of 3 from B to C, and why?

7.3 Income is Tk 450, notebooks cost Tk 50 and pens cost Tk 30. Find where the budget line meets each axis, and its steepness.

7.4 Income is Tk 400 and pens cost Tk 20. The price of a notebook rises from Tk 40 to Tk 50. Find the new ends of the budget line and say how the line moves.

7.5 Notebooks cost Tk 40 and pens Tk 20. At 4 notebooks and 12 pens, which lie on her budget line, her marginal rate of substitution is 4. Is she at consumer equilibrium? If not, which good should she buy more of?

7.6 At 6 notebooks and 8 pens her marginal rate of substitution is 2 pens per notebook, and a pen costs Tk 20. Find in taka the most she would pay for one more notebook. Compare it with the price of Tk 40 and say what this shows about consumers' surplus on that notebook.



A Victorian farm-cart arriving at market, loaded with sacks and crates

CHAPTER VIII

Supply and the Supply Curve

Why does a farmer bring more sacks to market when the price is high?



This chapter builds the seller's side of the market. It explains what sellers are willing and able to sell at each price, why a higher price brings a larger quantity, how to draw the supply curve, and what shifts it. It ends with a few exceptional curves that do not rise.

Two ideas from Chapter 4 and one appendix help with this chapter.

A farm-cart arrives at the market town, and its load depends on the price.

The same farmer brings different loads, and he cannot bring an unlimited load.

Chapter 4 told the story of buyers; this chapter tells the story of sellers.

Begin with a question about a seller: how many would you offer at each price?

Supply. The quantities of a good that sellers are willing and able to sell at each price in a period.

Supply is the whole list of prices and quantities supplied, not one number.

Supply. Word in use. In daily life, supply can mean a store of goods kept for later, as in a supply of rice, or the act of providing something. In economics, supply is the whole list of quantities sellers will sell at each price.

In this book, supply alone always means the whole list.

You will need two ideas from Chapter 4: what buyers are willing and able to buy, from Chapter 4 'Demand and the Demand Curve', section 4.1 'Quantity Demanded and the Law of Demand'; and the difference between a movement along a curve and a shift, from Chapter 4 'Demand and the Demand Curve', section 4.2 'Movement and Shift'. Appendix A1 'Maths You Will Use' helps with reading a graph and with rearranging an equation.

Picture a farm-cart arriving at a Victorian market town on a Saturday morning. It is loaded with sacks of potatoes and crates of eggs. The farmer has driven eleven miles in the dark, and he has chosen carefully what to bring. Last month, when potatoes sold poorly, he brought a few sacks and kept the rest to feed his pigs. This month the price is high, and the cart is piled so high that the horse leans into the harness.

The farmer, the farm and the horse are the same as last month. Only the price has changed. Yet he brings more sacks, and the market sees a larger load. Notice also what he does not do. Even at a very high price he does not bring a hundred sacks, because he has only so many fields, so many workers and so many hours. Why does a higher price bring a larger load, and what limits it?

Chapter 4 told the story of the crowd that buys. This chapter tells the story of the farmer who sells. The two stories mirror each other, and many ideas return in reverse form. That mirror makes the chapter easier, but it also makes a trap: students sometimes copy a rule from demand and forget to reverse it. We therefore build the seller's side step by step, and we mark each place where it differs from the buyer's side.

* * *

8.1 QUANTITY SUPPLIED AND THE LAW OF SUPPLY

Begin with a question from your own life. Suppose you could make bracelets at home and sell them to friends. At a price of Tk 20 each you might make a few, because the beads and the time are worth more than the price. At Tk 60 each you would make many more. The number of units that sellers choose to offer for sale at one particular price in a period is called the quantity supplied. It is one number tied to one price, in the same way as the quantity demanded.

Economists use the word **supply** for the larger idea. Supply means the quantities of a good that sellers are willing and able to sell at each price in a period. As with demand, two conditions are needed. A seller who wishes to sell but has no stock and no means of making more is not supplying anything. A seller with plenty of stock who refuses to sell is not supplying either. For the notebook at the shop, supply is the full list: so many notebooks at Tk 20, so many at Tk 30, and so on. One entry is a quantity supplied. The whole list is supply.

The everyday word needs care. In daily speech people say that "supply is short" when they mean that the stock is small today. This book keeps two words apart. Quantity supplied means one quantity at one price. Supply alone means the whole list of prices and quantities.

The simplest record of supply is a table. A **supply schedule** is a table showing the quantity supplied at each price. We now use the case of the shop's own wholesaler. The next paragraph is a record, and its figures are fixed.

Case file, illustrative. The wholesaler who sells the 200-page notebook to Meghna and Sons records its weekly supply. At a price of Tk 20 the quantity supplied is 20 notebooks; at Tk 30, 30; at Tk 40, 40; at Tk 50, 50; at Tk 60, 60. The quantity supplied equals the price in taka.

WEEKLY SUPPLY SCHEDULE FOR THE 200-PAGE NOTEBOOK

PRICE (TK)	QUANTITY SUPPLIED (NOTEBOOKS A WEEK)
20	20
30	30
40	40
50	50
60	60

To read a row, take the line for Tk 40. It says that at a price of Tk 40, the wholesaler chooses to offer 40 notebooks a week. Now read down the table. As the price rises by Tk 10 each time, the quantity supplied also rises by 10 each time. Compare this with the demand schedule of Chapter 4: there the quantity fell as the price rose, and here it rises.¹ The rule in the case file, quantity equals price, gives the same numbers.

The pattern has a name. The **law of supply** says that when the price of a good rises, the quantity supplied rises, if other conditions stay the same. The law works in both directions: when the price falls, the quantity supplied falls. As in the law of demand, the final clause matters. Costs, technology and taxes must stay fixed, or we cannot tell whether a change in the quantity offered came from the price.

Why do sellers behave in this way? Two reasons are enough for now. The first is the reward. A higher price makes each extra unit more profitable to offer, so a seller will use more hours, more materials and more space. The second is the expense of expanding. Making a large quantity usually costs more for each extra unit than making a small quantity, because the best fields, the best workers and the best machines are used first. A seller will accept the higher expense of the extra units only if the price is high enough to pay for them.

Notice how the word *cost* is used in the paragraph above. This book gives it a narrow meaning: the expense that a seller meets in making and delivering a good. What the buyer pays is called the price. A notebook may have a price of Tk 40 and cost the wholesaler less than that to supply. Keeping the two words apart avoids many confusions later.

PAUSE AND CHECK 8.P1 At a price of Tk 30 the wholesaler supplies 30 notebooks a week. Without drawing anything, say how many notebooks are supplied at Tk 50, and state in one sentence the law of supply that your answer shows.

Supply schedule. A table showing the quantity supplied at each price.

A supply schedule is a table of prices and the quantities supplied at them.

The case file records the weekly supply of the notebook by its wholesaler.

The same record is laid out as a table.

Each row pairs one price with one quantity supplied.

Law of supply. When the price of a good rises, the quantity supplied rises, if other conditions stay the same.

The pattern in the table is the law of supply.

Two reasons explain why a higher price brings a larger quantity supplied.

Cost. Word in use. In daily life, cost often means what a buyer pays for something. In economics, cost means the seller's expense in making the good, and what the buyer pays is the price.

In this book, cost means the seller's expense, and price means what the buyer pays.

¹ Recall: demand, Chapter 4, 'Demand and the Demand Curve', section 4.1 'Quantity Demanded and the Law of Demand'.

Supply curve. A line on a graph showing the quantity supplied at each price.

A supply curve draws the table as a line on a graph.

The supply curve rises from the bottom left to the top right.

The section closes with the four ideas in a few lines.

A tax on paper changes what the wholesaler offers, although the notebook's price has not moved.

A change in the price of the good itself moves us along one curve.

Four conditions decide where the whole supply curve lies.

Dearer inputs raise the seller's expense and reduce what is offered.

Better methods allow a seller to make more from the same inputs.

A tax raises the seller's expense, and a subsidy lowers it.

A table is exact, but a picture shows the pattern at a glance. A **supply curve** is a line on a graph showing the quantity supplied at each price. It is drawn in the same way as the demand curve.² Price is on the vertical axis, and the quantity supplied is on the horizontal axis. Each row of the table gives one point: the price in the row sets the height, and the quantity sets the distance from the vertical axis. Join the five points, and a line appears.

For the wholesaler, the five points lie on a straight line that rises from the bottom left to the top right. We call it a curve although it is straight, because the word covers lines of every shape. The upward slope is the law of supply drawn as a picture: a higher price goes with a larger quantity supplied. A demand curve slopes downward, and a supply curve slopes upward. That single difference marks the two sides of every market, and the chart in the next section shows both the curve and what moves it.

Before moving on. Quantity supplied is one number at one price. Supply is the whole list of prices and quantities, which the supply schedule records and the supply curve draws. The law of supply says that, other conditions staying the same, a higher price brings a larger quantity supplied.

8.2 DETERMINANTS OF SUPPLY

Suppose the government places a tax on paper, the main material in a notebook. The wholesaler's expenses rise at once. The price of the notebook has not changed, and yet the wholesaler now offers fewer notebooks at the same price. This event is different from a change in the price of the notebook, and it shows how supply has its own movements and shifts.³

First, the price of the good itself changes. If the notebook's price rises from Tk 40 to Tk 50, the quantity supplied rises from 40 to 50 notebooks a week. On the graph, the point moves along the same line. We call this a movement along the curve. The curve does not change position, because the whole list of prices and quantities is the same list as before. This is the mirror of the movement along a demand curve.

Second, something other than the price changes. The supply curve was drawn with other conditions held fixed, and four are most important. The first is the cost of inputs: the materials, labour, rent and power that a seller must buy. The second is technology, meaning the methods and machines used in making the good. The third is taxes and subsidies. The fourth is the number of sellers in the market. When any of these changes, sellers offer different quantities at every price.

Cost of inputs. If paper, wages or rent become dearer, each notebook is more expensive to make, and the wholesaler offers fewer at every price. If inputs become cheaper, the wholesaler offers more. The effect works through the seller's expense, which is why the word cost, as used in this book, stays on the seller's side.

Technology. A faster printing machine allows a firm to make more notebooks from the same paper and the same hours of work. Each notebook then costs less to make, and the firm offers more at every price. An improvement in technology raises supply.

Taxes and subsidies. A tax on a good, or on an input, raises the seller's expense, and supply falls. A subsidy is a payment from the government to sellers

² See Appendix A1 'Maths You Will Use', section A1.5 'Reading a Graph', for axes, points and slope.

³ Recall: shifts, Chapter 4, 'Demand and the Demand Curve', section 4.2 'Movement and Shift'.

to encourage them to produce, and it lowers the seller's expense, so supply rises. The tax on paper in our opening example is a tax on an input.

Number of sellers. If a second wholesaler begins to sell the notebook, the quantity offered to the market at each price rises. If a wholesaler leaves, the quantity falls. The table below lists the four determinants. We introduce it in words first: each row names a determinant, and the last column gives the effect on supply.

THE DETERMINANTS OF SUPPLY, WITH A NOTEBOOK EXAMPLE

DETERMINANT	EXAMPLE FOR THE NOTEBOOK	EFFECT ON SUPPLY
Cost of inputs	Paper becomes dearer	Falls
Technology	A faster printing machine is installed	Rises
Taxes and subsidies	A tax on paper is introduced	Falls
Taxes and subsidies	A subsidy on paper is introduced	Rises
Number of sellers	A new wholesaler enters the market	Rises

To read a row, take the third line: when a tax on paper is introduced, the wholesaler's expense rises, and supply falls. The word *falls* here means that fewer notebooks are offered at every price, and not only at one price.

When a determinant changes, the whole curve moves to a new position. We call this a shift of the supply curve, which some books call a change in supply. This book uses the word shift. The case file gives a precise example.

Case file, illustrative. A tax on paper lowers the wholesaler's weekly supply of the notebook by 10 at every price. At a price of Tk 20 the quantity supplied becomes 10; at Tk 30, 20; at Tk 40, 30; at Tk 50, 40; at Tk 60, 50.

Now we draw the numbers, and this is the first chart of the chapter. Read it as follows. The vertical axis shows the price in taka, and the horizontal axis shows the quantity supplied in notebooks a week. A point means: at this price, the wholesaler offers this quantity. The solid line is the original supply curve, and the dashed line is the supply curve after the tax. The bold part of the solid line shows a movement along the curve, from Tk 40 to Tk 50.

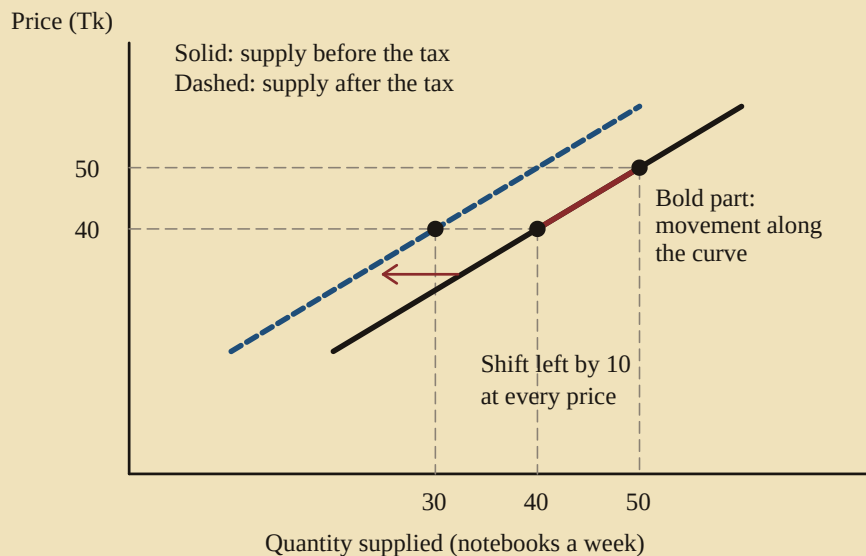


FIG. 8.1. The supply curve, with a movement along it and a shift to the left after a tax on paper

Read the picture along a horizontal line at Tk 40. The old curve meets this line at 40 notebooks, and the new curve meets it at 30. The gap between the two curves

More sellers in the market mean a larger quantity supplied at each price.

Read each row as a change of condition and the resulting change of supply.

A change in a determinant shifts the whole supply curve.

The case file records the shift caused by a tax on paper.

The chart shows the old curve, the new curve and a movement along the old curve.

The new curve lies to the left of the old curve.

is 10 notebooks at every price. The price on the vertical axis did not change, so the shift is not a movement along either curve. A move to the left means less supplied at every price. A move to the right means more supplied at every price.

PAUSE AND CHECK 8.P2 The government gives a subsidy to the makers of paper. Is the result a movement along the supply curve for notebooks or a shift? Say in which direction it goes, and give one reason.

A short test separates the two causes.

Before moving on. Ask one question: what changed? If the price of the good itself changed, there is a movement along the supply curve. If the cost of inputs, technology, a tax or subsidy, or the number of sellers changed, there is a shift of the whole curve. A fall in expenses or a rise in the number of sellers shifts the curve to the right, and the opposite changes shift it to the left.

8.3 EXCEPTIONAL SUPPLY CURVES

The law of supply describes most goods, but not all. Some supply curves do not slope upward, and economists have a name for them.

Not every supply curve rises, and the exceptions teach the reasons behind the rule.

Exceptional supply curve. A supply curve that does not slope upward, such as one that bends backward or stays fixed.

An exceptional supply curve does not rise.

A backward-bending curve appears when a higher price makes a seller supply less.

An **exceptional supply curve** is a supply curve that does not slope upward, such as one that bends backward or stays fixed. Two cases are worth knowing. Each is a limit of the law, and each has a reason that you can state in one sentence.

The first case is the backward-bending curve. The best example is the supply of working hours by one person. As the pay for each hour rises, the worker at first offers more hours, because work is now better rewarded. At a very high rate of pay, however, the worker can earn the income that he or she needs in fewer hours, and may prefer more rest. The hours offered then fall as the pay rises further. The curve rises, reaches a peak and then bends back.

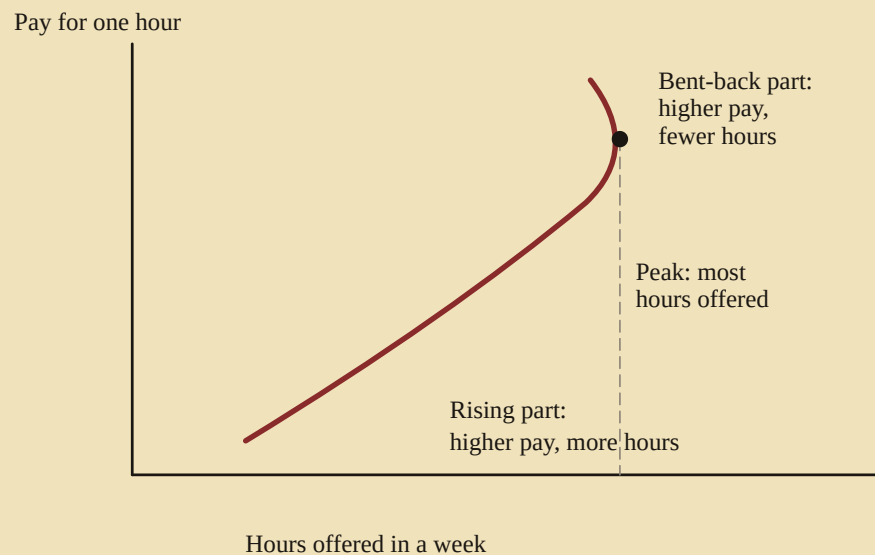


FIG. 8.2. A backward-bending supply curve of working hours

The curve rises at first and bends back after the peak.

Read the chart in the same way as before. The vertical axis shows the pay for one hour, and the horizontal axis shows the hours that one worker offers in a week. The solid line rises until the peak, which marks the most hours the worker will offer. Beyond the peak, the line goes back toward the vertical axis, so a higher pay brings fewer hours. This pattern is a matter of choice, and it applies to some people, not to all.

The second case is fixed supply. On a market day, a fishmonger who has caught a fixed number of fish cannot offer more, however high the price rises. A single famous painting, and a fixed number of seats in a hall, are other examples. The curve is a vertical line: the quantity supplied is the same at every price. In the short period before more can be made, the price cannot change what is on offer.

Before moving on. An exceptional supply curve does not slope upward. A backward-bending curve comes from a seller who chooses less as the price rises, and a fixed supply comes from a quantity that cannot change in the short period. Both are limits of the law of supply.

A fixed supply stays the same at every price.

The section closes with the two exceptional cases and their reasons.

WORKED CASE: USING THE WHOLESALE'S SUPPLY SCHEDULE

The case file gives a rule: the quantity supplied equals the price in taka. In symbols, $Q_s = P$, where Q_s is the quantity supplied in notebooks a week and P is the price in taka. After the tax on paper, the quantity supplied is 10 lower at every price, so the rule becomes $Q_s = P - 10$. Rearranged, the new rule gives $P = Q_s + 10$.⁴

Two links let us calculate any quantity or price from the supply rule.

Example 1. *Given:* $Q_s = P$. *Find:* the quantity supplied when the price is Tk 35, and the price at which 45 notebooks are supplied. *Formula:* quantity supplied equals the price in taka. *Steps:* at Tk 35, $Q_s = 35$; for 45 notebooks, $P = 45$. *Answer:* at Tk 35 the wholesaler supplies 35 notebooks a week, and 45 notebooks a week are supplied at Tk 45. *Check:* both answers fall between rows of the table, and the quantity rises with the price, as the law of supply says.

The first example is a clean calculation of a quantity and a price.

Example 2. *Given:* the old rule $Q_s = P$ and the new rule $Q_s = P - 10$ after the tax on paper. *Find:* the quantity supplied at Tk 50 under each rule, and the price at which 30 notebooks are supplied after the tax. *Formula:* $Q_s = P$ for the old rule, $Q_s = P - 10$ for the new rule, and $P = Q_s + 10$ for the price under the new rule. *Steps:* old rule, $Q_s = 50$; new rule, $Q_s = 50 - 10 = 40$; for 30 notebooks under the new rule, $P = 30 + 10 = \text{Tk } 40$. *Answer:* at Tk 50 the quantity falls from 50 to 40 notebooks a week, and the wholesaler needs a price of Tk 40 to offer 30 notebooks. *Check:* the table after the tax shows 30 notebooks at Tk 40, and the fall of 10 matches the case file.

The second example follows the shift caused by the tax.

Worked classification. *Situation:* in one month, six events happen. (a) The wholesaler raises the notebook price from Tk 40 to Tk 50. (b) A tax on paper is introduced. (c) A faster printing machine is installed. (d) A second wholesaler begins to sell the notebook. (e) Paper becomes dearer for another reason. (f) The notebook price falls from Tk 60 to Tk 50. *Question:* which events are movements along the supply curve and which are shifts? *Reasoning:* in (a) and (f) the price of the good itself changed. In (b) the tax changed, in (c) the technology changed, in (d) the number of sellers changed, and in (e) the cost of inputs changed. *Conclusion:* (a) and (f) are movements along the curve, and (b) to (e) are shifts, with (c) and (d) to the right and (b) and (e) to the left. *Check:* only (a) and (f) change the price on the vertical axis for the notebook.

Sorting events as price causes or non-price causes needs only one question.

Common slips. Students often call a price change a shift. They give the direction of a shift the wrong way, forgetting that a fall in supply moves the curve to the left. They treat the new rule as $Q_s = P + 10$ instead of $Q_s = P - 10$. And they use the word cost for what the buyer pays, when this book uses price for that.

Four common slips spoil these answers.

⁴ Appendix A1 'Maths You Will Use', section A1.3 'Rearranging an Equation', shows how to move a term across the equals sign.

SUMMARY

The chapter's main ideas in a few lines.

Supply is the set of quantities that sellers are willing and able to sell at each price, and a quantity supplied is one entry in that set. The law of supply says that, other conditions staying the same, a higher price brings a larger quantity supplied, so the supply curve slopes upward. A change in the price of the good is a movement along the curve. A change in the cost of inputs, technology, taxes or subsidies, or the number of sellers is a shift of the curve. An exceptional supply curve, such as a backward-bending curve or a fixed supply, does not slope upward.

THINK IT THROUGH

Two problems that apply the chapter to new settings.

8.T1 A worker is offered more and more pay for each hour, yet at a very high rate chooses to work fewer hours. Give two reasons for this choice, and say what shape of supply curve it makes.

8.T2 A fishmonger has 200 fish on a market day and no way of getting more before the market closes. Describe the supply curve for that day, and say what changes if the market is open for a week.

WHERE PEOPLE GO WRONG

A shift is often confused with a movement.

Error 1: "A rise in the price of the good shifts its supply curve." This is wrong. A rise in the price of the good moves the point along the same curve. Only a change in a determinant shifts the curve.

The direction of a shift is often reversed.

Error 2: "A tax on inputs shifts supply to the right." This is wrong. A tax raises the seller's expense, so fewer units are offered at every price, and the curve shifts to the left.

Supply is not the same as quantity supplied.

Error 3: "Supply and quantity supplied mean the same." This is wrong. Quantity supplied is one number at one price. Supply alone is the whole list of prices and quantities.

Cost and price are different ideas in this book.

Error 4: "The cost of a notebook is what the shop pays the wholesaler, so cost and price are the same." This is wrong in this book. Price is what a buyer pays. Cost is the seller's expense in making and delivering the good.

IN HISTORY

An English professor compared supply and demand to the two blades of a pair of scissors.

In 1890, Alfred Marshall, who lived from 1842 to 1924 and taught economics at Cambridge in England, published *Principles of Economics*. He had begun the book in 1881. Before him, economists had long argued about what governs the price of a good: the satisfaction it gives to buyers, or the expense of making it. Marshall answered with a comparison. We might as sensibly ask which blade of a pair of scissors cuts a piece of paper, he wrote, as ask which of the two decides the price.

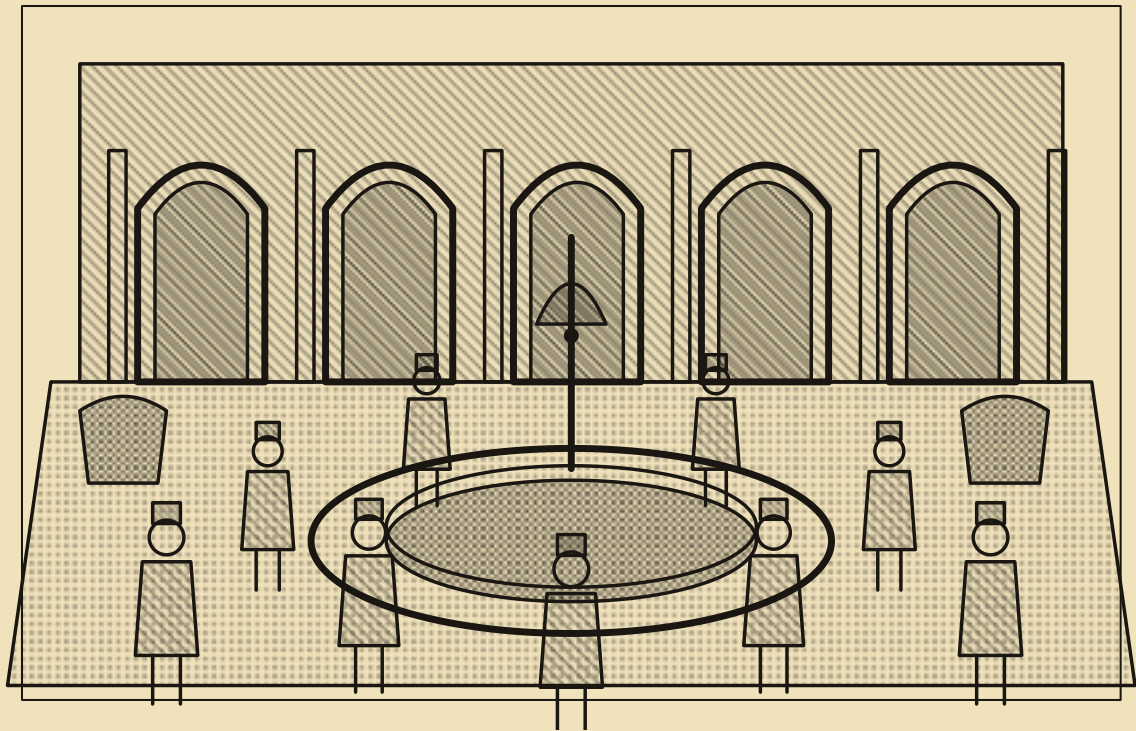
The picture has lasted, and modern economists still begin with it.

Marshall's point was that supply and demand work together, and both are needed. The book became the leading textbook in England for many years. To this day, an economist who wishes to explain why the price of a good has changed begins by asking whether the demand curve shifted, or the supply curve, or both. The next chapter, Chapter 9 'Elasticity of Supply and Market Equilibrium', puts the two blades together.

REVIEW QUESTIONS

- 8.1 Plot the supply schedule of the notebook from the case file on a graph, with price on the vertical axis. Say what the slope of the line shows.
- 8.2 After the tax on paper, the quantities supplied at Tk 20, 30, 40, 50 and 60 are 10, 20, 30, 40 and 50. Plot the new curve on the same graph as the old one. Then draw a third curve that shows an increase in supply of 15 at every price, and give its quantity at Tk 40.
- 8.3 Sort each cause as a price cause (movement along the curve) or a non-price cause (shift of the curve): (a) the notebook price rises from Tk 30 to Tk 40; (b) paper becomes cheaper; (c) a subsidy is paid to notebook makers; (d) the wholesaler lowers the notebook price from Tk 60 to Tk 50.
- 8.4 Using $Q_s = P$, find the quantity supplied at Tk 55 and the price at which 25 notebooks are supplied. Then, using $Q_s = P - 10$, find the quantity supplied at Tk 55.
- 8.5 State the law of supply, and name one condition that must stay the same for it to hold.
- 8.6 Name two kinds of exceptional supply curve, and give one example of each.
- 8.7 A new wholesaler enters the market, and the notebook's price is unchanged. Say what happens to supply, and name the determinant that changed.

Seven questions that check the chapter's ideas.



Merchants at a corn exchange, where a bell opens the day's bargaining

CHAPTER IX

Elasticity of Supply and Market Equilibrium

Who decides a price when no single person decides it?



This chapter measures how strongly sellers respond when the price changes. It then joins the buyers' side and the sellers' side of a market to find the price where the market settles, and shows what moves that price.

YOU WILL NEED

Chapter 4, 'Demand and the Demand Curve', section 4.1 'Quantity Demanded and the Law of Demand'; Chapter 8, 'Supply and the Supply Curve', section 8.1 'Quantity Supplied and the Law of Supply'; Chapter 5, 'Elasticity of Demand', section 5.1 'Price Elasticity of Demand'; and Appendix A1, 'Maths You Will Use'.

A seller must choose a price, yet no single seller truly decides it.

By evening the sellers charge almost the same price, and nobody announced it.

The chapter has two jobs: measure how sellers respond, then find where the market settles.

The direction of the response is known from Chapter 8; its size is not.

Elasticity of supply. The percentage change in quantity supplied divided by the percentage change in price.

A single number records how strongly sellers respond to a price change.

Recall: price elasticity of demand, Chapter 5, section 5.1 'Price Elasticity of Demand'.

Percentages let us compare goods that are measured in different units.

Early on a market morning, a seller unloads crates of notebooks and must choose a price. If the price is too high, shoppers walk past and the crates stay full. If the price is too low, the crates are empty by ten o'clock, and she wonders how much money she gave away. Every seller in the row faces the same worry, and each one watches the others.

By evening, a strange thing has happened. Nobody called a meeting and nobody gave an order, yet the sellers in the row charge almost the same price, and most of the crates are sold. Something has guided hundreds of separate choices to one number. What is that something? Why did the price settle at this number and not at another?

This chapter answers in two steps. First, we measure how strongly sellers react when the price changes; section 9.1 gives this measure its name. Second, we join the buyers' side and the sellers' side to find the price where the market settles, and we ask what moves that price. Many students find the second step easier once the first is secure, because both steps use the same idea of percentage change.

* * *

9.1 ELASTICITY OF SUPPLY

Chapter 8 taught the law of supply: when the price of a good rises, firms offer more of it. That law gives a direction and nothing more. Suppose the price of a notebook rises by 10%. Will firms offer 2% more, or 20% more? A baker with a hot oven can bake extra loaves by the afternoon. A farmer who planted onions in March cannot grow more by Friday. Both respond in the same direction, yet the size of the response differs greatly. To compare them, we need one number that records the size.

That number is called **elasticity of supply**. Elasticity of supply means the percentage change in quantity supplied divided by the percentage change in price. Suppose the price rises by 10% and the quantity supplied rises by 5%. Then the elasticity of supply is $5 \div 10 = 0.5$. Because supply slopes upward, price and quantity supplied move in the same direction, so the answer is never negative.

Why do we use percentages and not plain changes? A change of 10 notebooks is large for a small shop and tiny for a printing firm. A change of 10 kilograms means little for rice and much for saffron. A percentage compares the size of the move with the starting amount. The measure copies the one you met for buyers in Chapter 5, with the quantity supplied in place of the quantity demanded.

MEASURING IT

Measuring needs one decision: a percentage of which number? If the price moves from Tk 30 to Tk 40, the rise is 33% of Tk 30 but only 25% of Tk 40. The same journey gives two answers. The *midpoint method* removes the problem by dividing each change by the average of the start value and the end value.

The formula, in words: elasticity of supply equals the change in quantity supplied divided by the average quantity, all divided by the change in price divided by the average price. In symbols: $E_s = (\Delta Q \div \text{average } Q) \div (\Delta P \div \text{average } P)$. Here E_s is the elasticity of supply, ΔQ is the change in quantity supplied, and ΔP is the change in price.¹

Case file, illustrative. In the notebook market where Meghna and Sons sells its popular 200-page notebook, the quantity supplied equals the price in taka. At a price of Tk 30, firms offer 30 notebooks. At a price of Tk 40, firms offer 40 notebooks. We now calculate the elasticity of supply between these two prices.

Given. The price rises from Tk 30 to Tk 40. The quantity supplied rises from 30 to 40 notebooks. Find. The elasticity of supply between Tk 30 and Tk 40. Formula. $E_s = (\Delta Q \div \text{average } Q) \div (\Delta P \div \text{average } P)$, as defined above. Steps. (1) Change in quantity: $40 - 30 = 10$ notebooks. (2) Average quantity: $(30 + 40) \div 2 = 35$ notebooks. (3) Change in quantity as a fraction of the average: $10 \div 35 = 0.286$. (4) Change in price: $40 - 30 = \text{Tk } 10$. (5) Average price: $(30 + 40) \div 2 = \text{Tk } 35$. (6) Change in price as a fraction of the average: $10 \div 35 = 0.286$. (7) Divide: $0.286 \div 0.286 = 1.0$. Answer. The elasticity of supply between Tk 30 and Tk 40 is 1.0. Check. Both fractions are equal, so their ratio must be 1.

TYPES OF ELASTICITY OF SUPPLY

The size of the number places supply into one of five types. Read the table row by row: the first column gives the value, the second names the type, and the third says what firms do when the price rises.

FIVE TYPES OF ELASTICITY OF SUPPLY

VALUE	TYPE	WHAT FIRMS DO WHEN THE PRICE RISES
Greater than 1	Elastic supply	Quantity supplied rises by a larger percentage than the price
Equal to 1	Unit elastic supply	Quantity supplied rises by the same percentage as the price
Between 0 and 1	Inelastic supply	Quantity supplied rises by a smaller percentage than the price
Zero	Perfectly inelastic supply	Quantity supplied does not change; the supply curve is a vertical line
Without limit	Perfectly elastic supply	Firms offer any quantity at one price; the supply curve is a horizontal line

The two extreme rows are rare. Seats in a stadium on match day are close to perfectly inelastic, because the number of seats cannot change that afternoon. A straight supply line that starts at the origin, like the notebook line above, has an elasticity of exactly 1 at every point on it. Most goods lie between the extremes.

Percentage of which number? The midpoint answers this question.

The formula, first in words and then in symbols.

The case file gives a clean pair of numbers for the first calculation.

Each step of the first calculation is shown, in the standard order.

The size of the number places supply into one of five types.

The two extreme types are rare, and the notebook case sits at 1.

¹ Some books divide by the starting value instead of the average. That method gives a slightly different number when the change is large. This book uses the midpoint throughout.

Time is the strongest factor, followed by spare capacity and stocks.

The manager and the owner see the same factors from different positions.

Before moving on, recall the measure and the five types.

A puzzle: neither side can force the price on the other.

Market. Word in use. In daily life, a place where people meet to buy and sell, such as a street market. In economics, all the buyers and sellers of one good, wherever they are.

Competitive market. A market with many buyers and sellers, none of whom can set the price.

We begin with markets where nobody controls the price.

Equilibrium. Word in use. In daily life, a state of balance, such as a scale holding still. In economics, a price and quantity at which nothing pushes the price to change.

Equilibrium price. The price at which the quantity buyers want equals the quantity sellers offer.

The price where both sides agree is the equilibrium price.

The case file finds the equilibrium by setting the two quantities equal.

How to read the chart: axes, curves and the crossing point.

FACTORS

Four factors decide where a good lies in the table. *Time* matters most. In a day, a firm can only sell what it holds; in a year, it can build more machines. *Spare capacity* matters next: a printing firm with idle presses can raise output quickly. *Stocks* of finished goods allow a quick response, but perishable goods cannot be stored. *Inputs* matter too: if workers and materials are hard to find, output rises slowly.

A manager asks, "Can my workers and machines make more this week?" An owner asks, "Is it worth buying a new machine?" The first question concerns the short run and gives a low elasticity. The second concerns the long run and gives a higher one. Elasticity of supply is therefore never a fixed property of a good; it depends on the period we choose.

Before moving on. Elasticity of supply measures how strongly firms respond to a price change, using percentages and the midpoint. A value above 1 is elastic, 1 is unit elastic, and below 1 is inelastic. Time, spare capacity, stocks and inputs decide the value.

* * *

9.2 MARKET EQUILIBRIUM

Return to the market row. A seller wishes to charge Tk 60 for a notebook, but buyers will pay only Tk 30. Neither side can force the other. The price that finally appears comes from the meeting of both sides, and from neither side alone. To study that meeting, we must first say what kind of market we mean.

We study a **competitive market**. A competitive market means a market with many buyers and sellers, none of whom can set the price. A street of tea stalls selling the same tea is an example: if one stall raises its price, buyers walk to the next stall. A market where one seller does control the price is a different case, taught in Chapter 14, 'Monopoly and Imperfect Competition', section 14.1 'Monopoly'.

The price at which the market settles has a name: the **equilibrium price**. The equilibrium price means the price at which the quantity buyers want equals the quantity sellers offer. If buyers want 40 notebooks and sellers offer 40 notebooks, both sides are satisfied and nobody has a reason to change the price. The quantity bought and sold at that price is the *equilibrium quantity*.

Case file, illustrative. In the notebook market, the quantity demanded equals 80 minus the price, and the quantity supplied equals the price. At the equilibrium price the two quantities are equal, so $80 - \text{price} = \text{price}$. Then $80 = 2 \times \text{price}$, and the price is Tk 40. At Tk 40, buyers want $80 - 40 = 40$ notebooks and firms offer 40 notebooks. The equilibrium price is Tk 40 and the equilibrium quantity is 40.

To read the next chart, start with the axes. The horizontal axis shows the quantity of notebooks. The vertical axis shows the price in taka. A point on a line

shows one price and the quantity that goes with it. The demand curve slopes downward and the supply curve slopes upward. Where they cross, the quantity demanded equals the quantity supplied, and that crossing is the equilibrium.

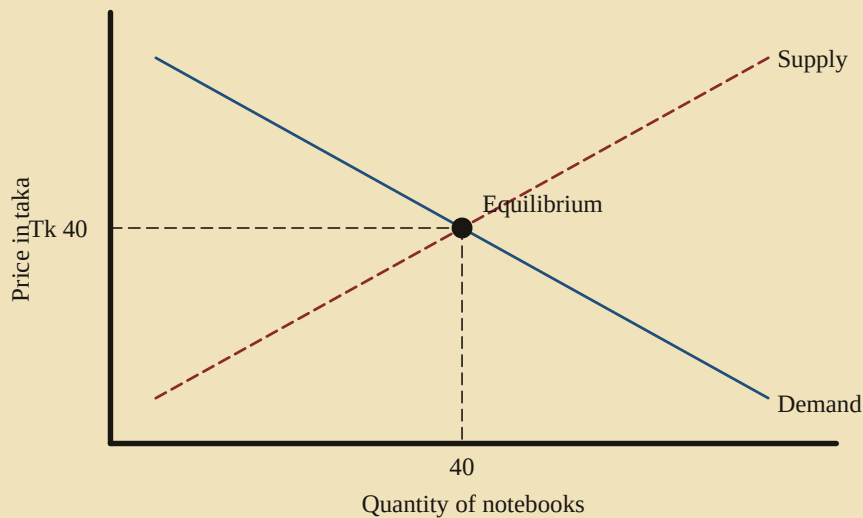


FIG. 9.1. Notebook market: demand and supply cross at the equilibrium price

PAUSE AND CHECK 9.P1 At the equilibrium price of Tk 40, how many notebooks do buyers want, and how many do firms offer?

Now ask what happens at a price other than Tk 40. The table below lists three prices from the notebook market. For each row, compare the two middle columns, which show how many notebooks buyers want and how many firms offer.

NOTEBOOK MARKET AT THREE PRICES

PRICE	QUANTITY DEMANDED	QUANTITY SUPPLIED	RESULT
Tk 30	50	30	Buyers want 20 more than firms offer
Tk 40	40	40	The two quantities are equal
Tk 50	30	50	Firms offer 20 more than buyers want

At Tk 30, buyers want 50 notebooks but firms offer only 30. This is **excess demand**. Excess demand means a situation in which buyers want more than sellers offer at a price, pushing the price up. The 20 buyers who find empty shelves offer more money to win the notebooks, and sellers notice that they can charge more. The price therefore rises towards Tk 40.

At Tk 50, firms offer 50 notebooks but buyers want only 30. This is **excess supply**. Excess supply means a situation in which sellers offer more than buyers want at a price, pushing the price down. Twenty notebooks stay unsold, and sellers cut the price to empty their shelves. The price therefore falls towards Tk 40.

GO SLOWLY HERE.

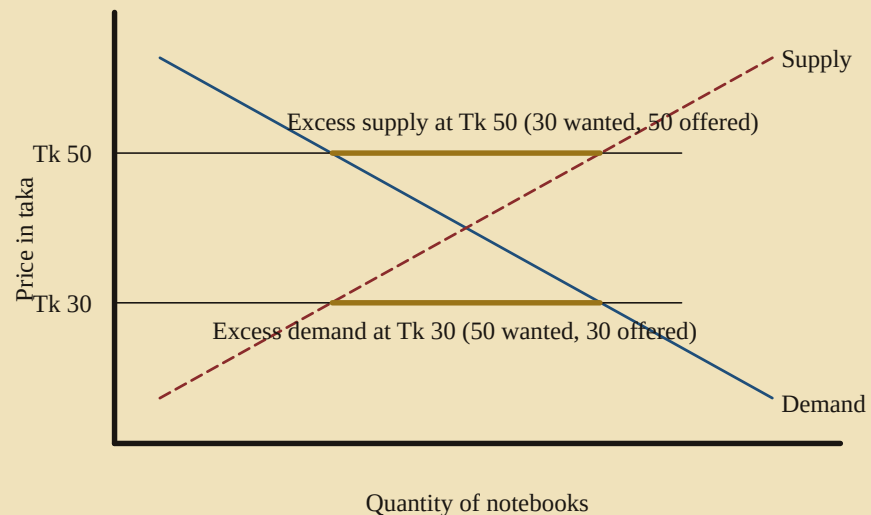
What happens at any other price is the work of the next two ideas.

Excess demand. A situation in which buyers want more than sellers offer at a price, pushing the price up.

Too low a price leaves buyers wanting more than is offered.

Excess supply. A situation in which sellers offer more than buyers want at a price, pushing the price down.

Too high a price leaves sellers with unsold goods.

FIG. 9.2. *Shortage and excess supply at wrong prices*

Both forces push towards the same price, which is why the market settles there.

The rule has limits: adjustment takes time, and a controlled price blocks it.

Before moving on, recall the three ideas of this section.

The two forces point to the same place. Below the equilibrium price, excess demand pushes the price up. Above it, excess supply pushes the price down. Only at Tk 40 do the pushes stop. That is the answer to the opening puzzle: nobody decides the price, but every buyer and seller adds a small push, and the pushes balance at the equilibrium price.

The rule has limits. Adjustment takes time, and in a market for a perishable good the price may fall before the sellers understand why. A law that fixes the price blocks the pushes altogether; the Think It Through section returns to this. The rule also assumes a competitive market. Finally, from the owner's view the price is a signal, and from the consumer's view it is a limit on what can be bought.

Before moving on. A competitive market has many buyers and sellers, none of whom set the price. The equilibrium price makes the quantity demanded equal the quantity supplied. Above it there is excess supply and the price falls; below it there is excess demand and the price rises.

* * *

9.3 CHANGES IN EQUILIBRIUM

A puzzle: a school opens nearby, and the equilibrium price may no longer hold.

GO SLOWLY HERE.

A shift moves a whole curve; a change of price moves us along a curve.

When demand shifts right, buyers want more at every price, so price and quantity both rise.

A school opens beside the shop, and a hundred more students need notebooks. The price of notebooks was Tk 40 yesterday. Will it still be Tk 40 today? The equilibrium price holds only while the conditions behind demand and supply stay the same. When a condition changes, a curve moves, and the market finds a new crossing point.

Two kinds of change must be kept apart. A *shift* means the whole demand curve or the whole supply curve moves, because something other than the price has changed, such as incomes or the price of materials. A *movement along* a curve happens when the price changes and the curve stays still. Section 9.2 showed movements along: as the price fell towards Tk 40, the quantity demanded rose along the demand curve.

Take a demand shift first. When more buyers arrive, the demand curve shifts to the right. At the old price of Tk 40, buyers now want more than firms offer, so there is excess demand. The price rises. As it rises, firms offer

more along the supply curve. The market settles at a higher price and a larger quantity.

Case file, illustrative. In the notebook market, demand rises to a quantity demanded of 100 minus the price, and supply stays at a quantity supplied equal to the price. At the new equilibrium, $100 - \text{price} = \text{price}$, so $100 = 2 \times \text{price}$ and the price is Tk 50. The quantity is 50 notebooks. The price has risen from Tk 40 to Tk 50, and the quantity from 40 to 50.

PAUSE AND CHECK 9.P2 If demand shifts to the right and supply does not shift, what happens to the equilibrium price and the equilibrium quantity?

Now take a supply shift. When the price of paper rises, firms make notebooks only at a higher cost to themselves, and the supply curve shifts to the left. At the old price, buyers want more than firms offer, so the price rises. As it rises, buyers move up the demand curve and buy fewer notebooks. The new equilibrium has a higher price and a smaller quantity.

Case file, illustrative. In the notebook market, supply falls to a quantity supplied of the price minus 10, and demand returns to the original quantity demanded of 80 minus the price. At the new equilibrium, $80 - \text{price} = \text{price} - 10$, so $90 = 2 \times \text{price}$ and the price is Tk 45. The quantity is 35 notebooks. The price has risen from Tk 40 to Tk 45, and the quantity has fallen from 40 to 35.

The table below collects the four single shifts. Read each row from the left: the first column names the shift, and the next two columns give the effect on the equilibrium price and the equilibrium quantity.

EFFECTS OF SINGLE SHIFTS ON EQUILIBRIUM

SHIFT	EQUILIBRIUM PRICE	EQUILIBRIUM QUANTITY
Demand shifts right	Rises	Rises
Demand shifts left	Falls	Falls
Supply shifts right	Falls	Rises
Supply shifts left	Rises	Falls

Sometimes both curves shift at once. Then the table gives two answers that may disagree. If demand shifts right and supply shifts left, both shifts push the price up, so the price certainly rises. But the shifts push the quantity in opposite directions, so the quantity depends on which shift is larger. When the effects disagree, we must calculate with the numbers; guessing is unsafe.

Before moving on. A shift moves a whole curve; a price change moves us along a curve. Demand shifting right raises both price and quantity. Supply shifting left raises the price and lowers the quantity. With two shifts, one effect is certain and the other depends on the sizes.

* * *

WORKED EXAMPLES

The first example uses figures that are not round, as real data seldom are. A tailoring firm sells shirts at Tk 200 and supplies 100 shirts a month. When the price rises to Tk 240, it supplies 110 shirts a month.

Given. Price rises from Tk 200 to Tk 240. Quantity supplied rises from 100 to 110 shirts a month. *Find.* The elasticity of supply. *Formula.* $E_s = (\Delta Q \div \text{average } Q) \div (\Delta P \div \text{average } P)$. *Steps.* (1) $\Delta Q = 110 - 100 = 10$ shirts. (2) Average $Q = (100 + 110) \div 2 = 105$ shirts. (3) $10 \div 105 = 0.0952$. (4) $\Delta P = 240 - 200 = \text{Tk } 40$.

The case file finds the new crossing after demand rises.

When supply shifts left, sellers offer less at every price, so price rises and quantity falls.

The case file finds the new crossing after supply falls.

Four single shifts have four predictable results, shown in one table.

When both curves shift together, one effect is certain and the other depends on size.

Before moving on, recall the shift rule and the two kinds of change.

Worked example 1 uses messier figures for the elasticity formula.

The steps of worked example 1, from data to check.

Worked example 2 combines a market with two curves and a shift.

The steps of worked example 2, before and after the shift.

Common slips: units, signs and mixed-up shifts cause most wrong answers.

The whole chapter in a few lines.

Two short problems apply the chapter to real choices.

Mistake one: treating a change of price as a shift.

(5) Average $P = (200 + 240) \div 2 = \text{Tk } 220$. (6) $40 \div 220 = 0.1818$. (7) $E_s = 0.0952 \div 0.1818 = 0.52$. *Answer.* The elasticity of supply is about 0.52, so supply is inelastic. *Check.* The quantity rose by about 9.5% and the price by about 18.2%. A smaller percentage over a larger one gives a value below 1, as found.

The second example joins the earlier techniques. It uses invented figures for a market in rice sacks. First we find the equilibrium; then demand shifts, and we find it again.

Given. Quantity demanded = $70 - \text{price}$. Quantity supplied = $2 \times \text{price} - 20$. Later, demand shifts to a quantity demanded of $100 - \text{price}$. *Find.* The equilibrium price and quantity before and after the shift. *Formula.* At equilibrium, quantity demanded = quantity supplied. *Steps.* Before: (1) $70 - \text{price} = 2 \times \text{price} - 20$. (2) Add 20 to both sides: $90 - \text{price} = 2 \times \text{price}$. (3) Add price to both sides: $90 = 3 \times \text{price}$. (4) $\text{Price} = 90 \div 3 = 30$. (5) Quantity = $70 - 30 = 40$. After: (6) $100 - \text{price} = 2 \times \text{price} - 20$. (7) $120 = 3 \times \text{price}$. (8) $\text{Price} = 40$. (9) Quantity = $100 - 40 = 60$. *Answer.* The equilibrium moves from a price of 30 and a quantity of 40 to a price of 40 and a quantity of 60. *Check.* At a price of 40, quantity demanded is $100 - 40 = 60$ and quantity supplied is $2 \times 40 - 20 = 60$. Demand shifted right, and both price and quantity rose, as the table says.

Three slips cause most wrong answers. First, forgetting to divide by the average and using the start value only. Second, writing the ratio upside down: the percentage change in quantity goes on top. Third, calling a price change a shift. Elasticity has no unit, because it divides one percentage by another; do not write taka or notebooks after it.

* * *

SUMMARY

Elasticity of supply is the percentage change in quantity supplied divided by the percentage change in price, found with the midpoint method. Values above 1 are elastic and below 1 are inelastic. In a competitive market, the equilibrium price makes the quantity demanded equal the quantity supplied. Excess demand pushes the price up; excess supply pushes it down. A shift of demand or supply moves the equilibrium to a new price and quantity.

THINK IT THROUGH

Use the notebook market of this chapter to answer the first question. Use the factors of section 9.1 for the second.

9.11 A law forbids any price above Tk 30 for a notebook. Using the notebook market, state the quantity demanded, the quantity supplied and the excess demand at Tk 30, and say who gains and who loses.

9.12 A farmer says, "Give me a year and I can supply far more rice, but this month I cannot." Which factor from section 9.1 does she describe, and what does it say about the elasticity of supply this month and next year?

WHERE PEOPLE GO WRONG

Mistake: "A fall in price shifts the demand curve." A change in the price of the good moves buyers along the same curve. A shift needs a change in something else, such as the number of buyers. The

corrected rule: price changes cause movements along; other changes cause shifts.

Mistake: "A flat supply curve is always elastic." Steepness depends on the units of the axes, but elasticity uses percentages and does not. A straight line from the origin has an elasticity of 1 at every point, however steep it looks.

Mistake: "At equilibrium everyone is satisfied." Equilibrium means only that no force pushes the price to change. Some buyers still cannot afford the good at that price, and some firms would like a higher price. The corrected rule: equilibrium describes balance, not fairness.

Mistake two: reading elasticity as the steepness of the line.

Mistake three: thinking equilibrium means a fair price or a happy market.

IN HISTORY

On 29 September 2019, the government of India banned the export of onions with immediate effect, to improve supply at home. Bangladesh imports about a quarter of the onions it uses, and the Commerce Minister of Bangladesh said that about nine-tenths of those imports came from India. The ban therefore shifted the supply of onions in Dhaka to the left. In the Kawran Bazar market, the price of imported onions rose from Tk 60 to Tk 100 a kilogram overnight, according to the *Business Standard*. In the following weeks, retail prices in Dhaka reached about Tk 250 a kilogram, according to the *Dhaka Tribune*.²

The case also shows the factors of section 9.1. In the short run, the supply of onions was inelastic: the harvest season had ended and the crop could not be grown again in a week. Buyers kept buying, because onions are a daily kitchen good. Over the following weeks, importers brought onions from other countries, including Pakistan and Egypt, and supply became more elastic. As the new onions arrived, the price eased.

In 2019, a sudden ban shifted supply and the price of onions in Dhaka jumped.

The case shows that short-run supply is inelastic and that time raises the response.

REVIEW QUESTIONS

9.1 The price of a good rises from Tk 20 to Tk 30, and the quantity supplied rises from 90 to 110 units. Calculate the elasticity of supply with the midpoint method, and name its type.

9.2 The price of mangoes rises from Tk 100 to Tk 130 a kilogram. The weekly quantity supplied rises from 400 to 500 kilograms. Calculate the elasticity of supply.

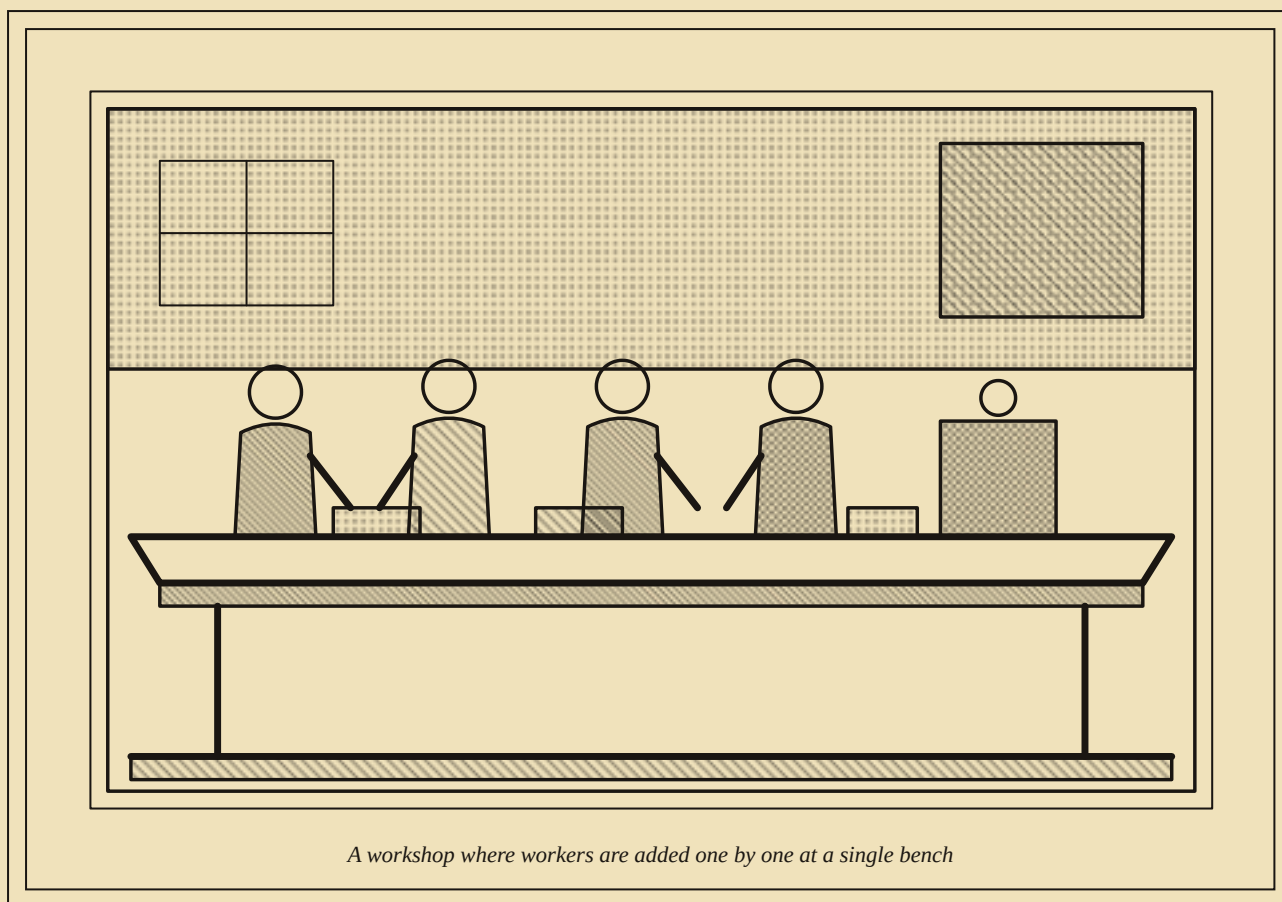
9.3 The price of a good falls from Tk 60 to Tk 50, and the weekly quantity supplied falls from 330 to 270 units. Calculate the elasticity of supply, and name its type.

9.4 In a market, quantity demanded = $60 - \text{price}$ and quantity supplied = $2 \times \text{price}$. Find the equilibrium price and quantity. Then demand shifts to a quantity demanded of $90 - \text{price}$; find the new equilibrium.

9.5 In the market of question 9.4, supply then shifts to a quantity supplied of $2 \times \text{price} - 30$, with demand at $90 - \text{price}$. Find the new equilibrium, and compare it with the equilibrium after the demand shift only.

Five questions follow the ladder from clean figures to a combined problem.

² Prices differed by market, by variety and by day, so reports give different peaks.



CHAPTER X

Production Function and Returns

Why does the sixth worker add less than the second?



This chapter shows how inputs become output, why each extra worker added to a fixed workshop eventually adds less than the one before, and what happens to output when every input grows together.

YOU WILL NEED

Chapter 2, 'Scarcity, Choice and Opportunity Cost', section 2.2 'Opportunity Cost', for the idea of giving up one use to gain another; Chapter 8, 'Supply and the Supply Curve', section 8.1 'Quantity Supplied and the Law of Supply', for supply; and Appendix A1, 'Maths You Will Use', for ratios, averages and simple graphs.

A workshop owner adds workers one by one and watches the output.

By the sixth worker the bench is crowded and the output no longer grows.

The puzzle is why extra workers first add more and then add less.

A bookbinder owns a small workshop in an old street. It has a single long bench and one sewing machine. On Monday the owner hires one worker, who folds the pages, sews them and fits the covers alone, and the day's output is modest. On Tuesday a second worker arrives. One folds while the other sews, and the output is more than double. Encouraged, the owner hires a third worker, then a fourth. The output keeps rising, but each new worker adds a little less than the last.

By the sixth day the bench is crowded. Workers wait for the machine, carry the same pile of paper from hand to hand and step round one another. The sixth worker is as willing as the first, and works as hard, yet the day's output does not rise at all. The owner has paid for one more worker and received nothing for it. Nothing about the workers has changed. Something about the bench has.

This is the puzzle of the chapter. Why did the second worker add more than the first, and why did the sixth add nothing? If extra workers always added as much as the first, one workshop could feed the whole city, and we would never need a second bench. Section 10.1 names the parts of the problem and defines the measures. Section 10.2 explains the pattern, which is among the most important in economics. Section 10.3 asks what happens when the owner builds a bigger workshop and raises every input at once.

* * *

10.1 INPUTS AND OUTPUT

THE RULE THAT LINKS THEM

Begin in a kitchen. A baker uses flour, water, an oven, fuel and time to make loaves. The things used are the inputs, and the loaves are the output. Inputs are the workers, machines, materials and land that a firm uses in production; output is the quantity of goods that results. A baker with more flour and a second oven can make more loaves, but not without limit, and not in any proportion.

Every input has another use, so using it has an opportunity cost, as Chapter 2 explained. A firm therefore wants the largest output from the inputs it has chosen. Engineers and owners can describe the relationship between inputs and output in a table, a graph or a rule. Economists give the rule a name.

A **production function** means a rule showing the largest output that can be made from each combination of inputs. Suppose a baker with 1 oven and 2 helpers can make at most 100 loaves a day. Then 100 is the output in the production function for that combination. The words 'largest output' matter. A careless baker might produce less from the same inputs, but the production function records only what the best method can achieve.

GO SLOWLY HERE.

A baker's kitchen shows how inputs become output.

Recall: opportunity cost, Chapter 2, section 2.2 'Opportunity Cost'.

Production function. A rule showing the largest output that can be made from each combination of inputs.

A production function links each combination of inputs to its largest output.

Change the combination, and the output changes. Two helpers and two ovens might give 180 loaves. One helper and one oven might give 60. The production function lists all such pairs. It does not say which combination to choose, because that depends on prices and costs, which later chapters study. It states only what is technically possible, and it assumes the best known methods.

Before moving on. Inputs are what a firm uses, and output is what it makes. A production function is the rule that shows the largest output for each combination of inputs. It describes what is possible, not what is wise.

THE SHORT RUN AND THE LONG RUN

Not every input can be changed at once. A baker can hire another helper by next week, but a new oven must be ordered, delivered and installed, which may take months. A firm cannot adjust all its inputs at the same speed, and the speed decides what choices it has.

The **short run** means a period so short that at least one input, such as a machine, cannot be changed. In the short run the baker may hire more helpers and buy more flour, but the oven stays. An input that cannot be changed in the period is called a fixed input, and an input that can be changed is a variable input. The short run is not a number of days or months. It is whatever period leaves at least one input fixed.

The **long run** means a period long enough for all inputs to be changed. Given enough time, the baker can build a new bakery, buy five ovens and move to another street. In the long run nothing is fixed. Different firms have different long runs: a roadside tea stall can change every input in a week, while a steel plant may need years. The two periods divide this chapter, because the law in section 10.2 belongs to the short run, and the idea in section 10.3 belongs to the long run.

Before moving on. In the short run at least one input is fixed. In the long run all inputs can be changed. The length of each depends on the firm and the industry.

TOTAL, MARGINAL AND AVERAGE PRODUCT

We now need three measures of output. They are needed to follow how output changes when the firm varies one input, such as workers, while all other inputs stay fixed. Take the smallest possible example, a baker with one oven. With 1 helper the baker makes 6 loaves a day, and with 2 helpers 14 loaves a day.

The first measure is the simplest. **Total product** means the whole quantity of output produced with a given amount of inputs. With 1 helper, total product is 6 loaves. With 2 helpers, it is 14 loaves. It is the number that the firm sells.

The second measure asks what the last helper added. **Marginal product** means the extra output from adding one more unit of an input, others held fixed. In words: marginal product equals the change in total product divided by the change in the input. In symbols, $MP = \Delta TP \div \Delta L$, where MP stands for marginal product, TP for total product, L for the number of workers, and Δ for 'change in'. When the baker adds the second helper, total product rises from 6 to 14, so $MP = (14 - 6) \div (2 - 1) = 8$ loaves.

The third measure asks what each helper produces on average. **Average product** means total product divided by the number of units of the input used. In symbols, $AP = TP \div L$. With 2 helpers, $AP = 14 \div 2 = 7$ loaves per helper. Many students find these three measures confusing, because they use the same units

Product. Word in use. In daily life, a good that is offered for sale. In this chapter, the quantity of output, as in total product.

Different combinations of inputs give different outputs.

Before moving on: a production function ties inputs to the largest output.

Some inputs can be changed quickly, and others cannot.

Short run. A period so short that at least one input, such as a machine, cannot be changed.

The short run is a period in which at least one input is fixed.

Long run. A period long enough for all inputs to be changed.

The long run is a period in which every input can be changed.

Before moving on: the short run has a fixed input and the long run has none.

GO SLOWLY HERE.

Three measures describe how output responds to one input.

Total product. The whole quantity of output produced with a given amount of inputs.

Total product is the whole output from the given inputs.

Marginal product. The extra output from adding one more unit of an input, others held fixed.

Marginal product is the extra output from one more unit of the input.

Average product. Total product divided by the number of units of the input used.

Average product is the output per unit of the input.

The measures can be read from a table of workers and output.

You should now be able to say what each of the three measures asks.

The binding workshop gives the pattern in numbers.

Recall: supply, Chapter 8, section 8.1 'Quantity Supplied and the Law of Supply'.

The schedule is a table that lists the three measures worker by worker.

Total product rises throughout, but by smaller and smaller steps after the second worker.

The workshop's pattern is an example of a general law.

Law of diminishing returns. Adding more of one input to fixed amounts of others will, after a point, add less and less to output.

The law of diminishing returns says extra units of one input eventually add less.

and sound alike. Remember the questions: total product asks how much altogether, marginal product asks how much the last unit added, and average product asks how much each unit produced on average.

PAUSE AND CHECK 10.P1 A baker with one oven makes 5 loaves with 1 helper, 12 with 2 helpers and 18 with 3 helpers. Find the marginal product of the third helper and the average product with 3 helpers.

You should now be able to say what total, marginal and average product ask: how much altogether, how much the last unit added, and how much per unit on average.

* * *

10.2 THE LAW OF DIMINISHING RETURNS

THE PATTERN IN THE WORKSHOP

Return to the bookbinder. The workshop has one machine, and the owner varies only the number of workers. The record shows what happened.

Case file, illustrative. A binding workshop has one machine. With 1 to 6 workers, daily output is 10, 24, 36, 44, 48 and 48. The marginal product is 10, 14, 12, 8, 4 and 0. The average product is 10, 12, 12, 11, 9.6 and 8.

To read the table below, go along each row. The first column gives the number of workers, the second gives total product, the third gives the marginal product of the last worker, and the fourth gives average product. Each marginal product is the difference between a total product and the one above it. Each average product is a total product divided by the number of workers on its row.

THE BINDING WORKSHOP'S PRODUCT SCHEDULE

WORKERS	TOTAL PRODUCT	MARGINAL PRODUCT	AVERAGE PRODUCT
1	10	10	10
2	24	14	12
3	36	12	12
4	44	8	11
5	48	4	9.6
6	48	0	8

Follow the marginal column. The first worker adds 10 and the second adds 14, because two people can share the tasks and keep the machine busy. The third adds 12, the fourth 8, the fifth 4 and the sixth 0. From the third worker onward every extra worker adds less than the one before, while total product still rises until the fifth worker. The sixth leaves total product at 48.

PAUSE AND CHECK 10.P2 In the schedule, what does the figure 8 in the marginal column mean, and what is the average product when 4 workers are used?

The pattern is general enough to be called a law. The **law of diminishing returns** means that adding more of one input to fixed amounts of others will, after a point, add less and less to output. The binding workshop shows it: with the machine fixed, each worker beyond the second adds less than the one before. The law does not say that output falls. It says that the extra output shrinks.

IN SHORT

Hold some inputs fixed, keep adding one input, and the extra output from each new unit will eventually become smaller and smaller.

PAUSE AND CHECK 10.P3 State the law of diminishing returns in your own words and name the fixed input in the binding workshop.

WHY RETURNS DIMINISH, AND WHEN THE LAW HOLDS

Why does the pattern appear? In the binding workshop the machine is fixed. The first worker must do every task alone, moving between folding, sewing and covering, and so wastes time. The second worker allows each person to specialise, and the machine is kept busy. This is why marginal product rises at first. Soon the machine is in use all day. Each new worker now has less machine time to share, must wait for a turn, or takes over a task someone else was doing well. By the sixth worker the machine can produce no more, and no addition of labour can change that.

The law holds under conditions that you should state each time. First, at least one input is fixed, so the law is a short-run law. Second, the methods of production stay the same; a better machine would move the whole schedule up. Third, the units of the variable input are alike. The workers in the workshop are equally skilled, so the fall in marginal product is not caused by hiring worse workers. It is caused by the crowding of a fixed input. If a student explains the fall by saying 'the later workers are lazier', the explanation is wrong, and the law is lost.

A common error concerns what diminishes. Total product does not have to fall, and in our case it rises until the fifth worker. What diminishes is the marginal product, the addition to output. An example from farming helps. Put one more labourer on a single field, and the harvest rises. Put a hundred more, and it hardly rises, because the soil, not the labour, limits the harvest. The extra output shrinks long before any harm is done.

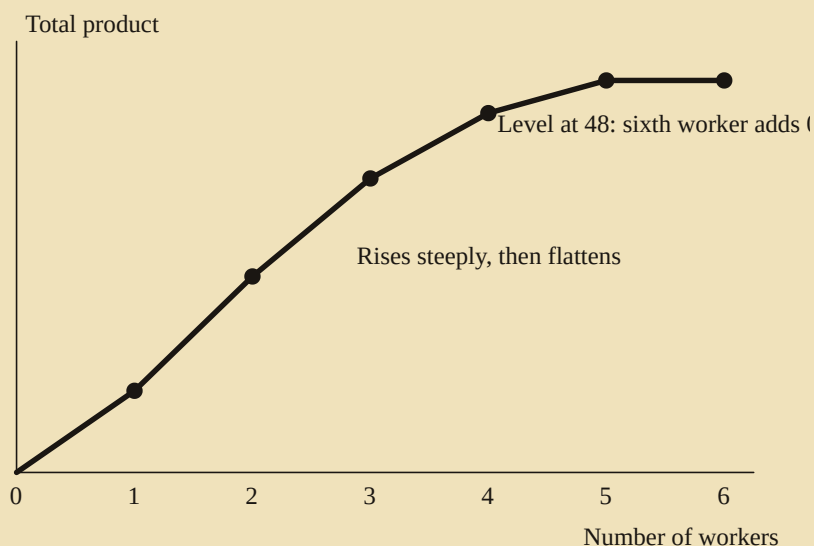
To read the first figure, look at the axes. The horizontal axis shows the number of workers; the vertical axis shows total product. Each point shows the output for one number of workers. The curve rises steeply for the first two workers, flattens through the fourth and fifth, and levels out at the sixth, where it no longer rises.

The cause is a fixed input shared among more and more workers.

The law holds only under stated conditions.

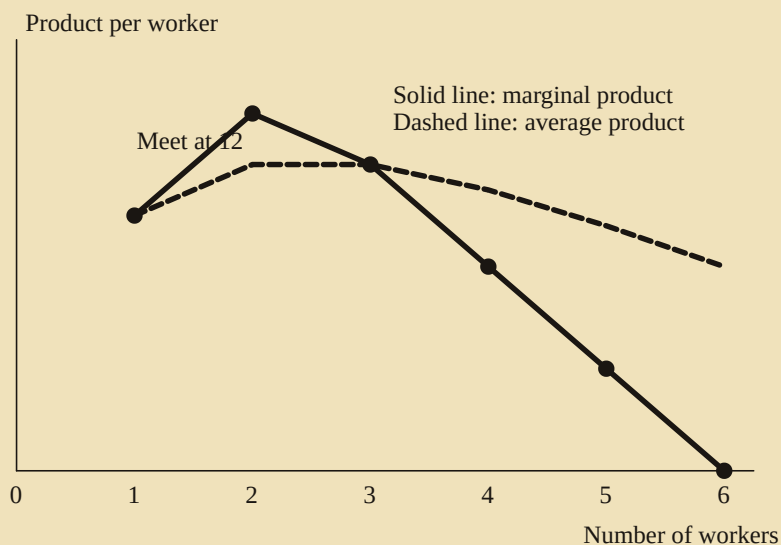
The law is about the extra output, not about the whole output.

The first figure shows total product rising, quickly at first, then slowly, then not at all.

FIG. 10.1. *Total product curve for the binding workshop*

The second figure shows the marginal and average curves, which meet at the highest average.

The second figure draws the other two measures. The solid line is marginal product and the dashed line is average product. Marginal product climbs to 14 at the second worker, then falls to 0. Average product climbs to 12, stays there for the third worker and then falls. The curves meet at the third worker, where both equal 12, and this is where average product is at its highest.

FIG. 10.2. *Marginal and average product curves for the binding workshop*

The marginal product pulls the average up or down.

The link between the two curves is a rule that deserves attention. When marginal product is above average product, the average rises; when it is below, the average falls; when the two are equal, the average neither rises nor falls. Check it against the table. The second worker adds 14, above the average of 10, so the average climbs to 12. The third adds 12, equal to the average, so the average stays at 12. The fourth adds 8, below 12, so the average falls to 11. The same rule governs the average mark of a class: a new student above the average raises it, and one below lowers it.

THREE STAGES OF PRODUCTION

The schedule divides into three stages.

The schedule can be divided into three stages. The first stage runs from the first worker to the point where average product is highest, here the third worker.

Average product is rising, so each worker added makes the others more productive. The second stage runs from that point to the point where marginal product reaches zero, here the sixth worker. Marginal product is positive but falling, and average product is falling. The third stage begins when marginal product turns negative, which means total product falls. The binding workshop has not reached it.

A careful firm avoids the third stage, since no firm wants to pay for a worker whose presence lowers output. It also has little reason to stop in the first stage, since extra workers raise the average output of all. This reasoning points a firm toward the second stage. Where in the second stage the owner stops depends on costs and prices, which Chapter 11 and later chapters study.

Before moving on. The law of diminishing returns says that with some inputs fixed, extra units of one input eventually add less and less. Marginal product rises at first, then falls; average product rises while marginal product is above it. The three stages divide the schedule at the peak of average product and at the point where marginal product reaches zero.

* * *

10.3 RETURNS TO SCALE

CHANGING EVERY INPUT

The law of diminishing returns changed one input and held the others fixed. The owner can ask a different question: if the workshop is rebuilt with twice as many workers and twice as many machines, will output double? The answer matters for the long run, when all inputs can change.

Returns to scale means the change in output when all inputs are increased in the same proportion. Double the workers and double the machines, and see what happens to output. There are three possible answers, and each has a name. The test is to compare the proportional rise in output with the proportional rise in inputs.

Output may rise by more than the inputs. This is increasing returns to scale. It may rise by exactly as much, which is constant returns to scale. Or it may rise by less, which is decreasing returns to scale. In symbols, if inputs are multiplied by k , output is multiplied by more than k , by exactly k , or by less than k . Here k stands for any number above 1, such as 2 or 1.5.

Case file, illustrative. In the binding workshop, 2 workers and 1 machine give an output of 24. With 4 workers and 2 machines the output is 60. With 6 workers and 3 machines it is 90. With 8 workers and 4 machines it is 108.

Given: the four outputs above. *Find:* the kind of returns to scale at each step. *Steps.* From 2 workers and 1 machine to 4 and 2, inputs are multiplied by $4 \div 2 = 2$, and output by $60 \div 24 = 2.5$. Output grows faster than inputs: increasing returns. From 4 and 2 to 6 and 3, inputs are multiplied by 1.5, and output by $90 \div 60 = 1.5$. They match: constant returns. From 6 and 3 to 8 and 4, inputs are multiplied by $8 \div 6 = 1.33$, and output by $108 \div 90 = 1.2$. Output grows more slowly: decreasing returns. *Answer:* increasing, then constant, then

A firm usually works in the second stage.

Before moving on: the law, the schedule and the stages describe one pattern.

GO SLOWLY HERE.

The long run raises every input together, which is a different question.

Scale. Word in use. In daily life, an instrument for weighing, or the ratio on a map. In economics, the size of a firm's operation, meaning the amount of all its inputs taken together.

Returns to scale. The change in output when all inputs are increased in the same proportion.

Returns to scale is the change in output when all inputs grow in the same proportion.

The three cases are compared by the size of the rise in output.

The record gives four outputs for four sizes of workshop.

The record can be classified step by step.

decreasing. *Check:* the pattern of 2.5, 1.5 and 1.2 agrees with the pattern of inputs.

CAUSES

Why increasing returns at first? A larger workshop allows each worker to specialise, can use larger machines that a small workshop cannot afford, and can buy materials in bulk at lower prices. Why decreasing returns later? A very large firm is hard to manage. Orders pass through more people, mistakes spread, and effort is spent on coordination rather than production. Constant returns lie between: the firm repeats what it already does. Many firms pass through all three as they grow.

Do not mix up the two ideas. The table below compares them. Read each row across: the first column names the feature, and the next two give the answer for each idea.

DIMINISHING RETURNS COMPARED WITH RETURNS TO SCALE

FEATURE	LAW OF DIMINISHING RETURNS	RETURNS TO SCALE
Inputs changed	One input; others fixed	All inputs together
Period	Short run	Long run
Question	How does the extra output change?	How does output change with the size of the firm?
Result	Extra output shrinks	Output may rise by more, the same or less

Before moving on: the long run asks about all inputs, the short run about one.

Before moving on. Returns to scale compares the proportional rise in output with the proportional rise in all inputs. Increasing means output rises faster, constant means equally, and decreasing means more slowly. The law of diminishing returns belongs to the short run, and returns to scale to the long run.

* * *

WORKED EXAMPLES: FOUR RUNGS

Four examples, from clean numbers to a combined problem.

Rung 1: a table with small clean numbers.

Rung 2: realistic data in kilograms are less tidy.

Rung 3: the unit and the sign change.

Each example below follows the order Given, Find, Formula, Steps, Answer, Check.

Given: a baker with one oven makes 5, 12, 18 and 20 loaves with 1, 2, 3 and 4 helpers. *Find:* the marginal and average product at each number of helpers. *Formula:* $MP = \Delta TP \div \Delta L$ and $AP = TP \div L$. *Steps:* MP: 5; $12 - 5 = 7$; $18 - 12 = 6$; $20 - 18 = 2$. AP: $5 \div 1 = 5$; $12 \div 2 = 6$; $18 \div 3 = 6$; $20 \div 4 = 5$. *Answer:* MP is 5, 7, 6, 2 and AP is 5, 6, 6, 5. *Check:* the marginal product is above the average product while the average rises, and equals it at the peak of 6 with 3 helpers.

Given: on one acre, 1 to 5 labourers harvest 420, 900, 1,300, 1,560 and 1,690 kg of rice. *Find:* the marginal and average product. *Steps.* MP: 420; 480; 400; 260; 130. AP: 420; 450; 433.33; 390; 338. *Answer:* marginal product peaks with the second labourer and falls after, while average product peaks at the second as well. *Check:* the third labourer's marginal product (400) is below the average (450), so the average falls, to 433.33, as it should.

Given: a workshop records output in dozens: 4 dozen with 5 workers, 4 dozen with 6 workers, and 3.75 dozen with 7 workers. *Find:* the marginal product of the sixth and seventh workers, in single items. *Steps.* Sixth: $4 - 4 = 0$

dozen. Seventh: $3.75 - 4 = -0.25$ dozen, and $-0.25 \times 12 = -3$ items. *Answer:* the sixth adds nothing and the seventh takes away 3 items. *Check:* a negative marginal product means total product is falling: the third stage.

Given: a bakery with 1 oven and 2 helpers makes 30 loaves a day. With 2 ovens and 4 helpers it makes 66. With 2 ovens and 5 helpers it makes 76. *Find:* the returns to scale from the first to the second combination, and the marginal product of the fifth helper. *Steps.* Inputs are multiplied by 2. Output is multiplied by $66 \div 30 = 2.2$. The fifth helper's marginal product, with the ovens fixed at 2, is $76 - 66 = 10$. *Answer:* increasing returns to scale, and a marginal product of 10 loaves. *Check:* the first answer changed all inputs; the second changed one input only.

Rung 4: a combined problem uses the law and returns to scale together.

* * *

SUMMARY

A production function shows the largest output from each combination of inputs. In the short run at least one input is fixed; in the long run none is. Total product is the whole output, marginal product is the extra output from one more unit, and average product is output per unit. The law of diminishing returns says that, with some inputs fixed, extra units of one input eventually add less and less. Returns to scale asks what happens when all inputs rise in the same proportion.

The chapter's main results in a few lines.

THINK IT THROUGH

10.T1 The sixth worker in the binding workshop adds nothing. Explain why a seventh worker is unlikely to add anything, without using the idea that later workers are less able.

10.T2 The workshop pays each worker the same daily wage. Why does the labour cost of each extra unit of output rise as more workers are hired?

WHERE PEOPLE GO WRONG

Many students think that diminishing returns means output falls; it means that the extra output shrinks. Some explain it by saying that later workers are less skilled, when the cause is the crowding of a fixed input. Others confuse the law with returns to scale, and the table in section 10.3 shows why they differ. The last error is to treat marginal and average product as the same measure; the first is the last unit's addition, and the second is the output per unit.

Four common misunderstandings, each corrected.

IN HISTORY

Thomas Robert Malthus (1766 to 1834) was an English clergyman and scholar. In 1798 his book *An Essay on the Principle of Population* appeared in London, first without the author's name. He wrote it in reply to the hopeful ideas of the writer William Godwin. His central claim was that population tends to grow faster than the food supply.

In 1798 Malthus applied the idea of shrinking extra output to food.

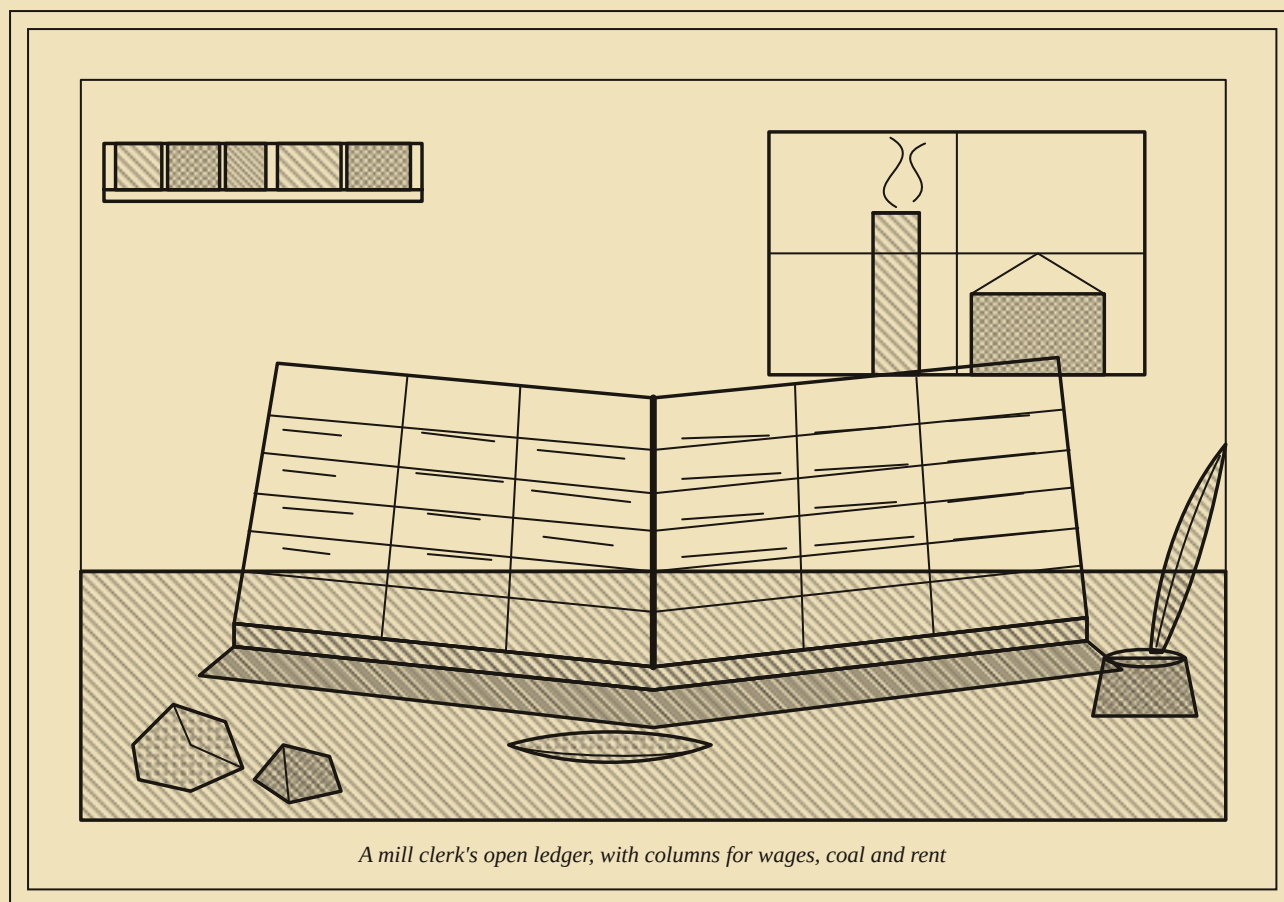
The argument rests on the idea of this chapter. Land is limited, and adding more labourers to the same land adds less and less food. If people multiply faster than food, hunger follows. Malthus later taught at the East India Company's college at Haileybury. His forecast has not held in most countries, because new seeds, tools and farming methods raised output beyond what the old methods

His argument is a famous use of diminishing returns, though it failed in one respect.

allowed. In the language of this chapter, improved methods moved the whole schedule upward.

REVIEW QUESTIONS

- 10.1 Define production function, and say why it concerns the largest output.
- 10.2 Distinguish the short run from the long run, and give an example of a fixed input.
- 10.3 A farm with fixed land has outputs of 30, 70, 100 and 120 kg with 1, 2, 3 and 4 labourers. Complete a table of marginal and average product.
- 10.4 In the binding workshop, explain why the average product stays at 12 for the third worker.
- 10.5 Inputs are multiplied by 3 and output is multiplied by 2.4. Classify the returns to scale.
- 10.6 Inputs are multiplied by 2 and output is multiplied by 2.6. Classify the returns to scale.
- 10.7 State two differences between the law of diminishing returns and returns to scale.



A mill clerk's open ledger, with columns for wages, coal and rent

CHAPTER XI

Costs of Production

Why can a shop pay every bill and still lose?



Rent, wages and paper are easy to count. Other costs hide, because nobody sends a bill for them. This chapter sorts costs into kinds, shows how to calculate each one by hand, and explains why the cost curves take the shapes they do.

YOU WILL NEED

Chapter 10, 'Production Function and Returns', section 10.1, 'Inputs and Output' (the production function, the short run and the long run); Chapter 10, section 10.2, 'The Law of Diminishing Returns'; and Appendix A1, 'Maths You Will Use'.

Two stalls show the same takings, yet one owner has chosen badly.

The answer depends on counting what the owner gave up, not only what the owner paid.

This chapter sorts costs into kinds, measures each kind and shows how cost changes with output.

A household garden shows that a resource can be costly even when nobody is paid for it.

GO SLOWLY HERE.

The first kind of cost is the one that appears in a receipt or a bank record.

Cost. Word in use. In daily life, cost often means the price a buyer pays. In economics, cost means the seller's expense of making a good, and it includes costs that nobody pays in money.

Explicit cost. A cost that involves a payment of money to outsiders.

Two friends open stalls in the same market. At the end of each day, each friend counts Tk 900 left after paying for goods and stall rent. Asha left a job that paid Tk 1,100 a day. Bina had no other offer of work. The two tills show the same figure. Yet one friend has made a poor choice, and the other has not. Which one, and why?

Most students first say that the two stalls are equal, because the tills agree. Look again at Asha. She could have earned Tk 1,100 by working elsewhere, so her stall leaves her Tk 200 worse off each day. Bina had nothing better to do, so for her the same Tk 900 is a pure gain. The puzzle shows that economists use the word *cost* more widely than a shop's account book does. Paying money is one kind of cost. Giving up a better use of your own time or money is another kind.

We do three things in this chapter. First, we separate costs that involve a payment from costs that do not. Next, we study how cost behaves when a firm makes more output in the short run, and we learn to calculate every kind of cost by hand. Last, we see why the cost curves have the shapes they have. Two examples run through the chapter: a small binding workshop, and the shop of Meghna and Sons.

* * *

11.1 EXPLICIT AND IMPLICIT COSTS

Think first of a household, not a firm. A family grows vegetables in its own garden. It pays Tk 300 a month for seeds and water. It pays nothing for the land, which it owns, and nothing for the hours its members work. Is the garden free, apart from the Tk 300? Not quite. The land could be rented to a neighbour, and the hours could earn wages elsewhere. These lost gains are real, even though no money changes hands. Economists call the best alternative given up the *opportunity cost* of a choice.¹

The first kind is easy to see. An **explicit cost** is a cost that involves a payment of money to outsiders. The word *explicit* means openly stated. For example, the rent that a shop pays its landlord each month is an explicit cost, because money leaves the shop and reaches another person. Wages, paper, electricity and interest on a bank loan are also explicit costs. Each one appears in a receipt or a bank record, and an accountant writes every one of them down.

¹ The idea of opportunity cost is taught in Chapter 2, 'Scarcity, Choice and Opportunity Cost'. In this chapter you need only the one-line meaning given in the text.

The second kind is easy to miss. An **implicit cost** is the value of the resources that the owner supplies to the firm, for which no payment is made. The word *implicit* means present but not openly stated. For example, an owner who works in her own shop without a salary supplies her time. That time has a value, because she could sell it to another firm. In the same way, money the owner has placed in the firm could have earned interest at a bank. Each of these has an opportunity cost, though nobody sends a bill.

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhanmondi, Dhaka, founded on 4 April 1996 by Nurul Meghna. Record for 2025: sales Tk 14,400,000; profit shown in the accounts Tk 750,000; capital employed Tk 4,100,000; rent Tk 60,000 a month; 7 staff with total wages Tk 2,940,000; bank loan Tk 600,000. Rent, wages and interest on the loan are explicit costs, and each appears in the shop's records. The case file values the owner's time at Tk 600,000 for the year. It also records forgone interest of 7 per cent on the capital employed, which is Tk 287,000. These two amounts are implicit costs. No bank record shows either one.

The test for an implicit cost is one question: what is the best use of this resource outside the firm? The answer gives its value. Nurul Meghna's time could earn a salary elsewhere, and her capital could earn interest. Each answer is the opportunity cost of one of her own resources. Many students find this step confusing, because the owner never feels the sacrifice as a payment. The confusion disappears when you ask the same question about every resource, whoever owns it. A rented room has an explicit cost. An owned room has an implicit cost. Both are costs.

The viewpoint changes the list. The accountant records explicit costs only. The owner, who must decide whether to carry on, counts both kinds, and economists take the owner's view. There are limits. An implicit cost is an estimate, not a receipt. Also, a resource with no other use has no implicit cost. This answers the puzzle at the start. Bina's time had no better use, so her implicit cost is nil, while Asha's is Tk 1,100 a day. Her stall leaves Tk 900 after paid costs, which is Tk 200 less than her old wage.

Before moving on. A cost is either explicit, involving a payment to outsiders, or implicit, the value of what the owner supplies without payment. We find an implicit cost by asking what the resource could earn elsewhere. The accountant records explicit costs only, and the economist counts both kinds. You should now be able to say, in your own words, why an owner's unpaid time is a cost.

PAUSE AND CHECK 11.P1 A tailor owns the building of her shop and could rent it to another person for Tk 5,000 a month. She pays rent to nobody. Is there a rent cost, and if so, of which kind?

* * *

11.2 SHORT-RUN COSTS

Begin with a market tea stall, because the numbers are small. The owner pays Tk 100 a day for the right to use her space. She pays that fee whether she sells no cups or fifty cups. Milk, tea leaves and sugar for each cup cost her Tk 4. If she sells no cups, she owes Tk 100. If she sells 10 cups, she owes Tk 100 plus 10 times Tk 4, which is Tk 140. If she sells 25 cups, she owes Tk 100 plus Tk 100, which is Tk 200. Two kinds of cost are at work: one that does not move, and one that grows with output.

The second kind of cost is the value of what the owner supplies without being paid.

Implicit cost. The value of resources the owner supplies, which involve no payment but have an opportunity cost.

The shop's records show explicit costs openly and implicit costs not at all.

To find an implicit cost, ask what the owner's own resource could earn elsewhere.

The accountant counts payments, while the owner counts everything given up.

The section closes with a recap and a test of understanding.

A tea stall shows how some costs stay put while others grow with every cup.

GO SLOWLY HERE.

Costs are studied in the short run, when at least one input cannot be changed.

A fixed cost stays the same whatever the output.

Fixed cost. A cost that does not change with the quantity produced in the short run, such as rent.

A variable cost rises and falls with output.

Variable cost. A cost that changes with the quantity produced, such as paper and wages for extra output.

Total cost joins the two kinds, so it is never smaller than the fixed cost.

Total cost. Fixed cost plus variable cost at a given output.

The three costs fit together in one short sum.

The workshop's records give fixed, variable and total cost at seven levels of output.

All the costs in this section belong to the short run. The short run is a period so short that at least one input, such as a machine, cannot be changed.² In the short run, therefore, the firm cannot avoid paying for its machine, however little it makes. Inputs that can be changed quickly, such as paper and extra workers, give the firm room to raise or cut its output.

The stall fee is the model. A **fixed cost** is a cost that does not change with the quantity produced in the short run, such as rent. A workshop's machine rent is fixed: it is owed on a day when ten books are bound and on a day when sixty are bound. Fixed cost is the cost of being ready to produce, not of producing. It is still a cost, and it must be paid.

The milk, tea leaves and sugar are the model for the other kind. A **variable cost** is a cost that changes with the quantity produced, such as paper and wages for extra output. If the stall sells more cups, it buys more milk. If it sells none, it buys none. In a workshop, paper, thread, glue and the wages of extra workers all rise as more books are bound.

Total cost is fixed cost plus variable cost at a given output. In symbols, $TC = FC + VC$, where TC is total cost, FC is fixed cost and VC is variable cost, all measured in taka for the same period. The tea stall at 10 cups has $TC = 100 + 40 = \text{Tk } 140$. Even at zero output, total cost equals fixed cost, because the fee is still owed.

Before moving on. Fixed cost does not move with output. Variable cost moves with output. Total cost is the two added together. You should now be able to say why total cost at zero output is not zero.

Case file, illustrative. A binding workshop uses one machine and a small team to bind books. Its fixed cost is Tk 600 a day. At daily outputs of 0, 10, 20, 30, 40, 50 and 60 books, its variable cost is Tk 0, 300, 520, 720, 960, 1,300 and 1,780. Adding the fixed cost to each variable cost gives total cost of Tk 600, 900, 1,120, 1,320, 1,560, 1,900 and 2,380. The table below sets these records in columns.

COSTS OF THE BINDING WORKSHOP, IN TAKA A DAY

OUTPUT (BOOKS A DAY)	FIXED COST	VARIABLE COST	TOTAL COST
0	600	0	600
10	600	300	900
20	600	520	1,120
30	600	720	1,320
40	600	960	1,560
50	600	1,300	1,900
60	600	1,780	2,380

To read a row, add the fixed and variable columns to get the total.

Read each row from left to right. In the row for 20 books, fixed cost 600 plus variable cost 520 gives total cost 1,120. Notice that the fixed column never changes, while the variable column climbs faster and faster. Notice also that the first row shows an output of nothing and a cost of Tk 600. The workshop pays for its machine even when it binds no book.

² Chapter 10, 'Production Function and Returns', section 10.1 'Inputs and Output', defines the short run and the long run.

Managers decide one step at a time, so they ask about the next unit. **Marginal cost** is the extra cost of producing one more unit of output. In words, marginal cost equals the change in total cost divided by the change in output. In symbols, $MC = (\text{change in TC}) \div (\text{change in Q})$, where MC is marginal cost and Q is the quantity of output. The tea stall's total cost rises from Tk 140 at 10 cups to Tk 144 at 11 cups. The change in TC is Tk 4 and the change in Q is 1 cup, so $MC = 4 \div 1 = \text{Tk } 4$.

The same rule works when output rises ten books at a time. In the workshop, total cost rises from Tk 900 at 10 books to Tk 1,120 at 20 books. The change in TC is Tk 220, and the change in Q is 10 books. So $MC = 220 \div 10 = \text{Tk } 22$ for each of those books. Fixed cost is Tk 600 in both rows, so it cancels in the subtraction. Marginal cost comes from variable cost alone. A fixed cost, however large, never changes marginal cost.

PAUSE AND CHECK 11.P2 At the workshop, total cost rises from Tk 1,320 at 30 books to Tk 1,560 at 40 books. What is the marginal cost of each of those books?

An owner who sets a price needs the cost of each unit on average. **Average total cost** is total cost divided by the number of units produced. In symbols, $ATC = TC \div Q$. For the tea stall at 25 cups, $ATC = 200 \div 25 = \text{Tk } 8$ a cup. For the workshop at 20 books, $ATC = 1,120 \div 20 = \text{Tk } 56$ a book. Average total cost measures the whole cost, fixed and variable, shared among all the units.

Average total cost splits into two parts. Fixed cost divided by output is the average fixed cost. It falls as output grows, because the same Tk 600 is shared among more books: Tk 60 a book at 10 books, and Tk 10 a book at 60 books. Variable cost divided by output is the average variable cost. At the workshop it is Tk 30, 26, 24, 24, 26 and 29.67 for outputs of 10 to 60 books. Adding the two averages gives ATC. At 10 books, 60 plus 30 gives Tk 90.

Case file, illustrative. For the same workshop, the marginal cost of each book in each ten-book step is Tk 30, 22, 20, 24, 34 and 48. Average total cost at outputs of 10 to 60 books is Tk 90, 56, 44, 39, 38 and 39.67. The table lists these figures beside the average variable cost.

PER-UNIT COSTS OF THE BINDING WORKSHOP, IN TAKA A BOOK

OUTPUT (BOOKS A DAY)	MARGINAL COST OF THE LAST TEN BOOKS	AVERAGE VARIABLE COST	AVERAGE TOTAL COST
10	30	30.00	90.00
20	22	26.00	56.00
30	20	24.00	44.00
40	24	24.00	39.00
50	34	26.00	38.00
60	48	29.67	39.67

Read down each column. Marginal cost falls from 30 to 20 and then climbs to 48. Average variable cost falls to 24 and then climbs to 29.67. Average total cost falls from 90 to 38 and then rises to 39.67. The lowest average total cost, Tk 38 a book, occurs at 50 books a day. The next section explains why the costs move in this pattern.

Marginal cost asks what one more unit adds to total cost.

GO SLOWLY HERE.

Marginal. Word in use. In daily life, marginal means small or barely enough, as in a marginal gain. In economics, marginal means extra: the change caused by one more unit.

Marginal cost. The extra cost of producing one more unit of output.

At the workshop, marginal cost is each rise in total cost divided by the rise in output.

Average total cost spreads total cost over every unit made.

Average total cost. Total cost divided by the number of units produced.

Average total cost is the sum of two smaller averages, one for each kind of cost.

The second table gives the per-unit costs at each output.

In the table, every cost per unit first falls and then rises.

IN SHORT

Total cost equals fixed cost plus variable cost. Marginal cost is the change in total cost divided by the change in output. Average total cost is total cost divided by output.

The section ends with a recap and a test of understanding.

Before moving on. Fixed cost stays still, variable cost grows, and total cost is their sum. Marginal cost measures the next unit and average total cost measures every unit. You should now be able to say why marginal cost and average total cost are different numbers.

* * *

11.3 THE COST CURVES

A cost curve draws a table as a picture, with one point for each output.

GO SLOWLY HERE.

A curve shows a table at a glance. The first figure below draws the workshop's fixed, variable and total cost. Here is how to read it. The horizontal axis, the line along the bottom, shows output in books a day. The vertical axis, the line up the side, shows cost in taka a day. Each point on a curve tells you the cost at one output. Start at an output on the bottom, move upward until you meet a curve, and read the cost from the side.

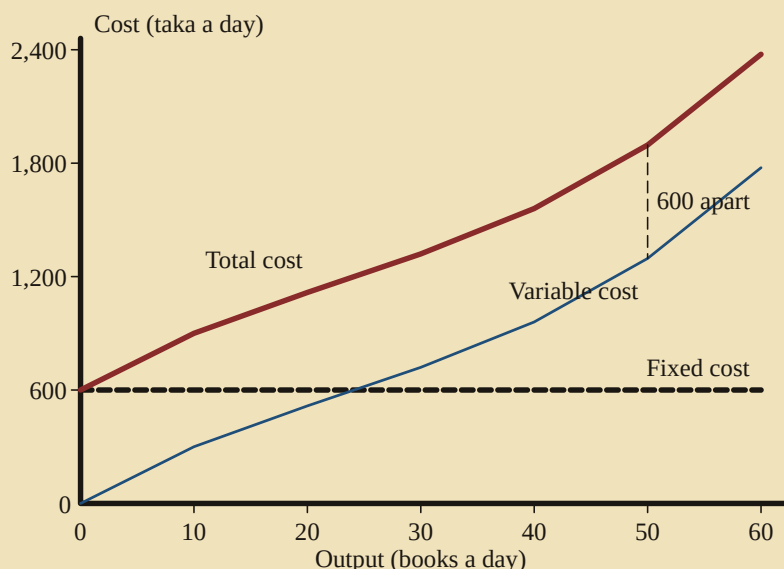


FIG. 11.1. Fixed, variable and total cost curves of the binding workshop

Fixed cost is a flat line, and total cost is the variable cost curve lifted by the fixed cost.

The fixed cost line is flat, because fixed cost does not change with output. The variable cost curve starts at zero, because nothing is spent on variable inputs when nothing is made. The total cost curve has exactly the shape of the variable cost curve, but it sits Tk 600 higher at every output. That gap is the fixed cost. At 30 books, for example, variable cost is 720 and total cost is 1,320.

Variable cost climbs slowly at first and quickly later, because extra workers add less and less to output.

The variable cost curve is gentle at first and steep later. The reason is the law of diminishing returns, which says that adding more of one input to fixed amounts of others will, after a point, add less and less to output.³ Suppose the workshop adds workers to its one machine. At first, each new worker adds a great deal of output, so each extra book needs little extra wage and paper. Later, each new worker adds little, so each extra book needs much more labour.

³ Chapter 10, 'Production Function and Returns', section 10.2 'The Law of Diminishing Returns', explains why the extra output from each new worker rises at first and then falls.

Marginal cost therefore falls at first and then rises. In the table it falls from 30 to 20 and climbs to 48.

The curve of average total cost is shaped like the letter U. Two forces explain this. The first force pulls the average down: fixed cost is shared among more books as output grows, so average fixed cost falls. The second force pulls the average up: after a point, rising marginal cost makes average variable cost rise. At low output the first force is stronger, and the average falls. At high output the second force is stronger, and the average rises. Near 50 books the forces balance, and average total cost is at its lowest, Tk 38.

Average total cost has a U shape because two forces pull on it in opposite directions.

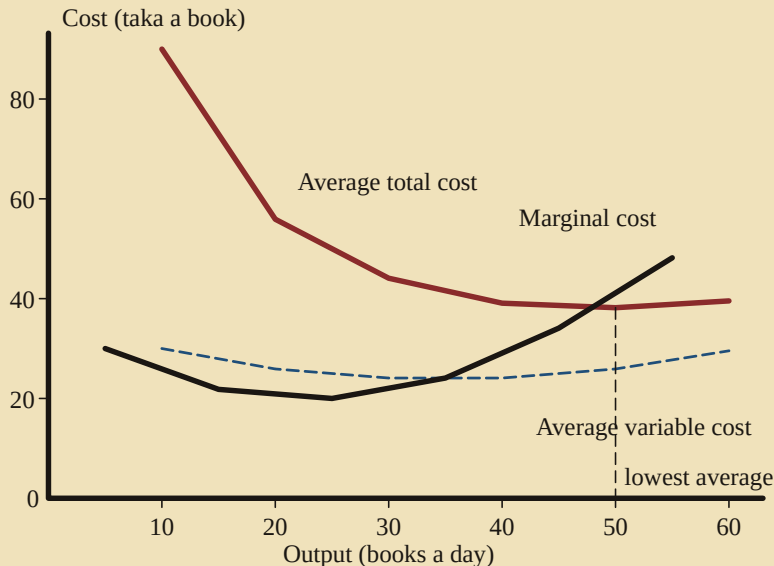


FIG. 11.2. Marginal, average variable and average total cost curves of the binding workshop

The second figure draws the three per-unit costs. The vertical axis now shows cost per book, not cost for the whole output. Look at where the marginal cost curve meets the average curves. Think of the average age of a class. A new student who is younger than the average lowers it, and a new student who is older raises it. The new student is the marginal addition. Costs follow the same rule. While the marginal cost of the next book is below average total cost, the average falls. When marginal cost is above it, the average rises. So marginal cost must cross average total cost at the bottom of the U.

Marginal cost crosses average total cost at the lowest point of the average.

The table confirms the rule. For the step from 40 to 50 books, marginal cost is Tk 34, below the average of Tk 39 at 40 books, so the average falls to Tk 38. For the step from 50 to 60 books, marginal cost is Tk 48, above the average of Tk 38, so the average rises to Tk 39.67. The same rule makes marginal cost cross average variable cost at its lowest point. Our table moves in steps of ten books, so the exact crossing lies between two rows.

The workshop figures show the crossing between two rows of the table.

So far one input, the machine, has been fixed. In the long run all inputs can be changed, so the firm has no fixed cost at all. It can choose a larger machine or a smaller one. The long-run cost curve shows the lowest average cost at which each output can be made when the firm is free to choose its size. It is usually also U-shaped, but for a different reason: the cause is returns to scale, taught in Chapter 10, and not diminishing returns.

In the long run every input can change, so the cost curve changes with it.

Before moving on. Cost curves draw the cost tables as pictures. Marginal cost falls and then rises because extra workers add less and less. Average total cost is U-shaped because fixed cost is shared more widely while variable cost per unit rises. Marginal cost cuts average total cost at its lowest point. You should now be

The section closes with a recap.

able to say why the bottom of the U is where marginal cost equals average total cost.

* * *

WORKED EXAMPLE: COSTS OF THE BINDING WORKSHOP

Two examples apply the formulas, one small and clean and one realistic.

The first example gives total, marginal and average total cost at 30 books.

Each line of arithmetic is shown, with its unit, so the method is the only new thing.

The answer is stated in a sentence and tested against the table.

The second example uses the shop's records to count implicit costs.

Every step is written out, so that nothing depends on guesswork.

The shop looks successful in its accounts but falls short once its own resources are valued.

Five common slips spoil these calculations, and each has a simple remedy.

We now work two examples in the same order each time: Given, Find, Formula, Steps, Answer and Check. The first uses the workshop table at an output of 30 books. The second uses the shop's own records and shows how implicit costs change the picture.

Given. Fixed cost is Tk 600 a day. Variable cost is Tk 520 at 20 books and Tk 720 at 30 books. *Find.* Total cost at 30 books, marginal cost of each book from 20 to 30 books, and average total cost at 30 books. *Formula.* In words, total cost is fixed cost plus variable cost. In symbols, $TC = FC + VC$. Marginal cost is $MC = (\text{change in } TC) \div (\text{change in } Q)$, and average total cost is $ATC = TC \div Q$.

Steps. Total cost at 30 books: $TC = 600 + 720 = \text{Tk } 1,320$. Total cost at 20 books: $TC = 600 + 520 = \text{Tk } 1,120$. Change in $TC = 1,320 - 1,120 = \text{Tk } 200$. Change in $Q = 30 - 20 = 10$ books. Marginal cost: $MC = 200 \div 10 = \text{Tk } 20$ a book. Average total cost: $ATC = 1,320 \div 30 = \text{Tk } 44$ a book.

Answer. At 30 books a day, total cost is Tk 1,320, the marginal cost of each book from 20 to 30 is Tk 20, and average total cost is Tk 44 a book. *Check.* Multiply back: $44 \times 30 = 1,320$, which equals total cost. The marginal cost of Tk 20 is below the average of Tk 44, so the average should be falling, and the table shows it falling from Tk 56 to Tk 44. Both figures agree with the table.

Given. For 2025, the profit shown in the accounts of Meghna and Sons is Tk 750,000.⁴ The owner's time is valued at Tk 600,000. Capital employed is Tk 4,100,000, and the interest rate the capital could earn elsewhere is 7 per cent a year. *Find.* The implicit costs, and the result after counting them. *Formula.* Forgone interest = rate $\times$ capital employed. Result after implicit costs = profit shown in the accounts – total implicit costs.

Steps. Write 7 per cent as $7 \div 100 = 0.07$. Forgone interest = $0.07 \times 4,100,000 = \text{Tk } 287,000$. Total implicit costs = $600,000 + 287,000 = \text{Tk } 887,000$. Result = $750,000 - 887,000 = -\text{Tk } 137,000$.

Answer. The implicit costs are Tk 887,000, and after counting them the shop falls Tk 137,000 short. *Check.* The implicit costs are larger than Tk 750,000, so the result must be negative. Adding back: $-137,000 + 887,000 = 750,000$, the figure in the accounts. What does the owner learn? Her resources would have earned Tk 137,000 more elsewhere, if other things were equal. She may still choose to carry on, because she may value the shop for other reasons. The calculation tells her the price of that choice.

Common slips are easy to name. First, forgetting fixed cost when adding up total cost. Second, dividing by the output when you need the change in output, or the reverse. Third, forgetting units, so that a cost for the whole output is mistaken for a cost per book. Fourth, mixing periods, such as a daily fixed cost with a yearly wage. Fifth, a wrong sign, such as subtracting the larger number from the smaller. Name the unit beside every number, and the first four slips are caught early.

⁴ Profit has several exact meanings, taught in Chapter 16, 'Interest and Profit'. Here it means only the money left after explicit costs are paid.

* * *

SUMMARY

Costs are explicit, when money is paid to outsiders, or implicit, when the owner supplies the resource. In the short run, total cost is fixed cost plus variable cost. Marginal cost is the change in total cost divided by the change in output, and average total cost is total cost divided by output. Marginal cost falls and then rises, average total cost is U-shaped, and marginal cost crosses average total cost at its lowest point.

The chapter's main points fit in a few lines.

* * *

THINK IT THROUGH

11.T1 In the workshop table, variable cost per book rises from Tk 24 at 40 books to Tk 26 at 50 books, yet average total cost falls from Tk 39 to Tk 38. Explain how this can happen.

11.T2 A bookseller says, "My own time is free, because I pay myself nothing." Reply using the idea of implicit cost, and say what question she should ask.

* * *

WHERE PEOPLE GO WRONG

Mistake: "A fixed cost never changes." The cost is fixed only in the short run, and only as a total. In the long run every input can change, so no cost stays fixed. Fixed cost per book also changes, because the same total is shared among more books.

A fixed cost is fixed only in the short run, and only as a total.

Mistake: "If nobody sends a bill, there is no cost." An implicit cost has no bill, but it has an opportunity cost. The owner's time and money could have been used elsewhere.

No bill does not mean no cost.

Mistake: "Marginal cost is total cost divided by output." That is average total cost. Marginal cost divides the change in total cost by the change in output. At 30 books, marginal cost is Tk 20 and average total cost is Tk 44.

Marginal cost and average total cost answer different questions.

Mistake: "Average cost is lowest where marginal cost is lowest." In the workshop, marginal cost is lowest at 30 books (Tk 20), but average total cost is lowest at 50 books (Tk 38). The average stops falling only when marginal cost rises to meet it.

The two lowest points are at different outputs.

* * *

IN HISTORY

In 1908, the Ford Motor Company of the United States began to sell the Model T, a car for ordinary families, at about USD 825. By 1925, the basic model cost about USD 260. In 1913, Ford introduced a moving assembly line at its factory in Highland Park, Michigan. The car moved past the workers, and each worker repeated one task, so more cars were built in less time.

A famous fall in the price of a car shows unit cost falling as output grows.

Higher output spread fixed costs and lowered cost per car, as the falling part of the U predicts.

The usual account is that the cost of making each car fell as output rose, and Ford passed much of the saving to buyers as lower prices. The figures above are prices, not costs. The fall in cost per car is the falling part of the average total cost curve, where large fixed costs are shared among very many units.

* * *

REVIEW QUESTIONS

The questions use a small print stall; the data are given here.

A print stall has a fixed cost of Tk 200 a day. At outputs of 0, 5, 10, 15, 20, 25 and 30 pages of posters a day, its variable cost is Tk 0, 100, 180, 240, 320, 430 and 560. Use these data for questions 11.2 to 11.6.

11.1 A tailor spends Tk 8,000 on cloth and Tk 3,000 on a helper's wage in a month. She works in her own shop, and her time is worth Tk 12,000 a month elsewhere. The building she owns could be rented for Tk 5,000. Find her explicit costs and her implicit costs.

11.2 Find the total cost of the print stall at each output from 0 to 30.

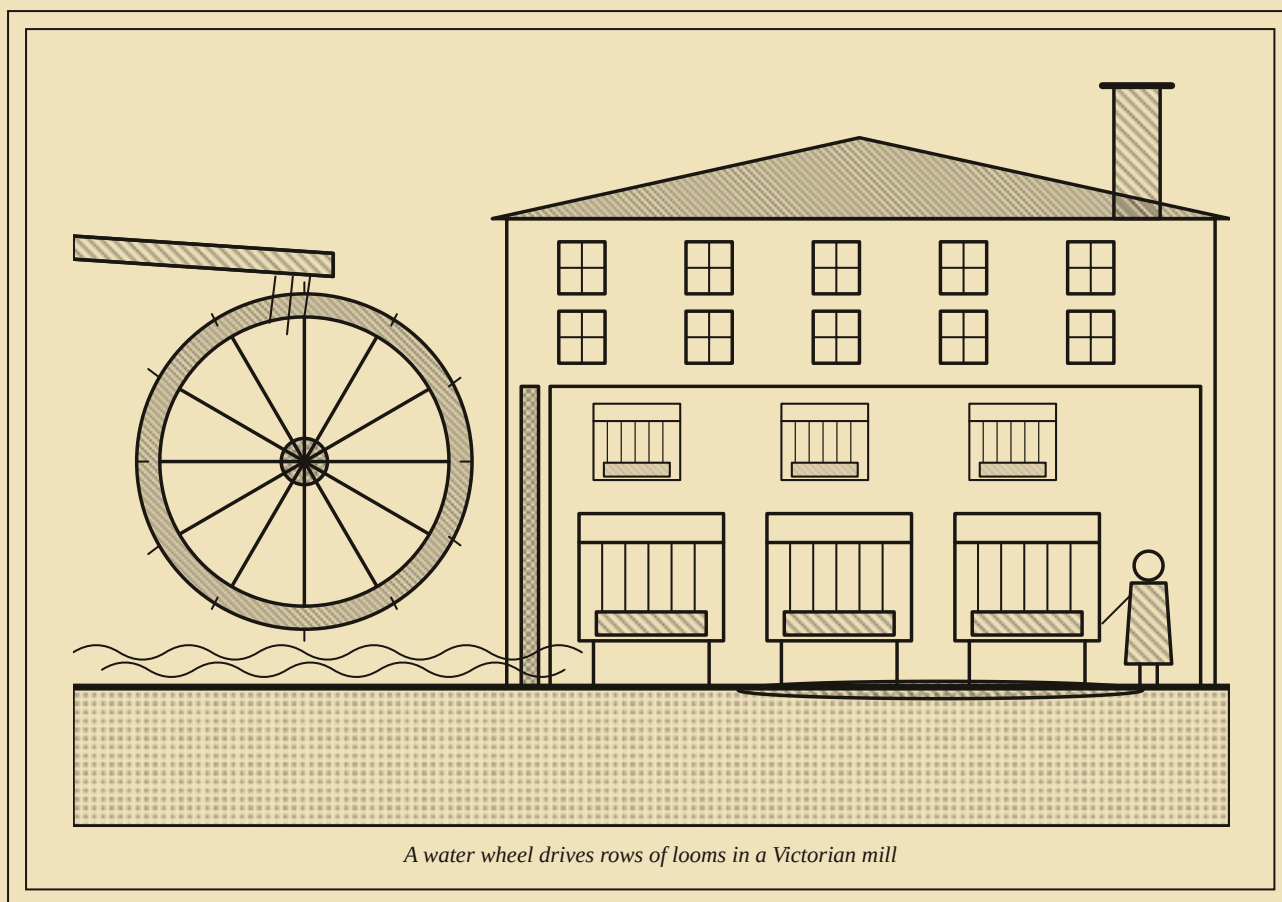
11.3 Find the marginal cost of each page in every step of five pages.

11.4 Find average total cost at each output from 5 to 30, to two decimal places. At which output is it lowest?

11.5 The stall works 6 days a week and makes 20 pages every day. What is its total cost for the week?

11.6 The stall's daily fixed cost rises from Tk 200 to Tk 260. State what happens to total cost at 20 pages, to marginal cost, and to average total cost at 20 pages.

11.7 A shopkeeper's accounts show a profit of Tk 500,000. Her own time is worth Tk 360,000 a year elsewhere. Her capital of Tk 3,000,000 could earn 8 per cent a year. Find her implicit costs and the result after counting them.



CHAPTER XII

The Least-Cost Combination of Inputs

How can a firm tell which mix of workers and machines is the cheapest way to make the same output?



This chapter shows how a firm chooses among the many ways of making the same output. You will learn to draw and read two lines, one for output and one for total cost, and to find the mix of inputs that costs least.

YOU WILL NEED

Chapter 10 'Production Function and Returns', section 10.1 'Inputs and Output';
 Chapter 11 'Costs of Production', section 11.1 'Explicit and Implicit Costs';
 Chapter 7 'Indifference Curve Analysis', section 7.2 'Marginal Rate of
 Substitution'; Appendix A1 'Maths You Will Use'.

A workshop owner must bind 100 books every day and can use workers, machines or both.

Each way has its own daily bill, so the owner needs a method to find the smallest.

Trying every way is slow and explains nothing; two lines give a better method.

Imagine the owner of a binding workshop who must bind 100 books every day. The books can be bound by hand, by machine, or by some of each. With hand work only, the workshop needs many workers and no machine. With machines only, it needs many machines and few workers. Between these two extremes lie many other ways to reach the same 100 books.

Each way has its own bill at the end of the day. Workers must be paid, and machines must be rented. More machines mean fewer workers, so one bill rises as the other falls. The owner is left with a question that has a definite answer: which way gives the same 100 books for the smallest total?

A beginner might try every way, one after another. That works when there are few ways, but it does not explain why one way wins. A better method uses two pictures. One line shows every way of making the output. Another line shows every way of spending a given sum. Where the two lines meet in the right manner, the cheapest mix is found. Many students find this idea new, but the thinking is familiar from the consumer's choice in Chapter 7, now applied to a firm. We start with the first line.

* * *

12.1 CHOOSING INPUTS

Recall: production function, Chapter 10, section 10.1 'Inputs and Output'.

A firm makes output from inputs, and a mix of inputs means the amounts used together.

Input. Word in use. In daily life, a contribution of ideas or advice, as in your input to a meeting. In economics, a thing a firm uses to make output, such as labour, a machine or a raw material.

Mix. Word in use. In daily life, to blend things together. In economics, the amounts of different inputs that a firm uses together, as in a mix of workers and machines.

Workers and machines can replace each other within limits, so many mixes give one output.

A firm makes its output by using inputs. Chapter 10 showed that a production function links the inputs to the output. In this chapter we study two inputs: workers and machines. Other inputs, such as paper and glue, are held fixed. Four workers with four machines is one mix. Five workers with three machines is another mix.

Workers and machines can replace each other, within limits. A machine can do the work of several people, and a skilled worker can do what a poor machine cannot. So the same output can come from different mixes. The production function tells us the output for each mix. Here we ask the reverse question: for a chosen output, which mixes reach it? The answer is a line.

The line has a name. An **isoquant** is a line showing all combinations of two inputs that produce the same output. The word *iso* means equal, and *quant* comes from quantity, so isoquant means equal quantity of output. If five mixes each give 100 books a day, all five lie on one isoquant.

Case file, illustrative. For an output of 100 books bound a day, the wage is Tk 600 per worker per day and the machine rent is Tk 400 per machine per day. Five combinations give this output, written as (workers, machines): (2, 10), (3, 6), (4, 4), (5, 3) and (6, 2). Joined together, they form one isoquant.

An isoquant has three properties. First, it slopes downward: to keep output the same, more workers mean fewer machines. Second, it curves toward the corner where the two axes meet, because each added worker replaces fewer machines than the one before. Third, two isoquants never cross, and an isoquant farther from the corner shows a larger output. These properties match those of an indifference curve and an indifference map, with one difference. Satisfaction cannot be counted, but output can be counted in books.

Now consider the owner's money. Suppose the owner decides to spend Tk 4,000 a day on inputs. With that sum the owner can hire many workers and rent few machines, or the reverse. Every division of Tk 4,000 between the two inputs lies on one line.

An **isocost line** is a line showing all combinations of two inputs that cost the same total amount. Here *cost* is the firm's expense. The formula in words is: total cost equals the wage times the number of workers, plus the machine rent times the number of machines. In symbols, $C = w \times L + r \times M$, where C is the total cost, w is the wage per worker, L is the number of workers, r is the rent per machine and M is the number of machines.

Take the case prices. If all of Tk 4,000 buys workers, the firm hires $4,000 / 600 = 6.67$ workers a day, which is an average over several days. If all of it buys machines, the firm rents $4,000 / 400 = 10$ machines. These are the two ends of the line. The mix (4, 4) lies on it, because $600 \times 4 + 400 \times 4 = 2,400 + 1,600 = \text{Tk } 4,000$. A higher total cost gives a line farther from the corner.

The slope of the line depends only on the two prices. One worker costs Tk 600 and one machine costs Tk 400, so a worker costs as much as $600 / 400 = 1.5$ machines. Along the line, each extra worker must be paid for by renting 1.5 fewer machines. If the wage rises, the line becomes steeper. If the machine rent rises, it becomes flatter.

PAUSE AND CHECK 12.P1 An isocost line is drawn for a total of Tk 4,800, with a wage of Tk 600 and a machine rent of Tk 400. How many workers could be hired if no machine is rented?

Chapter 7 gave the consumer's marginal rate of substitution, which keeps satisfaction the same. The firm has a twin, and it keeps output the same. The **marginal rate of technical substitution** is the amount of one input a firm can give up when it adds one unit of another, keeping output the same. We write it as MRTS. Here it is the number of machines a firm can save when it adds one worker.

The table gives the MRTS between each pair of neighbouring mixes on the case isoquant. Its columns show the move, the machines saved, the workers

Isoquant. A line showing all combinations of two inputs that produce the same output.

An isoquant joins all mixes that give the same output.

Case file: five mixes of workers and machines give 100 books a day.

Isoquants slope downward, curve toward the origin, and never cross.

A firm decides how much to spend and can divide the sum between workers and machines.

Isocost line. A line showing all combinations of two inputs that cost the same total amount.

An isocost line joins all mixes that have the same total cost.

The Tk 4,000 line meets the axes at 6.67 workers and 10 machines.

The slope of an isocost line is the price ratio: one worker costs as much as 1.5 machines.

Marginal rate of technical substitution. The amount of one input a firm can give up when it adds one unit of another, keeping output the same.

Recall: marginal rate of substitution, Chapter 7, section 7.2 'Marginal Rate of Substitution'.

The marginal rate of technical substitution is the slope of the isoquant.

The case isoquant gives MRTS values of 4, 2, 1 and 1.

added and the MRTS, which is the machines saved divided by the workers added.

MRTS BETWEEN NEIGHBOURING MIXES ON THE ISOQUANT FOR 100 BOOKS

MOVE	MACHINES SAVED	WORKERS ADDED	MRTS
(2, 10) to (3, 6)	4	1	4
(3, 6) to (4, 4)	2	1	2
(4, 4) to (5, 3)	1	1	1
(5, 3) to (6, 2)	1	1	1

The MRTS falls as workers replace machines, because each added worker is worth less.

The MRTS also equals the marginal product of a worker divided by that of a machine.

Recap: inputs, the isoquant, the isocost line and the MRTS.

Read the first row: adding one worker saves four machines, so the MRTS is 4. The values fall as we move down the table. This is called a diminishing MRTS. It happens because the more workers there are, the less each extra worker adds, while the fewer machines remain, the more each remaining machine matters. The falling MRTS is the reason the isoquant curves toward the corner.

There is a second way to write the MRTS. The marginal product of an input is the extra output from one more unit of it, a quantity met in Chapter 10, 'Production Function and Returns'. A worker replaces machines in proportion to what each adds, so $MRTS = \text{marginal product of a worker} / \text{marginal product of a machine}$. A worker who adds twice as much output as a machine replaces two machines.

Before moving on. An isoquant joins the mixes of two inputs that give the same output. An isocost line joins the mixes that have the same total cost, and its slope is the price ratio. The MRTS is the number of machines saved by adding one worker, and it falls as workers replace machines.

* * *

12.2 THE LEAST-COST RULE

Of five mixes that make 100 books, which costs least?

Least-cost combination. The mix of inputs that produces a given output at the lowest total cost.

The least-cost combination is the cheapest mix for a given output.

Case file: the cheapest of five mixes is 4 workers and 4 machines at Tk 4,000.

The table splits each total cost into a wage bill and a rent bill.

Return to the workshop. Five mixes give 100 books. Which of them costs least? The method has three steps. First, list the mixes on the isoquant. Second, find the total cost of each mix. Third, choose the mix with the smallest total cost.

The result has a name. The **least-cost combination** is the mix of inputs that produces a given output at the lowest total cost. The words *given output* matter. The firm does not choose how much to make here. It has fixed the output, and it asks only how to make it most cheaply.

Case file, illustrative. For an output of 100 books bound a day, with the wage at Tk 600 and the machine rent at Tk 400, the total costs are: (2, 10) = Tk 5,200; (3, 6) = Tk 4,200; (4, 4) = Tk 4,000; (5, 3) = Tk 4,200; (6, 2) = Tk 4,400. The least cost is (4, 4) at Tk 4,000. The MRTS moves from 2 to 1 around the price ratio $600 / 400 = 1.5$.

The table shows how each total is built. The wage bill is Tk 600 times the number of workers, and the rent bill is Tk 400 times the number of machines. The last column adds the two.

COST OF EACH MIX THAT GIVES 100 BOOKS A DAY

WORKERS, MACHINES	WAGE BILL (TK)	RENT BILL (TK)	TOTAL COST (TK)
2, 10	1,200	4,000	5,200
3, 6	1,800	2,400	4,200
4, 4	2,400	1,600	4,000
5, 3	3,000	1,200	4,200
6, 2	3,600	800	4,400

Read the last column from top to bottom. The total falls from 5,200 to 4,000 and then rises to 4,400. The lowest value is the least-cost combination, 4 workers and 4 machines. The cheapest mix is neither the one with most workers nor the one with most machines. The best mix lies in the middle, where the two bills are balanced.

Now we draw the same facts. On the chart, the horizontal axis shows the number of workers and the vertical axis shows the number of machines. A point is a mix. The curved line is the isoquant for 100 books. The straight lines are isocost lines, and a line farther from the corner means a higher total cost. The firm wants the lowest isocost line that still touches the isoquant. That line touches the curve at one point, without crossing it, which is called a tangent. The point of contact is the least-cost combination.

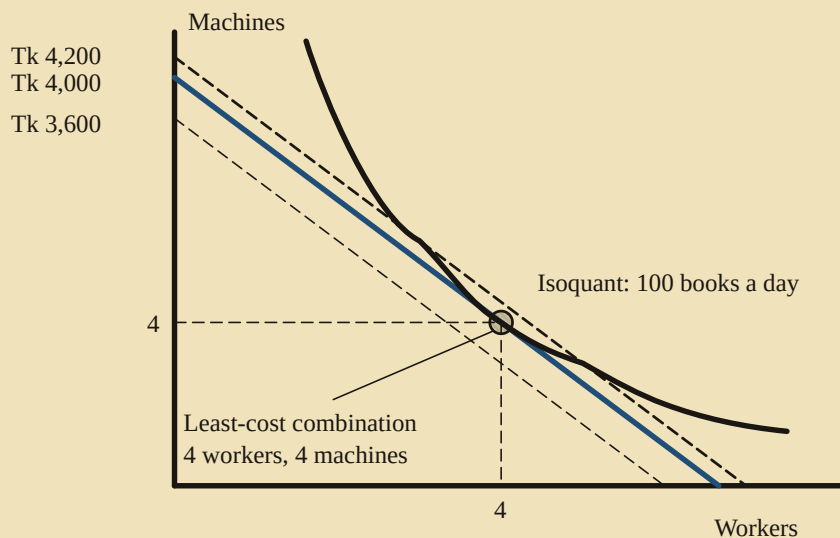


FIG. 12.1. Isoquant for 100 books with isocost lines, and the least-cost combination

At the point of contact the two lines have the same slope. The slope of the isoquant is the MRTS, and the slope of the isocost line is the price ratio. So the least-cost rule is: $MRTS = \text{wage} / \text{machine rent}$. In the case, $600 / 400 = 1.5$. Between (3, 6) and (4, 4) the MRTS is 2, which is above 1.5. Between (4, 4) and (5, 3) it is 1, which is below 1.5. The ratio 1.5 lies between them, at (4, 4).

The reason can be shown with numbers. At (3, 6), one more worker saves two machines worth Tk 800 and costs Tk 600, so the firm gains Tk 200. It should move. At (4, 4), one more worker saves one machine worth Tk 400 and costs Tk 600, so the firm loses Tk 200. It should stop. The firm adjusts until the gain from changing the mix falls to zero.

The rule has a second form. Divide each marginal product by the price of its input to find the extra output per taka. The least-cost combination has equal

Total cost falls and then rises, and the lowest point is the least-cost combination.

On the chart, the cheapest isocost line that touches the isoquant shows the best mix.

At the point of contact the slope of the isoquant equals the slope of the isocost line.

If the MRTS is above the price ratio, the firm should hire more workers and fewer machines.

The rule has a second form: each input must add the same output per taka spent.

A rise in the wage turns the isocost line steeper and moves the best mix toward machines.

Recap: the least-cost rule in two forms and the effect of a wage change.

What happens when the firm wants to make more than 100 books?

The expansion path joins the least-cost combinations for every output.

Cost-minimising asks how to make a given output; profit-maximising also asks how much to make.

In the short run some inputs are fixed, so the firm may be unable to reach the least-cost combination.

The rule assumes two inputs, steady prices and smooth substitution, and real firms differ.

A worker welcomes a higher wage, yet the rule shows a hidden effect on the number of workers hired.

marginal product per taka: marginal product of a worker / wage = marginal product of a machine / machine rent. If workers add more output per taka than machines, the firm should hire more workers and rent fewer machines. As it does so, the marginal product of a worker falls and that of a machine rises, until the two ratios are equal.

Now suppose the wage rises while the machine rent stays the same. The isocost line turns steeper, because a worker now costs as much as more machines. The point of contact moves along the isoquant toward fewer workers and more machines. The firm replaces workers with machines, because machines have become cheaper by comparison. Total cost for the same output still rises, but by less than it would if the firm kept the old mix. The worked examples give numbers.

Before moving on. The least-cost combination is found where the lowest isocost line touches the isoquant. There, MRTS equals the price ratio, which is the same as equal marginal product per taka. A rise in the wage steepens the isocost line and moves the best mix toward machines.

* * *

12.3 USING THE RULE

Suppose the firm wants to make more than 100 books a day. The same method applies at each output. Each output has its own isoquant, and a larger output has an isoquant farther from the corner. For each one, the lowest isocost line that touches it gives a new least-cost combination.

If we join these least-cost points, the line is called the *expansion path*. It shows how the best mix changes as output grows, while the input prices stay the same. Usually the path rises, because a larger output needs more of both inputs. The total cost at each point on the path is the least total cost of that output, which is the total cost schedule used in Chapter 11, 'Costs of Production'.

Two questions must be kept apart. A cost-minimising choice asks: for a given output, which mix costs least? A profit-maximising choice asks a larger question: which output should the firm make? A firm that wants the largest profit must answer both. It first finds the cheapest way to make each output, and then it chooses the output that gives the best gap between revenue and total cost. A firm can minimise cost and still lose money if it chooses a poor output. Later chapters deal with that second choice.

In the short run, some inputs cannot be changed, and in the long run all can. Suppose, as an illustration, that a firm has signed a contract for 6 machines. For 100 books it must use the mix (3, 6), which costs Tk 4,200 in the case. The least cost is Tk 4,000, so the contract adds Tk 200 a day. Only in the long run can the firm move to (4, 4).

The rule has limits. It assumes two inputs, steady prices and smooth substitution between them. Some inputs fit together in fixed amounts: a bus needs one driver, however cheap the fuel. Then the isoquant has a sharp corner and only one mix makes sense. Real managers also do not know their isoquants exactly. They estimate them from experience, and the rule shows the direction in which to adjust.

The viewpoints differ. The manager asks which mix costs least. A worker welcomes a higher wage, and those who keep their jobs gain. The rule, however, shows a hidden effect: after a wage rise, a firm may use fewer workers for the

same output. How large the effect is depends on how easily machines replace workers, which is the curvature of the isoquant.

Before moving on. Repeating the method for each output gives the expansion path. Minimising cost for a given output is not the same as maximising profit. In the short run a fixed input can prevent the firm from reaching the least-cost combination.

* * *

WORKED EXAMPLES: FINDING THE LEAST-COST MIX

We now apply the rule in full, in the order Given, Find, Formula, Steps, Answer and Check. The first example uses small numbers, so that the method is the only new thing to learn. The second uses the case and a change in the wage.

Example 1: output per taka. Given. The wage is Tk 20 per worker per day and the machine rent is Tk 10 per machine per day. At the firm's present mix, the marginal product of a worker is 40 books and that of a machine is 15 books. *Find.* Whether the mix is the least-cost combination and, if not, which way the firm should change it.

Formula. In words: the marginal product per taka equals the marginal product divided by the price of the input. The mix is least-cost when the ratio is equal for both inputs. In symbols: $MP \text{ of a worker} / w = MP \text{ of a machine} / r$, where MP means marginal product, w is the wage and r is the machine rent. *Steps.* Workers: $40 / 20 = 2$ books per taka. Machines: $15 / 10 = 1.5$ books per taka. Compare: 2 is greater than 1.5.

Answer. The mix is not least-cost. A taka spent on workers adds more output than a taka spent on machines, so the firm should hire more workers and rent fewer machines. *Check.* The MRTS is $40 / 15 = 2.67$, which is above the price ratio $20 / 10 = 2$. The rule of section 12.2 also says to hire more workers. The two tests agree.

Example 2: a wage rise. Given. The output is 100 books a day. The machine rent stays at Tk 400. The wage rises from Tk 600 to Tk 900. The mixes on the isoquant are (2, 10), (3, 6), (4, 4), (5, 3) and (6, 2). *Find.* The new least-cost combination and the rise in total cost.

Formula. $C = w \times L + r \times M$, with $w = 900$ and $r = 400$. *Steps.* (2, 10): $900 \times 2 + 400 \times 10 = 1,800 + 4,000 = \text{Tk } 5,800$. (3, 6): $2,700 + 2,400 = \text{Tk } 5,100$. (4, 4): $3,600 + 1,600 = \text{Tk } 5,200$. (5, 3): $4,500 + 1,200 = \text{Tk } 5,700$. (6, 2): $5,400 + 800 = \text{Tk } 6,200$. The smallest is Tk 5,100. The rise is $5,100 - 4,000 = \text{Tk } 1,100$.

Answer. The new least-cost combination is 3 workers and 6 machines, at Tk 5,100. Total cost rises by Tk 1,100. *Check.* The price ratio is now $900 / 400 = 2.25$. The MRTS between (2, 10) and (3, 6) is 4, which is above 2.25, so the firm moves. Between (3, 6) and (4, 4) it is 2, which is below 2.25, so the firm stops at (3, 6). Keeping the old mix (4, 4) would cost Tk 5,200, a rise of Tk 1,200, so changing the mix saves Tk 100.

Three slips are common. The first is to reverse the MRTS: divide the machines saved by the workers added, not the other way. The second is to mix periods: the wage and the machine rent must both be per day, or both per week. The third is to compare mixes that give different outputs. Total costs can be compared only along one isoquant.

Recap: the expansion path, the two kinds of choice and the short run.

Two examples apply the rule: the first with small numbers, the second with case data and a wage rise.

Example 1: test a mix by comparing output per taka from each input.

The test divides each marginal product by the price of its input.

Workers give more output per taka, so the firm should hire more workers and fewer machines.

Example 2: the wage rises from Tk 600 to Tk 900 in the case.

Total cost is recomputed for each mix at the new wage and the smallest is chosen.

The best mix moves to 3 workers and 6 machines, and the rise is Tk 1,100.

Three slips are common: reversed MRTS, mixed periods and comparing different outputs.

* * *

The main ideas of the chapter in a few lines.

A common belief is that the firm should use the input with the lower price; the rule compares output per taka.

A second belief is that least cost means fewest inputs; it means the lowest total bill.

A third belief is that the best mix spends the same amount on each input; it does not.

A fourth mistake is to read an isoquant like an indifference curve with measurable satisfaction.

In 1769 Richard Arkwright patented a spinning machine and ran it first in a horse-powered mill in Nottingham.

In 1771 he built Cromford Mill in Derbyshire, the first successful water-powered cotton spinning mill.

SUMMARY

An isoquant shows all mixes of two inputs that give the same output. An isocost line shows all mixes that cost the same total amount, and its slope is the price ratio. The marginal rate of technical substitution is the amount of one input that can be saved when another is added, keeping output the same. The least-cost combination is where the lowest isocost line touches the isoquant, so the marginal rate of technical substitution equals the price ratio. In the case, the least cost for 100 books is Tk 4,000, at 4 workers and 4 machines. A wage rise moves the best mix toward machines.

THINK IT THROUGH

12.T1 The wage rises from Tk 600 to Tk 900 and the machine rent stays at Tk 400. Without calculating, predict how the least-cost mix changes. Then explain why total cost for 100 books rises by less than $4 \times \text{Tk } 300$.

12.T2 A firm has signed a contract for 6 machines and cannot change it this month. Can it reach the least-cost combination for 100 books in the case? Explain.

WHERE PEOPLE GO WRONG

A common belief is that a firm should use more of the input with the lower price. Price alone does not decide. The rule compares the output added per taka, so a dear input that adds a great deal of output can still be the better choice.

A second belief is that least cost means the fewest inputs. It does not. The mix (3, 6) uses nine inputs in all and costs Tk 4,200, while the mix (4, 4) uses eight and costs Tk 4,000, so counting inputs is a poor guide. Only the total bill, found from the prices, decides.

A third belief is that the best mix spends equal amounts on workers and machines. At (4, 4), the wage bill is Tk 2,400 and the rent bill is Tk 1,600. The rule requires equal output per taka, not equal spending.

A fourth mistake is to treat an isoquant as if it were a consumer's indifference curve. The two have the same shape, but output can be counted in books and satisfaction cannot be counted. A larger output on a higher isoquant is a fact, not an opinion.

IN HISTORY

In 1769 Richard Arkwright, an English inventor and businessman, patented a spinning machine that later became known as the water frame. He first set up a horse-powered mill in Nottingham, a city in the English Midlands, to run his machines.

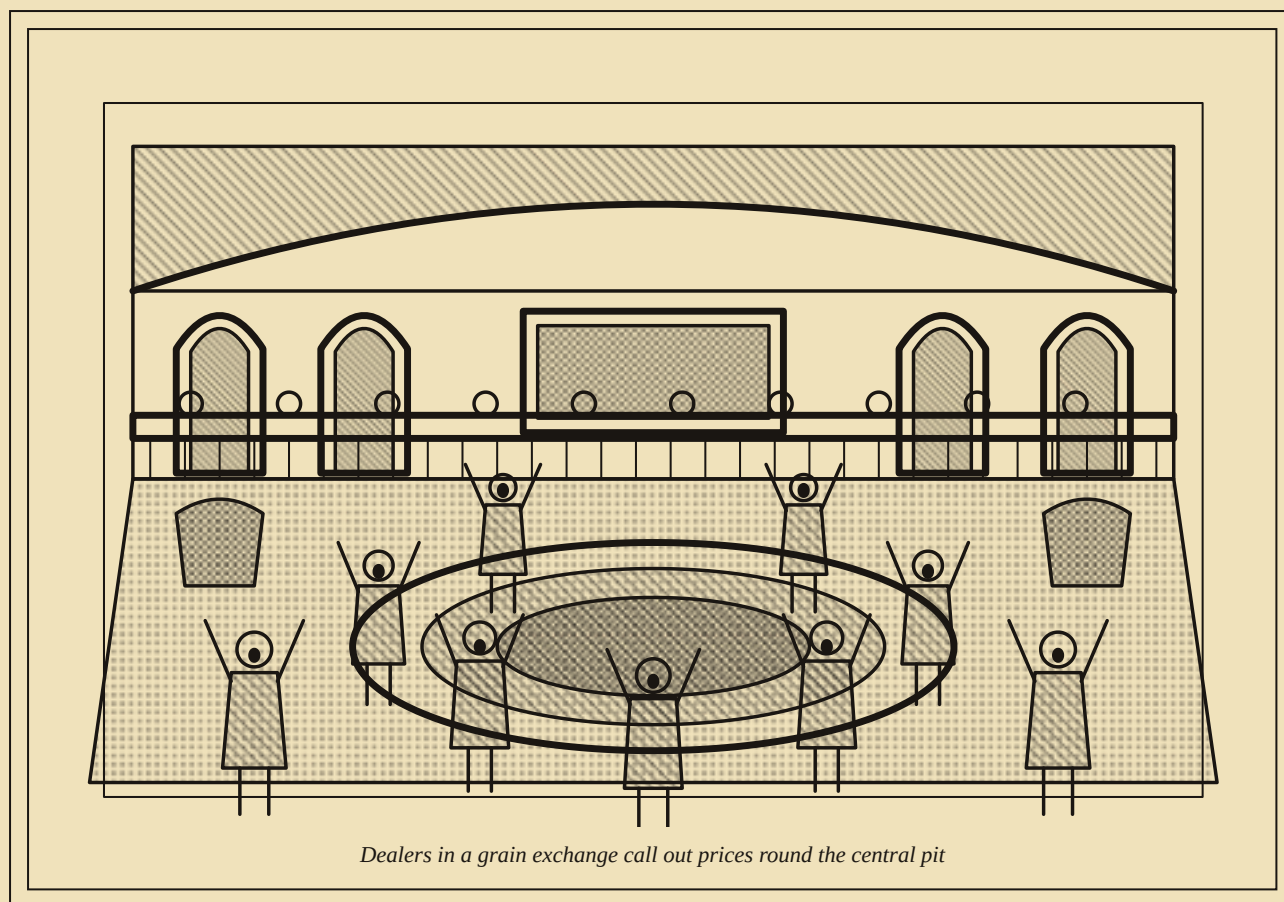
Horses gave too little power, so in 1771 Arkwright built a mill at Cromford, in Derbyshire, England, driven by a water wheel. Cromford Mill is regarded as the world's first successful water-powered cotton spinning mill. He chose the site because water flowed there all year, from the Bonsall Brook and the Cromford Sough, a drain that carried water out of nearby lead mines.

Before such mills, much cotton yarn was spun by hand on wheels in workers' homes. Arkwright gathered the spinning into one building, where a water wheel and machines did work that people had done by hand. In the language of this chapter, the mix moved toward machines and power. He also needed workers at the site, and in 1776 he began building houses for them in Cromford. We do not know what calculations he made, but his question was the one of this chapter: which mix of workers, machines and power makes the yarn at the lowest cost?

The mill moved spinning from workers' homes to machines, and workers needed housing at the new site.

REVIEW QUESTIONS

- 12.1 Define an isoquant and an isocost line in your own words, and state one difference between an isoquant and an indifference curve.
- 12.2 With a wage of Tk 600 and a machine rent of Tk 400, draw the isocost line for a total of Tk 4,800 by finding where it meets each axis. State its slope.
- 12.3 A firm must make 10 units. The mixes of (workers, machines) that give this output are (1, 8), (2, 4), (4, 2) and (8, 1). A worker costs Tk 30 a day and a machine costs Tk 10 a day. Find the least-cost combination.
- 12.4 Use the case mixes for 100 books and the wage of Tk 600, but let the machine rent fall from Tk 400 to Tk 300. Find the least-cost combination, and compare the MRTS between the cheapest mixes with the new price ratio.
- 12.5 In the case, find the wage bill as a percentage of total cost at the least-cost combination. Then find it again after the wage rises to Tk 900 and the firm moves to its new best mix. Explain the change.
- 12.6 The marginal product of a worker is 30 books and that of a machine is 10 books. The wage is Tk 600 and the machine rent is Tk 400. Which input adds more output per taka, and how should the firm change its mix?



CHAPTER XIII

Market Structure and Perfect Competition

Why must a grain dealer accept a price that the dealer did not choose?



This chapter sorts markets into four kinds by the number of sellers, the sameness of the good and the ease of entry. It then studies the firm in the most competitive kind of market, finds the output that gives that firm the largest profit, and follows the profit through time.

YOU WILL NEED

Chapter 9, 'Elasticity of Supply and Market Equilibrium', section 9.2 'Market Equilibrium'; Chapter 11, 'Costs of Production', section 11.2 'Short-Run Costs'; Chapter 11, 'Costs of Production', section 11.3 'The Cost Curves'; and Appendix A1, 'Maths You Will Use'.

Two sellers in one town choose their prices in very different ways.

The difference lies in the surroundings of each seller, not in the seller's character.

The chapter has three steps: sort markets, study one firm, then follow its profit over time.

Picture two sellers in one town. The first sells potatoes from a stall in a long row of stalls, and every stall offers the same potatoes at nearly the same price. The second runs the only pharmacy within ten kilometres. Both sellers wish to earn as much money as possible, yet they behave in opposite ways. The potato seller dares not charge one taka more than her neighbours, because her customers would move to the next stall. The pharmacist may raise a price and still keep many customers.

Why does the same wish lead to such different behaviour? The cause is not the character of the two sellers. It lies in the surroundings of each one: how many rivals there are, whether the goods are the same, and whether a new seller can arrive next month. When the surroundings change, the behaviour changes with them. Economists describe those surroundings in an exact way, so that they can predict what sellers will do.

This chapter takes three steps. First, we sort markets into kinds, so that we know where the potato seller and the pharmacist belong. Second, we study a firm that cannot choose its price and chooses only its output, and we find the output that gives it the largest profit, using the cost data of the binding workshop. Third, we ask what happens to that profit once other firms notice it. Many students find the second step surprising, because they expect every firm to name its own price.

* * *

13.1 KINDS OF MARKET

A question: which facts about a market would you ask for, to predict how its sellers behave?

Market structure. The set of features of a market, such as number of sellers and ease of entry, that shape how firms behave.

The list of features that shapes seller behaviour has a name: market structure.

Suppose you must predict how the sellers in a market will behave, and you may ask only a few questions. Most people begin with the number of sellers. That is a good first question, but not the only one. Economists have found that the answers to a small group of questions, taken together, explain much of what sellers do. A list of those answers describes the market.

The name for that list is **market structure**. A market structure means the set of features of a market, such as the number of sellers and the ease of entry, that shape how firms behave. The row of potato stalls and the village pharmacy have different structures. Once we know the structure, we can say much about the freedom a seller has over the price.

FEATURES THAT DISTINGUISH MARKETS

Four features separate one market from another.

Four features do most of the work. The first is the *number of sellers*: very many, a few, or one. The second is the *kind of good*: identical from every seller, or different in quality, brand or place. The third is *ease of entry*: whether a new firm can begin selling without being blocked by law, by cost or by an existing firm. The fourth is *information*: whether buyers and sellers know the prices and the

quality on offer. Each feature changes how much power a seller holds over the price.

The word *competition* needs care. In daily life it means a contest, and a seller who advertises loudly is said to compete hard. In economics the word is narrower. Economists judge competition by the number of rivals and the ease of entry, and not by the loudness of advertising. A market can contain fierce advertising and weak competition at the same time, for example when only two firms exist and no third firm may enter.

The next table sorts markets into four kinds. Read a row from the left: the first column names the kind, and the other columns give the number of sellers, the kind of good and the ease of entry. The last three kinds are taught in Chapter 14, 'Monopoly and Imperfect Competition'; the section on monopoly is 14.1 'Monopoly'. Here we use the table only to place the first kind among the others.

FOUR KINDS OF MARKET

KIND	SELLERS	GOOD	ENTRY
Perfect competition	Very many	Identical	Free
Monopolistic competition	Many	Different versions	Easy
Oligopoly	A few	Identical or different	Difficult
Monopoly	One	One good with no close substitute	Blocked

Markets and firms need a link word between them. The **industry** means all the firms that produce the same kind of good. All the firms that make notebooks form the notebook industry; all the wheat farmers of a country form the wheat industry. The quantity supplied by the industry is the sum of the quantities supplied by its firms. The difference between one firm and its industry matters greatly in section 13.3, where firms enter and leave.

The table has limits. Real markets rarely match one row exactly, and a market with five large firms and fifty small ones belongs partly to two rows. A map leaves out detail so that the main roads are easy to see. Use the table to ask which features a market has.

Before moving on. A market structure is the set of features that shape how firms behave: number of sellers, kind of good, ease of entry and information. There are four kinds of market. An industry is all the firms that produce the same kind of good.

* * *

13.2 THE PERFECTLY COMPETITIVE FIRM

Imagine a hall in which dozens of dealers trade the same grain. One price is called across the floor. A dealer who asks for more finds no buyer, and a dealer who accepts less loses money for no reason. In History gives a true account of such a hall.

That kind of market is called **perfect competition**. Perfect competition means a market with very many buyers and sellers, identical goods, free entry and full information. A hall of grain dealers comes near it, because wheat of one grade is the same whoever grew it. No real market meets all four conditions exactly.

Perfect competition is

Competition. Word in use. In daily life, a contest in which each person tries to win, such as a race. In economics, the pressure that sellers place on one another through price and quality when several offer the same good.

Competition has an everyday meaning and a narrower meaning in economics.

The table gives four kinds of market, sorted by the features above.

Industry. All the firms that produce the same kind of good.

We also need one word for all the firms that make the same good.

Real markets rarely match one row exactly, so the table is a map and not a photograph.

Before moving on, recall the three ideas of section 13.1.

A hook: in a grain hall, one price is called across the floor and every dealer must accept it.

Perfect competition. A market with very many buyers and sellers, identical goods, free entry and full information.

Perfect competition is the market with four strict features.

Each of the four conditions removes one kind of power from the seller.

Price taker. A firm that must accept the market price because its own sales are too small to affect it.

Because of these conditions, each firm must accept the price that the market gives.

How to read the first chart: two panels share a price scale but not a quantity scale.

therefore a *model*: a simple picture that shows the main forces in their cleanest form.¹

Each condition removes one kind of power from the seller. With very many sellers, each firm supplies a tiny part of the total. With identical goods, a buyer has no reason to prefer one seller. With free entry, new firms may start at any time. With full information, every buyer knows the price in the next stall and will not pay more.

The result is a firm of a special kind: a **price taker**. A price taker means a firm that must accept the market price because its own sales are too small to affect it. Suppose the market price of a notebook is Tk 40. A workshop that makes notebooks can sell any quantity it is able to make at Tk 40, and none at Tk 41. It has no reason to sell below Tk 40, since all its output finds a buyer at Tk 40. For this firm the price is a fact, like the weather, and it cannot be changed.

The first chart shows a firm and its industry side by side. In each panel the vertical axis shows the price in taka, and the horizontal axis shows quantity. A point on a line shows one price and the quantity that goes with it. The two panels use the same price scale, so the same height means the same price. Their quantity scales differ, because one firm makes only a tiny part of the industry's total. On the right, the industry faces a falling demand line and a rising supply line, which cross at the market price. On the left, the single firm faces a flat line at that price.

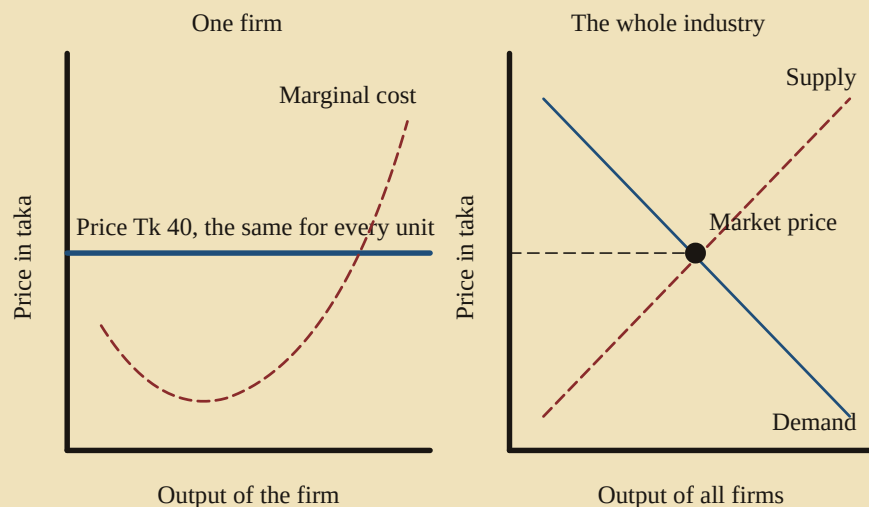


FIG. 13.1. *Firm and industry side by side: the market sets the price, and the firm accepts it*

PAUSE AND CHECK 13.P1 The market price is Tk 40. A firm in perfect competition asks Tk 42 for its good. What happens to its sales, and why?

GO SLOWLY HERE.

A price taker has one decision left: how much to make.

Recall: marginal cost, Chapter 11, section 11.2 'Short-Run Costs'.

Marginal cost is the extra cost of making one more unit of the good.

MARGINAL COST EQUALS PRICE

Because the market fixes the price, the firm has one decision left: how much to make. The answer uses a measure from Chapter 11.

Marginal cost is the extra cost of making one more unit. Each extra unit adds the market price to the firm's revenue, and it adds the marginal cost to the firm's cost. While the price is larger than the marginal cost, the extra unit adds to profit. Once the marginal cost is larger than the price, the extra unit reduces

¹ A model leaves out detail on purpose. Its value lies in the forces it shows, and not in an exact match with any one market.

profit. So the firm raises its output until marginal cost reaches the price. This is the rule *marginal cost equals price*.

Case file, illustrative. The market price is Tk 40, taken from the notebook market. Cost data of the binding workshop (Chapter 11): marginal cost for each 10-unit step is 30, 22, 20, 24, 34 and 48, at outputs 10 to 60. Each figure is the extra cost of one unit in that step. A step of 10 units therefore adds 10 times its marginal cost to total cost.

The table below applies the rule one step at a time. Read a row from the left: the first column names the step by its last unit, the second gives the extra revenue (10 units at Tk 40), the third gives the extra cost (10 times the marginal cost), and the fourth gives the difference. A positive difference means the step adds to profit.

EACH 10-UNIT STEP AT A PRICE OF TK 40

STEP ENDING AT OUTPUT	EXTRA REVENUE	EXTRA COST	EXTRA PROFIT
10	400	300	+100
20	400	220	+180
30	400	200	+200
40	400	240	+160
50	400	340	+60
60	400	480	-80

The case file gives the price and the workshop's marginal costs.

The table compares the extra revenue and extra cost of each 10-unit step.

Now read the second chart. The horizontal axis shows the output of the workshop, and the vertical axis shows the price and the cost in taka. The flat line is the price of Tk 40. Each dot shows the marginal cost at the end of one step. The hatched area lies between the price line and the cost line, where each unit adds to profit. The cost line passes above the price line after the fifth step, where each unit would add a loss.

How to read the second chart: the price line, the cost points and the best output.

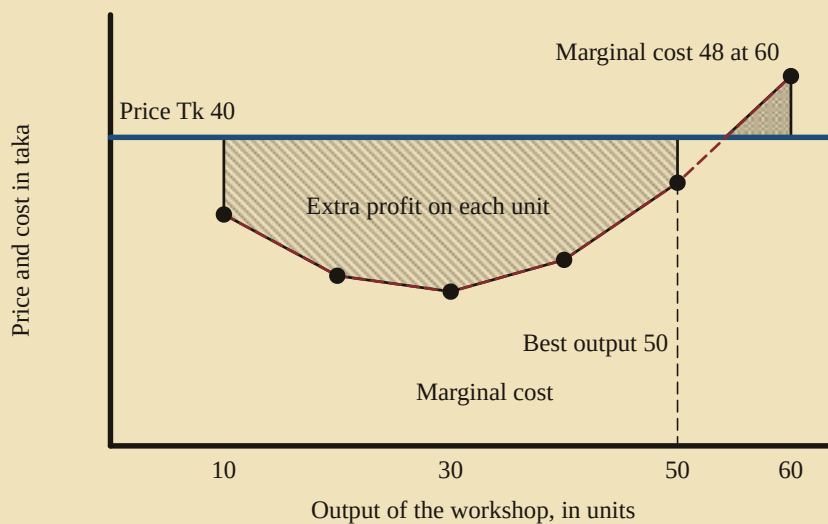


FIG. 13.2. Price taker at marginal cost equal to price: the workshop stops at 50 units

The first five steps all add to profit, and the sixth reduces it by 80. So the workshop makes 50 units. At 50 units, the marginal cost of the last step (34) is below the price (40), and the marginal cost of the next step (48) is above it. The output of 50 is therefore the best whole step.

The workshop adds every step whose extra profit is positive and stops at 50 units.

The rule has limits: marginal cost must be rising, and the data come in steps.

Before moving on, recall the price taker and the output rule.

A puzzle: the workshop earns a profit, so would that profit last?

Equilibrium of the firm. The output at which a firm earns the largest possible profit and has no wish to change.

The best output of a firm has a name: the equilibrium of the firm.

The case file states the workshop's revenue, cost and profit at its best output.

A test: at 40 units and at 60 units the profit is lower, so the workshop wishes no change.

Profit. Word in use. In daily life, any money gained, or the money left after the bills are paid. In economics, revenue minus all costs, including the value of the owner's own time and money; Chapter 16, 'Interest and Profit', section 16.2 'Profit' explains it fully.

In this chapter, profit means total revenue minus total cost.

In the short run, the price decides whether the firm gains or loses.

The rule has two limits. First, the marginal cost must be *rising* where it meets the price. A marginal cost that meets the price while it is still falling does not mark the best output, because the next units cost less and add more profit. The workshop's costs fall over the first three steps and rise afterwards, so only the rising part can give the answer. Second, the data come in steps of 10 units. A smooth curve would cross the price line between 50 and 60 units, but the workshop sells whole steps, and the best full step is 50.

Before moving on. Perfect competition is a market with very many buyers and sellers, identical goods, free entry and full information. A price taker accepts the market price and chooses only its output. The firm raises output while marginal cost is below the price and stops where marginal cost equals the price.

* * *

13.3 EQUILIBRIUM OF FIRM AND INDUSTRY

At a price of Tk 40 the binding workshop earns a profit. Other firms can see this, and in perfect competition they are free to enter. Will the profit last for ever? To answer, we first need to be exact about the workshop's best output, and about what the word profit means in this chapter.

The **equilibrium of the firm** means the output at which a firm earns the largest possible profit and has no wish to change. The idea is the same as in Chapter 9, where equilibrium meant a balance of forces, but now it describes one firm. At the best output, a larger output gives less profit and a smaller output gives less profit, so the firm stays where it is.

Case file, illustrative. At the market price of Tk 40, the binding workshop produces 50 units. Total revenue is $50 \times 40 = 2,000$. Total cost is 1,900. Profit is $2,000 - 1,900 = \text{Tk } 100$.

We can test the claim. At 40 units, total revenue is $40 \times 40 = 1,600$. Total cost is the cost at 50 units less the cost of the fifth step: $1,900 - 340 = 1,560$. Profit is $1,600 - 1,560 = \text{Tk } 40$. At 60 units, total revenue is $60 \times 40 = 2,400$, and total cost is $1,900 + 480 = 2,380$, so profit is Tk 20. Both figures are below Tk 100. The workshop has no wish to change, so 50 units is its equilibrium.

In this chapter, *profit* means total revenue minus total cost. Total revenue is the price times the quantity sold. If total revenue is larger than total cost, the firm earns a profit. If it is smaller, the firm makes a loss. The word has an everyday meaning too, and the two meanings differ in what they count as a cost. The note above gives both.

SHORT-RUN PROFIT AND LOSS

The short run is a period too short for new firms to enter or for old firms to leave. During it, the market price decides the result for the firm. When the price is high, the firm earns a profit, as the workshop does at Tk 40. When the price is low, the firm still chooses the output where marginal cost equals the price, but its total revenue may fall below its total cost, and it makes a loss. The Worked Examples show a price of Tk 30, where the workshop makes a loss.

Now consider what the owner must earn to continue. **Normal profit** means the least profit a firm must earn to stay in business, covering the owner's implicit costs. Implicit costs are the values of things the owner gives up and does not pay for in money, such as the owner's own time. A baker who could earn a good wage in a factory must earn at least that much from her own bakery, or she will close it.

In this chapter we count normal profit as part of total cost. A profit of Tk 100 is therefore a profit above normal. Chapter 16, 'Interest and Profit', section 16.2 'Profit' gives such a profit its own name and explains where it comes from. Here it is enough to say that a profit above normal attracts new firms, and that a loss drives firms away.

LONG-RUN EQUILIBRIUM

The long run is a period long enough for firms to enter and to leave the industry. Follow the workshop's profit through time in four steps. First, at Tk 40 the workshop earns Tk 100 above normal, and other firms notice it. Second, because entry is free, new firms begin to make notebooks, and the supply of the industry shifts to the right. Third, with more notebooks offered, the market price falls. Fourth, the fall continues until the profit above normal is gone and no new firm has a reason to enter.

The process also runs in reverse. If the price is so low that firms make losses, some firms leave. The supply of the industry shifts to the left, and the market price rises. The rise continues until the remaining firms earn normal profit. In both directions the price moves to the level where no firm earns more than normal profit, and no firm earns less.

The resting point is called *long-run equilibrium*. In it, every firm earns normal profit only, the price equals the lowest cost per unit at which a firm can produce, and no firm wishes to enter or to leave. The consumer gains, because the good sells at the lowest cost at which it can be made. The owner gains nothing above normal, and that is why owners prefer markets where entry is blocked.

The result has limits. It needs free entry, so a blocked market keeps its profit. It assumes that all firms have the same costs, which real firms seldom do. And it takes time: new firms must find money and build workshops, so a profit above normal can last for months. The result gives the direction of the change, and not its speed.

Before moving on. The equilibrium of the firm is the output with the largest profit, where marginal cost equals the price. In the short run a firm may earn a profit above normal or make a loss. In the long run, entry and exit leave every firm with normal profit, at a price equal to the lowest cost per unit.

* * *

WORKED EXAMPLES

The first example uses the workshop's cost data at a lower market price, where the result is a loss.

Given. The market price is Tk 30. The marginal cost of each 10-unit step is 30, 22, 20, 24, 34 and 48, at outputs 10 to 60. Total cost at 50 units is 1,900. *Find.* The best output, and the profit or loss at that output. *Formula.* In words: raise output while each step adds to profit, and stop before the first step that reduces it. Profit equals total revenue minus total cost, and the extra cost of a

Normal profit is the least payment that keeps the owner in this line of business.

Normal profit. The least profit a firm must earn to stay in business, covering the owner's implicit costs.

In this chapter normal profit counts as a cost, so a profit means a profit above normal.

In the long run, free entry removes profit above normal, and exit removes losses.

Recall: equilibrium price, Chapter 9, section 9.2 'Market Equilibrium'.

The same process runs in reverse when firms make losses.

The resting point is long-run equilibrium: normal profit, and a price equal to the lowest cost per unit.

The result holds under conditions: free entry, equal costs and enough time.

Before moving on, recall the short run, the long run and normal profit.

Worked example 1 finds the best output and the result at a lower price of Tk 30.

The steps of worked example 1, from the data to the check.

step equals 10 times its marginal cost. *Steps.* (1) Extra revenue of each step: $10 \times 30 = 300$. (2) Extra cost of the steps: 300, 220, 200, 240, 340, 480. (3) Extra profit of the steps: 0, +80, +100, +60, -40, -180. (4) The fifth step reduces profit, so the best output is 40 units. (5) Total revenue: $40 \times 30 = 1,200$. (6) Total cost at 40 units: $1,900 - 340 = 1,560$. (7) Profit: $1,200 - 1,560 = -360$. *Answer.* The workshop makes 40 units and a loss of Tk 360. *Check.* At 30 units the loss would be $(1,560 - 240) - 900 = 420$, and at 50 units it would be $1,900 - 1,500 = 400$. Both are larger than 360, so 40 units gives the smallest loss.

The second example joins this chapter to the cost ideas of Chapter 11. It supposes that other firms have the same costs as the workshop, and that entry is free.

Given. Total cost is 1,560 at 40 units, 1,900 at 50 units and 2,380 at 60 units. The market price is Tk 40, and the workshop earns Tk 100 above normal. *Find.* The price at which the profit above normal disappears. *Formula.* In words: cost per unit equals total cost divided by quantity. In the long run, the price equals the lowest cost per unit. *Steps.* (1) At 40 units: $1,560 \div 40 = 39$. (2) At 50 units: $1,900 \div 50 = 38$. (3) At 60 units: $2,380 \div 60 = 39.67$, to two decimal places. (4) The lowest cost per unit is Tk 38, at 50 units. (5) At Tk 38 the best output is still 50 units, since the marginal cost of the fifth step (34) is below 38 and that of the sixth (48) is above it. (6) Total revenue: $50 \times 38 = 1,900$, which equals total cost. *Answer.* The long-run price is Tk 38. The workshop makes 50 units and earns normal profit only. *Check.* A fall of Tk 2 in the price takes $2 \times 50 = \text{Tk } 100$ from the revenue. That equals the profit above normal at Tk 40, so no profit above normal remains.

Three slips cause most wrong answers. First, forgetting that the extra cost of a 10-unit step is 10 times its marginal cost. Second, stopping at the first output where the price equals the marginal cost, when the marginal cost is still falling. Third, comparing the price with total cost, when it should be compared with marginal cost.

* * *

SUMMARY

A market structure is the set of features of a market that shape how firms behave. Perfect competition has very many buyers and sellers, identical goods, free entry and full information. In it, each firm is a price taker, which accepts the market price and chooses only its output. The firm raises output until marginal cost equals the price, and that output is the equilibrium of the firm. In the short run the firm may earn a profit above normal or make a loss. In the long run, free entry and exit leave every firm with normal profit, at a price equal to the lowest cost per unit.

THINK IT THROUGH

Use section 13.3 for the first problem and section 13.1 for the second.

Recall: cost curves, Chapter 11, section 11.3 'The Cost Curves'.

Worked example 2 finds the price at which the profit above normal disappears.

The steps of worked example 2, from the data to the check.

Common slips: step costs, falling marginal cost and mixing totals with unit figures.

The chapter in a few lines.

Two short problems apply the chapter to the potato seller and the pharmacist.

13.T1 A firm in perfect competition earns a profit above normal this year. Explain why this profit is unlikely to last, using free entry and the shift of the industry's supply.

13.T2 The potato seller must accept the market price, but the only pharmacist for ten kilometres can raise a price and still keep customers. Name two features of their markets that explain the difference.

WHERE PEOPLE GO WRONG

Mistake: "To earn the most, the workshop should make as many units as it can." The sixth step of the workshop adds Tk 480 to cost and only Tk 400 to revenue, so it reduces profit. The corrected rule: raise output only while each step adds to profit, and stop where marginal cost reaches the price.

Mistake: "A price taker is any firm that is small." A price taker is a firm whose sales are too small *relative to the market* to affect the price. A very large firm in a very large market can be a price taker, while a small firm that is the only seller in a village is not. The corrected rule: judge the firm against its market.

Mistake: "In the long run the owner earns nothing." The owner earns normal profit, which pays for the owner's time and money. What vanishes is the profit above normal. The corrected rule: zero profit above normal means that the owner is paid exactly as much as the next best use of the owner's effort.

Mistake one: believing that the largest output gives the largest profit.

Mistake two: believing that a price taker is always a small firm in a small town.

Mistake three: believing that zero profit in the long run means the owner earns nothing.

IN HISTORY

In 1848, a group of businessmen in Chicago, in the United States, founded the Chicago Board of Trade. Their aim was to bring order to the grain market of the American Midwest, which had been disorderly. The exchange gave dealers one place to meet, and its trading produced prices that every member could see.² For much of its history, traders there called out their bids and offers face to face in trading rooms known as pits.

Economists use a grain exchange as an example because it comes close to the model of this chapter: many dealers traded the same kind of grain, and each price was called for all to hear. The match was not perfect. Trading on the floor was open to members, so entry was not entirely free, and no real market meets all four conditions.

In 1848 grain dealers in Chicago founded an exchange to bring order to a chaotic market.

Such an exchange comes close to perfect competition but meets not every condition.

REVIEW QUESTIONS

13.1 A firm in perfect competition sells its good at Tk 10. Its marginal cost for each 5-unit step is 4, 7, 9 and 12, at outputs 5 to 20. Find the best output and the extra profit of the steps that it makes.

13.2 The market price of a notebook rises to Tk 50. Using the binding workshop's marginal costs (30, 22, 20, 24, 34 and 48 for each 10-unit step, with total cost 1,900 at 50 units), find the best output and the profit.

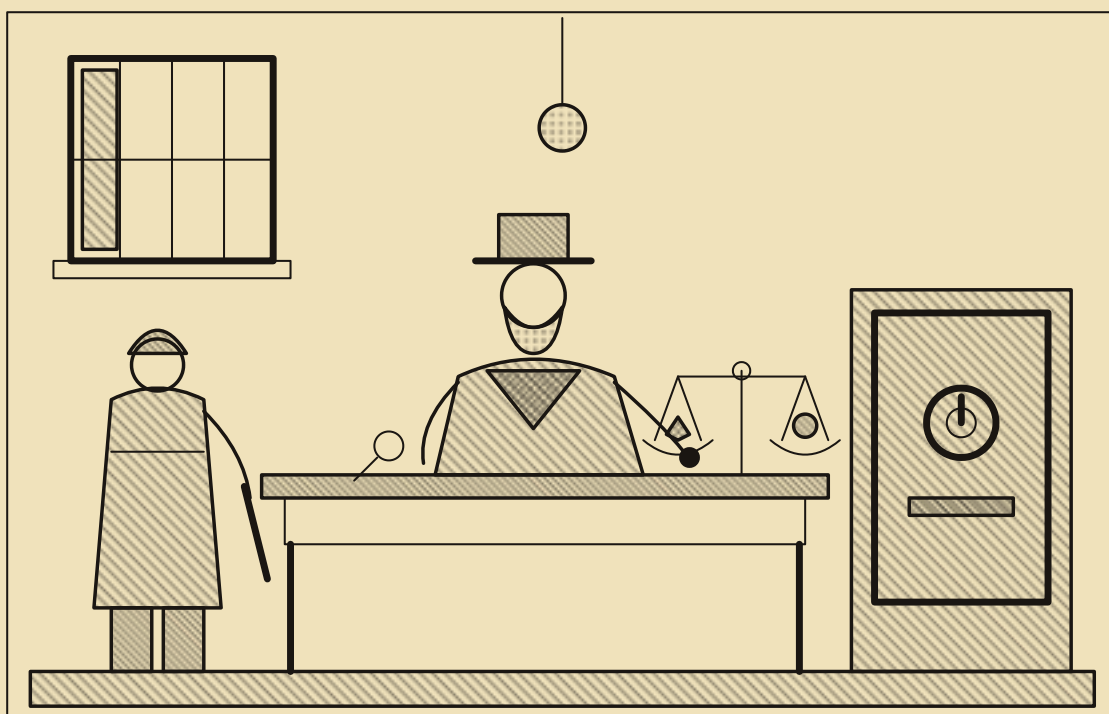
13.3 A rice farmer sells rice at Tk 20 a kilogram. Her marginal cost for each 100-kilogram step is Tk 18, 14, 16, 22 and 30 a kilogram, at outputs 100 to 500 kilograms. Find the best output and the extra profit of the steps she makes.

Five questions follow the ladder from a clean example to a combined problem.

² The exchange later added futures contracts and other products. Those later developments lie outside this chapter.

13.4 In the notebook market, the quantity demanded is 80 minus the price and the quantity supplied equals the price. Demand then shifts to 100 minus the price. Find the new market price, then find the workshop's best output and profit at that price, and say what you expect to happen in the long run.

13.5 A town has thirty tailors. Each offers clothes in a different style, and anyone may open a new shop. Using the table in section 13.1, name the kind of market and give the features that support your answer.



A Victorian diamond merchant weighs stones in a guarded room, as one seller guards his supply

CHAPTER XIV

Monopoly and Imperfect Competition

Who sets the price when there is no crowd of sellers?



Chapter 13 studied a firm so small that it must accept the market price. This chapter studies the other cases: the single seller, the many sellers whose goods differ, and the few large sellers who watch one another.

YOU WILL NEED

Chapter 13 'Market Structure and Perfect Competition', section 13.1 'Kinds of Market'; Chapter 13 'Market Structure and Perfect Competition', section 13.2 'The Perfectly Competitive Firm'; Chapter 11 'Costs of Production', section 11.2 'Short-Run Costs'; Appendix A1 'Maths You Will Use'.

One seller seems able to charge any price, yet something stops it.

The seller must weigh the price she asks against the quantity she sells.

Most real markets lie between the two extremes.

A town has one shop that sells a rare title which every student of a certain subject must read. No other shop stocks it. The owner thinks: "I am the only seller, so I may charge Tk 1,000 a copy." Why does she not do so? At Tk 1,000 almost no student would buy, and her shelf would stay full. At a very low price many students would buy, but she would earn little on each copy. Somewhere between the two lies a price that brings her the most profit. How does she find it?

Chapter 13 described the opposite case. A firm in perfect competition is a price taker: it accepts the market price and chooses only its output. A single seller faces a harder and more interesting choice. Every time she lowers her price to sell more, she earns less on the copies that she could already sell. So she must weigh price against quantity, and the weighing needs a new idea, which we meet in section 14.1.

Few markets look like either extreme. A street of tea stalls has many sellers, yet each stall is a little different. Mobile telephone service in many countries is supplied by a handful of large firms. This chapter therefore studies four markets in turn: the single seller, the many sellers whose goods differ, the few large sellers, and then all four side by side. Many students find the long list of market names confusing. We give each name one plain definition and one picture, and we compare the markets only at the end.

* * *

14.1 MONOPOLY

A village with one ferry shows the single-seller idea.

Monopoly. A market in which one seller supplies a good with no close substitute.

Monopoly. Word in use. In daily life, people call any very large firm a monopoly, and a board game carries the same name. In economics, it means one seller with no close substitute, however small the market.

One seller with no close substitute makes a monopoly.

Price maker. A firm that can choose its price because it faces the whole demand of the market.

A monopolist can choose its price, unlike a price taker.

Begin with a village on a wide river. One ferry carries people across, and no bridge or other boat exists. The ferry owner does not fear that a neighbouring ferry will take her passengers, because there is none. She can choose her fare, and the villagers must pay it or stay on their side. Her position is the heart of this section: a seller with the whole market to itself.

A **monopoly** is a market in which one seller supplies a good with no close substitute. The ferry across the river is a monopoly, and so is the only pharmacy on a remote island. The word *close* matters: a good has a close substitute if consumers would readily switch to it when the price rises. Swimming is not a real choice for the villagers, so the ferry has none. The size of the market does not matter; a monopoly may serve a village or a nation.

Because the monopolist is the only seller, the demand curve of the whole market is its own demand curve. That makes it a **price maker**: a firm that can choose its price because it faces the whole demand of the market. Compare the price taker of Chapter 13, which must accept the market

price.¹ The monopolist's power has a limit. If it raises its price, consumers buy less; if it lowers its price, they buy more. It can choose the price or the quantity, but not both: the demand curve fixes the other.

A monopoly lasts only while rivals cannot enter. Four common barriers stand in their way. The firm may own the only supply of a key resource, as one mine may hold all the stones. The law may give it an exclusive right, such as a patent or a licence. One large firm may serve the market at a lower cost for each unit than two smaller firms could, so a second firm would lose money. Or the firm may use threats to drive rivals away. The law differs by country: some governments grant exclusive rights, and others forbid firms to misuse their power, so a reader must learn the rules of the country concerned.

Return to the bookshop. To choose her output, the owner must know what each extra copy adds to her revenue. This idea is called **marginal revenue**, which means the extra revenue from selling one more unit of output. If she sells one more copy and her revenue rises by Tk 40, the marginal revenue is Tk 40. For a price taker, marginal revenue equals the price, because extra units sell at the same price. For a price maker it is smaller than the price, and the next paragraph shows why.

Take the case that follows. The seller sells 20 copies at Tk 60. To sell 30 copies she must cut the price to Tk 50, and the cut applies to all 30 copies, including the first 20. She gains 10 new copies at Tk 50, which is Tk 500. She loses Tk 10 on each of the first 20 copies, which is Tk 200. The net gain is Tk 300 for 10 extra copies, or Tk 30 a copy. So marginal revenue, Tk 30, lies well below the price of Tk 50.

Given. At 20 units the price is Tk 60 and total revenue is Tk 1,200; at 30 units the price is Tk 50 and total revenue is Tk 1,500. *Find.* The marginal revenue per unit for the step from 20 to 30 units. *Formula.* In words: the rise in total revenue divided by the rise in output. In symbols: $MR = X \div Y$, where MR is marginal revenue, X is the rise in total revenue and Y is the rise in output. *Steps.* $X = 1,500 - 1,200 = \text{Tk } 300$. $Y = 30 - 20 = 10$ units. $MR = 300 \div 10 = \text{Tk } 30$ per unit. *Answer.* Marginal revenue for this step is Tk 30 per unit. *Check.* The gain of Tk 500 less the loss of Tk 200 is Tk 300, which agrees; and Tk 30 is below the price of Tk 50, as the argument said it must be.

PAUSE AND CHECK 14.P1 A seller sells 20 units at Tk 60 and 30 units at Tk 50. Why is the marginal revenue of the step below Tk 50?

Case file, illustrative. A seller with the whole market for a specialised title: price Tk 70 at 10 units, Tk 60 at 20, Tk 50 at 30, Tk 40 at 40, Tk 30 at 50 (price = 80 – quantity); total revenue 700, 1,200, 1,500, 1,600, 1,500; marginal revenue per step 70, 50, 30, 10, –10; marginal cost per step 30, 22, 20, 24, 34. Output 30 at price Tk 50; total cost 1,320; profit Tk 180.

The table repeats the case file in a form that is easy to scan. Read across each row: the output, the price at which that output sells, the total revenue (price times output), the marginal revenue for the 10-unit step that ends at that output, and the marginal cost for the same step. Marginal revenue falls faster than price, and it becomes negative in the last row, where selling more lowers total revenue.

Barriers to entry protect a monopoly from rivals.

Marginal revenue. The extra revenue from selling one more unit of output.

To choose her output, the seller needs the extra revenue from one more unit.

To sell more, a monopolist must cut the price on every unit.

A worked example turns the idea into a formula.

The case file gives the seller's price, revenue and costs.

The table sets the case out in five rows.

¹ Recall: price taker, Chapter 13, 'Market Structure and Perfect Competition', section 13.2 'The Perfectly Competitive Firm'.

THE SELLER'S PRICE, REVENUE AND COSTS (TAKA)

OUTPUT	PRICE	TOTAL REVENUE	MARGINAL REVENUE	MARGINAL COST
10	70	700	70	30
20	60	1,200	50	22
30	50	1,500	30	20
40	40	1,600	10	24
50	30	1,500	-10	34

The best output is where marginal revenue meets marginal cost.

The figure draws the same facts as three lines.

How much should she sell? Compare, step by step, what a step adds to revenue with what it adds to cost. A step is worth taking while marginal revenue is at least marginal cost, because it then adds at least as much to revenue as to cost. Look down the table. The step to 30 units adds Tk 30 a unit to revenue and Tk 20 to cost, so it adds profit. The step to 40 units adds only Tk 10 to revenue but Tk 24 to cost, so it loses profit. The best output is 30 units, and the price comes from the demand curve: Tk 50. This is the rule that Chapter 13 used for the competitive firm, with marginal revenue in place of price.²

The figure shows the same facts. The horizontal axis counts output in units, and the vertical axis counts taka. The falling straight line is the demand curve: each point gives the price at which that output sells. The steeper line below it is marginal revenue, which falls twice as fast and reaches zero at 40 units. The curved line is marginal cost. Where marginal revenue meets marginal cost, at about 30 units, we climb to the demand curve to read the price, Tk 50. The shaded rectangle is the profit: the gap between the price of Tk 50 and the average cost of Tk 44 ($1,320 \div 30$) is Tk 6, and Tk 6 times 30 units is Tk 180.

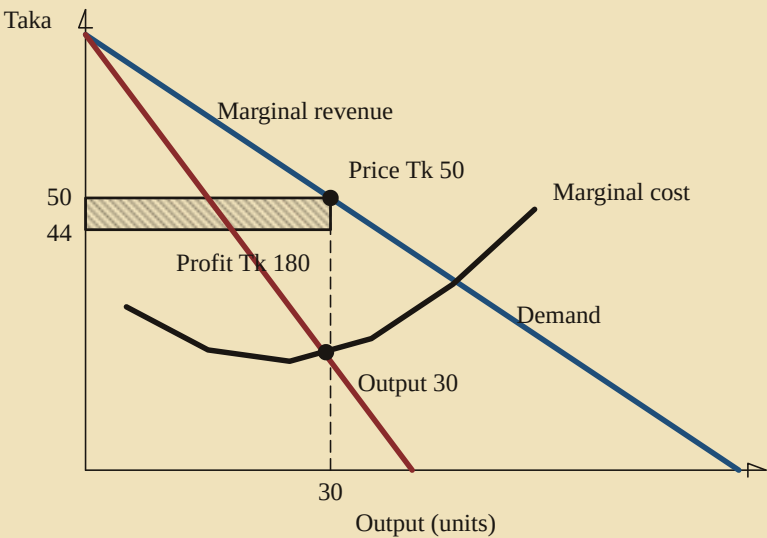


FIG. 14.1. The monopolist's demand, marginal revenue and marginal cost, with the best output and its profit

A monopoly is not sure of profit, and its position may not last.

The theory has exceptions. First, a monopolist may lose money if no price on the demand curve covers its average cost; being the only seller does not create buyers. Second, a monopoly seldom lasts for ever. A high profit tempts rivals to find a way round the barrier, and governments may control the price or break up the firm. From the consumer's viewpoint, the monopoly price is

² Recall: marginal cost, Chapter 11, 'Costs of Production', section 11.2 'Short-Run Costs'; the output of largest profit is taught in Chapter 13, 'Market Structure and Perfect Competition', section 13.3 'Equilibrium of Firm and Industry'.

usually higher and the output lower than in a competitive market, as section 14.4 shows.

Before moving on. A monopoly has one seller and no close substitute, so the seller is a price maker. Marginal revenue lies below price, and the seller chooses the output at which marginal revenue meets marginal cost, then reads the price from the demand curve. *You should now be able to say why a monopolist's marginal revenue is smaller than its price.*

Recap of section 14.1.

14.2 MONOPOLISTIC COMPETITION

Walk down a busy street in Dhaka lined with small stalls that sell tea and biscuits. There are many sellers, and anyone may open another stall. Yet no two stalls are the same. One has a shaded seat, another brews a stronger tea, a third stands where the buses stop. Regular consumers prefer one stall and will pay a little more to keep going there. Each stall therefore holds a small power over its own price, although many rivals are close by.

A street of tea stalls shows many sellers whose goods differ.

This market has a name. **Monopolistic competition** is a market with many sellers of similar but not identical goods, and free entry. The tea street is one example; hair salons and small restaurants in one neighbourhood are others. The name joins two ideas. Each seller is a little like a monopolist, because its good differs from rivals' goods. Each seller also competes, because rivals offer close, though not perfect, substitutes, and anyone may enter.

Monopolistic competition. A market with many sellers of similar but not identical goods, and free entry.

Many sellers of similar but different goods, with free entry, make monopolistic competition.

The source of each seller's small power is **product differentiation**, which means making a good different from rivals' goods in quality, style or brand, to build loyal buyers. A stall that serves tea in clean glasses with a friendly greeting has differentiated itself. The *brand* of a good is the name and style by which consumers recognise it. The differences may be real, such as a stronger tea, or only believed, such as a famous name. Either way, a loyal consumer finds the good hard to replace.

Product differentiation. Making a good different from rivals' goods in quality, style or brand, to build loyal buyers.

Brand. Word in use. In daily life, a name printed on a good, or a mark burned on cattle. In economics, a name and style that make a firm's good known and different from rivals' goods.

Differences in quality, style or brand give each seller loyal consumers.

Because the good is different, a seller can raise its price without losing every consumer, so its demand curve slopes downward, as a monopolist's does. Because close substitutes exist, a rise in its price still sends many consumers to rivals, so the curve is flatter than a monopolist's would be. Like the monopolist, the seller finds that marginal revenue lies below price, and it chooses the output at which marginal revenue equals marginal cost. We call that choice the equilibrium of the firm, a term taught in Chapter 13, section 13.3 'Equilibrium of Firm and Industry'.

Its demand curve slopes down but is flatter than a monopolist's.

In the short run, a differentiated seller may earn more than the least profit it needs to stay in business. That profit attracts new sellers, who open new stalls with similar goods. Each new rival takes some consumers from the existing sellers, so every existing seller's demand curve shifts inwards. The process continues until a typical seller earns only enough to stay in business. This is the same ending as in perfect competition, but with one difference: each seller still has loyal consumers, so its price stays a little above marginal cost.

Free entry removes extra profit over time.

Differentiation has costs. Sellers spend on advertising, packaging and decoration to build a brand, and these costs enter the price. Consumers gain variety, which many value. Whether the gain outweighs the cost is a matter of judgement, and economists differ. From the owner's viewpoint, the aim is loyalty.

Differentiation costs money, and whether it helps consumers is a matter of judgement.

Recap of section 14.2.

Three bus firms on one road show how few sellers watch each other.

Oligopoly. A market dominated by a few large sellers, each aware of what the others do.

A few large sellers, each watching the others, make an oligopoly.

Each firm's best choice depends on its rivals' choices.

Kinked demand explains why prices in an oligopoly often stay fixed.

Firms may compete on features or agree to act together, and the law often forbids agreement.

Recap of section 14.3.

From the consumer's viewpoint, it is choice. From the employee's viewpoint, a stall with loyal consumers may offer steadier work.

Before moving on. Monopolistic competition has many sellers, free entry and goods that differ. Product differentiation gives each seller a downward-sloping demand curve, flatter than a monopolist's, and free entry removes extra profit over time. *You should now be able to say what separates it from both perfect competition and monopoly.*

14.3 OLIGOPOLY

Imagine a long road served by only three bus firms. If one firm cuts its fare, it will take many riders from the other two, so they will probably cut theirs too. If one raises its fare, the others may keep theirs low and take its riders. Before changing a fare, each firm must guess how the others will react. A seller among many sellers never worries about this. A seller among a few worries about it all the time.

An **oligopoly** is a market dominated by a few large sellers, each aware of what the others do. Mobile telephone service, flights on a busy route and cement in a country are common examples. The word comes from Greek words meaning *few* and *to sell*. Entry is hard, because the firms are large and a newcomer needs a great deal of money, equipment or consumer loyalty to compete with them.

The key feature is *interdependence*: the best price or output of each firm depends on what its rivals do. In a competitive market a firm ignores its rivals, because each is too small to matter. In a monopoly there are no rivals to watch. In an oligopoly the firms are few enough that every move is noticed. This makes the market harder to predict, and economists offer several theories rather than one.

One theory explains why oligopoly prices often stay the same for long periods. Suppose a firm believes that its rivals will copy a price cut but ignore a price rise. If it cuts its price, rivals match the cut, and it gains few extra consumers. If it raises its price, rivals do not follow, and it loses many consumers. So its demand curve bends, or has a kink, at the present price: flat above that price and steep below it. The firm sees little reward in changing its price in either direction, so it leaves the price alone. The theory explains why a price stays where it is; it does not explain how the price was first set.

PAUSE AND CHECK 14.P2 Why might an oligopolist leave its price unchanged even when its costs fall a little?

Oligopolists have two broad paths. They may compete without cutting prices, by advertising, quality and service. Or they may agree with one another on prices or output, acting together like a single monopolist. Many countries forbid such agreements through competition law, but the rules and the penalties differ from country to country. Agreements are also fragile, because each firm can gain by secretly selling below the agreed price, and when many firms do this the agreement fails.

Before moving on. An oligopoly has a few large sellers, hard entry and interdependence: each firm must guess how rivals will react. Kinked demand explains why prices often stay fixed, and agreements among firms are fragile and often unlawful. *You should now be able to say why one firm's choice affects the others.*

14.4 COMPARING MARKETS

The table below sets the four markets side by side. Read one column at a time to see what defines each market, and read one row at a time to see how a single feature changes as we move from many sellers to one. The figure that follows shows the number of sellers as dots, and the shapes show whether the goods are alike or different.

A table compares the four markets on four features.

THE FOUR MARKET STRUCTURES COMPARED

FEATURE	PERFECT COMPETITION	MONOPOLISTIC COMPETITION	OLIGOPOLY	MONOPOLY
Sellers	Very many	Many	Few	One
Goods	Identical	Similar but different	Identical or different	No close substitute
Entry	Free	Free	Hard	Blocked
Price	Price taker	Price maker, small power	Price maker, watches rivals	Price maker, large power

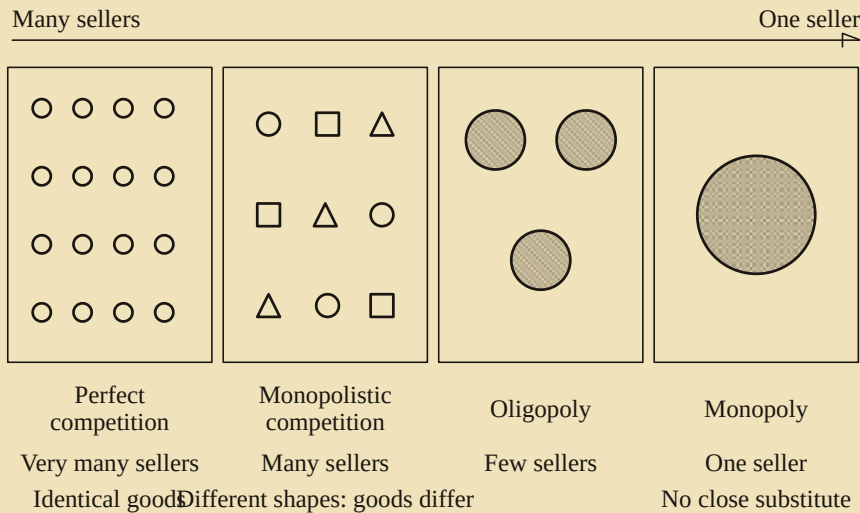


FIG. 14.2. Four market structures: very many, many, few and one seller

Suppose, as an illustration, that a competitive market faced the same demand and the same marginal cost as the seller of the case file. Competitive sellers push output to the point where price equals marginal cost. At 40 units the price is Tk 40 and the marginal cost of the step is Tk 24, so output rises. At 50 units the price is Tk 30 and the marginal cost is Tk 34, so output stops at 40. The competitive market would sell 40 units at Tk 40, while the monopolist sells 30 units at Tk 50: ten fewer units, at a price Tk 10 higher.

In the case file, the monopoly sells less at a higher price than a competitive market would.

Is a lower price always better? Not always. A competitive market tends to give the lowest price for a given good, but one large firm may serve everyone at a lower cost for each unit than many small firms could. Monopolistic competition gives consumers variety that a market of identical goods cannot. Price is one measure of how well a market serves consumers; quantity, quality, variety and the pace of new ideas are others. The reader should weigh them all.

Lower price is one measure of a market, not the only one.

Before moving on. The four markets differ in the number of sellers, the kind of goods and the ease of entry, and these differences decide how much power a seller

Recap of section 14.4.

has over price. *You should now be able to say which market a given description fits.*

WORKED CASE: THE SELLER'S BEST OUTPUT

The case file gives the data for the worked case.

Given. From the case file of section 14.1: demand price = $80 - \text{quantity}$; total revenue 700, 1,200, 1,500, 1,600, 1,500 at outputs 10 to 50; marginal cost per unit for each 10-unit step 30, 22, 20, 24, 34; total cost Tk 1,320 at 30 units. *Find.* The output of largest profit, the price, and the profit. *Formula.* In words: raise output while marginal revenue is at least marginal cost; read the price from demand; profit is total revenue minus total cost. In symbols: $\text{profit} = P \times Q - C$, where P is price, Q is output and C is total cost.

The steps compare marginal revenue with marginal cost for each step.

Steps. (1) Marginal revenue per unit for each step: $700 \div 10 = 70$; $(1,200 - 700) \div 10 = 50$; $(1,500 - 1,200) \div 10 = 30$; $(1,600 - 1,500) \div 10 = 10$; $(1,500 - 1,600) \div 10 = -10$. (2) Compare with marginal cost: 70 against 30, take the step; 50 against 22, take it; 30 against 20, take it; 10 against 24, stop. (3) Output is 30 units. (4) Price = $80 - 30 = \text{Tk } 50$. (5) Total revenue = $50 \times 30 = \text{Tk } 1,500$. (6) Profit = $1,500 - 1,320 = \text{Tk } 180$. *Answer.* The seller sells 30 units at Tk 50 and earns a profit of Tk 180. *Check.* Total cost at 20 units is $1,320 - 10 \times 20 = \text{Tk } 1,120$, so profit there is $1,200 - 1,120 = \text{Tk } 80$. Total cost at 40 units is $1,320 + 10 \times 24 = \text{Tk } 1,560$, so profit there is $1,600 - 1,560 = \text{Tk } 40$. Both are below Tk 180.

A second example uses a new seller and less tidy numbers.

Given. Price = $60 - \text{quantity}$; fixed cost Tk 100; marginal cost per unit for the 10-unit steps ending at 10, 20, 30 and 40 units: 20, 22, 25, 30. *Find.* Output, price and profit. *Formula.* As above. *Steps.* Total revenue at 10, 20, 30, 40 units: 500, 800, 900, 800. Marginal revenue per unit: 50, 30, 10, -10. Against marginal cost, the first two steps are taken (50 over 20, 30 over 22) and the third is not (10 under 25). Output is 20 units and price is $60 - 20 = \text{Tk } 40$. Total cost = $100 + 10 \times 20 + 10 \times 22 = \text{Tk } 520$. Profit = $800 - 520 = \text{Tk } 280$. *Answer.* 20 units at Tk 40, profit Tk 280. *Check.* At 30 units, total cost is $520 + 10 \times 25 = 770$ and revenue is 900, so profit is 130, which is less than 280.

Four common slips spoil these calculations.

Common slips deserve a warning. Some students compare price with marginal cost instead of marginal revenue with marginal cost. Some read the price from the marginal revenue line, when it must come from the demand curve at the chosen output. Some forget that a step of 10 units must be divided by 10 to give a figure per unit. Others choose an output where marginal revenue is negative, which always loses money when marginal cost is positive.

Recap of the worked case.

Before moving on. To find a monopolist's best output, compute marginal revenue for each step, take steps while it is at least marginal cost, read the price from demand, and then find profit. *You should now be able to say how each of the six steps leads to the answer.*

* * *

SUMMARY

The chapter's four markets and one rule fit together.

A monopoly has one seller and no close substitute, so the seller is a price maker whose marginal revenue lies below its price. It chooses the output at which marginal revenue meets marginal cost, and it reads the price from demand. Monopolistic competition has many sellers, free entry and product differentiation. An oligopoly has a few large sellers who must guess one another's moves. Compared with competition, a monopoly tends to sell less at a higher price.

THINK IT THROUGH

14.T1 Is a lower price always better for consumers? Give one reason from the chapter why the answer may be no.

14.T2 Why will a seller with positive marginal cost never choose an output at which marginal revenue is negative? Use the 50-unit row of the table to test your answer.

WHERE PEOPLE GO WRONG

Some students think a monopolist can charge any price. It cannot. It controls its price only up to the limit set by the demand curve: a higher price brings fewer buyers. The seller in the case file chooses Tk 50, not Tk 500.

Some students use the price as the marginal revenue of a monopolist. For a price taker the two are equal, but for a price maker marginal revenue is smaller, because a cut in price applies to every unit sold.

Some students treat monopolistic competition as a weak form of monopoly. It differs in kind. It has many sellers and free entry, so extra profit does not last, while a monopoly is protected by barriers.

Some students think firms in an oligopoly must agree with one another. The defining feature is interdependence, not agreement. Firms may compete hard, and agreements are often unlawful and fragile.

A monopolist cannot charge any price it likes.

Marginal revenue is not the same as price for a price maker.

Monopolistic competition is not a mild monopoly.

Oligopoly does not mean that firms agree.

IN HISTORY

At Kimberley, in what is now South Africa, many separate owners worked diamond claims in the 1870s and 1880s. Cecil Rhodes, a British businessman who had come to South Africa as a young man, set out to buy the claims and join them under one owner. His chief rival was Barney Barnato, another Briton, whose firm owned much of the Kimberley mine.

In March 1888 the rival holdings were merged to form De Beers Consolidated Mines, with Rhodes as its first chairman. In July 1889 the firm paid £5,338,650, then an unprecedented sum, for the Kimberley mine, and in the same month it won control of two more mines, Bultfontein and Dutoitspan. It then controlled most of the world's supply of diamonds, the first barrier of section 14.1. That barrier did not hold for ever: later, new mines in other places came into production and loosened the firm's hold.

A British businessman joined many diamond owners into one firm.

In 1888 the rival holdings merged into De Beers Consolidated Mines.

REVIEW QUESTIONS

14.1 Total revenue is Tk 1,500 at 30 units and Tk 1,600 at 40 units. Find the marginal revenue per unit for the step from 30 to 40 units.

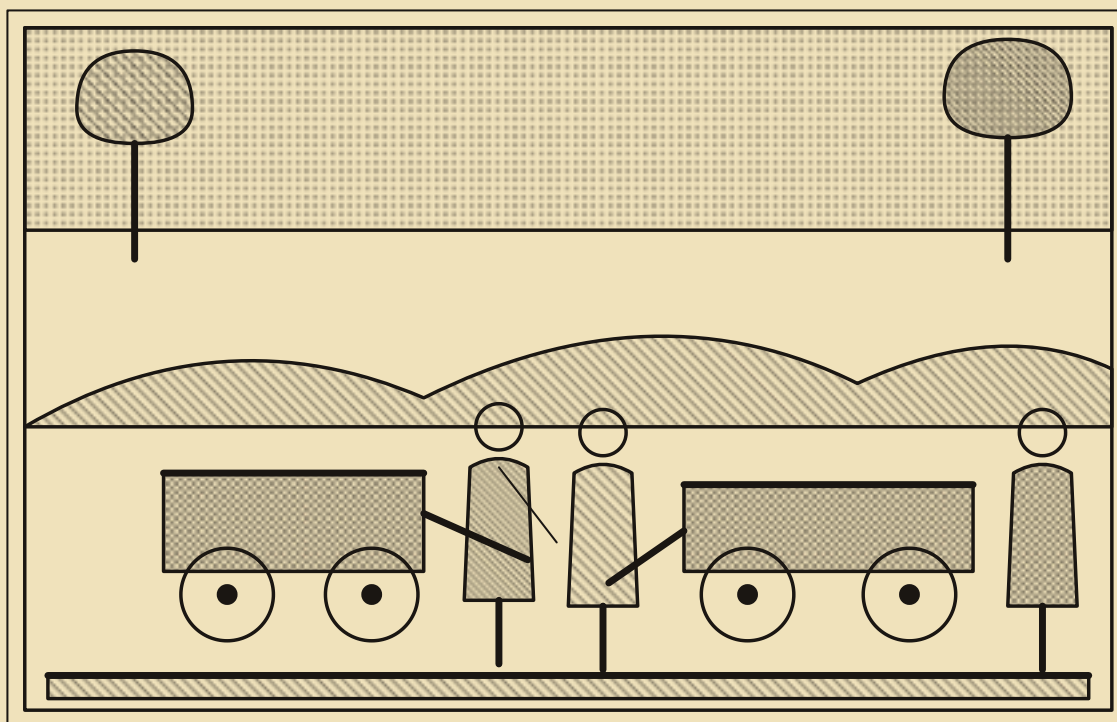
14.2 A seller faces $\text{price} = 60 - \text{quantity}$ and has a fixed cost of Tk 100. Total revenue is 500, 800, 900 and 800 at 10, 20, 30 and 40 units, and the marginal cost per unit for the four 10-unit steps is 20, 22, 25 and 30. Find the output, the price and the profit.

14.3 In the case file the marginal revenue of the step to 30 units is Tk 30 per unit. A manager says it is Tk 300. Show that both figures agree, and say which to compare with a marginal cost of Tk 20 per unit.

14.4 Suppose a competitive market faced the demand and marginal cost of the case file, so that price equals marginal cost. Find its output and price, and compare them with the monopoly's.

14.5 Name the market: (a) many tailors, each with a different style, free to open new shops; (b) three firms supply all the cement in a country; (c) one firm holds the only licence to supply a town with water.

14.6 A tea stall's total revenue is Tk 800 at 40 cups and Tk 900 at 50 cups, and its marginal cost is Tk 12 a cup for those 10 cups. Find the marginal revenue per cup and say whether the stall should sell the extra 10 cups. Is marginal revenue below the price of Tk 18 at 50 cups?



A hiring fair, with workers waiting beside the farm carts

CHAPTER XV

Rent and Wages

Why is one field paid for, and one worker paid more than another?



This chapter explains why land earns a payment that no one needs to be tempted to give, how the payment to labour is measured in money and in goods, and how demand and supply set a wage and explain why wages differ.

YOU WILL NEED

Chapter 14, 'Monopoly and Imperfect Competition', section 14.1 'Monopoly', for monopoly; Chapter 9, 'Elasticity of Supply and Market Equilibrium', section 9.3 'Changes in Equilibrium', for changes in equilibrium; Chapter 11, 'Costs of Production', section 11.1 'Explicit and Implicit Costs', for explicit and implicit costs; and Appendix A1, 'Maths You Will Use', for percentages and index numbers.

A hiring fair brings together owners of land and sellers of labour.

Two puzzles stand beside each other: the payment to land and the payment to workers.

The chapter answers both puzzles by looking at demand and supply.

On a market day in an old English county, farmers drive their carts into the square and workers stand beside them, waiting to be hired for the year. A farmer with two fields looks over the crowd. Everyone at the fair knows that the farmer will pay the labourers, for they have to be tempted to work. Few ask a stranger question: why does the farmer's field earn a payment at all? Nobody had to be tempted to make the field.

A second puzzle sits close to the first. In the crowd, a ploughman asks for less than a skilled stonemason, and the stonemason asks for less than a surgeon would in the town. They are all people. Why do their payments differ so widely? And what stops a farmer from offering half the going rate, or a worker from demanding double?

The answers come from the tools you already have. Section 15.1 studies rent, the payment to land, and shows that a whole class of payments share its character. Section 15.2 turns to wages and distinguishes the wage in money from the wage in goods. Section 15.3 uses demand and supply to explain how wages are set and why they differ.

* * *

15.1 RENT

RENT AND ECONOMIC RENT

Everyday speech uses the word rent for many payments.

Rent. Word in use. In daily life, any payment for the use of a house, a room, a car or a machine. In economics, the payment for the use of land.

Rent. The payment to the owner of land for its use, in everyday talk and in economics.

Rent is the payment to the owner of land for its use.

The supply of land as a whole is fixed, which makes its payment unusual.

Begin with a familiar payment. A family pays a landlord each month to live in a flat. A farmer pays an owner each year to plough a field. In both cases a person pays for the use of land that belongs to someone else. This payment is **rent**, which means the payment to the owner of land for its use, in everyday talk and in economics. A tenant who pays Tk 60,000 a month for a shop site is paying rent. In everyday speech the word is also used for cars and machines, so this book keeps it for land unless it says otherwise.

Land has a feature that most goods lack. Nobody can make more of it. The whole supply of land in a country is fixed, whatever the payment. In words you know from Chapter 9, the elasticity of supply of land as a whole is zero. Because supply cannot respond, the payment to land is set by demand alone: when many people want land, its payment rises, and when few want it, the payment falls. The landowner does not have to be tempted to supply it.

This observation leads to a wider idea. **Economic rent** means the payment to any factor above the least needed to keep it in use. A factor is any resource used in production, such as land, labour or a machine. Suppose a famous singer is paid Tk 100,000 for an evening and would have agreed to sing for Tk 40,000. The least needed is Tk 40,000, and the economic rent is Tk 60,000. For land as a whole the least needed is nothing, since it exists whatever is paid, so the whole payment is economic rent.

For one plot, the position differs. A plot of farmland has another use, and its owner could sell it for building. The least needed to keep it in farming is what that other use would pay, which Chapter 11 would call an implicit cost. Only the part above that is economic rent. This is a common source of confusion, and it is why economists speak of economic rent for any factor and not only land.

Viewpoints differ. For the owner, rent is income from a fixed asset that needs no extra effort to supply. For the tenant, rent is a payment that must be covered by the sales of the shop or the crop. For the buyer of the goods, rent is not a reason for a high price, since the site's advantage creates the rent and not the other way round. A tenant who pays a high rent for an excellent site is paying for its advantage, and could not make the same profit from a poor site at a low rent.

Before moving on. Rent is the payment to the owner of land for its use. Land as a whole is fixed in supply, so its whole payment is economic rent. Economic rent is the part of any payment above the least needed to keep a factor in use.

RICARDIAN THEORY OF RENT

Why does one plot pay more rent than another? The English economist David Ricardo gave the classic answer in 1817. His theory rests on a simple picture: land differs in quality, and the best land is used first.

Imagine a district with three grades of land. A farmer using one acre of Grade A land harvests 3,000 kg of rice. On Grade B the same work and seed yield 2,500 kg, and on Grade C they yield 2,000 kg. Suppose that Grade C is the poorest land still worth farming, so it pays no rent: it is the marginal land. A farmer will not pay more for Grade B than the extra 500 kg it gives over Grade C, because Grade C is available for nothing. For the same reason Grade A will rent for the extra 1,000 kg. Rent equals the difference in yield between a plot and the marginal land.

To read the first figure, look at the stacked bars. Each bar is one grade of land, and its height is the yield of one acre. The lower, hatched part of each bar is the yield of the marginal land, which every grade produces. The dotted part above it is the extra yield, and that is the rent. Grade C has no dotted part, because it is the marginal land and pays no rent.

Economic rent. The payment to any factor above the least needed to keep it in use.

Economic rent is the part of a payment above the least needed to keep a factor in use.

Recall: explicit and implicit costs, Chapter 11, section 11.1 'Explicit and Implicit Costs'.

The owner, the tenant and the buyer of the goods each see rent differently.

Before moving on: rent is a payment for land; economic rent is a surplus above the least needed.

Differences in the quality of land explain why some plots pay more.

Rent arises because better land produces more at the same cost.

The figure shows rent as the part of the yield above the marginal land's yield.

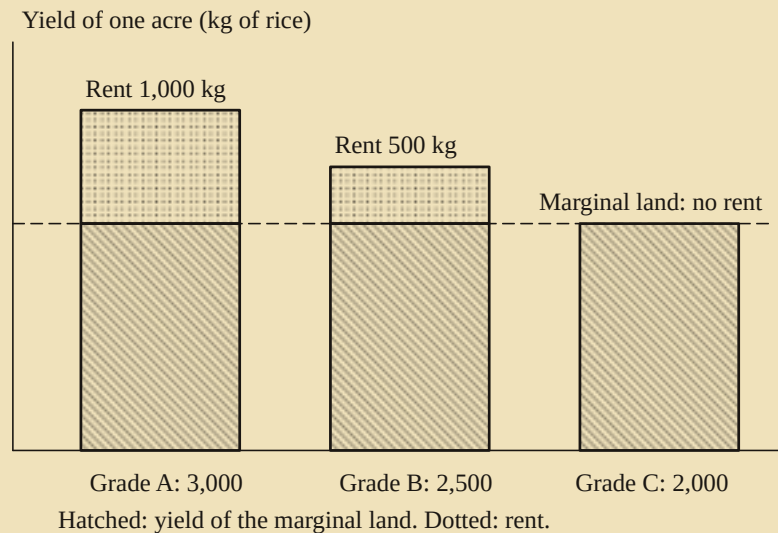


FIG. 15.1. Differential rent as stacked grades of land

Ricardo's theory applies to shop sites as well as farms.

The record gives two profits and the rent they imply.

The case file confirms that rent equals the gain of the better site over the marginal one.

Before moving on: Ricardian rent is the gain of a better plot over the marginal plot.

Man-made assets can earn a surplus for a short time.

Quasi-rent. A temporary surplus earned by a man-made asset while its supply is fixed in the short run.

Quasi-rent is a short-run surplus earned by a man-made asset.

Before moving on: rent is lasting, quasi-rent is temporary.

Ricardo's theory applies to shop sites as well as to farms, and the next record shows it.

Case file, illustrative. Meghna and Sons trades at a site in Dhanmondi, Dhaka. The shop's profit before rent at Dhanmondi is Tk 1,470,000 a year. At the marginal site, which pays no rent, the profit before rent would be Tk 750,000. The difference is Tk 720,000, which equals the rent paid: Tk 60,000 $\times$ 12 = Tk 720,000.

The record shows the theory at work. A tenant would not pay more than Tk 720,000 a year for the Dhanmondi site, because the marginal site yields Tk 750,000 for no rent. The rent also cannot fall far below Tk 720,000, because other shopkeepers would bid for such a profitable site. Competition between users therefore pushes the rent to the difference. Note that rent does not cause the price of the goods sold at the site. It is the result of the site's advantage.

PAUSE AND CHECK 15.P1 Three grades of land yield 90, 70 and 60 bags an acre, and the 60-bag land is the marginal land. Find the rent of each grade, in bags.

Before moving on. In Ricardo's theory, land of different quality earns differential rent, equal to the difference between its yield and that of the marginal land. The marginal land pays no rent. The same reasoning applies to shop sites.

QUASI-RENT

Land is a gift of nature. Is there a similar surplus for things people make? Yes, but it does not last. The English economist Alfred Marshall, in his *Principles of Economics* (1890), gave it a name.

Quasi-rent means a temporary surplus earned by a man-made asset while its supply is fixed in the short run. Suppose the demand for printed books rises. The number of printing presses cannot rise overnight, so those that exist earn more than the least needed to keep them running, namely their variable costs. The surplus above variable cost is what pays the fixed cost and leaves a gain. In the long run, new presses are built, supply rises and the surplus is competed away. That is why it is called quasi, meaning 'as if' rent: it behaves like rent for a short time only.

Before moving on. Rent belongs to land, whose supply is fixed for good. Quasi-rent belongs to man-made assets, whose supply is fixed only in the short run, and it disappears when new assets are built.

* * *

15.2 WAGES

THE WAGE

Labour is the work of people, and it is paid. A **wage** means the payment to labour for its work over a period. It may be stated by the hour, the day or the month. A shop assistant paid Tk 35,000 a month earns a wage, and so does a surgeon paid Tk 400,000 a month. In daily speech a salary and a wage are different; in economics the two are the same kind of payment.

The word wage is used loosely in daily life.

Wage. Word in use. In daily life, the weekly pay of a manual worker, as distinct from a salary. In economics, any payment to labour, whatever its name.

Wage. The payment to labour for its work over a period.

A wage is the payment to labour for its work over a period.

NOMINAL WAGE AND REAL WAGE

A wage is first stated in money, but money is only useful for what it buys. If the prices of goods double, the same money buys half as much. Economists therefore separate the money amount from its purchasing power.

A wage in money can hide what the worker can actually buy.

The **nominal wage** is a wage measured in money, without allowing for changes in prices. Tk 35,000 a month is a nominal wage. It is the figure on the pay slip.

Nominal wage. A wage measured in money, without allowing for changes in prices.

A nominal wage is the money amount, unadjusted for prices.

The **real wage** is a wage measured by the goods and services it can buy. To find it, divide the nominal wage by a price index, which is a number showing the level of prices compared with a base year set at 100. In words: real wage = nominal wage $\div$ (price index $\div$ 100). In symbols, real wage = $W \div (P \div 100)$, where W stands for the nominal wage and P for the price index.

Real wage. A wage measured by the goods and services it can buy.

A real wage measures the goods and services the wage buys.

Case file, illustrative. A shop assistant at Meghna and Sons earns a nominal wage of Tk 35,000 a month. The price index rises from 100 to 112. The real wage is $35,000 \div 1.12 = \text{Tk } 31,250$.

The next record sets out the figures.

Given: a nominal wage of Tk 35,000 a month, and a price index that rises from 100 to 112. *Find:* the real wage, in the prices of the base year. *Formula:* real wage = nominal wage $\div$ (price index $\div$ 100). *Steps.* The index divided by 100 is $112 \div 100 = 1.12$. Then $35,000 \div 1.12 = 31,250$. *Answer:* the real wage is Tk 31,250 a month, in base-year prices. *Check:* prices have risen 12%, so the wage should buy less than Tk 35,000 did before, and Tk 31,250 is lower.

The steps follow the usual order, from the given figures to a check.

The result carries a lesson. Suppose the assistant is given a rise of 10%, to Tk 38,500, while the price index rises to 112. The real wage is $38,500 \div 1.12 = \text{Tk } 34,375$ in base-year prices, which is below the first figure of Tk 35,000 in the same prices. The assistant has more money and less purchasing power. A common slip is to judge a wage by the nominal figure alone.

The nominal wage can rise while the real wage falls.

The employee cares about the real wage, because bills are paid in goods. The employer cares about the nominal wage, because the payroll is paid in money, but a careful employer also watches the real wage, since workers who see their purchasing power fall may leave or bargain for more. A rise in the nominal wage that only matches rising prices leaves the worker no better off, and economists call confusion over this point a mistake of treating money as if it were purchasing power.

A viewpoint check shows why the real wage matters to each side.

Before moving on: the nominal wage is in money and the real wage is in goods.

Before moving on. A wage is the payment to labour for its work over a period. The nominal wage is the money amount. The real wage divides it by the price level, so it shows what the wage buys.

* * *

15.3 HOW WAGES ARE SET

WHAT A WORKER ADDS

A firm hires a worker only if the worker brings in more than the worker costs.

Return to the shop. The owner is thinking of hiring one more assistant at a wage of Tk 35,000 a month. A sensible owner asks one question: how much more money will the shop receive because of this assistant? If the answer is more than Tk 35,000, hiring pays. If it is less, hiring loses money.

Marginal revenue product. The extra revenue a firm gains from employing one more unit of labour.

The measure has a name. **Marginal revenue product** means the extra revenue a firm gains from employing one more unit of labour. It combines two things: the extra output the worker produces, and the extra revenue that each unit of output brings, which is the marginal revenue you met earlier. In words: marginal revenue product = extra output from the worker $\times$ marginal revenue per unit. For a worker who produces 20 more books a week, where each extra book brings Tk 500, the marginal revenue product is Tk 10,000 a week.

Marginal revenue product is the extra revenue from one more unit of labour.

The rule for the owner follows. Hire one more worker while the marginal revenue product is at least as large as the wage, and stop when it falls below the wage. The extra output from each added worker eventually shrinks, since the workers share the same shop, shelves and counter, and so the marginal revenue product falls as the firm hires more.

The hiring rule compares the marginal revenue product with the wage.

Case file, illustrative. At Meghna and Sons, the first extra assistant adds Tk 40,000 a month of revenue, and the second adds Tk 30,000. The wage is Tk 35,000 a month.

The record gives the revenue that two extra assistants would add.

Apply the rule. The first assistant brings in Tk 40,000 and costs Tk 35,000, so the shop gains Tk 5,000 a month. The second brings in Tk 30,000 and costs Tk 35,000, so the shop would lose Tk 5,000 a month. The owner hires one extra assistant and stops. Since the figures fall as more are hired, the marginal revenue product schedule works as the firm's demand for labour: it shows how many workers the firm will hire at each wage.

The case file shows the rule in use: hire the first assistant but not the second.

DEMAND AND SUPPLY OF LABOUR

A labour market has a demand side and a supply side, like any other market.

The wage for a whole occupation is set where the demand for labour meets the supply of labour. Demand comes from all the firms that hire; it slopes downward because the marginal revenue product falls as more workers are hired. Supply comes from workers; it slopes upward because a higher wage draws more people into the work, as it draws people from other jobs, from study and from leisure. As in other markets, the equilibrium wage is the wage at which the quantity of labour demanded equals the quantity supplied.

To read the second figure, find the wage and the number of workers where the lines meet.

To read the second figure, look at the axes. The vertical axis shows the wage, and the horizontal axis shows the number of workers. The solid line, sloping down, is the demand for labour. The dashed line, sloping up, is the supply of labour. Where they cross, the wage and the number employed are set. At a higher wage, more workers want jobs than firms will fill; at a lower wage, firms want more workers than offer.

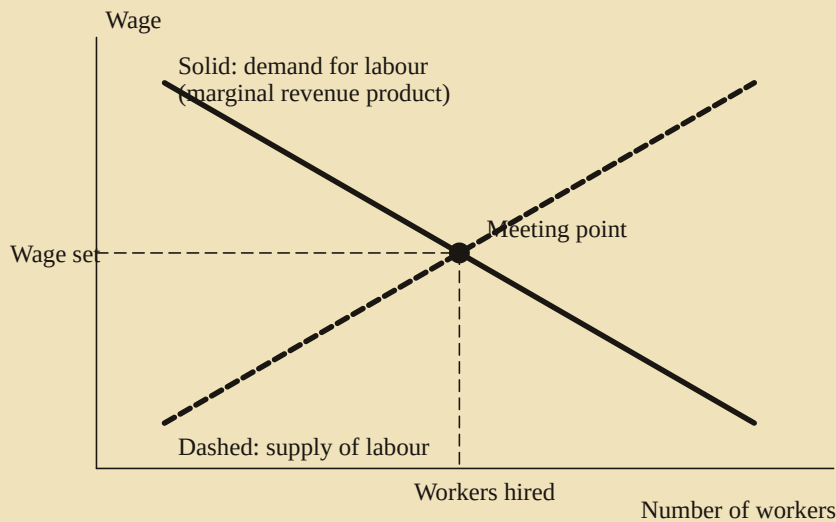


FIG. 15.2. Wage set by the demand for and supply of labour

Changes follow the pattern of Chapter 9. If firms become more productive or the price of their goods rises, the marginal revenue product rises and demand for labour moves to the right, so the wage and the number employed both rise. If more people enter the occupation, supply moves to the right and the wage falls. A wage fixed by law above the equilibrium wage leaves more workers seeking jobs than firms will fill, though it raises the pay of those who keep their work.

WHY WAGES DIFFER

The same tools explain why wages differ. A surgeon earns more than an assistant for reasons on both sides of the market. On the demand side, a surgeon's work brings in or saves a very large amount, so the marginal revenue product is high. On the supply side, few people can qualify, since the training is long, costly and demanding, so the supply is small. A high demand meeting a small supply gives a high wage.

Other causes follow the same logic. Dangerous or unpleasant jobs pay more, because few people accept them at an ordinary wage. Workers with scarce skills are in short supply. Trade unions raise wages by limiting supply or by bargaining. A firm that is the only large employer in a town, a position like the monopoly of Chapter 14, may pay less than a competitive market would, because workers have nowhere else to go. Differences in wages are therefore not random; each can be traced to demand, supply or both.

Wages are sometimes fixed by law. A minimum wage set above the equilibrium wage raises the pay of workers who keep their jobs, but at that higher wage firms demand fewer workers than seek work. Whether jobs are lost, and how many, depends on how strongly firms respond to the higher wage; the wider the gap between the legal wage and the equilibrium wage, the larger the likely loss. Governments weigh this against the gain to the workers who are paid more.

Before moving on. A firm hires while the marginal revenue product is at least the wage. The wage for an occupation is set where demand for labour meets supply. Wages differ because demand and supply differ across occupations.

Recall: changes in equilibrium, Chapter 9, section 9.3 'Changes in Equilibrium'.

Wages differ because the demand and supply differ between occupations.

Other causes work through the same two sides.

Recall: monopoly, Chapter 14, section 14.1 'Monopoly'.

A minimum wage fixed above the equilibrium wage helps those hired and may leave others without work.

Before moving on: wages follow the marginal revenue product and the supply of workers.

* * *

WORKED EXAMPLES: FOUR RUNGS

Four examples, from clean numbers to a combined problem.

Rung 1: a real wage with clean numbers.

Rung 2: differential rent with three sites and less tidy numbers.

Rung 3: the period changes from a year to a month.

Rung 4: hiring and economic rent in one problem.

Each example below follows the order Given, Find, Formula, Steps, Answer, Check.

Given: a nominal wage of Tk 20,000 a month, and a price index that rises from 100 to 125. *Find:* the real wage. *Formula:* real wage = nominal wage ÷ (price index ÷ 100). *Steps.* $125 \div 100 = 1.25$, and $20,000 \div 1.25 = 16,000$. *Answer:* Tk 16,000 a month in base-year prices. *Check:* prices rose by a quarter, so the wage should buy only four-fifths as much: $20,000 \times 0.8 = 16,000$.

Given: three shop sites have profits before rent of Tk 1,470,000, Tk 1,100,000 and Tk 750,000 a year, and the last is the marginal site. *Find:* the rent that each site can bear. *Formula:* rent = profit before rent at the site - profit before rent at the marginal site. *Steps.* First site: $1,470,000 - 750,000 = 720,000$. Second site: $1,100,000 - 750,000 = 350,000$. Marginal site: 0. *Answer:* Tk 720,000, Tk 350,000 and nothing. *Check:* the better the site, the higher the rent, and the first figure agrees with the Dhanmondi record.

Given: an annual nominal wage of Tk 420,000, and a price index that rises from 100 to 112. *Find:* the real wage per month. *Steps.* The monthly nominal wage is $420,000 \div 12 = 35,000$. The real monthly wage is $35,000 \div 1.12 = 31,250$. *Answer:* Tk 31,250 a month. *Check:* the annual real wage is $420,000 \div 1.12 = 375,000$, and $375,000 \div 12 = 31,250$, the same. The slip to avoid is to divide by 12 at one stage and forget it at another.

Given: a firm pays Tk 35,000 a month. Successive extra workers add Tk 52,000, Tk 41,000, Tk 36,000 and Tk 30,000 a month of revenue. The third worker would have accepted Tk 28,000 a month. *Find:* how many to hire, the gain, and the third worker's economic rent. *Steps.* Compare each marginal revenue product with 35,000: the first three exceed it, the fourth (30,000) does not, so hire three. The revenue added is $52,000 + 41,000 + 36,000 = 129,000$, and the wages are $3 \times 35,000 = 105,000$, a gain of 24,000. The economic rent of the third worker is $35,000 - 28,000 = 7,000$. *Answer:* hire three, gain Tk 24,000 a month, and the third worker's economic rent is Tk 7,000. *Check:* the last worker hired adds 36,000, slightly above the wage of 35,000, as the rule requires.

* * *

SUMMARY

The chapter's main results in a few lines.

Rent is the payment to the owner of land for its use. Land as a whole is fixed in supply, so its whole payment is economic rent, which is any payment above the least needed to keep a factor in use. Ricardo's theory makes rent the difference between a plot's yield and that of the marginal land. Quasi-rent is a temporary surplus earned by a man-made asset in the short run. A wage is the payment to labour, the nominal wage is in money and the real wage is in goods. A firm hires while the marginal revenue product is at least the wage, and wages are set by demand and supply.

THINK IT THROUGH

15.T1 Explain why surgeons earn more than shop assistants, using both the demand side and the supply side of the labour market.

15.T2 A new road makes a distant farm easier to reach. What do you expect to happen to the rent of that farm, and why?

WHERE PEOPLE GO WRONG

Many students think that rent raises the price of the goods grown on the land. In Ricardo's theory it is the other way: the marginal land fixes the cost, and rent is the surplus of better land. They often judge a wage by its nominal figure alone, when the real wage shows what the wage buys. They think that economic rent belongs only to land, when any factor can earn it. The fourth error is to expect a firm to hire until the marginal revenue product is zero; it stops when the marginal revenue product falls below the wage.

Four common misunderstandings, each corrected.

IN HISTORY

David Ricardo (1772 to 1823) was an English businessman and economist. In 1817 he published *On the Principles of Political Economy and Taxation*, which set out the theory of rent that this chapter describes. At the time, England had high food prices and a fierce argument about the corn laws, which taxed imported grain. Ricardo argued that rent followed from the price of food, and that the poorest land in use paid none.

In 1817 Ricardo explained rent as the gain of better land over the poorest land in use.

Wages can also be set by law. In April 2013 the Rana Plaza building, which housed garment factories near Dhaka, collapsed, and more than a thousand people died. After this disaster and a wave of strikes, a government wage board raised the monthly minimum wage for entry-level garment workers from Tk 3,000 to Tk 5,300, effective from December 2013. The minimum wage was revised again in later years. The case shows a wage fixed above the level that the market had set, as discussed in section 15.3.

In 2013 Bangladesh raised the garment minimum wage after protests and a factory disaster.

REVIEW QUESTIONS

15.1 Define rent, and explain why the supply of land as a whole is said to be fixed.

15.2 A singer is paid Tk 80,000 for an evening and would have accepted Tk 50,000. Find the economic rent.

15.3 Three grades of land yield 120, 100 and 80 kg an acre, and the 80 kg land is the marginal land. Find the differential rent of each grade.

15.4 A nominal wage of Tk 24,000 a month meets a price index that rises from 100 to 120. Find the real wage.

15.5 A nominal wage rises by 5% while prices rise by 8%. Does the real wage rise or fall?

15.6 A firm pays Tk 30,000 a month. Successive workers add Tk 45,000, Tk 34,000, Tk 29,000 of revenue. How many should it hire?

15.7 Distinguish rent from quasi-rent in one sentence each.



CHAPTER XVI

Interest and Profit

Why is money paid to wait, and who deserves what is left at the end?



Lenders are paid for letting others use their money, and owners are paid, sometimes, for taking chances. This chapter explains what sets the rate of interest, separates two meanings of profit, and shows why profit arises at all.

YOU WILL NEED

Chapter 15, 'Rent and Wages', section 15.1, 'Rent'; Chapter 11, 'Costs of Production', section 11.1, 'Explicit and Implicit Costs'; Chapter 13, 'Market Structure and Perfect Competition', section 13.3, 'Equilibrium of Firm and Industry'; and Appendix A1, 'Maths You Will Use'.

A neighbour with spare money must decide whether to lend it for nothing.

A second puzzle asks whether the money left after repayment is truly a gain.

This chapter studies the price of money and the reward of the owner.

In spring, a farmer needs Tk 100,000 to buy seed and tools. His neighbour has exactly that sum saved. She has three choices. She can keep the money in a box, give it to him for nothing, or hand it over and ask for more money back after the harvest. Most neighbours choose the third. Why do they ask for more, and how much more is fair?

A second puzzle follows the farmer to the autumn. He sells the harvest, repays the loan with the extra amount, and has Tk 20,000 left. Is that Tk 20,000 a gain? Not necessarily. If he could have earned more by working for another farmer, the Tk 20,000 may be less than he gave up. Both puzzles ask the same thing: what is a fair payment for money, and what is a fair reward for running a business?

We study two payments in this chapter. The first is the payment for the use of money, which is called interest. The second is the reward of the person who runs the firm and takes the chances, which is called profit. We use the shop of Meghna and Sons and a famous lender of Bangladesh as our examples.

* * *

16.1 INTEREST

A friendly loan shows why a lender wants more back than was lent.

GO SLOWLY HERE.

Interest is the price paid for the use of money over time.

Interest. Word in use. In daily life, interest often means curiosity or attention, as in an interest in music. In economics, it means a payment for the use of money.

Interest. The payment a borrower makes to a lender for the use of money over time.

The bank's loan to the shop is an example of interest in the case file.

Begin with a loan between friends. You lend a friend Tk 1,000 today, and he promises to repay in one year. During that year you cannot spend the money. You also run a small risk that he will not repay. Most lenders will not accept Tk 1,000 back, because they would be worse off than if they had kept the money. They ask for a little more, perhaps Tk 1,100. The extra Tk 100 is the lender's reward for waiting and for taking the risk.

The extra Tk 100 has a name. **Interest** is the payment a borrower makes to a lender for the use of money over time. The payment is stated as a rate, which is the interest for one year divided by the sum borrowed, written as a per cent. In the friend's loan, Tk 100 on Tk 1,000 is a rate of 10 per cent a year. Interest is like the rent of land, which is a payment for the use of something owned by another person.¹

Case file, illustrative. Meghna and Sons holds a bank loan of Tk 600,000. The rate of interest is 11 per cent a year. Interest for one year is therefore Tk 66,000. The loan is an explicit cost of the shop, because the money is paid to an outsider, the bank.

¹ Rent as a payment for the use of land is taught in Chapter 15, 'Rent and Wages', section 15.1 'Rent'. Interest is the matching payment for the use of money.

Why is the rate 11 per cent and not 5 or 20? One answer treats money as a good that is bought and sold. **Loanable funds** are the money that savers offer and borrowers demand at each interest rate. Savers are people and firms with money they do not need now. Borrowers are people and firms that want money to spend now. The loanable funds theory says that the rate of interest is the price at which the funds offered equal the funds demanded.

The first figure below shows the idea. To read it, look at the horizontal axis, which measures loanable funds in taka, and the vertical axis, which measures the rate of interest in per cent a year. The falling line shows that borrowers want more funds when the rate is low, because borrowing is cheap. The rising line shows that savers offer more funds when the rate is high, because waiting is better rewarded. The lines cross at one rate. At that rate, the funds that borrowers demand equal the funds that savers offer.

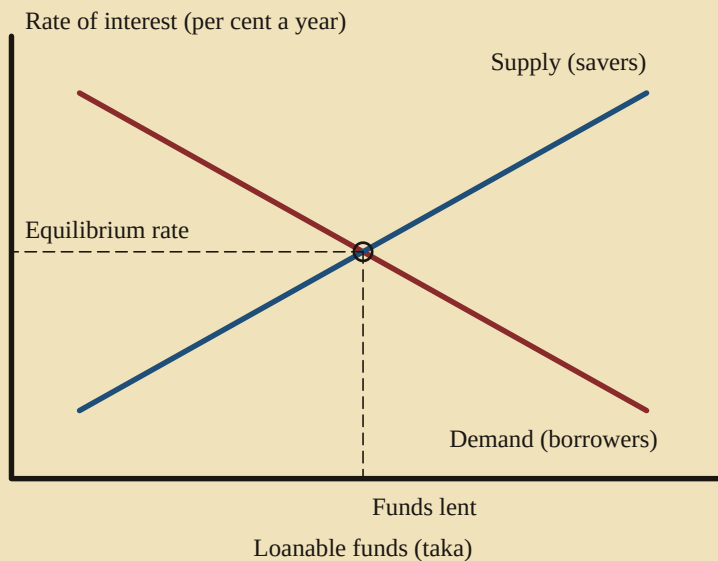


FIG. 16.1. Demand for and supply of loanable funds, and the rate of interest where they meet

The theory predicts how the rate moves. If people begin to save more, the rising line moves to the right, and the rate falls. If firms want to borrow more to build factories, the falling line moves to the right, and the rate rises. Many students find it odd that a rate can be a price. The oddity disappears when you remember that the thing being bought is the use of money for a year.

A second theory looks at the other side of the choice. **Liquidity preference** is the wish of people to hold money in cash form rather than lend it. Cash can be spent at once, which is useful. John Maynard Keynes, a British economist, argued in 1936 that people hold cash to pay for daily purchases, to be ready for emergencies, and to wait for better chances to buy assets. A lender who gives up cash loses these benefits. Interest is the reward that persuades people to part with cash.

Both theories have limits. The loanable funds theory stresses savers and borrowers. The liquidity preference theory stresses the supply of money and the wish to hold it. A country's central bank, such as Bangladesh Bank, also influences rates through its policy. In practice, economists use both theories together.

Rates differ from loan to loan, even on the same day. A loan to a small trader with no property carries more risk than a loan to a large firm, so the lender asks a higher rate. A loan for ten years ties up money longer than a loan for ten weeks. A very small loan costs almost as much to administer as a large one, which raises its rate. In some countries, including Bangladesh, Islamic banks

The rate of interest is set where the wishes of savers and borrowers meet.

Loanable funds. The money that savers offer and borrowers demand at each interest rate.

Borrowers want more at a low rate, and savers offer more at a high rate.

If savers offer more, the rate falls, and if borrowers want more, the rate rises.

A second theory says that people hold money because holding it is useful.

Liquidity preference. The wish of people to hold money in cash form rather than lend it.

Both theories explain interest, and neither is complete.

Different loans carry different rates, because they differ in risk, length and cost.

Interest looks different to the lender, the borrower and the owner of a firm.

The section closes with a recap and a first test.

A harvest sale shows that the money left over can be counted in two ways.

GO SLOWLY HERE.

The accountant's measure subtracts only the costs that were paid in money.

Profit. Word in use. In daily life, profit means any gain or money left in the till. In economics, it has two exact meanings, which differ by the owner's implicit costs.

Accounting profit. Revenue minus explicit costs only.

The economist's measure also subtracts what the owner's own resources could have earned.

Economic profit. Revenue minus all costs, explicit and implicit, including normal profit.

Normal profit is the least profit that keeps an owner in the business.

avoid a fixed rate of interest and share the results of the venture with the borrower instead.

The viewpoint changes the meaning. To the lender, interest is income for waiting. To a household that borrows, interest is a cost of buying now instead of later. To the manager of a firm, interest on a loan is an explicit cost, and the manager compares it with the extra output that the borrowed money can buy. A loan is worth taking only when the money earns more than the interest it costs.

Before moving on. Interest is the payment for the use of money, stated as a yearly rate. The loanable funds theory sets the rate where savers' offers equal borrowers' demands. The liquidity preference theory says interest rewards people for giving up cash. Rates differ with risk, length and size. You should now be able to say why a lender asks for more than was lent.

PAUSE AND CHECK 16.P1 A bank lends Tk 200,000 for one year at 9 per cent a year. How much interest does the borrower pay, and how much does the borrower repay in total?

* * *

16.2 PROFIT

Return to the farmer. He sells his harvest for Tk 300,000. This money received from sales is called revenue. He pays Tk 250,000 for seed, labour, fuel and the interest on his loan. He has Tk 50,000 left. A bookkeeper would call that his gain. But the farmer also worked for months on his own land, and his capital was tied up in the farm. The Tk 50,000 does not yet say whether the year was worth his effort. We need two measures, not one.

The accountant's measure is the simpler one. **Accounting profit** is revenue minus explicit costs only. An explicit cost is a cost that involves a payment of money to outsiders, such as wages, rent and interest.² The farmer's accounting profit is Tk 300,000 minus Tk 250,000, which is Tk 50,000. This is the profit that appears in a firm's accounts and on which taxes are usually based. Rules for tax differ by country, so a student should always name the country whose rules apply.

The economist's measure goes further. **Economic profit** is revenue minus all costs, explicit and implicit, including normal profit. An implicit cost is the value of resources the owner supplies, such as the owner's time and money. If the farmer could have earned Tk 60,000 as a hired worker, that sum is an implicit cost. His economic profit is Tk 50,000 minus Tk 60,000, which is a loss of Tk 10,000. The same year looks like a success in the accounts and a poor choice in the economist's reckoning.

The words *normal profit* need a reminder. Normal profit is the least profit a firm must earn to stay in business, covering the owner's implicit costs.³ When we subtract all implicit costs, we subtract normal profit. What remains is economic profit, which may be above, equal to, or below zero. Economic profit above zero is called supernormal profit, because it is more than normal. The table below sets out the three cases.

² Explicit and implicit costs are taught in Chapter 11, 'Costs of Production', section 11.1 'Explicit and Implicit Costs'.

³ Normal profit is taught in Chapter 13, 'Market Structure and Perfect Competition', section 13.3 'Equilibrium of Firm and Industry'.

THE THREE RESULTS OF ECONOMIC PROFIT

ECONOMIC PROFIT	WHAT IT MEANS	LIKELY RESULT
Above zero	Supernormal profit: more than the owner could earn elsewhere	New firms are attracted
Equal to zero	Normal profit only: the owner earns exactly what is needed	The owner stays
Below zero	Less than the owner could earn elsewhere	The owner may leave

Read the table one row at a time. In the first row, the owner earns more than the best alternative, so other people want to enter the same line of business. In the second row, the owner is neither tempted nor driven away. In the third row, the owner would do better to use time and money elsewhere, although some owners stay for reasons that money does not measure.

The measure matters because owners compare their own results with the best alternative. When economic profit is above zero in an industry, other people notice and open similar firms. When it is below zero for long, owners move their time and money to other uses. Accounting profit cannot give this guidance, because it leaves out the alternatives. A limit: implicit costs are estimates, so economic profit is an estimate too.

Case file, illustrative. For 2025, Meghna and Sons recorded accounting profit of Tk 750,000. The case file counts two implicit costs: Tk 600,000 for the owner's time, and Tk 287,000 for the interest forgone on capital employed of Tk 4,100,000 at 7 per cent. Together, the implicit costs are Tk 887,000. Economic profit is therefore 750,000 minus 887,000, which is -Tk 137,000.

The shop shows the difference between the two measures. The accounts report a gain of Tk 750,000, and any lender or tax office would see a firm with a positive profit. Yet the owner's own resources could have earned Tk 887,000 elsewhere. From the owner's viewpoint, the shop earns Tk 137,000 less than the best alternative. The employee and the customer see nothing of this. Only the owner pays the implicit costs.

PAUSE AND CHECK 16.P2 A firm has revenue of Tk 900,000 and explicit costs of Tk 700,000. Its owner's time and capital could earn Tk 150,000 elsewhere. Find the accounting profit and the economic profit.

Before moving on. Accounting profit subtracts explicit costs only. Economic profit subtracts explicit and implicit costs, so it also subtracts normal profit. Economic profit above zero is supernormal profit. You should now be able to say why a firm can have a positive accounting profit and a negative economic profit.

* * *

16.3 WHY PROFIT OCCURS

Interest has a clear reason: lenders wait. Profit is harder to explain, because it is what remains after every cost, including the owner's, has been paid. Why should anything remain? Economists give four answers. Each one fits some cases and fails in others, so we study them as four views and not as one rule.

Each row of the table pairs one result with its likely consequence.

Economic profit guides decisions to enter, stay in or leave a line of business.

The shop's records contain all the figures needed for both measures.

The shop has a positive accounting profit and a negative economic profit.

The section ends with a recap.

If profit can be negative, we must ask why it exists at all.

Profit rewards the owner for taking on the chance of loss.

Risk bearing. The act of taking on the chance of loss in business, which profit rewards.

A second view treats profit as a payment for the owner's own contribution.

A third view says that new methods create profit until rivals copy them.

A fourth view says that power over price can protect profit.

The first view is the oldest. **Risk bearing** is the act of taking on the chance of loss in business, which profit rewards. A firm that makes goods before knowing whether anyone will buy them may lose its money. Frank Knight, an American economist, argued in 1921 that some chances cannot be known in advance or insured, and that profit is the reward for facing them. Limit: the view explains why a risky venture should earn more on average, not why every owner who takes a risk is paid.

The second view is called the marginal productivity view. It treats the owner, who organises the land, labour and capital, as one more input. The owner's pay is the extra output that the owner's organising adds, as with any other input. Limit: the contribution of one owner is hard to measure, so the view is hard to test.

The third view comes from Joseph Schumpeter, an Austrian economist who wrote in 1911. He argued that profit is the reward of innovation, which means introducing a new good or a new method. The innovator's costs are lower, or the good is new, so economic profit appears. Rivals then copy the idea, and the profit shrinks to zero. Limit: many firms earn profit without any new idea.

The fourth view is monopoly. A firm that is protected from rivals, because entry is difficult, can keep its price above its costs for a long time. Its supernormal profit lasts, because no new firm can enter to share it. Limit: this explains profit that is protected, but not profit in markets with many rivals.

FOUR VIEWS OF WHY PROFIT OCCURS

VIEW	SOURCE OF PROFIT	MAIN LIMIT
Risk bearing	Reward for facing chance of loss	Does not pay every risk-taker
Marginal productivity	Pay for the owner's own contribution	Hard to measure for one owner
Innovation	New methods and goods, until copied	Many profits need no new idea
Monopoly	Protection from rivals	Does not explain profit among many rivals

The table pairs each view with the case it explains best and the case it misses.

Supernormal profit tends to attract rivals unless something blocks them.

The four views disagree only because they describe different firms.

The section closes with a recap.

To read a row, take the view in the first column and ask what source it names in the second. Then check the third column for its weakness. No row is complete alone. A careful economist uses all four views, and asks which fits the firm being studied.

Two of the views end in the same place. Supernormal profit signals that an activity pays more than its alternatives, and it invites new firms to enter. As they enter, the extra supply lowers the price, and the supernormal profit shrinks. Only protection from rivals, such as a new idea that others cannot copy or a market that new firms cannot enter, can keep it for long. This is why innovation and monopoly are linked with lasting profit.

The views are better understood as answers to different questions. A farmer who plants before the rains faces risk. A manager of a large firm supplies organising skill. A new software firm relies on innovation. A single seller of a protected good relies on monopoly. A student who meets a profit in any case should ask which source applies.

Before moving on. Profit has four main explanations: risk bearing, the owner's own contribution, innovation and monopoly power. Each fits some firms and fails for

others. You should now be able to name the four views and give one limit of each.

* * *

WORKED EXAMPLE: INTEREST AND ECONOMIC PROFIT OF THE SHOP

We work two examples in the same order each time: Given, Find, Formula, Steps, Answer and Check. The first finds the interest on the shop's bank loan. The second finds the shop's economic profit.

Given. The shop owes Tk 600,000 to its bank. The rate of interest is 11 per cent a year, and the loan runs for one year. *Find.* The interest for the year. *Formula.* In words, interest equals the sum borrowed multiplied by the rate. In symbols, $I = S \times r$, where I is the interest, S is the sum borrowed and r is the rate written as a decimal.

Steps. Write 11 per cent as $11 \div 100 = 0.11$. Then $I = 600,000 \times 0.11 = \text{Tk } 66,000$. *Answer.* The shop pays Tk 66,000 in interest for the year. *Check.* One per cent of 600,000 is Tk 6,000. Eleven per cent is eleven times that, which is $11 \times 6,000 = \text{Tk } 66,000$. The two routes agree.

Given. Sales were Tk 14,400,000, and accounting profit was Tk 750,000. The owner's time is valued at Tk 600,000. Capital employed was Tk 4,100,000, and it could have earned 7 per cent a year elsewhere. *Find.* The economic profit. *Formula.* Forgone interest = rate $\times$ capital employed. Economic profit = accounting profit – total implicit costs. This is the same as revenue minus all costs, because accounting profit is revenue minus explicit costs.

Steps. Write 7 per cent as 0.07. Forgone interest = $0.07 \times 4,100,000 = \text{Tk } 287,000$. Total implicit costs = $600,000 + 287,000 = \text{Tk } 887,000$. Economic profit = $750,000 - 887,000 = -\text{Tk } 137,000$.

Answer. The economic profit is $-\text{Tk } 137,000$. The shop earns Tk 137,000 less than its owner's time and capital could earn elsewhere. *Check.* The implicit costs of Tk 887,000 exceed the accounting profit of Tk 750,000, so the answer must be negative. Adding back: $-137,000 + 887,000 = 750,000$, the figure in the accounts. A negative result does not force the owner to close the shop. It tells her the price of staying.

Common slips are easy to name. First, using 11 instead of 0.11, which gives an answer one hundred times too large. Second, mixing periods, such as a yearly rate with a loan held for three months. Third, forgetting implicit costs, which gives accounting profit and not economic profit. Fourth, a wrong sign, such as subtracting the smaller number from the larger. Write the unit beside every number, and check the sign by asking whether the implicit costs exceed the accounting profit.

* * *

SUMMARY

Interest is the payment for the use of money, stated as a yearly rate. The loanable funds theory sets it where savers' offers equal borrowers' demands, and the liquidity preference theory says it rewards people for giving up cash. Accounting profit subtracts explicit costs only. Economic profit subtracts all costs, including normal profit. Profit is

Two examples apply the ideas, one small and clean and one realistic.

The first example finds a year's interest on the bank loan.

Each line of arithmetic is shown, with its unit, so the method is the only new thing.

The second example finds the economic profit of the shop for 2025.

Every step is written out, so nothing depends on guesswork.

The shop is below normal profit by Tk 137,000.

Four common slips spoil these calculations.

The chapter's main points fit in a few lines.

explained by risk bearing, the owner's contribution, innovation and monopoly power.

* * *

THINK IT THROUGH

16.T1 Two ventures have the same expected sales and costs, but one is far more risky. Using the risk bearing view, explain why the risky venture should be expected to earn more on average.

16.T2 A bank charges 8 per cent a year to a large firm and 18 per cent a year to a small trader. Give two reasons for the difference.

* * *

WHERE PEOPLE GO WRONG

Profit in the accounts is not the same as economic profit.

Mistake: "A profit of Tk 750,000 means the shop is doing well." That is accounting profit. After the owner's implicit costs of Tk 887,000, the shop's economic profit is –Tk 137,000.

Interest is a price, not a punishment.

Mistake: "Interest is only an unfair charge on borrowers." Interest pays the lender for waiting and for risk, and it rations scarce funds among borrowers who want them. Whether a particular rate is fair is a separate question.

Normal profit is a cost, not a gain.

Mistake: "Normal profit is a form of profit left for the owner." Normal profit is part of costs, because it covers the owner's implicit costs. Only economic profit above zero is a surplus above all costs.

Each view of profit fits some firms only.

Mistake: "Profit has one cause." Risk, the owner's contribution, innovation and monopoly power each explain some profits and fail for others.

* * *

IN HISTORY

A village experiment showed that very poor borrowers could repay small loans.

In 1976, Muhammad Yunus, a Bangladeshi economist then teaching at Chittagong University, began to lend small sums to poor villagers in Jobra, a village near the university. Banks at that time refused to lend to people who owned no property. Yunus found that the borrowers repaid, and he went on to found the Grameen Bank in 1983.

The bank charged interest, and it showed that risk and cost are not the only forces that fix its level.

The Grameen Bank grew into a large lender to poor households in Bangladesh, and in 2006 Yunus and the bank shared the Nobel Peace Prize. The story links to this chapter in two ways. Loans to people without property carry more risk and cost more to administer, which usually raises the rate. Yet repayment turned out better than lenders had expected, so the chance of loss was smaller than they feared.

* * *

REVIEW QUESTIONS

The questions move from one loan to the full measure of economic profit.

16.1 A loan of Tk 50,000 is made at 10 per cent a year for one year. Find the interest and the total repayment.

- 16.2 A loan of Tk 450,000 is made at 9.5 per cent a year for one year. Find the interest.
- 16.3 A loan of Tk 600,000 at 11 per cent a year is held for three months. Interest is charged on the sum borrowed only and is not added to it. Find the interest.
- 16.4 An owner's accounting profit is Tk 420,000. Her time is worth Tk 300,000 a year elsewhere, and her capital of Tk 2,000,000 could earn 6 per cent a year. Find the economic profit and say which of the three results in the table it is.
- 16.5 A firm has an accounting profit of Tk 380,000 and implicit costs of Tk 450,000. Find the economic profit and say what it tells the owner.
- 16.6 A trader has sales of Tk 2,000,000 and explicit costs of Tk 1,700,000. Her time is worth Tk 200,000 a year elsewhere, and her capital of Tk 1,000,000 could earn 8 per cent. Find her accounting profit and economic profit, and name the result.
- 16.7 Name the four views of why profit occurs, and give one limit of each.

GLOSSARY

Accounting profit. Revenue minus explicit costs only. Revenue is the money a firm receives from sales. Accounting profit appears in a firm's accounts. A farmer who sells a harvest for Tk 300,000 and pays Tk 250,000 in costs has an accounting profit of Tk 50,000. (*Chapter XVI*)

Average product. Total product divided by the number of units of the input used. It shows the output per unit. Example: $44 \div 4 = 11$ with 4 workers. (*Chapter X*)

Average total cost. Total cost divided by the number of units produced, so that fixed and variable costs are shared among all units. It is the cost of each unit on average. A total cost of Tk 1,120 for 20 books gives Tk 56 a book. (*Chapter XI*)

Bandwagon effect. A rise in demand for a good because many other people buy it. A buyer wants what friends and neighbours have, as when a mobile phone becomes more useful because all of one's friends use it. (*Chapter IV*)

Budget line. A line on a graph showing every combination of two goods that a consumer can buy with a given income at given prices. Its ends are income divided by each price, and its steepness is the price ratio. Example: Tk 400 at Tk 40 and Tk 20 gives ends of 10 notebooks and 20 pens. (*Chapter VII*)

Ceteris paribus. A Latin phrase meaning all other things stay the same, used to study one change at a time. It is a method of thinking, not a claim that nothing else changes. To study the effect of price on notebook sales, we hold incomes and other prices fixed. (*Chapter I*)

Competitive market. A market with many buyers and sellers, none of whom can set the price. Each one must accept the price that the whole market produces. A street of tea stalls selling the same tea is a small example. (*Chapter IX*)

Complementary goods. Goods used together, so a price rise for one lowers demand for the other. Pens and notebooks are used together. If pens become dearer, fewer notebooks are wanted at every price. (*Chapter IV*)

Consumer equilibrium. The point where a consumer gets the greatest satisfaction possible from a given income. It lies on the budget line, on the highest reachable indifference curve, where the marginal rate of substitution equals the price ratio. Example: 6 notebooks and 8 pens. (*Chapter VII*)

Consumers' surplus. The gap between what buyers are willing to pay for a good and what they actually pay. Added over all units bought, it measures the gain consumers keep. In the notebook case, a price of Tk 40 leaves three customers a surplus of Tk 30 in total. On a chart it is the triangle under the demand curve and above the price. (*Chapter VI*)

Consumption. The use of goods and services by people to satisfy their wants. Eating a meal, riding a bus and filling a notebook with notes are all consumption. It takes place when the good is used, which may be later than the day of purchase. A firm that uses cloth to make shirts does not consume the cloth. (*Chapter VI*)

Cross elasticity. The percentage change in quantity demanded of one good divided by the percentage change in the price of another. Positive for substitutes, negative for goods used together. Example: a rival notebook dearer by 10% and 4% more quantity give 0.4. (*Chapter V*)

Demand. The quantities of a good that buyers are willing and able to buy at each price in a period. Both the wish and the money are needed. One entry is a

quantity demanded; the whole list is demand. For the notebook: 40 at Tk 40, 50 at Tk 30, and so on. (*Chapter IV*)

Demand curve. A line on a graph showing the quantity demanded at each price. Price is on the vertical axis and quantity on the horizontal axis. A point on the line means that at this price buyers choose this quantity. Usually the line slopes downward. (*Chapter IV*)

Demand schedule. A table showing the quantity demanded at each price. Each row pairs one price with one quantity. The shop's table lists Tk 60 with 20 notebooks, Tk 50 with 30, and continues down to Tk 20 with 60. (*Chapter IV*)

Economic law. A statement of how people usually behave, true when other conditions stay the same. It describes a tendency, not a certainty, because people choose. When the price of a good rises, consumers usually buy less of it. (*Chapter I*)

Economic profit. Revenue minus all costs, both explicit and implicit, including normal profit. It is accounting profit less the value of what the owner's time and money could earn elsewhere. A positive result is supernormal profit, and a negative result is below normal. (*Chapter XVI*)

Economic rent. The payment to any factor above the least needed to keep it in use. For land as a whole the least needed is nothing, so the whole payment is economic rent. Example: a singer paid Tk 100,000 who would sing for Tk 40,000 earns Tk 60,000. (*Chapter XV*)

Economics. The study of how people, firms and governments choose to use limited resources to satisfy wants. It exists because wants are many and means are few. A student with Tk 100 who must choose among wants worth Tk 150 meets the problem on a small scale. (*Chapter I*)

Elastic demand. Demand for which the percentage change in quantity is larger than the percentage change in price, so price elasticity of demand is above 1. A price cut raises total revenue. Example: the notebook between Tk 50 and Tk 40, at 1.29. (*Chapter V*)

Elasticity of supply. The percentage change in quantity supplied divided by the percentage change in price. It tells us how strongly firms respond to a price change. If a 10% rise in price raises quantity supplied by 5%, the elasticity of supply is 0.5. (*Chapter IX*)

Equilibrium of the firm. The output at which a firm earns the largest possible profit and has no wish to change. A price taker finds it where marginal cost equals the price. For the binding workshop at Tk 40, it is 50 units. (*Chapter XIII*)

Equilibrium price. The price at which the quantity buyers want equals the quantity sellers offer. At this price neither side has a reason to change its plans, so the price stays still. In the notebook market it is Tk 40. (*Chapter IX*)

Exceptional supply curve. A supply curve that does not slope upward, such as one that bends backward or stays fixed. A worker who offers fewer hours at very high pay gives a backward-bending curve. Fish caught for one market day give a fixed supply. (*Chapter VIII*)

Excess demand. A situation in which buyers want more than sellers offer at a price. Some buyers cannot obtain the good, so they offer more money and push the price up. In the notebook market at Tk 30, buyers want 50 and firms offer 30. (*Chapter IX*)

Excess supply. A situation in which sellers offer more than buyers want at a price. Some goods stay unsold, so sellers cut the price and push it down. In the notebook market at Tk 50, firms offer 50 and buyers want 30. (*Chapter IX*)

Explicit cost. A cost that involves a payment of money to people outside the firm, such as rent, wages, paper or interest on a loan. It appears in a receipt or bank record. A shop's monthly rent paid to its landlord is an example. (*Chapter XI*)

Fixed cost. A cost that does not change with the quantity produced in the short run, because it is owed even when nothing is made. It is the cost of being ready to produce. A workshop's machine rent is an example. (*Chapter XI*)

Giffen good. A rare, very basic good for which quantity demanded rises when its price rises. Marshall wrote that Giffen noticed it among very poor households buying bread. It is a special case and the law of demand holds for nearly all other goods. (*Chapter IV*)

Implicit cost. The value of resources that the owner supplies to the firm, for which no money is paid. These resources could have earned money elsewhere, so they have an opportunity cost. An owner's unpaid time is an example. (*Chapter XI*)

Income elasticity. The percentage change in quantity demanded divided by the percentage change in consumers' income. A positive result means more income raises the quantity demanded. Example: income up 10% and fiction up 15% give 1.5. (*Chapter V*)

Indifference curve. A line on a graph that joins every combination of two goods giving a consumer the same satisfaction. Because the satisfaction is equal along the line, she would not mind which combination she received. Example: 4 notebooks with 14 pens and 6 notebooks with 8 pens may lie on one curve. (*Chapter VII*)

Indifference map. A set of indifference curves drawn on one graph. A curve further from the corner where the axes meet shows greater satisfaction, so the map ranks every combination of the two goods. Example: 6 notebooks with 14 pens lies on a higher curve than 6 notebooks with 8 pens. (*Chapter VII*)

Industry. All the firms that produce the same kind of good. The quantity supplied by the industry is the sum of what its firms supply. All the firms that make notebooks, for example, form the notebook industry. (*Chapter XIII*)

Inelastic demand. Demand for which the percentage change in quantity is smaller than the percentage change in price, so price elasticity of demand is below 1. A price cut lowers total revenue. Salt is a likely example. (*Chapter V*)

Interest. The payment a borrower makes to a lender for the use of money over time. It is stated as a yearly rate, which is the year's interest divided by the sum borrowed. Tk 100 paid on a loan of Tk 1,000 for a year is a rate of 10 per cent. (*Chapter XVI*)

Isocost line. A line showing all combinations of two inputs that cost the same total amount. Its slope is the ratio of the two input prices. With a wage of Tk 600 and a machine rent of Tk 400, every mix on the Tk 4,000 line, such as 4 workers and 4 machines, costs Tk 4,000. (*Chapter XII*)

Isoquant. A line showing all combinations of two inputs that produce the same output. The word means equal quantity. In the case, the mixes (2, 10), (3, 6), (4, 4), (5, 3) and (6, 2) of workers

and machines all give 100 books a day, so they lie on one isoquant. (*Chapter XII*)

Law of demand. When the price of a good rises, the quantity demanded falls, if other conditions stay the same. The same law says a fall in price raises the quantity demanded. It is drawn as a downward-sloping line. (*Chapter IV*)

Law of diminishing marginal utility. As a person consumes more units of a good, the satisfaction added by each extra unit falls. It holds when units are alike, tastes are steady and consumption has no long gaps. Marginal utilities of 10, 8, 5, 2 and 0 show the law. (*Chapter III*)

Law of diminishing returns. Adding more of one input to fixed amounts of others will, after a point, add less and less to output. It belongs to the short run and does not say that total output falls. Example: each worker beyond the second in the workshop. (*Chapter X*)

Law of supply. When the price of a good rises, the quantity supplied rises, if other conditions stay the same. The same law says a fall in price lowers the quantity supplied. It is drawn as an upward-sloping line. (*Chapter VIII*)

Least-cost combination. The mix of inputs that produces a given output at the lowest total cost. It lies where the lowest isocost line touches the isoquant. In the case, 4 workers and 4 machines make 100 books for Tk 4,000, less than any other mix on the isoquant. (*Chapter XII*)

Liquidity preference. The wish of people to hold money in cash form instead of lending it, because cash can be spent at once. Interest is the reward that persuades people to give up cash. Keeping money for emergencies is an example. (*Chapter XVI*)

Loanable funds. The money that savers offer to lend and that borrowers demand to borrow at each rate of interest. Borrowers want more when the rate is low, and savers offer more when it is high. The rate where the two amounts are equal is the equilibrium rate. (*Chapter XVI*)

Long run. A period long enough for all inputs to be changed. Its length differs between firms. Example: given years, a baker can build a new bakery and buy more ovens. (*Chapter X*)

Macroeconomics. The branch of economics that studies the whole economy, including output, prices and employment. Its unit is a total or an average for a country. A question about the price level of a whole country belongs to macroeconomics. (*Chapter I*)

Marginal cost. The extra cost of producing one more unit of output, found by dividing the change in total cost by the change in output. Fixed cost does not affect it. If total cost rises by Tk 220 for 10 more books, marginal cost is Tk 22 a book. (*Chapter XI*)

Marginal product. The extra output from adding one more unit of an input, others held fixed. It equals the change in total product divided by the change in the input. Example: the fourth worker adds $44 - 36 = 8$. (*Chapter X*)

Marginal rate of substitution. The number of units of one good a consumer will give up to gain one more unit of another good, while her satisfaction stays the same. It is the steepness of the indifference curve. Example: giving up 3 pens for each extra notebook. (*Chapter VII*)

Marginal rate of technical substitution. The amount of one input a firm can give up when it adds one unit of another, keeping output the same. It is the

machines saved divided by the workers added along an isoquant, and it falls as workers replace machines. At the least-cost combination it equals the price ratio. (*Chapter XII*)

Marginal revenue. The extra revenue from selling one more unit of output. For a price maker it is smaller than the price, because a cut in price applies to every unit sold. Example: selling 10 more copies, from 20 to 30, adds Tk 300, or Tk 30 a copy. (*Chapter XIV*)

Marginal revenue product. The extra revenue a firm gains from employing one more unit of labour. A firm hires while it is at least equal to the wage. Example: an assistant who adds Tk 40,000 a month. (*Chapter XV*)

Marginal utility. The extra satisfaction a person gets from consuming one more unit of a good. It is the height of the last step, not the height of the whole staircase. If the third cup adds 5 utils, its marginal utility is 5 utils. (*Chapter III*)

Market structure. The set of features of a market, such as the number of sellers, the kind of good, the ease of entry and the information buyers hold, that shape how firms behave. A row of identical potato stalls and a lone village pharmacy have different structures. (*Chapter XIII*)

Microeconomics. The branch of economics that studies choices made by individual consumers and firms. Its unit is one consumer, one firm or one market. A question about how one bookshop chooses its price belongs to microeconomics. (*Chapter I*)

Monopolistic competition. A market with many sellers of similar but not identical goods, and free entry. Each seller has a small power over its price, but new entrants take away extra profit over time. Example: a street of tea stalls. (*Chapter XIV*)

Monopoly. A market in which one seller supplies a good with no close substitute. Entry is blocked by barriers such as sole ownership of a resource, an exclusive legal right or a large cost advantage. Example: the only ferry across a river. (*Chapter XIV*)

Nominal wage. A wage measured in money, without allowing for changes in prices. It is the figure on the pay slip. Example: Tk 35,000 a month. (*Chapter XV*)

Normal good. A good for which demand rises when buyers' incomes rise. The word normal does not mean usual. If families with more money buy more notebooks, the notebook is a normal good. (*Chapter IV*)

Normal profit. The least profit a firm must earn to stay in business, covering the owner's implicit costs, such as the value of the owner's own time. A baker who could earn a wage in a factory must earn at least that from her bakery. (*Chapter XIII*)

Normative economics. Statements about what ought to be, which depend on values and cannot be tested by facts alone. Words such as should and ought often signal them. Textbooks should be cheaper is a normative statement. (*Chapter I*)

Oligopoly. A market dominated by a few large sellers, each aware of what the others do. Entry is hard, and each firm's best choice depends on its rivals' choices. Example: three bus firms on one road. (*Chapter XIV*)

Opportunity cost. The value of the best alternative given up when a choice is made. Only the best alternative counts, not

all alternatives added. Example: Rafiq's extra parcel gives up 2 books. (*Chapter II*)

Perfect competition. A market with very many buyers and sellers, identical goods, free entry and full information. No real market meets all four conditions, so it is a model. A grain exchange comes near it, because wheat of one grade is the same whoever grew it. (*Chapter XIII*)

Positive economics. Statements about what is, which can be tested against facts. A positive statement may be true or false, but records or measurement can settle which. The shop sold 41,000 copies in 2025 is a positive statement. (*Chapter I*)

Price elasticity of demand. The percentage change in quantity demanded divided by the percentage change in price, each found by the midpoint method. A result of 2 means quantity changes twice as fast as price. The size of the result is reported, without its sign. (*Chapter V*)

Price maker. A firm that can choose its price because it faces the whole demand of the market, as a monopolist does. It cannot choose both price and quantity, since the demand curve fixes one once the other is chosen. Example: the seller in the case file, who chooses Tk 50. (*Chapter XIV*)

Price taker. A firm that must accept the market price because its own sales are too small to affect it. If the market price is Tk 40, such a firm sells any quantity at Tk 40 and nothing at a higher price, so it chooses only its output. (*Chapter XIII*)

Product differentiation. Making a good different from rivals' goods in quality, style or brand, to build loyal buyers. The differences may be real or only believed. Example: a stall that serves tea in clean glasses with a friendly greeting. (*Chapter XIV*)

Production function. A rule showing the largest output that can be made from each combination of inputs. It records what is technically possible with the best known methods. Example: 1 oven and 2 helpers can make at most 100 loaves a day. (*Chapter X*)

Production possibility curve. A line showing the largest combinations of two goods that can be produced with all resources used fully. Points inside waste resources; points outside cannot be reached. Example: Rafiq's 800 books or 400 parcels. (*Chapter II*)

Quasi-rent. A temporary surplus earned by a man-made asset while its supply is fixed in the short run. It disappears when new assets are built. Example: printing presses when demand for books suddenly rises. (*Chapter XV*)

Real wage. A wage measured by the goods and services it can buy. It equals the nominal wage divided by the price index over 100. Example: $35,000 \div 1.12 = \text{Tk } 31,250$. (*Chapter XV*)

Rent. The payment to the owner of land for its use, in everyday talk and in economics. In this book the word is kept for land. Example: Tk 60,000 a month paid for a shop site. (*Chapter XV*)

Returns to scale. The change in output when all inputs are increased in the same proportion. It may be increasing, constant or decreasing. Example: doubling workers and machines raised output from 24 to 60. (*Chapter X*)

Risk bearing. The act of taking on the chance of loss in business, which profit rewards. A firm that makes goods before knowing whether buyers will come bears risk. One view of

profit says owners are paid more on average for facing this chance. (*Chapter XVI*)

Scarcity. The condition that resources are limited while wants are unlimited, so choices must be made. It applies to rich and poor alike, and it is the reason economics exists. Example: a family cannot buy everything on its list. (*Chapter II*)

Short run. A period so short that at least one input, such as a machine, cannot be changed. It is not a fixed number of days; it is whatever period leaves an input fixed. Example: a baker can hire helpers but cannot install a new oven. (*Chapter X*)

Snob effect. A fall in demand for a good because many other people now buy it. A buyer who values rarity leaves a good that has become common, as when a once-rare scarf is worn by everyone. (*Chapter IV*)

Substitute goods. Goods that can replace each other, so a price rise for one raises demand for the other. If a rival notebook becomes dearer, some buyers move to ours, and demand for our notebook shifts to the right. (*Chapter IV*)

Supply. The quantities of a good that sellers are willing and able to sell at each price in a period. Both the stock and the wish to sell are needed. One entry is a quantity supplied; the whole list is supply. For the notebook: 40 at Tk 40, 50 at Tk 50, and so on. (*Chapter VIII*)

Supply curve. A line on a graph showing the quantity supplied at each price. Price is on the vertical axis and quantity on the horizontal axis. A point on the line means that at this price sellers offer this quantity. Usually the line slopes upward. (*Chapter VIII*)

Supply schedule. A table showing the quantity supplied at each price. Each row pairs one price with one quantity. The wholesaler's table lists Tk 20 with 20 notebooks, Tk 30 with 30, and continues up to Tk 60 with 60. (*Chapter VIII*)

Total cost. The whole cost of producing a given output, found by adding fixed cost and variable cost. At zero output it equals fixed cost. A workshop with fixed cost of Tk 600 and variable cost of Tk 720 has total cost of Tk 1,320. (*Chapter XI*)

Total product. The whole quantity of output produced with a given amount of inputs. Example: the binding workshop with 4 workers has a total product of 44. (*Chapter X*)

Total revenue. Price multiplied by quantity sold; the money a firm receives from its sales. Example: 40 notebooks sold at Tk 40 give total revenue of Tk 1,600. The total revenue test uses it to judge the effect of a price change. (*Chapter V*)

Total utility. The whole satisfaction a person gets from all the units of a good consumed in a period. It is found by adding the marginal utility of each unit. If three cups gave 10, 8 and 5 utils, total utility is 23 utils. (*Chapter III*)

Trade-off. Giving up some of one thing in order to have more of another. It is the exchange itself, while opportunity cost measures what is given up. Example: planting more jute and less rice. (*Chapter II*)

Unit elastic demand. Demand for which the percentage change in quantity equals the percentage change in price, so price elasticity of demand equals 1. Total revenue does not change when the price changes. It is the dividing line between elastic and inelastic demand. (*Chapter V*)

Util. An imaginary unit used to count satisfaction in economic examples. A util is not a real measurement; it lets us reason exactly about more and less for one person. A cup worth 10 utils pleases more than a cup worth 8 utils. (*Chapter III*)

Utility. The satisfaction a person gets from using a good or service. It is a feeling, not a judgement of whether the good is wise or healthy. A cold drink on a hot day gives utility to the person who drinks it. (*Chapter III*)

Variable cost. A cost that changes with the quantity produced. It is zero when nothing is made and rises as output rises. Paper, thread, glue and the wages of extra workers are examples in a binding workshop. (*Chapter XI*)

Wage. The payment to labour for its work over a period. It may be stated by the hour, the day or the month. Example: Tk 35,000 a month for a shop assistant. (*Chapter XV*)

Want. Something a person would like to have, however little or much the person can pay for it. A want may be a good, a service or a condition. Example: a bicycle, a haircut or a free afternoon. (*Chapter II*)

Willingness to pay. The highest price a buyer would pay for one more unit of a good. It is a limit held in the mind, not a price on a tag. It falls as a consumer buys more units and it differs between consumers. A consumer buys a unit only when the price is at or below this limit. (*Chapter VI*)

ANSWERS TO QUESTIONS

CHAPTER I · WHAT ECONOMICS IS

1.p1 (a) Positive: the shop's staff records can confirm the number 7. (b) Normative: whether the shop ought to employ an eighth person depends on values and judgement, and facts alone cannot settle it.

1.t1 The claim ignores that another condition changed in the same month, since free notebooks reduce the need to buy. *Ceteris paribus* requires all other things to stay the same, and here they did not. We cannot tell how much of the fall came from the price.

1.t2 Economic laws describe how people usually behave, so they state tendencies and not certainties. Exactness is not the test of usefulness; a usual pattern can still guide a seller or a government. Used with *ceteris paribus*, such a law separates one cause from many.

1.1 (a) Positive: staff records can test it. (b) Normative: it says what ought to be. (c) Positive: the accounts can confirm the figure. (d) Normative: should signals a value judgement.

1.2 (a) Micro: one firm. (b) Macro: the general price level. (c) Micro: one consumer. (d) Macro: a total for the whole economy.

1.3 Her wants cost Tk 60 + Tk 60 + Tk 30 = Tk 150, but she has only Tk 100. Wants exceed means by Tk 50, so she must choose and give up something.

1.4 Consumption, production, exchange and distribution. Examples: a reader takes a book home (consumption); a printer makes the book (production); the shop buys it from the printer (exchange); the shop's income pays staff and owner (distribution).

1.5 *Ceteris paribus* means all other things stay the same. An economic law states a usual tendency of people, while a law of nature, such as the fall of a stone, describes things that do not choose and holds exactly.

CHAPTER II · SCARCITY, CHOICE AND OPPORTUNITY COST

2.p1 The best alternative given up, which is the Tk 300 earned. Sleeping late is a lesser alternative and is not added.

2.t1 The cost is the Tk 120 that could have been earned, the best alternative. For a student with no job offer, or with a different alternative such as a tutoring fee, the cost would differ.

2.t2 The maximum is 1,000 books or 500 parcels. 600 books needs 30 hours and 300 parcels needs 30 hours, so 60 hours are needed; it is still not attainable.

2.1 A want is something a person would like to have, whatever the price. One want may be met, but others appear in its place, so the whole list is without limit.

2.2 Scarcity is the condition that resources are limited while wants are unlimited, so choices must be made. A rich person has limited time; a poor person has limited money.

2.3 The Tk 400 from working, as the best alternative. Resting is a lesser alternative and is not counted.

2.4 One pair of trousers costs $8 \div 4 = 2$ shirts. One shirt costs $4 \div 8 = 0.5$ pair of trousers.

2.5 Shelving gives $30 \times 20 = 600$ books; packing gives $10 \times 10 = 100$ parcels. All 40 hours are used, and at 600 books the curve allows 100 parcels, so the point lies on the curve.

2.6 First step: 10 buses lost for 5 cars, 2 per car. Second step: 20 buses lost for 5 cars, 4 per car. The opportunity cost rises.

2.7 Resources differ in their suitability, so moving them costs more and more. More resources, or better methods, would shift the curve outward.

CHAPTER III · UTILITY AND DIMINISHING MARGINAL UTILITY

3.p1 Higher. The second cup added some satisfaction, so the total rose, even though the second cup pleased her less than the first.

3.p2 2 utils (25 minus 23). The marginal utility column gives it directly.

3.t1 By the fifth cup her want for tea has been met by the first four cups. She is no longer cold or thirsty, so a further cup adds nothing.

3.t2 Total utility = $25 + (-3) = 22$ utils. She should refuse the sixth cup, because it lowers her total satisfaction.

3.1 Utility is the satisfaction a person gets from using a good or service. Example: the relief of a cold drink on a hot afternoon.

3.2 Totals: 50, then $50 + 35 = 85$, then $85 + 20 = 105$, then $105 + 5 = 110$ utils.

3.3 Marginal utilities: 20, then $36 - 20 = 16$, $48 - 36 = 12$, $54 - 48 = 6$, $56 - 54 = 2$ utils.

3.4 As a person consumes more units of a good, the satisfaction added by each extra unit falls. It holds when the units are alike, tastes stay steady and consumption has no long gaps.

3.5 The marginal utility of the 5th book is 0 utils. Its opportunity cost is the other good the money could have bought, which would have given some satisfaction, so the student gives up something for nothing.

3.6 Falling marginal utility means each addition is smaller, not that the total shrinks. The total keeps rising while marginal utility is above zero.

CHAPTER IV · DEMAND AND THE DEMAND CURVE

4.p1 50 notebooks. A lower price (Tk 30 against Tk 50) brings a larger quantity demanded, which is the law of demand.

4.p2 A movement along the curve, because the price of the good itself changed, and the quantity demanded falls from 40 to 30.

4.t1 Buyers may take a high price as a sign of quality or rank, or may expect the price to rise further. Their tastes or expectations then change with the price, so other conditions are not equal. A ring is not a very basic good, so it is not a Giffen good.

4.t2 The fare rise is a movement along the bus demand curve, with a smaller quantity demanded. Cheaper petrol makes car travel cheaper, and cars are a substitute for buses, so demand for bus travel shifts to the left.

4.1 The points are (20, Tk 60), (30, Tk 50), (40, Tk 40), (50, Tk 30), (60, Tk 20), with quantity on the horizontal axis. The line slopes downward: each Tk 10 fall in price adds 10 notebooks, so a lower price goes with a larger quantity demanded.

4.2 The new curve lies 5 notebooks to the left of the old one at every price. A rise of 10 on the old curve gives $Q = 90 - P$, so the quantity at Tk 40 is 50 ($40 + 10$).

4.3 (a) movement; (b) shift, to the right, since pens and notebooks are used together; (c) shift, to the left; (d) movement.

4.4 At Tk 35, $Q = 80 - 35 = 45$ notebooks. For $Q = 55$, $P = 80 - 55 = \text{Tk } 25$.

4.5 Substitute goods: tea and coffee; a price rise for tea raises demand for coffee. Complementary goods: pens and notebooks; a price rise for pens lowers demand for notebooks.

4.6 A Giffen good is a rare, very basic good for which quantity demanded rises when its price rises. It is not the usual case because for almost every good a higher price lowers the quantity demanded.

4.7 Demand rises, so the curve shifts to the right. The number of buyers has changed, and that is a determinant of demand.

CHAPTER V · ELASTICITY OF DEMAND

5.p1 $\% \Delta P = 4 \div 10 \times 100 = +40\%$; $\% \Delta Q = -10 \div 25 \times 100 = -40\%$; $E = 40 \div 40 = 1$, so demand is unit elastic.

5.t1 Quantity demanded falls by about $0.3 \times 10\% = 3\%$. Demand is inelastic, so total revenue rises, because the 10% higher price outweighs the 3% fall in quantity.

5.t2 With time, consumers find substitutes and change habits, so demand becomes more elastic. A large rise may also lose customers who watch the price closely, so the gain in revenue may not last.

5.1 It is the percentage change in quantity demanded divided by the percentage change in price. Percentages remove units and starting levels, so goods of different size can be compared.

5.2 The base is the average of the old and new values, which is the same whichever is called old. The percentage and so the elasticity do not depend on the direction.

5.3 $\% \Delta Q = -20 \div 50 \times 100 = -40\%$; $\% \Delta P = 10 \div 25 \times 100 = +40\%$; $E = 1$. Demand is unit elastic.

5.4 Quantity rises from 40 to 50: $\% \Delta Q = 10 \div 45 = 22.22\%$; price falls from 40 to 30: $\% \Delta P = 10 \div 35 = 28.57\%$; $E = 0.78$, inelastic. Total revenue falls from Tk 1,600 to Tk 1,500, as the test predicts.

5.5 Income elasticity = $10\% \div 20\% = 0.5$.

5.6 Cross elasticity = $-3\% \div 5\% = -0.6$. The negative sign shows that tea and sugar are used together.

5.7 A brand of tea has many close substitutes, namely other brands, so a consumer can switch when its price rises. Tea in general has fewer substitutes.

CHAPTER VI · CONSUMPTION AND CONSUMERS' SURPLUS

6.p1 No. The price of Tk 12 is above the willingness to pay of Tk 9, so the consumer does not buy the pen.

6.t1 Each old buyer keeps Tk 10 more, because the price fell by Tk 10, and $3 \times 10 = \text{Tk } 30$. A fourth customer joins but adds Tk 0. The total rises by Tk 30, from Tk 30 to Tk 60.

6.t2 The toll is only what users pay. Many would have paid more, and the gap between what they would pay and the Tk 20 is their surplus. The benefit is the toll plus this surplus, so it is larger than the tolls collected.

6.1 Consumption is the use of goods and services by people to satisfy their wants. Example: drinking water, or travelling by bus.

6.2 Price is what the consumer must pay, Tk 40 in the notebook case. Willingness to pay is the most the consumer would pay, for instance Tk 60 for the first customer. The gap, Tk 20, is that customer's surplus.

6.3 At Tk 50: $(60 - 50) + (50 - 50) = \text{Tk } 10$. At Tk 20: $40 + 30 + 20 + 10 + 0 = \text{Tk } 100$.

6.4 At Tk 16: $(30 - 16) + (22 - 16) + (16 - 16) = 14 + 6 + 0 = \text{Tk } 20$. At Tk 10: $20 + 12 + 6 + 0 = \text{Tk } 38$. The rise is $38 - 20 = \text{Tk } 18$.

6.5 Willingness to pay falls with each unit, because marginal utility falls. The last unit bought has a willingness to pay close to the price, so the gap between them is close to zero.

CHAPTER VII · INDIFFERENCE CURVE ANALYSIS

7.p1 No. Q has the same notebooks as P but 2 more pens, so Q is better. Two combinations on one curve must please equally.

7.p2 As she gives up pens, those that remain become more precious to her, and as notebooks pile up each extra one is worth less. So she parts with fewer pens for each extra notebook.

7.t1 The old combination now costs $6 \times \text{Tk } 40 + 8 \times \text{Tk } 10 = \text{Tk } 240 + \text{Tk } 80 = \text{Tk } 320$, which is less than Tk 400, so she can still buy it and has Tk 80 left. She can therefore reach a higher curve, so she is better off. The exact new best point depends on her curves.

7.t2 Suppose two curves cross at X. Take A on one curve to the right of X, and B on the other curve directly above A, with the same notebooks and more pens. B is better than A. But A and B each please equally with X, so they would be equal. The two statements clash, so the curves cannot cross.

7.1 $\text{Tk } 200 \div \text{Tk } 40 = 5$ notebooks, or $\text{Tk } 200 \div \text{Tk } 20 = 10$ pens. Two others: 2 notebooks and 6 pens ($80 + 120 = 200$); 3 notebooks and 4 pens ($120 + 80 = 200$).

7.2 Pens given up: $8 - 5 = 3$. Notebooks gained: $8 - 6 = 2$. Rate = $3 \div 2 = 1.5$ pens per notebook. It is smaller than 3, because she now holds fewer pens and more notebooks, so she gives up fewer pens for each notebook.

7.3 Notebooks: $450 \div 50 = 9$. Pens: $450 \div 30 = 15$. Steepness: $50 \div 30 = 1.67$ pens per notebook. Check: $9 \times 50 = \text{Tk } 450$.

7.4 Notebooks: $400 \div 50 = 8$, down from 10. Pens: $400 \div 20 = 20$, unchanged. The line turns inwards around the pen point and becomes steeper: $50 \div 20 = 2.5$ pens per notebook.

7.5 No. The price ratio is 2, but her rate is 4: she would give up 4 pens for one more notebook while the market asks only 2. She gains by buying more notebooks and fewer pens, moving towards 6 notebooks and 8 pens.

7.6 $2 \text{ pens} \times \text{Tk } 20 = \text{Tk } 40$. This equals the price of Tk 40, so her consumers' surplus on that last notebook is $\text{Tk } 40 - \text{Tk } 40 = 0$.

CHAPTER VIII · SUPPLY AND THE SUPPLY CURVE

8.p1 50 notebooks. A higher price (Tk 50 against Tk 30) brings a larger quantity supplied, which is the law of supply.

8.p2 A shift to the right, because a subsidy lowers the seller's expense, so more notebooks are offered at every price. The price of notebooks itself did not change.

8.t1 At a high rate of pay the worker can earn the income needed in fewer hours, and may value rest more. The curve rises, reaches a peak and bends backward.

8.t2 For the day, the curve is a vertical line at 200 fish, since more cannot be had at any price. Over a week more fish can be brought, so the quantity supplied can rise with the price and the curve slopes upward.

8.1 The points are (20, Tk 20), (30, Tk 30), (40, Tk 40), (50, Tk 50), (60, Tk 60), with quantity on the horizontal axis. The line slopes upward: each Tk 10 rise in price adds 10 notebooks, so a higher price goes with a larger quantity supplied.

8.2 The new curve lies 10 notebooks to the left of the old one at every price. An increase of 15 on the old curve gives $Q_s = P + 15$, so the quantity at Tk 40 is 55 ($40 + 15$).

8.3 (a) movement; (b) shift, to the right; (c) shift, to the right; (d) movement.

8.4 At Tk 55, $Q_s = 55$ notebooks. For $Q_s = 25$, $P = \text{Tk } 25$. With $Q_s = P - 10$, at Tk 55 the quantity is $55 - 10 = 45$.

8.5 When the price of a good rises, the quantity supplied rises, if other conditions stay the same. Costs of inputs, technology, taxes and subsidies, or the number of sellers must stay fixed.

8.6 A backward-bending curve, such as hours of work offered at a very high rate of pay; and a fixed supply, such as the fish caught for one market day.

8.7 Supply rises, so the curve shifts to the right. The number of sellers has changed, and that is a determinant of supply.

CHAPTER IX · ELASTICITY OF SUPPLY AND MARKET EQUILIBRIUM

9.p1 Buyers want $80 - 40 = 40$ notebooks, and firms offer 40 notebooks. The two quantities are equal.

9.p2 Both rise. The shift creates excess demand at the old price, the price climbs, and firms offer more along the supply curve.

9.t1 At Tk 30, buyers want 50 notebooks and firms offer 30, so excess demand is 20. Buyers who obtain a notebook gain a lower price; the 20 who cannot buy, and the firms who cannot charge more, lose.

9.t2 She describes time. This month supply is inelastic because the crop cannot grow faster; next year it is more elastic because she can plant more.

9.1 $\Delta Q = 20$, average $Q = 100$, so 0.2. $\Delta P = 10$, average $P = 25$, so 0.4. Elasticity = $0.2 \div 0.4 = 0.5$. Supply is inelastic.

9.2 $\Delta Q = 100$, average $Q = 450$, so 0.222. $\Delta P = 30$, average $P = 115$, so 0.261. Elasticity = $0.222 \div 0.261 = \text{about } 0.85$.

9.3 $\Delta Q = 60$, average $Q = 300$, so 0.2. $\Delta P = 10$, average $P = 55$, so 0.182. Elasticity = $0.2 \div 0.182 = 1.1$. Supply is elastic.

9.4 Before: $60 - \text{price} = 2 \times \text{price}$, so price = 20 and quantity = 40. After: $90 - \text{price} = 2 \times \text{price}$, so price = 30 and quantity = 60.

9.5 $90 - \text{price} = 2 \times \text{price} - 30$, so $120 = 3 \times \text{price}$, $\text{price} = 40$, $\text{quantity} = 50$. Compared with the demand shift only ($\text{price } 30$, $\text{quantity } 60$), the price is higher and the quantity is lower.

CHAPTER X · PRODUCTION FUNCTION AND RETURNS

10.p1 Marginal product of the third helper = $18 - 12 = 6$ loaves. Average product with 3 helpers = $18 \div 3 = 6$ loaves.

10.p2 The 8 means the fourth worker adds 8 to total product ($44 - 36$). Average product with 4 workers is $44 \div 4 = 11$.

10.p3 With some inputs fixed, extra units of one input eventually add less and less to output. The fixed input is the machine.

10.t1 The machine is already used to its limit and the bench is crowded, so the extra worker has no machine time to use. The cause is the fixed input, not the skill of the worker.

10.t2 Each extra worker adds less output for the same wage, so more worker-days, and more pay, are needed for each extra unit of output.

10.1 A production function is a rule showing the largest output that can be made from each combination of inputs. It records the best method known, not what a careless firm might produce.

10.2 In the short run at least one input is fixed; in the long run all inputs can be changed. A machine or an oven is a fixed input.

10.3 Marginal product: 30, 40, 30, 20. Average product: 30, 35, 33.33, 30.

10.4 The third worker's marginal product is 12, equal to the average of 12 for 2 workers, so the average does not change.

10.5 Output rises by less than the inputs (2.4 against 3): decreasing returns to scale.

10.6 Output rises by more than the inputs (2.6 against 2): increasing returns to scale.

10.7 The law changes one input and holds others fixed, and it belongs to the short run; returns to scale changes all inputs together, and it belongs to the long run. The law concerns the extra output; returns to scale compares proportional changes.

CHAPTER XI · COSTS OF PRODUCTION

11.p1 Yes. It is an implicit cost, because the building has a best alternative use worth Tk 5,000 a month, although she pays nothing in money.

11.p2 Total cost rises by $1,560 - 1,320 = \text{Tk } 240$ for 10 books, so marginal cost is $240 \div 10 = \text{Tk } 24$ a book.

11.t1 Average variable cost rises by Tk 2, but average fixed cost falls from Tk 15 to Tk 12, a fall of Tk 3. The net change is $-\text{Tk } 1$. Marginal cost of Tk 34 is also below the old average of Tk 39, which pulls the average down.

11.t2 Her time is an implicit cost, because it could earn a wage elsewhere. She should ask: what is the best use of my time outside the shop, and what is it worth?

11.1 Explicit costs: $8,000 + 3,000 = \text{Tk } 11,000$. Implicit costs: her time, Tk 12,000, plus the rent forgone on the building, Tk 5,000, which is Tk 17,000.

11.2 Total cost = $200 + \text{variable cost}$: Tk 200, 300, 380, 440, 520, 630 and 760 at outputs of 0, 5, 10, 15, 20, 25 and 30.

11.3 Marginal cost per page for each step of five pages: Tk 20, 16, 12, 16, 22 and 26. For example, $(300 - 200) \div 5 = 20$.

11.4 $ATC = TC \div Q$: Tk 60.00, 38.00, 29.33, 26.00, 25.20 and 25.33 at 5 to 30 pages. It is lowest at 25 pages, at Tk 25.20.

11.5 Total cost for one day at 20 pages is Tk 520, which includes the fixed cost. For 6 days, $520 \times 6 = \text{Tk } 3,120$.

11.6 Total cost at 20 pages rises by Tk 60, from 520 to 580. Marginal cost does not change, because fixed cost cancels in the subtraction. Average total cost at 20 pages is $580 \div 20 = \text{Tk } 29$.

11.7 Forgone interest = $0.08 \times 3,000,000 = \text{Tk } 240,000$. Implicit costs = $360,000 + 240,000 = \text{Tk } 600,000$. Result = $500,000 - 600,000 = -\text{Tk } 100,000$.

CHAPTER XII · THE LEAST-COST COMBINATION OF INPUTS

12.p1 $4,800 / 600 = 8$ workers.

12.t1 The firm moves toward machines: fewer workers and more machines. At the new wage the least-cost mix is (3, 6) at Tk 5,100. Keeping (4, 4) would cost Tk 5,200, a rise of $4 \times 300 = \text{Tk } 1,200$. Changing the mix limits the rise to Tk 1,100.

12.t2 No. With 6 machines fixed, the mix for 100 books is (3, 6), which costs Tk 4,200. That is Tk 200 above the least cost of Tk 4,000 at (4, 4). Only workers can be changed in the short run.

12.1 An isoquant joins all mixes of two inputs that give the same output. An isocost line joins all mixes that cost the same total amount. Output can be counted in units, but the satisfaction shown by an indifference curve cannot be counted.

12.2 All workers: $4,800 / 600 = 8$ workers. All machines: $4,800 / 400 = 12$ machines. The slope is the price ratio $600 / 400 = 1.5$.

12.3 Costs: (1, 8) = $30 + 80 = \text{Tk } 110$; (2, 4) = $60 + 40 = \text{Tk } 100$; (4, 2) = $120 + 20 = \text{Tk } 140$; (8, 1) = $240 + 10 = \text{Tk } 250$. The least-cost combination is (2, 4) at Tk 100.

12.4 Costs: (2, 10) = $1,200 + 3,000 = \text{Tk } 4,200$; (3, 6) = $1,800 + 1,800 = \text{Tk } 3,600$; (4, 4) = $2,400 + 1,200 = \text{Tk } 3,600$; (5, 3) = $3,000 + 900 = \text{Tk } 3,900$; (6, 2) = $3,600 + 600 = \text{Tk } 4,200$. Two mixes tie at Tk 3,600. The MRTS between them is 2, which equals the new price ratio $600 / 300 = 2$.

12.5 At (4, 4): $2,400 / 4,000 = 60\%$. At (3, 6) with the wage at Tk 900: $2,700 / 5,100 = 52.9\%$. The share falls because the firm replaces workers with machines.

12.6 Workers: $30 / 600 = 0.05$ books per taka. Machines: $10 / 400 = 0.025$ books per taka. Workers add more per taka, so the firm should hire more workers and rent fewer machines.

CHAPTER XIII · MARKET STRUCTURE AND PERFECT COMPETITION

13.p1 Its sales fall to nothing. Buyers can obtain an identical good at Tk 40 from other firms and know that price, so no one pays Tk 42.

13.t1 Other firms see the profit and, since entry is free, they begin to sell the good. Industry supply shifts to the right, the price falls, and the fall stops when no profit above normal remains.

13.t2 The potato seller faces very many sellers of an identical good, and customers can easily move to the next stall. The pharmacist is the only

seller for ten kilometres, so customers have little choice and entry of rivals is unlikely.

13.1 Extra revenue per step is $5 \times 10 = 50$. Extra costs are 20, 35, 45, 60, so extra profits are +30, +15, +5, -10. The best output is 15 units, and the steps made add $30 + 15 + 5 = \text{Tk } 50$.

13.2 Extra revenue per step is 500; extra costs are 300, 220, 200, 240, 340, 480, so every step adds profit and the best output is 60 units. Total revenue is $60 \times 50 = 3,000$, total cost is $1,900 + 480 = 2,380$, and profit is Tk 620.

13.3 Each step earns extra revenue of $100 \times 20 = 2,000$ against extra costs of 1,800, 1,400, 1,600, 2,200, 3,000. Extra profits are +200, +600, +400, -200, -1,000. The best output is 300 kilograms, and the steps made add Tk 1,200.

13.4 New demand: $100 - \text{price} = \text{price}$, so the price is Tk 50. The workshop's best output is 60 units and its profit is Tk 620, against Tk 100 at Tk 40. Profit above normal attracts new firms, so industry supply shifts right and the price falls in the long run.

13.5 Monopolistic competition: many sellers, different versions of the good (each tailor has a style) and easy entry.

CHAPTER XIV · MONOPOLY AND IMPERFECT COMPETITION

14.p1 To sell 30 units she must cut the price on all units. She loses Tk 10 on each of the first 20 units (Tk 200) and gains 10 new units at Tk 50 (Tk 500). The net gain is Tk 300, or Tk 30 a unit, which is below Tk 50.

14.p2 Each firm fears that rivals would copy a price cut but not a price rise, so a change brings it little gain either way. It leaves the price alone.

14.t1 No. One large firm may serve everyone at a lower cost for each unit than many small firms, and a market of differentiated goods gives variety that identical goods cannot. Price is only one measure of how well a market serves consumers.

14.t2 At 50 units marginal revenue is -10, so the extra units lower revenue while adding cost. Output of 40 units gives revenue Tk 1,600 against Tk 1,500 at 50, and cost falls from Tk 1,900 to Tk 1,560, so profit rises from -Tk 400 to Tk 40.

14.1 $(1,600 - 1,500) \div (40 - 30) = 100 \div 10 = \text{Tk } 10$ per unit.

14.2 Marginal revenue per unit: 50, 30, 10, -10. Take the first two steps (50 over 20, 30 over 22); stop at the third (10 under 25). Output 20 units, price $60 - 20 = \text{Tk } 40$. Total cost = $100 + 200 + 220 = \text{Tk } 520$. Profit = $800 - 520 = \text{Tk } 280$.

14.3 Tk 300 for the whole step of 10 units is $\text{Tk } 300 \div 10 = \text{Tk } 30$ per unit, so the figures agree. Compare the per-unit figure, Tk 30, with the marginal cost of Tk 20 per unit; marginal revenue is higher, so the step is worth taking.

14.4 At 40 units the price is Tk 40 and the marginal cost is Tk 24, so output rises; at 50 units the price is Tk 30 and the marginal cost is Tk 34, so it stops. Competitive output is 40 units at Tk 40, against 30 units at Tk 50 for the monopoly: ten more units at Tk 10 less.

14.5 (a) Monopolistic competition. (b) Oligopoly. (c) Monopoly.

14.6 Marginal revenue = $(900 - 800) \div 10 = \text{Tk } 10$ per cup. This is below the marginal cost of Tk 12, so the stall should not sell the extra 10 cups. Yes, Tk 10 is below the price of Tk 18.

CHAPTER XV · RENT AND WAGES

15.p1 The 60-bag land pays no rent. The 70-bag land pays $70 - 60 = 10$ bags, and the 90-bag land pays $90 - 60 = 30$ bags.

15.t1 Demand: a surgeon's work brings in or saves a great deal, so the marginal revenue product is high. Supply: few people can qualify because training is long and costly. High demand and small supply give a high wage.

15.t2 The farm's yield net of transport falls less, or its sale price rises, so its advantage over the marginal land grows and its rent rises. Demand for its use rises while its supply is fixed.

15.1 Rent is the payment to the owner of land for its use. Nobody can make more land, so the total supply does not respond to payment.

15.2 Economic rent = $80,000 - 50,000 = \text{Tk } 30,000$.

15.3 Marginal land: 0. The 100 kg grade: $100 - 80 = 20$ kg. The 120 kg grade: $120 - 80 = 40$ kg.

15.4 $120 \div 100 = 1.2$, and $24,000 \div 1.2 = \text{Tk } 20,000$.

15.5 The real wage falls. Money rises by 5% but prices rise by 8%, so purchasing power is lower.

15.6 Hire two. The first two add 45,000 and 34,000, above the wage of 30,000. The third adds 29,000, which is below it.

15.7 Rent is the lasting payment to the owner of land, whose supply is fixed. Quasi-rent is a temporary surplus of a man-made asset while its supply is fixed in the short run, and it disappears as new assets are built.

CHAPTER XVI · INTEREST AND PROFIT

16.p1 Interest = $200,000 \times 0.09 = \text{Tk } 18,000$. Total repayment = $200,000 + 18,000 = \text{Tk } 218,000$.

16.p2 Accounting profit = $900,000 - 700,000 = \text{Tk } 200,000$. Economic profit = $200,000 - 150,000 = \text{Tk } 50,000$, which is supernormal profit.

16.t1 The risky venture may end in loss, and owners will not accept that chance unless the average reward is larger. If it earned no more, few people would take the risk. Profit is the reward for facing the chance of loss.

16.t2 The small trader is a riskier borrower, because the chance of non-repayment is higher. A small loan also costs almost as much to administer as a large one, so the cost per taka lent is higher.

16.1 Interest = $50,000 \times 0.10 = \text{Tk } 5,000$. Total repayment = $50,000 + 5,000 = \text{Tk } 55,000$.

16.2 9.5 per cent is 0.095. Interest = $450,000 \times 0.095 = \text{Tk } 42,750$.

16.3 Interest for a year is $600,000 \times 0.11 = \text{Tk } 66,000$. Three months is $3 \div 12$ of a year, so interest = $66,000 \times 3 \div 12 = \text{Tk } 16,500$.

16.4 Forgone interest = $0.06 \times 2,000,000 = \text{Tk } 120,000$. Implicit costs = $300,000 + 120,000 = \text{Tk } 420,000$. Economic profit = $420,000 - 420,000 = \text{Tk } 0$. This is normal profit only.

16.5 Economic profit = $380,000 - 450,000 = -\text{Tk } 70,000$. The owner could earn Tk 70,000 more elsewhere, so the result is below normal profit.

16.6 Accounting profit = $2,000,000 - 1,700,000 = \text{Tk } 300,000$. Forgone interest = $0.08 \times 1,000,000 = \text{Tk } 80,000$. Implicit costs = $200,000 + 80,000 = \text{Tk } 280,000$.

280,000. Economic profit = $300,000 - 280,000 = \text{Tk } 20,000$, which is supernormal profit.

16.7 Risk bearing, limit: it does not pay every risk-taker. Marginal productivity, limit: one owner's contribution is hard to measure. Innovation, limit: many profits need no new idea. Monopoly, limit: it does not explain profit among many rivals.

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